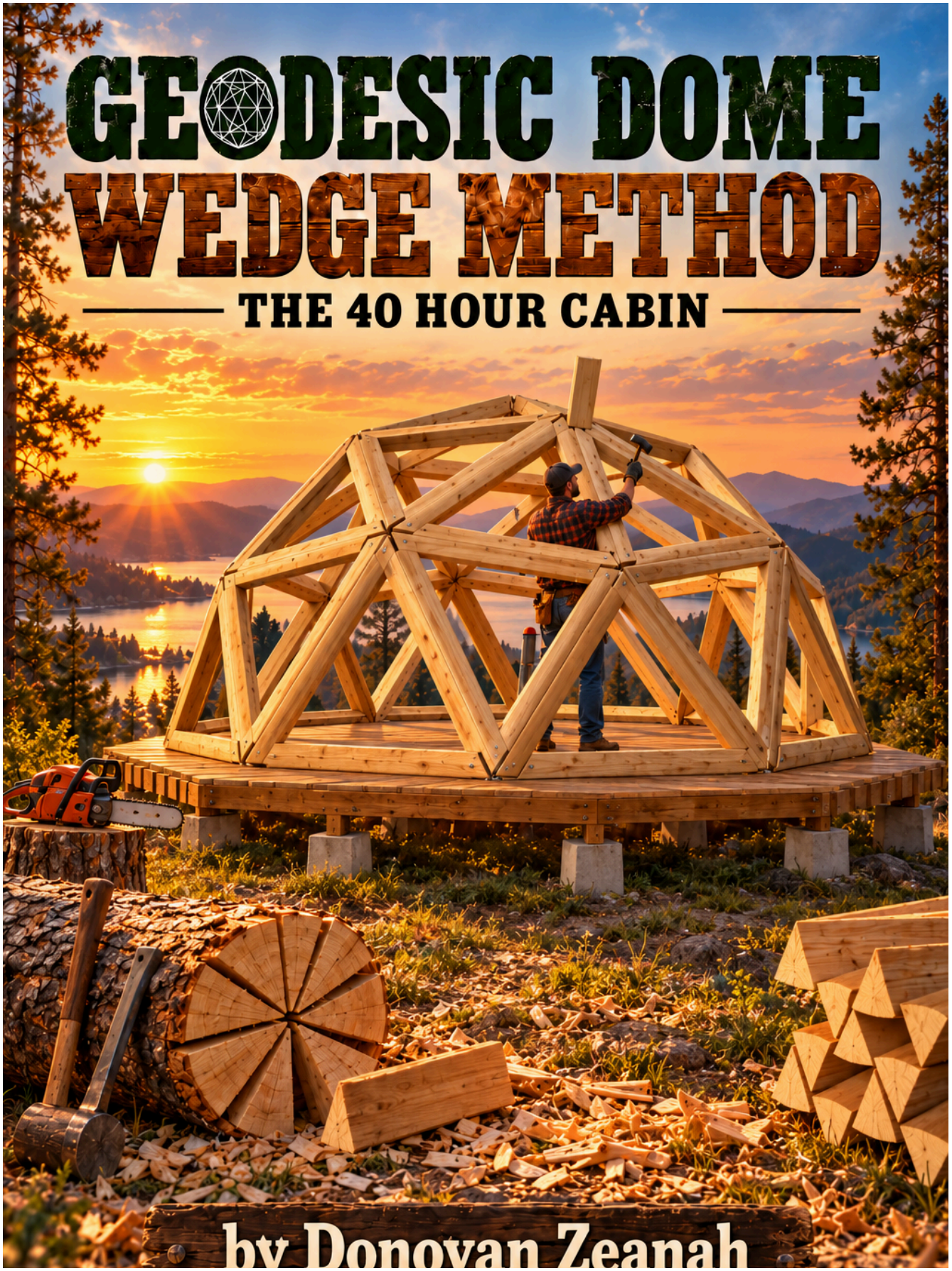
GEODESIC DOME
WEDGE METHOD
THE 40 HOUR CABIN
by Donovan Zeanah

Geodesic Dome Wedge Method

The 40 Hour Cabin: 3 Trees

One small chainsaw, three trees, forty hours or less: a geodesic wedge cabin, winched up between the trees, skinned and floored in any order

Built 2026-10-03. Every figure in this book was computed at build time from the geometry in the DomeSim project.

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Title pages

Title page

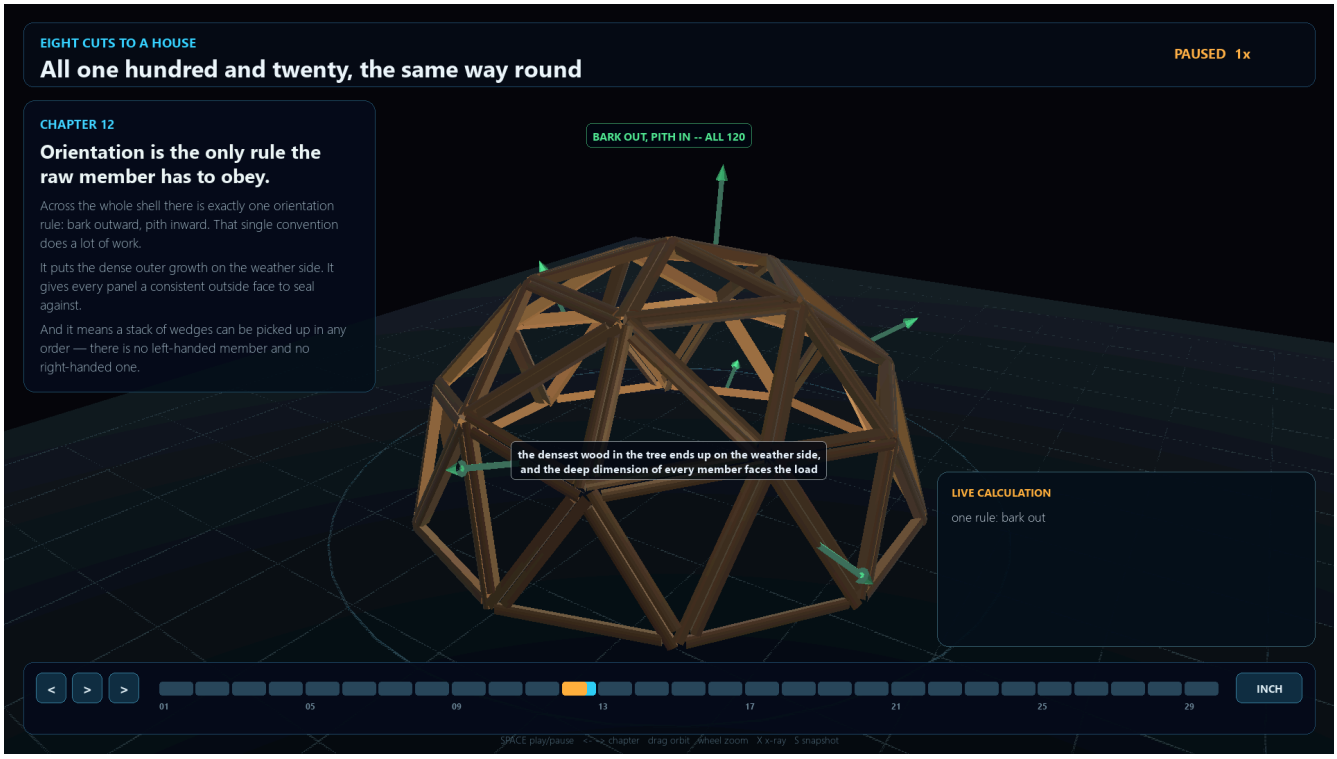
Geodesic Dome Wedge Method

The 40 Hour Cabin: 3 Trees

One small chainsaw, three trees, forty hours or less: a geodesic wedge cabin, winched up between the trees, skinned and floored in any order.

A builder's manual: the geometry, fabrication, construction and variations of a hubless timber geodesic system, documented from one standing prototype — raised between two trees on three cables, because that is how this one was actually built.

By the builder. Built with the tools included with this book, which are named in the back.



The frame, day fourteen.

That is the whole claim in one picture: a frame of 120 members, cut from 3 trees, standing on day fourteen of the fortnight this book teaches — and the fortnight's hands-on total is the title's other claim, counted below.

The forty hours

The title is an arithmetic, not a slogan. The clock is the one the companion wedge book prints, and every rate in it is a measured session figure rather than a process estimate:

At the log. The frame's 120 members, split at the measured 30 wedges per 4-hour session: **16 hours** of splitting.

From wedge to standing. Each member erected at 15 minutes: **30 straight hours** of assembly.

With the error allowance. A 25 per cent allowance on the erection, because a first build always pays its tuition: **37.5 hours** — inside the 40-hour week the title is named after, with 2.5 hours to spare.

The honest reading, stated once and used everywhere after: the forty hours is the *building* — the frame, from split wedges to standing, in forty hours or less. The harvest that produces the wedges is a further 16 hours at the log, and the skin, the floor and the ground are added later, in any order, which is the point of the next page and of the whole book.

The part you are allowed to get wrong

Your skull is not symmetrical either, and it protects the brain just fine. This book builds on that sentence.

A frame's job is to be a *frame*: forty triangles whose lengths set their angles, standing because the shape cannot fall without stretching a stick. It does not have to be pretty, milled, graded or symmetrical — the lumpy skull protects the brain, and the lumpy frame holds the cabin. The parts that must be right are the two that touch the weather and the earth: the **skin**, and the **ground**. Those are the chapters this book treats with the most care, and they are the parts that can be added, replaced or removed in any order — the frame is hot-swappable by design, because every panel, every layer, every floor and every pad is a separate object that bolts on and off.

The build order this book actually used is the order its title page promises: frame first, winched between the trees, then whatever the winter asks for — skin before floor, floor before skin, the ground last of all. The book is written so that any order works.

Copyright, and the engineer's note

Copyright © 2026, the author.

This book documents construction methods, geometric principles, prototype builds and design possibilities. It is not a substitute for site-specific structural engineering, and nothing in it is offered as code-compliant design. Codes, loads, materials, soil conditions and permitting requirements vary by place and by site. A structure intended for permanent occupancy – and anything carrying snow, wind or people you are not prepared to lose – should be reviewed by a qualified engineer and the local authority before it is built. The floating rig of the closing chapters is a design possibility, priced but not rated: the trees, the cables, the mast and the hoist all carry real loads, and none of those ratings is in this book.

Three kinds of claim run through this book, and each is labelled so they cannot be mistaken for one another:

Known geometry. A fact of the solved frame, true at every size: a 2V dome of this configuration has this many members, at these ratios. These are calculations, and the book shows the calculation behind each one.

Tested construction. Something that happened on the standing prototype this book is written around: the joint that was used, the fortnight that was clocked, the winch raise that put the cabin between the trees, the mistake that was made and corrected. These are reports, not promises.

Design possibilities. Things that are priced or drawn but not yet built: the rotating pad, the iris, the network. Each is marked as such where it appears, and the closing chapter restates every claim in the book as the condition it rests on.

A method whose three kinds of claim are kept apart is a method a reader can check. That separation is the point of this book, and the rest of it is the working out.

What this book is

3 trees, one chainsaw off a hardware-store shelf, one hundred and twenty structural members, and a geodesic dome standing in fourteen days — 37.5 hours of hands-on work, inside the forty-hour week the title is named after.

That is the whole claim, and everything in this book exists to either explain it, prove it, or teach you to do it.

It is a real build

The dome in the photographs was not rendered, proposed, or costed on a spreadsheet. It was cut, and the process of cutting it produced a number of surprises, most of which are in here. Several chapters exist specifically to report a figure that does not flatter the method — Chapter 13 revises the recovery claim downward, Chapter 46 halves an hourly rate this project publicised, and the errata pages list everything else that has had to be taken back. A book about a new building method that contains no corrections is a brochure.

It is three books at once

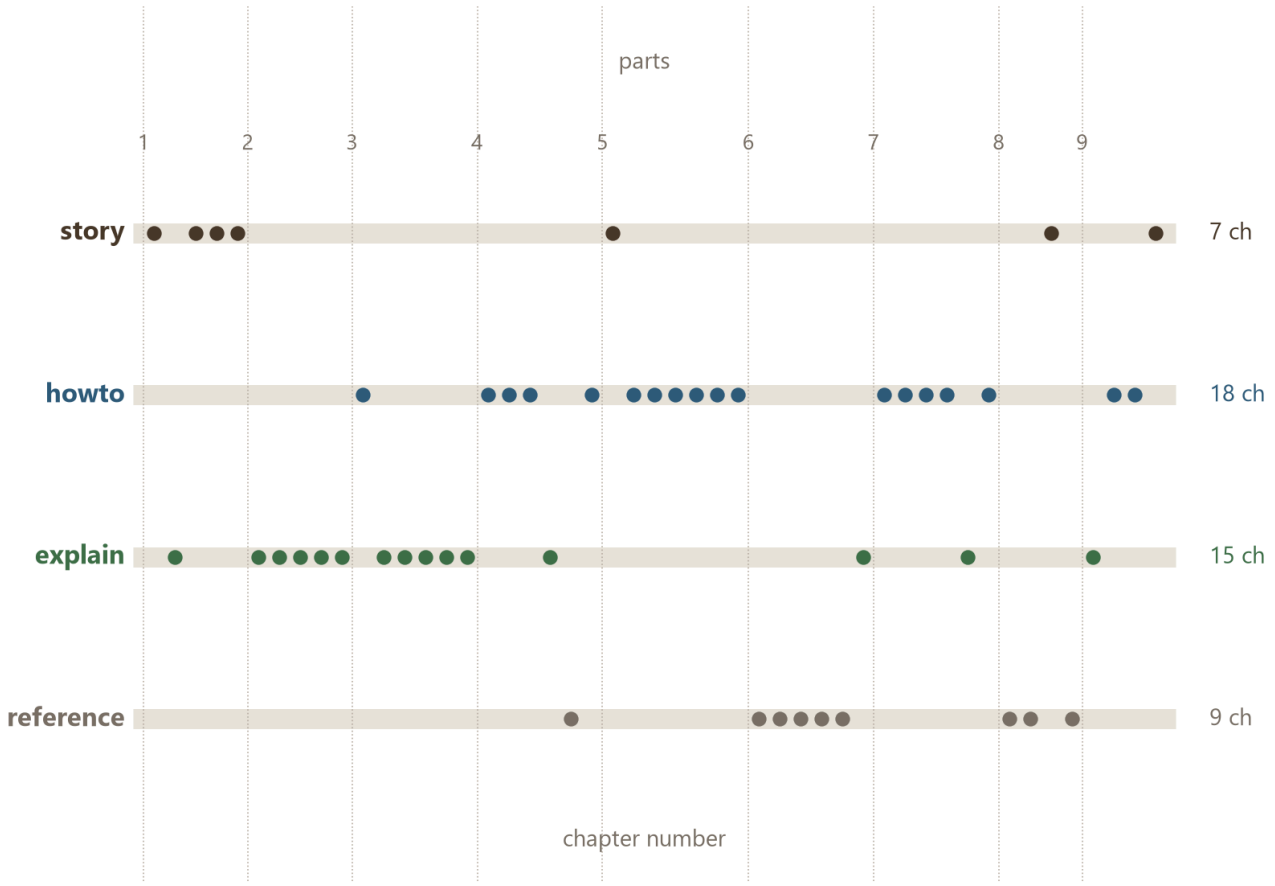
The reader I had in mind wanted three different things, and they read at three different speeds, so rather than blend them into a mush each chapter declares which one it is.

The story is what happened, in order, to somebody with a saw: the frankendome that came first, the afternoon the wedge was found, and the week lost to a problem that turned out not to exist. Seven chapters, and you can read them straight through as a narrative.

The manual is do this, then this. Eighteen chapters, and if you follow only those you get a dome. It is a complete set of instructions on its own.

The explanation is why any of it works — why a triangulated frame will accept a stick no mill would sell, why splitting a round log is free and squaring it is not, and why this method does not transfer to a rectangular house. Fifteen chapters, all skippable, all there because a method you do not understand is a method you cannot adapt.

The rest is reference: the tables you come back to at the saw rather than read in an armchair.



Three ways through this book.

Every number here is computed

This is the unusual part, and it is worth a paragraph.

Not one figure in this book was typed into a sentence. Every dimension, count, angle, percentage and price is a token in the manuscript that gets filled in, at the moment the book is made, by the same software that draws the three-dimensional models. When it says the dome is 21.6 feet across, that number came out of a solver about a second before it reached this page.

The reason is not showing off. It is that a number typed into a paragraph is correct on the day it is typed and silently wrong from the first time anybody changes anything — and there is no way to find those. Nothing fails. The book just quietly becomes untrue in places nobody remembers.

So the arithmetic in this book is a test suite. It runs, it checks itself against the geometry, and where two ways of computing something disagree the book prints both and says which is which. Chapter 23 is nothing but a proof that the

two methods in Part IV are the same calculation, and it prints the residual — 0.0e+00 inches — rather than asking you to take it on trust.

Every tool that produced these numbers is named in the back, and every one of them is included with this book. You can change the tree and rebuild the whole thing around your own.

What it is not

It is not an engineer's stamp. Nothing here has been checked by a licensed structural engineer, and where a figure depends on species, grade or moisture this book says so and tells you who to ask instead of guessing.

It is not a code compliance document. Your jurisdiction has opinions about round buildings and they are not in here.

And it is not a promise that you will do this in a fortnight. I did, on the second attempt, having already made every mistake the errata pages record. The forty hours is the honest claim — the frame, split wedges to standing — and the calendar around it is yours: a fortnight of part days, or a long weekend of full ones, with the skin, the floor and the ground added later, in any order.

PART 1

How to Build One

“Your skull is not symmetrical either, and it protects the brain just fine. The frame is the part you are allowed to get wrong. The skin and the ground are not.”

The method, in the order it is actually done: what a dome is, why the shape carries load, how three trees become forty panels, and the forty hours that raise those panels into a cabin winched between the trees -- with the skin, the floor and the ground added later, in any order.

CHAPTER 1 · STORY

The Tree That Was Already Down

A windfall pine, a small saw, and a question nobody had asked out loud

The Tree That Was Already Down

This book starts with one specific tree, because every number after this page descends from it, and because the story does not make sense if the tree stays abstract.

A pine came down in a storm on land I could walk to. Not a tree anybody planted for the purpose. Not a timber sale. A windfall: sixty or so feet of pine that had stood in a line of others, gone over in one night, root plate up, crown in the brush, and the trunk lying where it fell with the ground still pressed into its bark.

The useful part of a windfall like that is the straight part between the root flare and the first big limb. On this tree that ran 48 feet: 12 inches across at the butt, 8 at the top. The flare below it and the crown above it were firewood. In the middle was a cylinder of pine the length of a shipping container, and there was no way to use it.

That sentence is the whole book in miniature. The wood was not missing, rotten, or in some faraway forest. It was on the ground, within carrying distance of where the building now stands. What was missing was every machine, service, and middleman that stands between a fallen trunk and a usable stick of wood — and it is those, not the tree, that decide what a piece of wood is worth. Chapter 2 walks that chain one step at a time. This chapter is about the tree at the near end of it, before anybody got paid.

Sixty feet of pine, and no way to use it

Here is the problem stated plainly, because most of the ways it gets solved are ways of moving the problem somewhere more expensive.

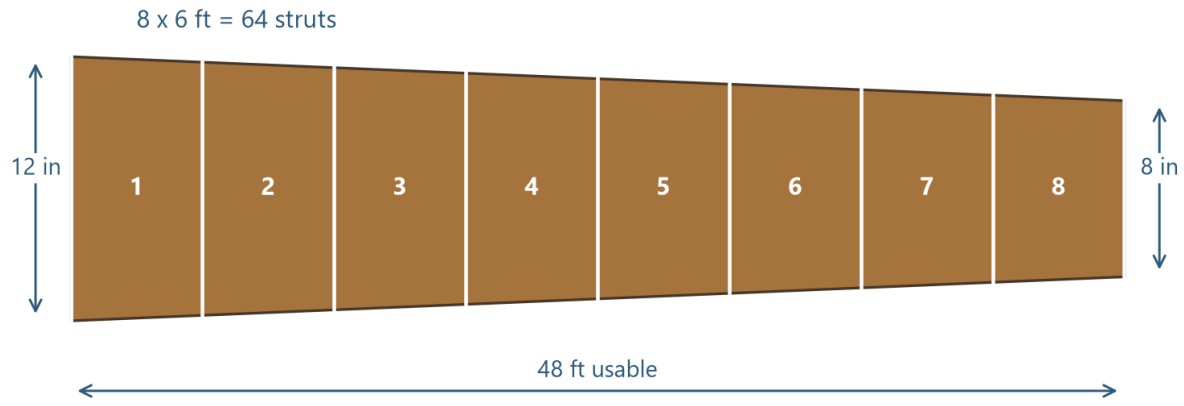
A standing tree is worth almost nothing until somebody converts it. A windfall is worth even less: it cannot wait for a forester, it will not be part of any sale, and it will rot where it lies. Convert it the normal way and the trunk has to travel — dragged to a landing, loaded, trucked to a mill, and then hauled again as boards to wherever the building is. Every one of those journeys is paid by the mile and the hour, and a mill will not touch a two-tree job without a setup charge that makes the exercise pointless. The honest summary of the conventional route is that the tree was worth less than the diesel to move it, which is why the tree stayed where it fell.

The saw that changed that was a 38.2-cc chainsaw with a 16-inch bar that cost 199 dollars on a shelf. The whole shelf price of the machine that eventually turned this tree into a building was less than what hauling the trunk to somebody else's mill would have cost before a single board was cut — not an exaggeration, a comparison, and it is the comparison the method lives on. The industrial route charges for handling, and handling is most of the route. A saw you already own, on the site where the wood already lies, removes the route and charges only fuel, a chain, and time.

But on the day I am describing, none of that was obvious. What was obvious was a 48-foot trunk that no mill wanted, cut from a tree no buyer would bother with, on ground nobody was being paid to stand on. The saw sat on its shelf because a saw alone does not make the trunk usable. A saw makes cuts. What I did not yet know was which cut made the tree stop being a problem and start being a frame.

Measured properly, the trunk held 318 board feet of solid wood. That figure is not a boast; it is what the tape said, and the way the book counts it is explained in full in Chapter 13. The same trunk, sawn into two-by-fours the way a mill would, keeps about 45 per cent of itself as saleable lumber. Split the way this book will split it, it keeps 87.4 per cent — 1.93 times as much — without a second machine ever touching it. All of that is later. Here it only matters as the reason the question in the next paragraph was worth asking, because the difference between 45 and 87.4 per cent of a 318-board-foot trunk is the difference between a load of firewood and a building.

The book's tree, measured



48 feet of usable trunk, 12 inches at the butt and 8 at the top, bucked into 8 sections of 6 feet. Every number in this book descends from these three measurements.

The book's tree: 12 inches at the butt, 8 at the top, 48 feet of usable trunk.

The tree, measured

The drawing matters more than it looks. Every chapter that follows — the recovery arithmetic, the two sizing methods, the cut list, the fortnight — takes its numbers from this same trunk, and if the book ever disagrees with itself about the tree,

the disagreement is a bug, not an editorial position. So the tree gets its own page, drawn to scale, with the three measurements the whole method hangs on.

The butt. 12 inches. The widest point that still counts as trunk; below it is the flare, which splits badly and gets bucked off.

The top. 8 inches, where the straight stem runs out at the first limb. Above it is crown, which is firewood.

The usable length. 48 feet between the two. A trunk tapers from 12 to 8 inches along that run, and everything about how the log gets cut is decided by that taper: buck it into 6-foot sections and each section is close enough to a cylinder that one jig fits them all.

Those are the only three numbers a tree-first build needs from the tree. Not species, not grade, not a timber buyer's walkaround. Butt, top, length. Chapter 22 turns exactly these three into a dome diameter, and the round trip in Chapter 23 proves it is the same calculation a designer runs from the other direction. The drawing here is not decoration. It is the input.

What I actually wanted

It is worth saying what the tree was *for*, because the method is only interesting against the thing it was asked to build.

I wanted a round building, big enough to live in, built alone.

Round, not square: a dome, 21.6 feet across with 365 square feet of floor when the arithmetic was done — a serious room rather than a shed, and the largest size these trees would fill. Chapter 22 shows that 3 trees are what it takes, and that the frame only actually consumes 1.88 of them; the rest of the pile is spares, blocking, floor stock and firewood, and the book says so rather than hiding it.

Built alone, because that is the constraint that shaped every decision after it. A crew can lift a sixteen-foot stick; one person cannot. So the members are 6 feet, the panels are lifted one at a time, and the whole frame is designed around what a single person can carry, set, and fasten without a second pair of hands. The book never pretends this is the fastest way to build. It is the way that works when the only labour budget is your own, and the fortnight in the middle of this part is what that constraint actually costs.

And from what was on the ground, because the economics only work that way. Buy the timber and the frame costs what timber costs; Chapter 46 does that arithmetic both ways and prints the number that does not help the argument. Here it is enough to say the goal was never to build cheaply in money alone. It was to build with the tree that was already down, one saw, and two weeks — and to find out what that made possible that the lumberyard price could not.

That is the book. The rest of it is the tree answering.

CHAPTER 2 · EXPLAIN

Fifteen Middlemen

Why a two-by-four costs what it costs, and how much of that is wood

Fifteen Middlemen

Between a standing tree and a board on a rack there are, depending on how you slice the chain, something like fifteen separate businesses, machines, and handoffs — and every one of them gets paid. The board does not cost what the tree costs. The board costs what the tree costs *plus fifteen wages, fifteen fuel bills, fifteen margins and fifteen sets of handling*, and almost none of that is wood.

This chapter walks the chain, because the whole wedge method is a way of skipping it, and you cannot skip something intelligently without first knowing what it is.

The claim to hold in your head while reading it: a two-by-four's price is mostly distance and patience. The tree spent a lifetime standing still; the board has to be moved, processed, dried, graded, and waited on, and each of those is a service somebody sells. The method in this book removes fourteen of the fifteen steps — the diagram on the next page names them — and keeps exactly one. That is not a metaphor. It is the count, and the rest of the book is the consequence of it.

From stump to rack

Walk it once, end to end, and count the hands.

Fell. Somebody drops the tree. Trained, insured, standing in the weather.

Skid. The trunk is dragged to a landing where a truck can reach it.

Haul. A log truck moves it to the mill, paid by the mile and the hour, and the round trip is empty one way.

Scale. Somebody measures and grades the log, because the mill buys by volume and quality, and the measurement is a negotiation.

Saw. The mill breaks it down: slabs off the round, boards out of the cants. This is the step people picture, and it is one step of 14.

Edge. Every board gets its wany edges ripped off square.

Trim. Ends squared to length.

Dry. The boards are stickered and air-dried, or kilned — weeks to months of time, and the kiln burns fuel.

Plane. The rough board is dressed to its finished size. A nominal two-by- four has now become a dressed one-and-a-half by three-and-a-half, and the missing wood is not an accident; it is this step.

Grade. Somebody looks at every board and assigns it a grade, because graded lumber can be sold by phone and ungraded cannot.

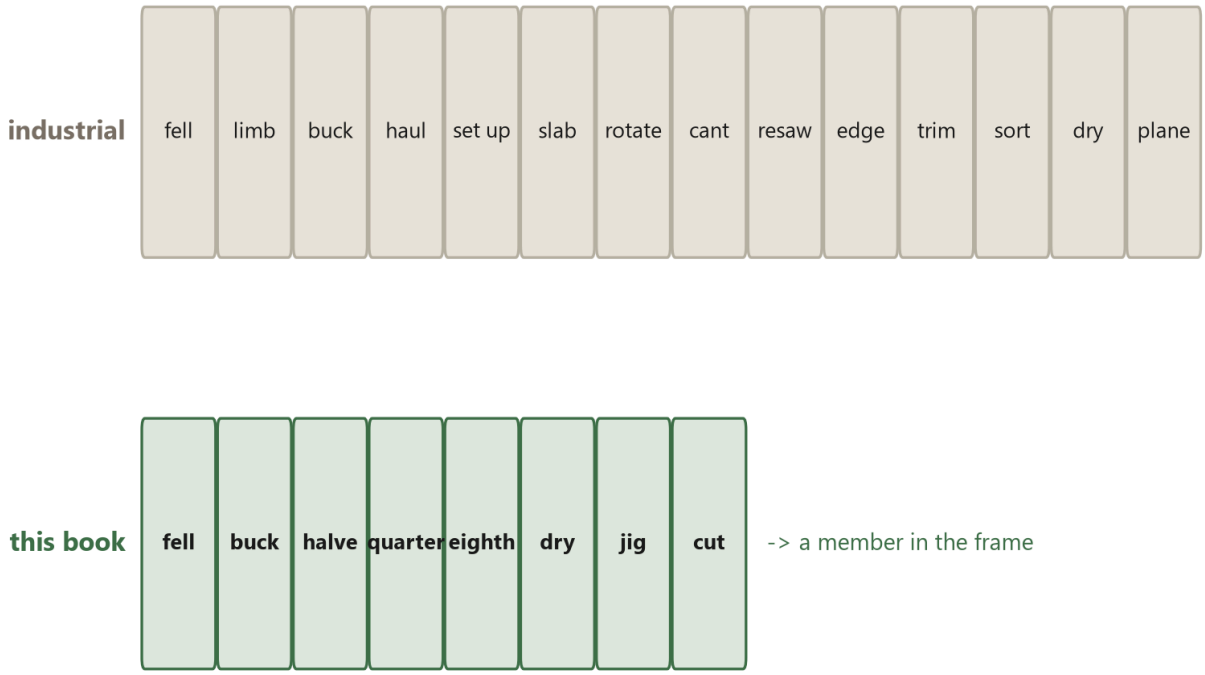
Bundle, ship, stock. It is unitised, moved again, and shelved in a yard that rents its ground, its racks and its fork trucks.

Sell. A desk takes an order and a margin.

That is the honest list — 14 operations between the tree and the graded board, and every one of them has fuel, wages, rent and a margin in it. None of them is a scam. Each one exists because the customer for a board is a contractor forty miles from the forest who needs a straight, dry, graded, predictable rectangle *tomorrow*, and the chain delivers exactly that. The chain is good at its job. Its job is just mostly not wood.

Now count the method this book uses: fell, buck, split, jig, raise — 8 operations from standing tree to member standing in a frame. The middle of that list — skid, haul, scale, saw, edge, trim, dry, plane, grade, bundle, ship, stock — is gone, not because it was unfair but because it was unnecessary. The wedge does not need the tree to become a rectangle, so it does not need the machinery whose entire purpose is producing rectangles.

Two routes from a standing tree to a structural member



14 operations against 8. The wedge route does not do any of it better -- it skips it, and it can only skip it because the dome will accept the shape that falls out of a split log. The two lists do not stop in the same place, and that is stated rather than hidden: the mill route ends at a graded board, which still has to be bought, carried and jointed. The wedge route already includes its joint cuts.

Every step between a standing tree and a board on a rack.

The chain, and where the wood goes

The drawing tells the story the walk could not, because the story is not only who gets paid. It is also what disappears.

Every processing step removes material, and the industrial chain removes it at both ends and in the middle. The slabs and edgings off the round log. The planer shavings off the dressed board. The trim cuts. The degrade — boards that crack, warp or stain while drying and get down-graded or scrapped. When the same trunk this book uses is sawn into two-by-fours, roughly 45 per cent of the solid wood survives as saleable lumber. The other 87.4 per cent of a good log — minus the kerf — survives when it is split into wedges instead. Chapter 13 does that arithmetic in full, including the parts that are less flattering than the headline. Here it belongs in the chain picture for a simpler reason: the waste is not an accident of carelessness. It is the price of the rectangle, paid in wood, at a chain of stations that each exists to pay it.

Look at the diagram with that in mind. The industrial route is long because each station exists to solve a problem the previous station created. The sawmill exists to cut rectangles, so the edger exists to clean their edges, so the planer exists to smooth them, so the kiln exists to dry them flat, so the grader exists to sort what all of that did to the wood. The wedge

route is short because it never creates the rectangle problem. There is nothing to edge, nothing to plane flat, and a green wedge that dries is allowed to move a little, because the frame was designed for a stick that is already the shape the tree made.

The price of a 2x4, taken apart

The shelf price this book uses is 8.88 dollars for a sixteen-foot two-by-four — a real sticker price, declared up front in the constants table at the front of the book, because it is a price and prices are looked up, not derived.

Now take it apart, and the first thing to notice is how little of it is the tree.

The tree's share of that board, at the stump, is a fraction of the sticker. A sixteen-foot two-by-four is nominally 8.00 square inches by sixteen feet — a little under eleven board feet of wood. At a stumpage price of a few cents a board foot standing, the wood in that board left the forest worth pocket change, and arrived on the rack worth 8.88 dollars. Everything in between — the fell, the haul, the sawing, the drying, the planing, the grading, the shipping, the yard, the desk — is what the price actually buys. The wood is the smallest line on the receipt.

That is not a complaint about the lumber industry. It is a measurement of what the industry is: a logistics and processing business that happens to have trees as its raw material. And it is the exact reason the wedge method works economically. When you stand next to a tree that is already down, with a saw in your hand, you are not competing with the industry's skill. You are simply not buying its product. The 14 steps are replaced by 8, and the price of the building material collapses to fuel, chain oil, and your own hours.

Chapter 46 does that arithmetic both ways — what the frame costs bought, against what it costs cut — and prints the hourly rate it implies, including the version that flatters the method least. This chapter only needed to establish why the two numbers are so far apart. They are far apart because one of them is a price for wood and the other is a price for fourteen services, and the services are what the saw on your shelf makes unnecessary.

Nominal, dressed, and the missing third

A two-by-four is not two inches by four inches. It is sold as if it were, and the difference matters enough that the book keeps both numbers on the page wherever wood is counted.

The nominal board — the name on the ticket — is two by four: 8.00 square inches of section.

The dressed board — the thing you actually carry out of the yard — is one and a half by three and a half: 5.25 square inches. The rest went to the planer.

That is a thirty-five per cent difference in section between what the lumber is called and what it is, and it is not a trick; it is the planing step from the chain above, made visible. The book always says which board it means when it compares, because comparing a wedge against a nominal two-by-four flatters the wedge by a third, and a book that flatters its own material by accident is not worth reading. There is also a third board the industry does not sell and the book does not pretend to: the rough-sawn two-by-four, planer-skipped, at something between the two numbers. The money chapter (Chapter 46) and the head-to-head in Chapter 7 both state their board explicitly, and where the choice matters they print the comparison against both.

Why the book cares this much: because the wedge's honest number — the figure that decides whether the whole idea is worth anything — changes by that thirty-five per cent depending on which board you compare it against. The book's answer, stated once here and used everywhere after: compare against the dressed board. It is what you would have bought, and it is the smaller, less flattering target. The method is strong enough to be measured against it.

The Frankendome

Forty triangles, no two struts alike, and the failure that taught the method

The Frankendome

Before the dome this book is about, there was another dome, and it matters that it existed, because every decision in the method that followed was a reaction to what that first one got wrong.

It was built from whatever wood was lying around — offcuts, salvaged sticks, mixed sections, boards that had been other things. Forty triangles, joined edge to edge, each strut whatever shape the pile happened to supply. The idea was deliberately forgiving: do not sort, do not select, do not machine. Let the connector absorb the difference between one stick and the next.

The idea worked, which is the worst thing that can happen to an idea that is secretly bad. The frame stood. It stood for six months, outside, through weather, and it taught more about how a dome behaves than any drawing had. And it was awful: lumpy, twisting, no two joints alike, and impossible to reproduce. Every lesson in this book is downstream of that specific combination — *it worked, and it was awful* — because it proved both halves of the method at once. A triangulated frame will stand on almost any consistent-ish sticks, which is the permission. And nothing about a frame built that way repeats, which is the trap.

This chapter is the story of the trap.

Strut-agnostic, and what that cost

The design rule of the first build was *strut-agnostic*: no stick is special, no stick is measured, any piece of wood of roughly the right length can be any member. Assemble the triangles from the top of the dome down, one at a time, and wherever two edges meet, a folded bracket joins them. Eight screws a bracket. Two bolts a seam. Tighten, move on.

The appeal was real. The pile was free. Nothing needed grading. A stick that was two inches wide where the next was three could still be a strut, because the bracket's flaps could be bent to whatever the wood wanted, and the screws found the wood where the wood was. Building that way feels like freedom — every decision made at the moment of assembly, by the piece in your hand — and it took the whole build to learn what that freedom cost.

It cost everything that comes after a build. The dome stood, but:

Nothing repeated. No two members were alike, so no operation could be jigged. Every cut, every hole, every bracket bend was a one-off, made to fit the exact stick in hand. The fortieth panel took as much thinking as the first.

The connector did all the work. Because the sticks agreed about nothing, the bracket had to absorb every difference: width, thickness, angle, wander. That is a lot to ask of a folded piece of metal, in forty places, for the life of the building —

and Chapter 4 is the afternoon I spent learning exactly how much.

It could not be drawn. There was no cut list, because there was no standard member. The frame existed only as itself; you could not hand the build to anyone, and you could not build a second one.

It was never finished. A frame that is improvised joint by joint is a frame you keep adjusting. It stood, but it never stopped being the thing you had to keep an eye on.

The photograph on the next page is that dome. It is in this book because the book's method is best understood as the answer to it, and because it is honest to show the before. A book about a better way to build owes the reader the worse way it is better than.

The frankendome: it stood, and no two members matched.

The frankendome standing

Look at it. Lumpy where a solved dome is smooth. Every member its own section, its own bow, its own idea of straight. The seams pinch and gap where the sticks disagreed. And yet the shape is recognisably a dome, and it is standing, which is the part that is easy to forget when the tidy version of this story gets told: the thing did work. A frame of forty triangles, joined any-old-how, will carry its own weight and a skin. Triangulation is that forgiving — Chapter 6 explains why — and the method in this book inherits that forgiveness as its foundation.

What the photograph cannot show is the labour inside it. Ten days to stand it, not because the work was hard but because none of it repeated. Every bracket was folded to a different angle. Every screw row was laid out fresh. The second dome took as long as the first would have, because there was no first dome there was only a sequence of one-offs.

The wedge method exists because of what this picture makes obvious: the geometry of a dome does not care what shape its sticks are, but the person building it absolutely does, and the difference between a frame you build once and a method you can build with is whether anything in it repeats.

The ten-piece pentagon

One part of the frankendome was genuinely good, and it is worth separating it from the rest, because it was the only piece carried forward.

The base — the ring where the dome meets the ground — was not improvised. It was ten pieces, cut to a plan, joined into a decagon, and it went together in an afternoon and never moved. The contrast with the chaos above it was the whole lesson of the build in miniature: the one part that was drawn before it was cut, the one part where every piece had a name and a number, was the one part that behaved.

That decagon became the footing of the method. The frame this book builds also meets the ground as a ten-sided ring — 10 rim edges, each a member in its own right — and the ring is the one place the book tells you to set out with a string and a check rather than a shrug. Chapter 39 is that ring, given the treatment the frankendome proved it deserved.

The pentagon, too, came forward in spirit. The frankendome's triangles were joined edge to edge in whatever arrangement the assembly order produced; the method's panels are assembled into the same two triangle shapes every dome has — the same count, the same shapes, the same corners. What changed is not the geometry. It is that in the method, every panel of a given shape is built on the same fixture, from sticks of the same section, with the same holes. The frankendome proved the shape. The method made the shape repeatable.

Why I stopped

The frankendome was not abandoned because it fell down. It was abandoned because of a sentence I could not unthink, after watching the tenth improvised joint go in:

A thing you build twice is only worth doing by hand once.

Nothing in that dome repeated, so nothing could be jigged, so nothing could get faster, and nothing could get more accurate, and no second dome could ever be the same as the first. The strut-agnostic idea had removed the need for precision at the cost of removing the possibility of it. Every bracket was a small negotiation between two sticks that had never met and never would again. That is not a build method. It is a sculpture process that happens to produce a room.

The question the wedge method answers is the question the frankendome left on the table: what would the sticks have to look like for one fixture to build every panel? The answer turned out to be in the next chapter — not *which sticks to buy*, but *which way to open the log*. And the moment that answer landed, the frankendome stopped being a failed building and became what it is in this book: the control group. Same dome. Different pile.

Chapter 5 is where the pile changed.

CHAPTER 4 · STORY

The V-Bracket Afternoon

Trying to join shapes that have nothing in common

The V-Bracket Afternoon

Between the frankendome and the wedge there was an afternoon spent trying to make the frankendome's problem go away by making a better bracket. That afternoon matters because it is the step most builders would have stopped at, and because the bracket it produced is a genuinely good answer to the wrong question.

The problem was stated clearly by then: the frame's sticks did not match, so the joint had to. The fix seemed obvious — stop improvising the joints one at a time and design one connector good enough to join any stick to any stick. Do the tolerance in metal, once, instead of in thinking, forty times.

So I sat down and designed the V-bracket: two flat flaps meeting at the angle the panels make, four holes in each flap, the flaps long enough to run half the length of a member so the screws spread along the stick instead of bunching at its end. Fold one from flat stock in a minute on a bench brake, and it will hold any two pieces of wood that arrive at the corner, whatever their sections, whatever their bows.

And it worked. That is the trap, again. A bracket that absorbs every difference works on the first panel and the fortieth. What it costs does not show up until you look at what the bracket has become.

A bracket that does not care

Here is why the V shape is good at its job, because the goodness is real and the method that follows borrows from it.

A V-bracket does not assume a member's width. The flaps are wider than any stick the build will use, so a two-inch board and a three-inch board both land on the same plate, and the screw rows find whatever wood is there. It does not assume a thickness: the bend that forms the V is made to the panel's fold angle at assembly, not stamped to a tolerance at a factory. And it does not assume straightness: the screws are what pull the stick to the plate, and a bowed stick just gets pulled harder.

That is a beautiful property, and it is worth saying what it is in one sentence, because the wedge method is, in a way, this sentence taken to its conclusion: **the connector absorbs the difference, so the wood does not have to be made the same.**

The warning sign was in the same sentence. When the connector absorbs every difference, the connector is now doing all the work of the frame. Every joint is a negotiation, held by screws that are the only thing in the whole structure that know the shape they are enforcing. Eight screws a bracket, two bolts a seam, all of it site-bent, site-drilled, site-tightened — the connector was not a detail. It had become the building.

And a connector that does everything has to be *right* everywhere. The flaps were long, which spread the load, which was honest engineering — and it meant forty-odd brackets, each consuming stock, each needing folding, drilling, fitting, checking. The frankendome had replaced the problem of matching wood with the problem of manufacturing forgiveness, in metal, at forty corners. Same labour, moved.

Four joints that nearly worked

The drawing shows the four joints that happened along the way, because the V-bracket was not arrived at in one step, and each of the earlier ones failed in a way that named a piece of the final answer.

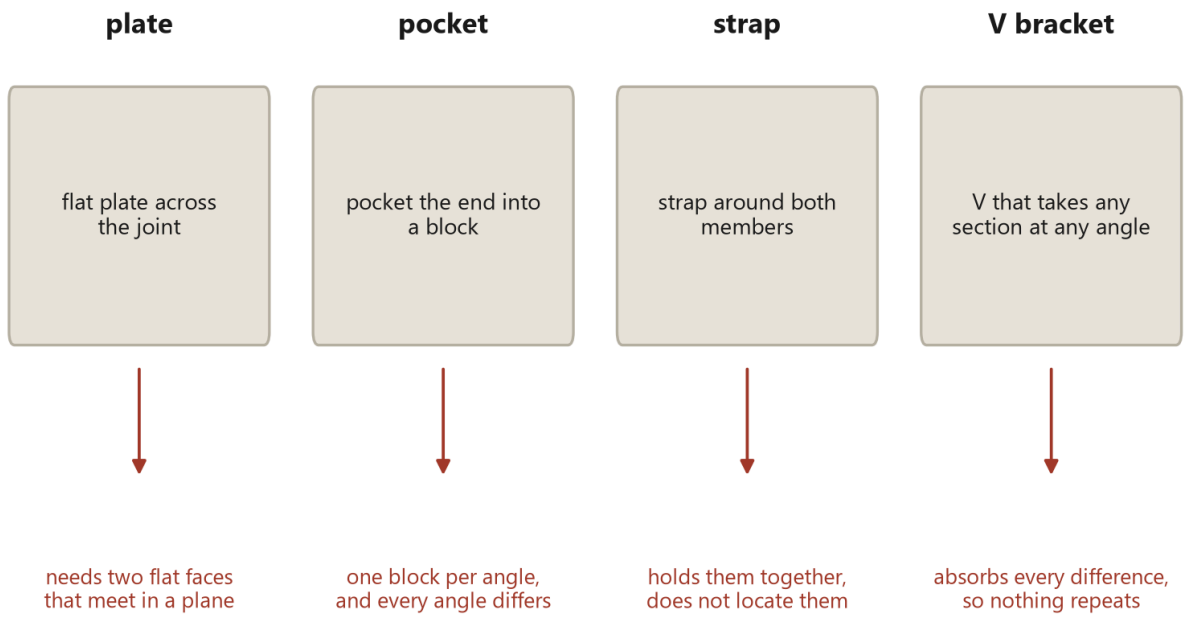
The hub. A flat plate with every member bolting to its own face — the classic geodesic hub. It assumes all members arrive at the same point at the same angle, and they never do when the sticks are salvaged. Every attempt needed shims, washers, longer bolts. The hub is perfect for a kit of identical struts, and useless the moment the struts differ.

The angle plate. A single fold, made to the panel's angle. It joins two sticks in a line but cannot hold the third that closes the triangle, and the third is the one that turns three sticks into structure.

The wrap. A strip of steel wrapped around both members and screwed through. It holds anything, for a while, but it crushes nothing into alignment — the joint stays as crooked as the wood, and every wrap is a one-off that has to be cut to the pair it is joining.

The V-bracket. The survivor, and it failed too — not at holding, but at repeating. It can join anything to anything, which is exactly why no two of them are alike, which is exactly why the frame never gets faster to build.

Four joints that nearly worked



The V bracket was the best of them, and that was the problem: a connector that tolerates any difference removes the pressure to make the parts the same. The wedge went the other way -- make the parts identical and the joint gets to be simple.

Four joints, and the one thing each could not do.

The pattern across all four is the same. Each joint failed not by being weak but by being asked to solve, in metal, a problem the wood kept re-creating. The hub assumed sameness. The plate assumed a line. The wrap assumed nothing and therefore had to be made for everything. The V accepted all of it and paid in labour. None of them asked the question the next chapter does — not *how do I join whatever arrives*, but *what would have to be true for the sticks to arrive the same*.

The question that ended it

The afternoon ended with the bracket in my hand doing its job, and a question that would not go away. The bracket worked by absorbing difference, and it had taken me all afternoon to make a bracket good at that. But every flap length, every screw, every bend was the cost of a difference that did not need to exist.

So: if the bracket no longer had to absorb any difference — if the sticks arrived at the corner already the same section, already at the same angle, already the same length — what would the joint collapse to?

The answer, worked out later, is Chapter 15: a member end landing on a member side, one key, a few screws, nothing folded, nothing measured twice. But the more important answer arrived that same afternoon, and it is the sentence that turned the project around:

Do not build a joint that tolerates the difference. Build sticks that do not have one.

That is the wedge method's founding sentence. The bracket was a good idea pushed past its natural size; it remains in the method only as a small, fixed part where a fold is genuinely the right tool. What it taught was that the wood was the thing to fix, and the wood was the one thing I had not tried changing — because a stick of wood seemed like something you buy, not something you decide. Chapter 5 is the afternoon that stopped being true.

The bracket attempts earned their page in this book for one more reason: they are the roadmap of every wrong turn a builder can make at a dome's corner. Somebody, somewhere, is designing a hub right now. The book's answer is the drawing above, and the sentence under it: make the sticks match, and the joint stops being the hard part.

CHAPTER 5 · STORY

The Cut That Changed It

Split the trunk like a cake, and the stick is already the right shape

The Cut That Changed It

I want to tell this one without hindsight tidying it up, because the tidy version — *I realised the dome would accept any consistent section, therefore I split the log radially* — is not what happened and it teaches nothing. What happened was that I was tired of brackets.

Half, half, half again

The frankendome had taught me that a frame of forty triangles will stand up even if no two of its sticks match. The V-bracket afternoon had taught me that making a connector tolerant enough to join anything to anything means the connector is now doing all the work, forever, in forty places.

So I had stopped asking how to join dissimilar sticks and started asking the opposite question: what would it take for them not to be dissimilar?

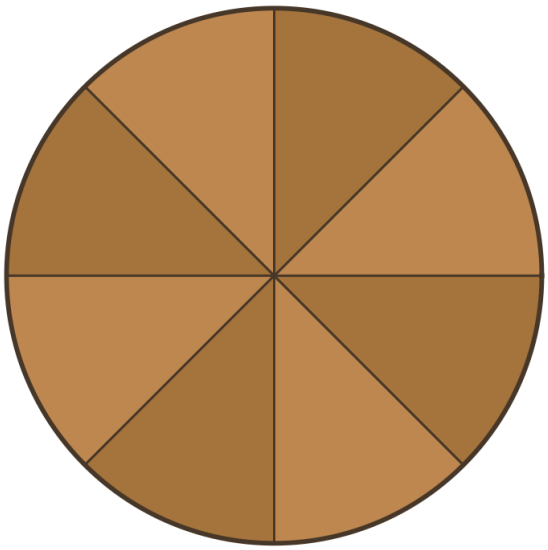
The answer that everybody gives is *mill them*. Square them up, run them through a planer, and now they are all 2x4s. That answer costs a mill, a planer, a shed to keep them in, and about half the tree.

I was sitting on a section of trunk at the time. Not thinking about it. Sitting on it.

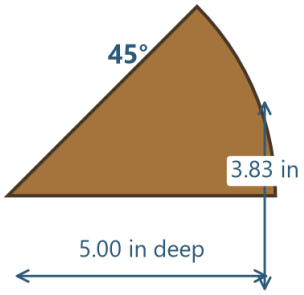
And the thing about a round section of trunk is that it is already rotationally symmetric. Every radius is the same as every other radius. If you split it through the middle you get two identical halves, not because you were careful but because the log was round before you touched it. Split those and you get four. Split those and you get 8.

Three passes. Eight sticks. Every one the same, to whatever tolerance the tree was round to, which is a tolerance nobody had to hold and nobody had to check.

One trunk, eight sticks



8 sectors, 3 saw passes



one member, as it comes off the saw

A 10-inch section split 8 ways gives 8 members of 3.83 by 5.00 inches, 9.8 square inches each. Nothing is squared, edged or planed. The bark face stays curved.

Three cuts. Eight structural members.

I did it that afternoon on a piece I had been about to burn.

What became obvious immediately

Four things, in about a minute, and I want to record that they were obvious because the ones in the next section took a week and I do not want to flatter the story.

One section, everywhere. Every stick in the frame has the same cross-section: 3.83 inches across the bark face, 5.00 inches from pith to bark, 9.82 square inches of wood. So one jig fits all of them. One fixture, one set of stops, one procedure repeated a hundred and twenty times. That is what makes this a fortnight instead of a summer.

Depth where you want it. A sector is deep — it runs the whole way from the pith to the bark — and depth is where bending stiffness lives. The stick is not a compromise between what the tree gave and what a mill wanted. It is the deepest member you can cut from that log, which is a strange thing to get for free.

No machine between the tree and the frame. Fell, buck, split, build. The chainsaw does all of it. There is no second machine, no second setup, no second building to keep the second machine in.

Four ways up. This one I noticed while stacking them. The sector is not symmetric about its own long axis — it has a point and a curved back — so you can rotate it four sensible ways in the wall and get four different things: which face the weather sees, which face the next stick lands on, whether the points of a pair aim together or apart. That is Chapter 14, and it is the most genuinely new thing in the method.

And what took another week

Now the part the tidy version leaves out.

I assumed, because the sticks were identical and the panels were identical, that the *joints* would be identical. Forty triangles, all the same, all meeting each other the same way.

They do not.

A dome is a sphere approximated by flat panels, and the angle at which two panels meet is not the same everywhere on it. It cannot be — the sphere curves differently in different places, and a 2V hemisphere has two classes of edge for exactly that reason. So the seam between two panels folds by one amount here and a different amount there, and a key cut to fit one seam will not fit the other.

I did not want this to be true. I spent the better part of a week trying to make it not be true: fudging the panel outlines, trying a compressible gasket, trying to talk myself into the idea that the difference was small enough to force closed. It is not small enough. Forty panels of *nearly* closing is a dome with a gap in it.

What actually resolved it was measuring instead of arguing. Across all 55 seams there are only **2** fold angles. Not forty. Not a continuum. Two: 18.03 degrees and 22.46.

That was the moment the method became buildable rather than merely elegant. 2 angles means 2 key profiles: cut one of each until it fits, then repeat it 25 and 30 times. The difference between “the seams vary” and “the seams take 2 values” is the difference between a research project and a fortnight, and I lost a week to not knowing which one I was in.

The chart is in Chapter 16, where it belongs.

The bit I got wrong in public

While all this was going on I was saying, out loud and on camera, that nothing in the dome needed a mitre.

That is not true and it was never true. The end of each stick that lands on the side of the next one — the butt end — is a compound cut: a mitre and a bevel together. What is true, and what I should have been saying, is that no end is ever mitred *to another end*. Every joint is an end landing on a flat side, which means each stick is cut once, on one end, before it goes anywhere near the assembly, and the other end is left long and sawn off in place.

That is a much better claim than the one I made, and it is a shame I spent several months making the wrong one. Chapter 15 has the geometry, and the errata pages in this part carry the rest of what this project has had to take back.

What it is worth

The number that convinced me it was not just a neat trick: splitting this way keeps 87.4 per cent of the solid wood in the trunk, against about 45 per cent for the same log sawn into two-by-fours the way a mill would. That is 1.93 times the usable wood, from a cheaper process, with one machine instead of four.

Chapter 13 does that arithmetic properly, including the parts of it that are less flattering than they first appeared.

But sitting on that log, what I had was a much simpler observation, and I still think it is the whole book:

A round thing splits into identical pieces for free. A square thing has to be made square, and you pay for that in wood, in machines, and in time. Domes are the buildings that will accept the free pieces.

Triangles Do Not Care What Shape Your Stick Is

Why a triangulated frame can accept a member no mill would sell

Triangles Do Not Care

This chapter is the permission slip for the whole method, so it goes first in the explanation strand and it is short enough to be unfair to nobody.

Start with the sentence the title page made its argument out of: **your skull is not symmetrical either, and it protects the brain just fine.** The frame is the part of this building you are allowed to get wrong — the lumpy, split, self-labelled frame that the frankendome proved will stand. The parts that must be right are the skin and the ground, and the book treats those with the care the title page promised. This chapter is the *reason* the frame is allowed to be lumpy, stated as geometry rather than as reassurance.

The wedge method proposes to build a building's frame out of sticks that are not rectangular — a fifth of a round log, sawn faces, bark back, a point at one edge. No lumberyard sells that. No building code assumes it. And yet the frame stands, because of one property of triangles that is easy to state and takes the rest of this chapter to spend:

A triangle does not care what shape its sticks are. It only cares how long they are, and that they reach each other.

Everything the mill does — square the edges, flatten the faces, dress the section to a standard — exists because rectangles ask their pieces to be beams: flat boards, loaded across their faces, whose stiffness and strength depend on the exact shape of the section. A triangulated frame asks something else entirely of its pieces. It asks them to be *struts*: pushed or pulled along their own length, ends only, never sideways, never flat. For a strut, what matters is how much wood is in the section and how far the load is from pith to bark — not whether the section is a rectangle.

So the central question of the book is not "is a wedge as good as a 2x4?" in general — Chapter 7 runs that comparison honestly and prints the column the wedge loses. The right question is narrower and much more useful: *inside a triangulated frame, where members are loaded along their length, is a wedge a good member?* This chapter builds the frame that question assumes, so the later chapters can answer it.

Axial, not bending

The difference between a strut and a beam is the difference between the two ways a building can load a piece of wood, and most of conventional framing is the second way.

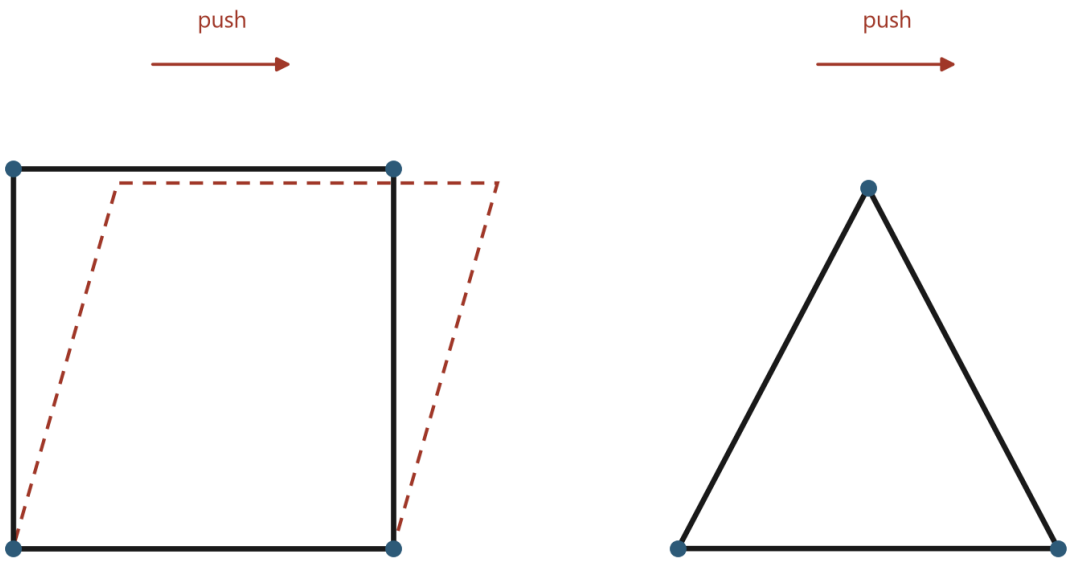
A beam is loaded across its length. A floor joist carries the floor on its narrow face; a rafter carries the roof on its wide one; a stud carries weight down its length but braces against sideways push. In all of them the load is trying to bend the piece, and bending is cruel: the material at the middle of the section does almost nothing, which is why an I-beam has

its steel at the top and bottom and air in between. Bending strength lives in the extreme fibres, far from the section's centre, so the shape of the section — rectangle, and which way round — decides how strong the piece is. This is why a 2x4 on edge is dramatically stiffer than the same 2x4 laid flat, and why the mill bothers to dress boards to a true rectangle at all. A rectangular building is a system for turning wood into beams, and beams need rectangles.

A strut is loaded along its length. Squeeze the ends of a stick toward each other, or pull them apart, and every fibre of the section shares the load equally. Shape stops mattering; the amount of wood matters. A round stick, a square stick, a wedge — for axial load, they differ only in section area (and, for a long thin strut in compression, in the way the section resists buckling, where depth matters more than shape).

Now look at a triangulated frame. Every load that arrives at a joint is carried to the next joint along the members, as push or pull along their lengths. The frame does not bend its members; it moves force through them. That is the entire trick of geodesic structures, and it is why a dome's struts can be slender for their span, and why the frame stands up without a single piece of sheet material keeping it square. A rectangle needs a diagonal, or a sheet of plywood, or it racks over and lies down. A triangle is already its own diagonal. The drawing on the next page is that sentence.

The same load, two frames



A rectangle of four pinned members is a mechanism: it folds. It needs a sheet, a brace or a stiff joint to stand. A triangle of three cannot change shape without one of them changing length, so the joint is only asked to hold the ends together -- which is why the shape of the stick stops mattering.

A rectangle needs a sheet to stand. A triangle does not.

The same load, two frames

Look at the two frames side by side.

On the left, a rectangle. Push on its top corner the way wind pushes on a wall, and the rectangle does what rectangles do: the corners stop being ninety degrees, the frame leans, and unless a sheet of plywood or a diagonal brace is holding it, it goes over. The joints are doing everything the shape cannot. That is why a conventional wall is a frame *plus* sheathing — the studs are beams, and the sheet is the triangle they never built.

On the right, the same corner, triangulated. Push on it. The load cannot change the angles, because the angles are fixed by the lengths of the three sticks, and a stick's length cannot change without stretching or squashing the stick itself. So the push travels down the members as pure compression and pure tension, along the grain, exactly where wood is strongest and exactly where shape does not matter. Nothing bends. Nothing needs a sheet. The frame is its own bracing, at every corner, forever.

The whole method follows from that picture. Because members are loaded axially, the frame will accept a member of any section that has enough area and enough depth — which is precisely what a split log provides. Because nothing bends, the frame does not ask for the flat, dressed, graded faces a beam economy demands. And because the triangle's angles are fixed by its lengths, the frame enforces its own geometry: get the lengths right and the shape has no choice. The frankendome stood on that fact alone, with no two members matching, and it stands on the same fact with every member identical. Chapter 3 was the demonstration. This chapter is the reason.

What the wedge is good at

If the frame wants area and depth, the wedge should be measured by area and depth — and on a modest log, it beats the store-bought board at both.

A dressed two-by-four carries 5.25 square inches of wood, planed down from its nominal size — a rectangle, because it must be a beam. The wedge this book cuts is 3.83 inches across its bark face and 5.00 inches from its point to its bark, for 9.82 square inches of section. The comparison that matters is in Chapter 7, computed properly on an 8-inch log so the wedge gets no advantage from a fat tree. Here, two qualitative points are enough to set the direction:

Depth is where stiffness lives. In bending, stiffness rises with the *cube* of depth; double the depth and a beam gets eight times stiffer. A wedge runs from the pith to the bark — the full radius of the log. For the same width of stick, that is more depth than any board a mill cuts from the same tree, because the mill's rectangle has to fit inside the circle, and the wedge does not.

Area is where axial strength lives. The wedge is a sector: one eighth of the log's circle. Measured as a strut — which is what the frame asks it to be — it carries more wood in its section than the dressed board the same log would have produced, because it never gave up the corners to the edger and the planer. Chapter 13 counts those corners; the count is the difference between 87.4 per cent and 45 per cent of the tree.

The honest note, saved for the full table: a wedge is not a better *beam*. Lay it flat and ask it to be a shelf and the missing rectangle matters. The claim is narrower and it is the only one the book makes — in this frame, loaded this way, the wedge is a better use of the tree than the board, and the shape the tree already had is not a defect the mill forgot to remove. It is the feature the frame was waiting for.

One eighth of a tree

The section, stated plainly, because the next chapters argue with it:

A 6-foot section of this book's tree, split into 8 equal sectors, yields sticks of 9.82 square inches each — 3.83 inches across the bark face, 5.00 inches deep from point to bark.

The dressed two-by-four the same tree would have been sold as carries 5.25 square inches. The nominal one — the name on the ticket, which no longer exists in the wood — carries 8.00.

So the wedge is a bigger section than the board the mill would have cut from the same log, in the one measure that matters for a strut, and it cost a chainsaw's fuel instead of a mill, a kiln, a planer and a yard. That single line — 9.82 against 5.25, from the same tree — is the economic engine of the whole book, and everything after this page is the machinery that turns it into a standing frame.

CHAPTER 7 · EXPLAIN

Not a Worse Two-by-Four

The head-to-head nobody runs honestly, run honestly

Not a Worse Two-by-Four

Everybody asks the same question, and they are right to ask it.

Is a split log actually as good as a board?

The honest answer has three parts, and only two of them are flattering. This chapter gives all three, because a method defended only with its good numbers is a method nobody should trust with a roof over their head.

The rectangle is not the tree's idea

Modern construction has trained us to think of wood as something that naturally arrives in rectangles. A tree goes into a sawmill and a stack of 2x4s comes out, and the form is so familiar that it stops looking like a choice.

It is a choice. The rectangle is not a property of wood. It is the output of a manufacturing process designed around a particular kind of building — one made of vertical studs, horizontal plates, flat walls, right-angle corners and rectangular openings. For that building the rectangle is close to perfect, and the industry that produces it is very good indeed at what it does.

A geodesic dome is almost the opposite of that building. Its structural language is triangles, changing face planes, radial geometry and repeated angular joints. Nothing in it is a right angle. Nothing in it is a flat wall.

So the question reopens, and it is worth asking properly rather than assuming the answer in either direction: *for this building, is the rectangle still the right thing to make out of the tree?*

More wood, from a modest tree

Start with the simplest measure: how much wood is actually in the stick.

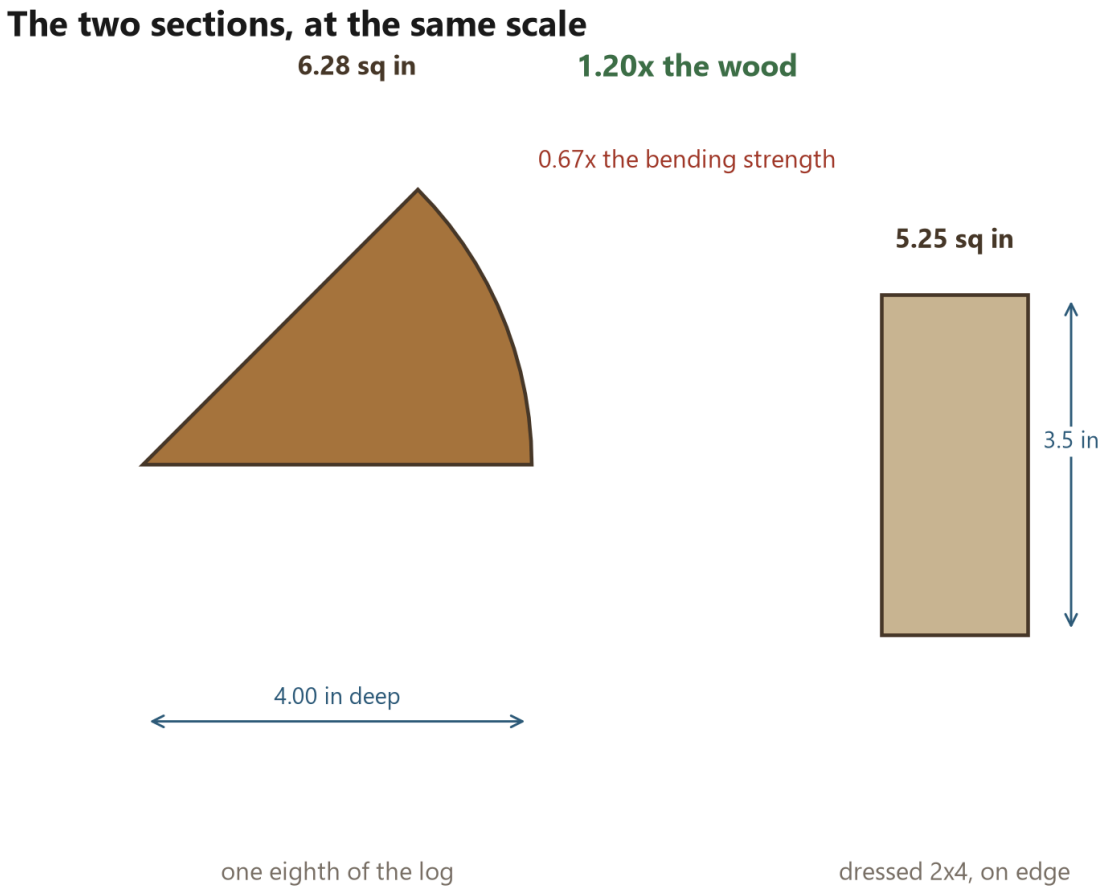
Take an 8-inch log — not a monster, a tree most people could fell and move without help. Split it 8 ways and each sector holds **6.28 square inches** of wood.

A dressed 2x4 — the board actually on the rack, which is 1.5 x 3.5 inches whatever it says on the label — holds **5.25**.

That is **+20 per cent more wood** in the wedge, from a tree you could carry out of a wood lot in sections.

And it is the conservative version of the claim. A larger log widens the gap sharply, because area grows with the square of the radius while a 2x4 stays a 2x4. The book quotes the small log deliberately: a claim is worth more made at its weakest.

The wedge is also deeper. Pith to bark is 4.00 inches, against 1.5 x 3.5 for the board — and depth is where bending resistance comes from.



An eighth of an 8-inch log holds 6.28 square inches of wood against the dressed 2x4's 5.25 -- +20 per cent. It is also deeper: 4.00 inches pith to bark against 3.5. What it is not is stronger in bending, because so much of that wood sits near the pith, where it does the least work.

The two sections, at the same scale.

And now the number that does not help

Here is where an honest chapter parts company with a sales pitch.

Area is not strength. How much a section resists bending depends not only on how much wood is in it but on *where that wood sits* — material far from the middle does most of the work, material near the middle does almost none.

And a wedge puts a great deal of its wood near the pith, in the narrow part of the pie slice, where it contributes very little.

Run the numbers properly and the comparison comes out like this:

	wedge	2x4 on edge
Wood in the section	**+20%**	—
Resistance to bending (stiffness)	**-1%**	—
Bending strength	**-33%**	—

Read that middle row again. The wedge has +20 per cent more wood in it and is **not stiffer**. It comes out within a per cent of a 2x4 stood on edge — effectively identical, having spent twenty per cent more material to get there.

And the bottom row is worse: in bending *strength* the wedge is -33 per cent against the same board.

That is the number this method's advocates do not usually print, and it is printed here, in a table, near the front.

One eighth of an 8" log against a dressed 2x4

	wedge	2x4 on edge	ratio
Cross-section area	6.28 in2	5.25 in2	1.20x
Depth	4.00 in	3.50 in	1.14x
Width	3.06 in	1.50 in	2.04x
Second moment, strong	5.33 in4	5.36 in4	0.99x
Section modulus, strong	2.05 in3	3.06 in3	0.67x
Same board laid flat	**5.33 in4**	**0.98 in4**	**5.41x**

The wedge wins on area and loses on bending strength, and both are printed. Area is what an axial member trades on; section modulus is what a beam trades on. In a triangulated frame the members are mostly axial, which is the argument -- but panels do see wind and snow, and there the last two rows are the ones that matter. Against the same board laid flat, which is how a lot of framing ends up loaded, the wedge is far stiffer.

The comparison in full.

One consolation, before the argument

There is a caveat to the caveat, and it is a fair one.

The comparison above puts the 2x4 **on edge** — the orientation in which a board is strongest, and the way a carefully framed floor joist is installed. A great deal of real framing does not get that. Sheathing battens, blocking, purlins and plenty of cheerfully improvised structures end up loading boards the flat way, and against a 2x4 laid flat the wedge is **5.4 times stiffer**.

So the honest summary is: worse than a well-oriented board, dramatically better than a badly-oriented one.

That is still not the argument.

Which is the right question

The argument is that *the comparison itself is the wrong shape*.

Asking “is this wedge as good a beam as a 2x4?” assumes that being a beam is the job. In a rectangular building it usually is: a stud, a joist and a rafter all spend their lives resisting bending, largely alone.

A triangulated frame does not work that way. That is the entire point of triangulating it. A properly designed triangle converts an external load into forces running *along* its members — compression down one, tension along another — instead of asking any single member to bend. The network carries the load; the sticks mostly just push and pull.

And for a member being pushed or pulled along its length, the property that matters is **cross-sectional area**. Not section modulus. Area.

Which is the column the wedge wins, by +20 per cent.

So the useful question is not:

> *Is this wedge as good a rectangular beam as a 2x4?*

but:

> *Is this wedge an effective axial member inside this triangulated system?*

And there the answer is straightforwardly yes, with more wood in the section than the board it replaces, from a tree that never went near a mill.

Mostly

One more piece of honesty, because “mostly axial” is doing real work in that paragraph.

Panels are not purely axial. Wind pushes on the skin between the members and snow sits on it, and those loads reach the frame as bending in the individual sticks. It is not the dominant case and it is not nothing, and it is exactly where the

wedge's -33 per cent deficit is a real deficit rather than an academic one.

Two things about that:

- The wedge is deep. 4.00 inches pith to bark is more depth than the board has, and depth is what resists the bending it does see — which is why the stiffness came out level rather than behind despite the shape being worse.
- Panel span is short. The distance between supports in a 2V panel is a few feet, not the fifteen a floor joist crosses, and bending demand falls away fast as spans shorten.

None of which is a substitute for somebody competent checking your particular building for your particular snow load. Chapter 10 says the same thing at more length and names who to ask.

What this chapter does not claim

It is worth fencing the argument, so nobody quotes it doing work it cannot do.

This chapter does **not** claim that a split-log wedge is better than a 2x4 as a wall stud, a floor joist, a roof rafter, a shelf, a header or a piece of furniture. For all of those the board wins, often easily, and the reasons are in the table above.

It claims something narrower and much more defensible:

> *For a purpose-designed geodesic dome whose geometry, connectors and jigs are built around radially split timber, the wedge is a better use of the tree than first converting that same tree into ordinary dimensional lumber.*

That is the whole claim. It depends on the building being designed around the material from the beginning — which is what the rest of this book is about.

We are not asking an unconventional piece of wood to imitate a 2x4.

We are building something in which it does not have to.

CHAPTER 8 · EXPLAIN

Forty Panels, One Hundred and Twenty Members

The 2V hemisphere, counted

Forty Panels

This chapter names the frame itself, because every number after it — the cut list, the jig, the fortnight, the seams — is a count of something that exists on this frame, and the frame deserves to be introduced before it is counted.

It is a two-frequency geodesic dome: a hemisphere of 40 flat triangular panels, held up by 120 wooden members. There are 65 distinct edges in its geometry, of which 55 are seams where two panels meet and 10 run around the base where the dome meets the ground and join nothing. Those are the four numbers the whole book hangs on, and none of them was chosen. All of them fell out of the geometry — an icosahedron, subdivided once, projected onto a sphere, cut in half at its widest line — and the rest of this chapter shows where they come from, because a count you can retrace is a count you can trust, and a count you cannot is a story.

The frame in the picture is the actual solved dome this book builds — the same meshes the simulator solves, pushed apart along each panel's own normal so you can see that it really is forty separate triangles and not one continuous surface. That separation is not just a drawing trick. It is how the thing gets built, one finished triangle at a time, on the ground, and it is the subject of Chapter 18.

Where 2V comes from

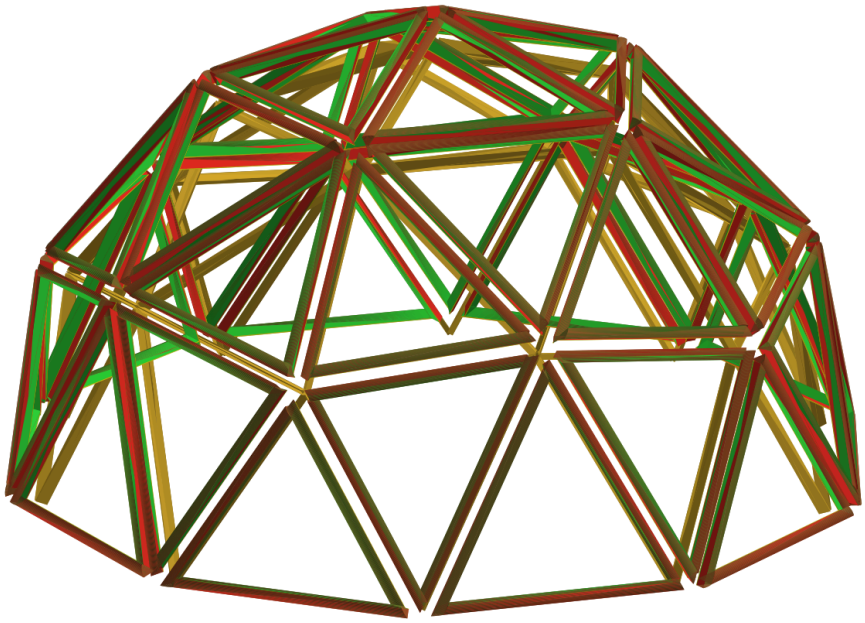
The geometry is older than the dome and worth knowing once, because everything the frame does follows from it.

Start with an icosahedron: twenty identical equilateral triangles, twelve vertices, the roundest of the five Platonic solids. It is already close to a sphere, but its faces are big flat slabs. Subdivide: split every edge in half and connect the midpoints, and each of the twenty faces becomes four smaller triangles — eighty faces in all. Now the move that makes it a dome: push every vertex outward until it lies exactly on a sphere of the same radius. The triangles stop being identical — those nearer the original vertices stretch differently from those near the midpoints — but they all lie on one spherical surface, which is what "geodesic" means here: the straight lines of the original faces become the chords of a sphere.

Cut that sphere at its equator and keep the top half. What remains is the frame: 40 triangles, because a full subdivided icosahedron has eighty faces and the hemisphere keeps half.

That one operation — subdivide once — is what "2V" names: the frequency two subdivision of the icosahedron. It is the smallest subdivision that produces a useful building, and its gift is the number in Chapter 9: all 65 edges of the hemisphere come in only two lengths, because the subdivision only ever produces two kinds of edge — the original icosahedron edges, now halved, and the new edges between midpoints. Two lengths, not forty, is what makes one person

with one jig able to build this at all. Chapter 48 shows what higher frequencies buy and cost; the whole of the rest of this book stays at 2V, because 2V is where the trade stops being worth paying.



Forty panels, 120 members, 55 seams between them.

The frame, exploded

The plate is the frame with its 40 panels pushed apart along their own normals, each triangle floating a little way out from where it will finally sit. Read it as an inventory before you read it as a picture.

Forty triangles. Two shapes, in a fixed ratio: the triangles that sit around the five-fold points and the triangles that fill between them. Not forty shapes — two — which is the second great simplification of the frame, after the two lengths.

Three members each. Every triangle is built complete, with its own three sticks, its own joints, its own edge. Three times forty is 120: the member count of the whole frame, and the number the tree chapter will chase in wood.

The seams where they will meet. Where two panel edges lie against each other there will be a seam — a key between the two members, a gasket, a line on the outside of the shell. The frame has 55 of them. The 10 edges around the base are different: nothing meets them from outside, so they are rim, not seam, and the book keeps the two counts apart because a builder closing 55 seams must not be told to close 65.

What the frame contains

	count
Panels (triangles)	40
Members (struts)	120
Members per panel	3
Unique edges	65
Seams between two panels	55
Rim edges, joining nothing	10

Counted from the solved dome, not chosen. The distinction between edges and seams matters: you close 55 joints, not 65.

Counted from the solved dome, not chosen.

What the frame contains

The table counts the frame two ways and they agree, which is the whole point of printing it.

By panels: 40 panels times 3 members each is 120 members.

By edges: 55 seams each holding two members — one from each neighbouring panel — plus 10 rim edges holding one, is 120 again.

The two routes give the same number because the frame is panelised: every panel owns all three of its edges, so an interior seam is always a sandwich of two members and a key, never one shared strut. That costs wood — 1.85 members for every unique edge — and Chapter 18 is the full accounting of what the extra wood buys. For this chapter the table's job is narrower: the frame is not a number somebody remembered, it is a number the geometry produces, and the geometry produces it twice.

Why hemispheres and not spheres

One honest question deserves an answer before the build chapters start: why is the book about half a sphere, when the geometry naturally makes a whole one?

A hemisphere has a floor. Cut a sphere at its equator and the cut line is a flat circle, lying in a plane, exactly where a building wants its ground. The dome meets the earth along its 10 base edges, all in one plane, and every vertex of the base ring sits on the ground at once. A full sphere has no such line; it touches the ground at a point and the entire weight of the building hangs from that point's anchorage. The hemisphere is the shape a sphere becomes when you give it somewhere to stand.

The rim is the honest part. Those 10 base edges are the one place the geometry does not close on itself — the place where the frame's seams run out. The book treats the rim as its own piece of the build: Chapter 39 gives it its own footing, Chapter 41 puts its doors in it, and Chapter 38 names it as the one place error can accumulate. A hemisphere is not a sphere with the bottom missing. It is the shape that has a bottom, and that is why it can be a building at all.

And the crown is the trade. Half a sphere sacrifices the volume above the equator — the part of the ball nobody could stand in anyway — and keeps everything from the widest line down, which is where the room is. The floor area, the headroom, and the standing-height map are all counted from that choice in Chapter 44. For now the frame simply is what it is: 40 triangles, 120 members, 55 seams, 10 rim edges — the count the geometry gives, printed whole, before anything is built from it.

Two Lengths, Not Forty

The whole reason a dome can be built by one person

Two Lengths

Chapter 8 counted the frame: 40 panels, 120 members, 55 seams. This chapter is about the number inside that count that makes the whole thing buildable by one person with one jig, and it is the smallest simplification in the entire method:

Every one of the dome's 65 edges comes in two lengths. Not forty. Two.

A room of forty different panels sounds like forty different kinds of stick, forty different cuts, forty different chances to be wrong. The 2V geometry quietly removes that nightmare before it starts. Because the frame is an icosahedron subdivided exactly once, its edges inherit only two families: the halves of the original icosahedron edges, and the new edges that connect midpoints. Project them onto the sphere and they stretch, but they stretch into two values, not forty. Two lengths of edge, in a fixed ratio of 1.1308 to one — and everything else in the build method follows from that sentence.

A two-length frame is a frame you can jig. A forty-length frame is a frame you can only improvise, which is exactly the failure Chapter 3 documented with ten days of labour. This chapter states the two lengths, shows the four actual cut lengths that fall out of them, and explains the surprising part: that four is not a mistake and not a regression. Four is what the pinwheel joint does to two, and the reason is worth its own section.

Short and long

The two edge families have names, and the names matter because they travel all the way to the cut list and the jig.

Short. The edge between two midpoints — the new edge the subdivision created. It spans a smaller arc of the sphere, 31.72 degrees, and its length is 0.5465 times the sphere's radius.

Long. Half of an original icosahedron edge — the older edge, spanning 36.00 degrees, 0.6180 times the radius.

Those two factors are not something a designer chose to keep the saw settings down. They fall out of the icosahedron's coordinates, and four independent routes — straight coordinates, central angles, the law of cosines, and a measured drawing — all land on the same pair, which is the book's idea of proof for a number this important. At this book's dome, the two lengths are 72.00 inches and 62.73 inches before the joint takes its bite: a difference of less than a foot across the whole frame.

The triangle shapes follow the same logic. The hemisphere contains 10 equilateral triangles — three long edges, sitting around the five-fold points — and 30 isosceles ones, two long edges over a short base. Two triangle shapes, two edge lengths, one dome. The forty panels of the previous chapter are forty *copies* of two designs, not forty designs.

Every stick in the dome

class	count	each (in)	each (ft) / all (ft)
SHORT	30	62.731	5.228
SHORT	30	63.267	5.272
LONG	30	71.241	5.937
LONG	30	72.000	6.000
total	**120**		**673.1**

Two edge classes become four lengths because the pinwheel inset differs at each end. Cut to these and the dome closes. The last column is each member's length in feet, except on the total row, where it is all of them added together.

Every stick in the dome, in four lengths.

The member schedule

The table is the bridge from geometry to lumberyard, and it is generated, not remembered: every row comes out of the same solved frame the simulator builds, so the table and the dome cannot disagree.

The headline is the first column of it: the whole frame reduces to 4 cut lengths. The longest member is 72.00 inches; the shortest 62.73. Between them sit the other two, and the spread across all four is the width of one hand. That is the entire vocabulary of the build: four numbers on a board beside the jig, and every stick that passes through the bench is one of them.

Why four and not two, when the frame has only two edge families? Because the pinwheel joint insets every member from the vertex it serves, and the inset depends on which side of the corner a member lands on. A long edge becomes two slightly different long members — one for each side of the triangle it borders — and the same for short. The table prints all four, with their counts, because a cut list that hides the difference produces forty panels that almost fit, which is the most expensive kind of fit there is. Chapter 15 is the joint's geometry in full; here the point is only that two became four *for a reason*, and the reason is printed.

That four-length schedule is what the model prints. What the members *measure* — the master cut list, 3 resultant lengths that make a panel the right size once it is together — is shorter: 72.0, 63.5 and 61.5 inches. The equilateral panels take three of the long member; the isosceles panels take the long at the base, the middle on the left and the short on the right. The build's list is the measured one and the model's is the approximation, and Chapter 24 prints both side by side rather than quietly picking one — a builder needs the master list at the saw and the seating schedule at the bench.

Why the pinwheel splits two into four

The step from two lengths to four is the one place a sceptical reader should slow down, because it looks like the dome is quietly getting more complicated, and the honest answer is that it is — by design, and the design buys more than it costs.

A hub dome cuts every strut to end at the vertex, so two edge families really do mean two stick lengths, and the saw never needs a third setting. The price is the hub itself: a machined connector at every corner, drilled to a tolerance, bolted to the end of every strut, and — as Chapter 4 recorded — useless the moment the sticks stop being factory-identical.

The pinwheel throws the hub away by making every member end land on the *side* of the next member instead of at the corner. That single decision is what lets rough, split, slightly irregular timber frame a dome at all: a butt end on a flat sawn face does not need the precision a hub demands. But it has a geometric consequence, because a member that stops short of the corner stops at a different distance on each side of the triangle. A long edge needs a long-left and a long-right; a short edge needs a short-left and a short-right. Two becomes four.

So the honest ledger reads: the pinwheel costs one extra saw setting per edge family, and in exchange it deletes the hub, the mitre on every end, and the requirement that rough timber behave like milled stock. Four cut lengths instead of two is not the dome getting complicated. It is the entire bill for the joint that makes the method possible — and it is a bill paid once, on a jig, where a fence and a stop make the four lengths as repeatable as two. The next chapter spends that simplicity on the question of what a stick of this shape can actually carry.

Compression, Tension, and One Eighth of a Tree

What a wedge strut can actually carry

One Eighth of a Tree

The last three chapters built the case for the wedge as a shape: a triangulated frame loads its members along their length, and along their length all that matters is how much wood is in the section. This chapter turns the case into numbers — and then, because a book is not an engineer, hands the rest of the question to one.

The unit of the whole method is one eighth of a tree: a 45-degree sector of the trunk, 3.83 inches across its bark face, 5.00 inches from pith to bark, 9.82 square inches of wood. That stick will be pushed and pulled along its grain for the life of the building, and its ends will press sideways into the sides of its neighbours. Those are the two load directions this chapter cares about — along the grain, and across it at the joint — and they behave so differently that the entire honesty of the method depends on keeping them apart.

The chapter ends with the one number the book will actually give you, and the reason it will not give you the number you probably wanted. The two are related: the book can compute geometry, and geometry is what a wedge's bearing area is. What a wedge's wood can *take* is a property of your tree's species, grade and moisture, and any book that prints that number for you is guessing about your tree while pretending to know it. This one does not.

Along the grain, and across it

Wood is not one material. It is two, depending on direction, and the difference is roughly an order of magnitude.

Along the grain — end to end, the direction the tree grew — wood is the material engineers mean when they say pine is strong. A stick squeezed along its length carries load through the fibres themselves, every one of them sharing. A stick pulled along its length is even stronger: the fibres take tension along their length the way rope does. This is the direction a triangulated frame uses, which is why Chapter 6 could say that shape does not matter and mean it: for push and pull along the member, the section's area is the count that matters, and the wedge has 9.82 square inches of it.

Across the grain — sideways, pressing into the flat of the wood — the same stick is a different material. Load the side of a pine stick and the fibres are asked to squash between their neighbours; they yield far sooner, they dent rather than break, and the load the joint may carry is a fraction of what the same member carries along its length. Every wooden building lives with this fact; it is why a post stands on a plate instead of a point, and why bolts through timber are spread out rather than clustered.

The wedge frame meets the problem head-on, and the design decision is in the next section. Here the point is the discipline: this book's claims about what a wedge can carry are claims about the *along-grain* direction — the direction the frame actually loads — and its numbers at the joint are numbers about *area*, which is geometry and computable, not

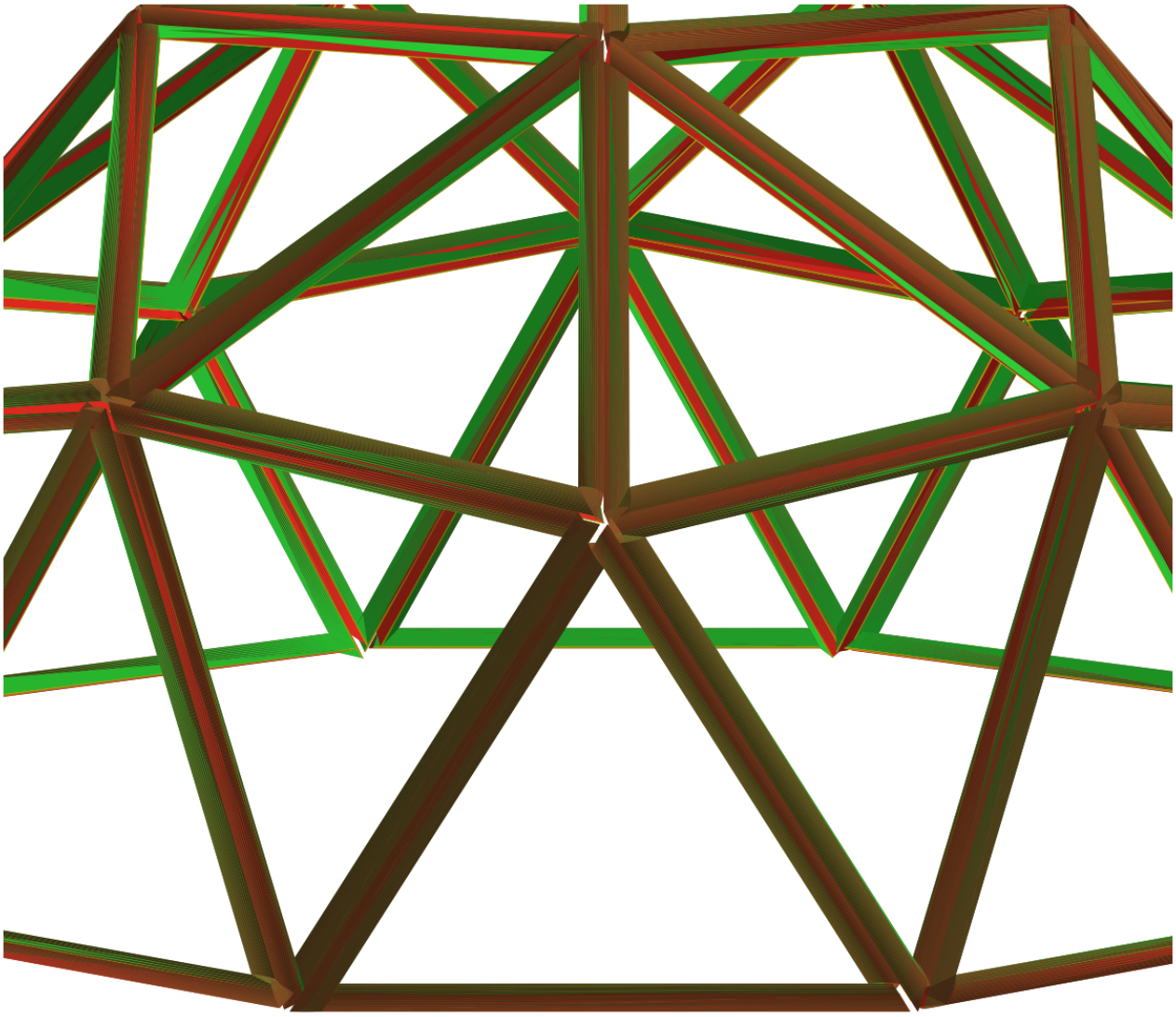
about permissible pressure, which is not. The safety page further down exists because the two are easy to swap, and swapping them is how dome books get people hurt.

The end-to-side bearing

The pinwheel joint's load path is worth stating precisely, because its elegance is real and its limits are real, and both live in the same sentence.

Every member's butt end — sawn square, once, on the jig — lands on the flat sawn face of its neighbour. The contact patch is the end of the stick, and the end of a sector is the sector itself: 9.82 square inches, nothing notched away, nothing reduced to a tongue or a dowel. A conventional joint would cut away wood to make the connection and then bolt what is left; the pinwheel does the opposite. The whole section bears, because the whole section is what the log produced, and the joint gets its strength from refusing to throw any of it away.

The bearing patch, stated as geometry: the butt end is a sector of a 10.0-inch circle — the depth of the member, doubled — cut at 45.0 degrees, which is the same sector shape the tree was split into.



The bearing, at a seam: one member's butt on the next member's side.

So the book's honest number for the joint is 9.82 square inches of end grain against side grain, per member, at every interior corner. The frame has 120 members, each with one loaded butt, and the arithmetic ends there, because the next number — how many pounds that patch may carry — belongs to your tree, not to this book.

What the book *can* say, and does: the bearing is generous by the standards of light framing, the load spreads across the full width of the sector rather than a bolt row, and the key between the members carries the seam's alignment while the wood carries the building. The chapter on the pinwheel (Chapter 15) and the one on the connector (Chapter 17) show how

that division of labour is built. This chapter's job was the load path, and it is this: along the grain through the members, across the grain at the butt, area the whole way.

What this book will not tell you

This page is a warning, and it is not decorative.

This book will not tell you what your tree's wood is allowed to carry. Allowable stresses depend on species, grade, moisture content, and the direction of load; they come from published tables tied to specific species grown in specific regions, and they are applied by someone who can look at the actual timber. A pine from a plantation is not a pine from a windbreak, a knot at a bearing face is not a knot mid-span, and green wood carries less than dry wood while it dries. None of that can be looked up by a book that has never seen your log.

It follows that nothing in this book is a structural sign-off. The frame described here is a prototype method, demonstrated by a build, not an engineered standard. Where a load figure matters — a floating dome hung from trees, a floor hung from a mast, a roof carrying snow — the book names the number as an engineer's job and leaves it there. Chapter 11 repeats this in its accounting of the costs nobody puts in a dome book, and when this project forgot its own rule – printing a load-ish claim it could not stand behind – it said so in public, on the page rather than in a footnote.

The rule, once, plainly: **compute the geometry, check the code, and ask an engineer about the loads.** Everything this book computes is geometry. Everything an engineer decides is the rest.

The number I will give you

One sidebar, one number, so the distinction above has a concrete object.

The bearing area of one butt end against one side: 9.82 square inches. That is geometry — the sector's area, computed from the tree's own taper — and the book will stand behind it, because it is the same arithmetic that produced the cut list, and it can be checked with a ruler on the stick in your hand.

What that patch is *allowed* to carry, in pounds per square inch, is a property of the species, the grade and the moisture — a table entry, not a computation — and this book will not print one. It will tell you which direction to look (across the grain, at the butt, at the wet case), and it will tell you that the frame's own redundancy is the reason the method tolerates being built from timber nobody graded: 40 panels mean a member can be watched, and a member can be swapped, which is how the frame was kept honest through its first year.

The book gives you the geometry and the method. The engineer gives you the stamp. Neither one can do the other's job, and this book is built so that the boundary between them is a page you cannot miss.

What a Dome Costs You

The chapter that argues against the book

What a Dome Costs You

Every book about a building method owes its reader one chapter that tries to talk them out of it. This is that chapter. It is not here for balance; it is here because the costs it lists are real, they arrive late, and they are the ones the rest of the dome literature leaves out — which is exactly how people end up surprised by them.

The argument of this book is that a wedge dome is a remarkable way to get a strong, cheap, buildable frame out of 3 trees. The argument of this chapter is that a frame is not a house, and between the two sit three costs the frame does nothing about: the room is round, the roof is many angles, and the world of permits and appraisals expects a rectangle. Each one is priced in time and money, each one is worse than the same problem in a box, and none of them is fatal. They are the admission price of the shape, and the honest builder pays it knowingly or pays it twice.

The chapter ends with the cases where the rectangle wins outright, because those exist and the book's method is not a religion. If you read this chapter and still want the dome, the rest of the book is yours. If you read it and do not, the book has just saved you a year.

Furniture does not fit a curve

The first cost shows up on moving day, and it never fully leaves.

A wall that curves is a wall that refuses the rectangle. Beds, dressers, bookshelves, kitchen cabinets, a sofa — the furniture economy is built on straight walls and ninety-degree corners, and against a dome's wall every straight piece leaves a wedge of dead air behind it. That dead wedge is real floor you paid for and cannot use, in every direction, at standing height. The honest floor number for a dome is smaller than the plan-area number, and Chapter 44 computes the standing-height map so the loss is visible before you build rather than after you move in.

The fixes are real and none of them is free. A pony wall under the dome buys back vertical wall at the edge — Chapter 58 prices it, and it is the cheapest square footage in the building — but it is still a cost the box never pays. Built-in storage shaped to the curve costs custom work. Loose furniture loses its footprint to the geometry, piece by piece. And the centre of a round room, where the headroom is highest, is also where the room most wants its furniture to be — which is how a round room ends up furnished as a ring around an empty middle, with the good space in the middle spent on walking.

None of this is hidden. It is just not in the frame's price, and a frame's price is what dome books quote. Count the round-room tax as real: a fraction of the floor, a custom-furniture bill, or both, from the first night you sleep in it.

Roofing a compound surface

The second cost is the one that decides whether the dome survives its third winter, and it is the reason dome builders talk about skin more than they talk about anything else.

A dome's roof is one surface made of many flat panels meeting at 55 interior seams, every one of them a line where water can get in. A shingled rectangle sheds water downhill in sheets; a dome sheds it down forty facets and fifty-five valleys, and the valleys are joints. The frame chapters show the seam doing its job with a key and a gasket, and the weather chapter (Chapter 42) says the number out loud — 55 of them — because the honest way to think about a dome's roof is not "one roof" but "55 small roofs, all meeting".

What that costs: the seam system is not optional and not trivial. The key must fit its fold angle, the gasket must take up the tolerance, and the whole assembly must still shed water after the wood has moved through a year of drying. The skin options are priced in Chapter 40, and the pattern across them is the same — the cheap skin costs more later, and the good skin costs now. A rectangle with standing-seam metal and two valleys is a solved problem with a hundred years of practice behind it; a compound surface is a solved problem only if you do the work this book spends its middle chapters describing. The roof is not where the dome is weak. It is where the dome is *particular*, and particular costs attention, which costs time, which is the currency a two-week frame is supposed to save.

Permits, appraisals and resale

The third cost never touches the building and still moves the price.

Permits assume a rectangle. The building department's checklists, the prescriptive code tables, the inspector's mental library — all of them are written around conventional framing, and a hubless dome of split wedges is not in any of them. That does not make it unpermittable; it makes it *unusual*, and unusual means the plan review takes longer, the official may want an engineer's letter (Chapter 10 tells you what the engineer's letter does and does not cover), and the fee schedule may have no line for you. Somewhere, the system will charge for being surprised.

Appraisals assume a comparable. An appraiser prices a house by finding three similar recent sales; a dome has no comparables, so the value comes out of the method instead of the market, and the number is whatever the method defends. That does not make the dome worth less — it makes its worth *harder to finance*, and financing difficulty is a real cost for anyone who is not paying cash.

Resale is where the two meet. The buyer's lender wants the same appraisal the seller's did, the buyer's inspector has never seen a wedge, and the pool of buyers who want a round house is smaller than the pool who want a rectangular one. The network chapters in Part Two (Chapter 68 especially) exist because this project's answer to resale is to make the dome *movable* — value it as a building that can leave its pad — but that is a proposal, not a market, and the chapter says so. The honest line for this chapter: a dome you build yourself and keep is not exposed to any of this. A dome you build hoping to sell like a house is exposed to all of it.

When to build a rectangle

The closing test, stated as plainly as the method allows.

Build a rectangle when the building is a shed, a garage, a workshop, or a storage room — anything where the square shape, the standard sheet goods, the off-the-shelf door and the appraiser's familiarity are worth more than the dome's savings. The catalogue this project prices shows the border: the cheapest and dearest per-square-foot numbers in it are 7 and 68 dollars, and the box shed it costs beside the dome shed comes out at 12,426 dollars against 9,444 — close enough that the shed decision is about doors and shelving, not about the frame. Build a rectangle when the site, the code, or the lender is already fighting you, and the fight would eat the savings. Build a rectangle when you need the work done by a crew whose language is studs and trusses, not jigs and seams.

Build the dome when the floor is round and wants to be, when the frame must be cheap enough to build twice, when the tree is already down, or when the building is a home and the home is yours. The finished-home comparison in the same catalogue — 84,983 dollars against 87,378 — shows the dome's frame advantage persisting into the finished house, and Chapter 46 takes that number apart line by line, including the lines that do not help.

The rectangle is a fine building. It is the right building in most cases. This book exists for the cases where it is not, and the method it describes is the strongest version of the alternative — measured against the rectangle's own numbers, printed here, in the chapter that was supposed to talk you out of it.

Three prices for the same building

The chapter has spent its length on costs that have nothing to do with the dome: furniture, roofing, permits, appraisals. Close it with the three prices the project itself quotes for one building, because a reader who has met all of them in different chapters is entitled to see them on one page.

the same building, three ways	price	per square foot	what is in it
Built cheaply, by you	6,943	22	pressure-treated frame, insulation, glass, basic fittings, nothing salvaged
Built as a product	10,030	—	the standard article: frame, cap, column, panels, labour at a wage
Sold as a product	12,036	43.45	the same, plus margin, warranty and overhead

Read those three rows downward and the shape of this book's whole economic argument appears. The gap between the first and the second is the labour of building it yourself, priced at a market wage — the hours the money chapter counted. The gap between the second and the third is what it costs to make a product of it: overhead, warranty, and a 20 per cent margin. Neither gap is a saving to be claimed from the geometry; both are decisions about who does the work.

Three notes a reader should carry away from that table.

The cheapest row is the one with the most caveats. 6,943 buys a finished, insulated, glassed and fitted shell *if you do the work*, on a pad that is priced separately, with the modules bought separately. It is a real number and it is not a quotation; the chapter's honest paragraph about the 7-dollar catalogue rows applies to it exactly.

The product row is not the method's failure. A reader who wanted to buy this building rather than build it is paying for somebody else's hours, and the product's 43.45 a square foot is genuinely cheap for a finished dwelling — it is only expensive next to the row above it, which represents your own evenings.

And the cap is the line that changes. The watertight cap over this shell is 631 square feet of membrane, and it is the one part of the building that cannot be made at home and cannot be skipped. Every price above includes it; a reader comparing a dome to a shed and finding the dome more expensive has usually just found the cap.

Split Like a Cake

Three passes, eight sectors, and why eight

Split Like a Cake

This is the chapter with the actual cut in it — the one motion the whole method hangs on, reduced to a procedure you could hand to somebody at the log.

The idea was introduced in Chapter 5 as a discovery. Here it is a technique. A round section of trunk, bucked to 6 feet, gets split along its length into 8 equal sectors — like a cake — by seven cuts, all of them straight rips through the pith. Every stick that comes off is the same shape: a 45-degree wedge, 3.83 inches across its bark face, 5.00 inches deep from point to bark, 9.82 square inches of section. There is no squaring, no planing, no second machine. The saw that felled the tree makes the members.

The rest of the chapter is the order of the cuts, the reason the order matters, and the reason it is eight and not six or twelve. The chapter is short because the operation is. That is the point, and it is worth remembering when the pile is knee-deep: the method's entire conversion step is seven cuts per section, and each cut is a straight line.

The three passes

The procedure, in order. The order is not tradition; it is what keeps the section stable while you cut it, and a cut made out of order costs you a pinched bar and a ruined pass.

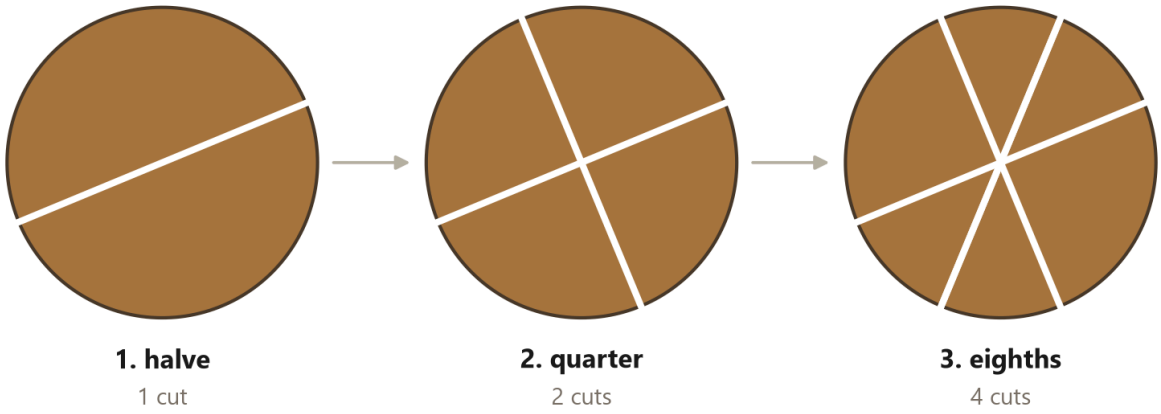
Pass one: halve it through the pith. Stand the section on end or lay it in a cradle, aim the bar down the centre of the end grain, and rip from one end to the other. You now have two halves, each with one flat sawn face and one round bark face. Do this pass first, every time, because it is the only cut in the whole sequence where the section is still round and wants to roll; once it is halved, every piece has a flat to sit on, and everything after is stable.

Passes two and three: quarter each half. Lay each half on its flat face and rip it into two quarters, again through the pith line. Four quarters, each 45 times two degrees wide, each with one flat and one round face still on it. The flat sits on the cradle; the saw follows the centre line; nothing moves.

Passes four through seven: eighths. Each quarter gets one more rip, from the remaining flat face through to the bark, and becomes two eighths — the final sector. Seven cuts in total: one halving cut, two quartering cuts, four eighth cuts. 8 sticks, all the same sector, from a section of trunk that no longer exists.

Two things make the sequence forgiving rather than fussy. First, every cut after the first is made on a piece that already has a flat face to sit on, so the workholding is the piece itself. Second, you are never measuring an angle: the cut list is "through the middle", repeated, and the tree supplied the symmetry. A sector is 45 degrees because a circle divided in half three times is; nobody held a protractor to it.

Three passes, in this order



The halving pass goes first every time: it is the only cut with a flat reference on both sides afterwards. Cut the eighths first and the last two sectors have nothing to hold them.

The order of the passes. Halve, quarter, eighth.

Three cuts, eight sticks

The drawing shows the sequence end-on: one circle, then two halves, then four quarters, then eight sectors, with every cut a diameter. Read it as a production plan rather than a diagram, because the sequence is what makes the work fast.

The first observation is that every pass doubles the number of pieces — one becomes two becomes four becomes 8. The second is that the work per piece falls as the pieces multiply: the halving cut is the whole width of the log, the quartering cuts are half that each, the eighth cuts a quarter of the log's width each. The third is that the sequence never asks a decision. Nothing in it requires choosing where the next stick comes from or how thick it should be; the geometry decided all of that when the log was round. A procedure with no decisions is a procedure you can repeat forty times without drifting, and a procedure that doubles its output per pass is a procedure whose rhythm pays you as you go.

Chapter 29 puts the sequence on a clock — 5.0 sticks an hour, sustained, once the cradle and the rhythm are settled — and Chapter 33 shows what happens to the pile after. Here the sequence stands alone, because it is the one thing in the method that cannot be substituted: no mill, no planer, no router table will produce this member, because the member *is* the cut.

One log, two, four, eight

The production hierarchy deserves its own paragraph, because it is the part of the method that looks obvious and is quietly load-bearing.

One log becomes two halves becomes four quarters becomes 8 members. Every major cut doubles the piece count, which means the whole tree converts in three layers of work: one cut at the top of the tree, two in the middle, four at the bottom. That shape — wide at the bottom, narrow at the top — is exactly what one person with one saw can exploit. The slow cuts are few and wide; the fast cuts are many and shallow. And because the pieces at each layer are identical to their siblings, the workholding at each layer is one cradle, one stop, one posture, repeated.

This is also the shape a guide rail, a cradle or a splitting fixture would be built around, if you ever stop doing it freehand — the geometry already knows where every cut goes, so a fixture that knows it too is just the sequence, made of steel. The book keeps the hand version, because the hand version is the whole point: the method exists so that a person with a chainsaw and 3 trees needs nothing else.

Why eight and not six or twelve

The question gets asked, usually at the first split pile: why eight? The answer has three parts, and each one is a hard constraint.

The sector angle has to leave the seam key workable. A sector is 45 degrees, and the seam between two panels folds by 18.03 to 22.46 degrees depending on where it sits — Chapter 16 prints the full schedule. The gap the key has to fill is the difference between the sector angle and the fold, and at eight splits that difference is a real, makable, compressible thing. Split the log into six and the sector is sixty degrees, the key becomes a structural wedge in its own right, and the seam starts deciding the geometry instead of the other way round. Split it into twelve and the sector is thirty degrees — a sliver, hard to handle and hard to keep straight over six feet.

A sixth is too fat to handle alone. Six sectors from a 12-inch butt are heavy, wide pieces with a sixty-degree point; they are awkward to cradle, awkward to carry, and awkward to fasten. Eight lands the member at 3.83 inches across and 9.82 square inches of section — the size a single person moves, lifts and sets all day, which is the constraint Chapter 28 keeps coming back to.

Eight is the halving count. Two, four, eight — the sequence above only needs cuts “through the middle”, never a measured angle. Six needs a sixty-degree first cut, held by eye or by fixture; twelve needs a thirty-degree refinement. Eight is the finest split you can reach with nothing but halving, and halving is the only cut a saw can aim by symmetry alone.

The table on the next page runs the alternatives side by side so the choice is visible rather than asserted.

Splitting a 10-inch section other ways

splits	angle	width in	depth in	sq in	= 2x4s
4	90	7.07	5.00	19.63	2.45
6	60	5.00	5.00	13.09	1.64
8	45	3.83	5.00	9.82	1.23
10	36	3.09	5.00	7.85	0.98
12	30	2.59	5.00	6.54	0.82
16	22	1.95	5.00	4.91	0.61

Depth does not change with the split count -- it is always pith to bark. Only the width does. Eight is the book's choice: a sixth is too heavy to handle alone and a twelfth is too slender to bear on its own end.

Sector width and depth against split count.

What other split counts give you

The table is the arithmetic of the choice: six, eight, ten, twelve, each with its sector width, depth and section. Eight is the reference row, and the reason is not that eight is magic — it is that eight is the row where the key fits, the stick is carryable, and the cut list is still “halve it”. Every other row trades one of those three away, and the table makes you look at which one.

One more note for the pile, because it will come up: the split count is a *build decision*, set before the first section is opened, and it travels. The member width it produces sets the pinwheel inset, which sets the cut list, which sets the jig. So the decision is made once, from this table, and every number downstream follows. Change it later and you are not adjusting a cut; you are starting the frame again. So the table gets its own page, and the choice gets its own warning: read the row, then commit.

Eighty-Seven Percent

The recovery arithmetic, with the kerf paid for

> ***This chapter corrects something.*** *The 88% recovery figure in the project brief, which was computed for a 15-inch butt rather than this book's 12-inch tree.*

Eighty-Seven Percent

The headline number of the whole method, stated once with its caveats attached, because the caveats are where the number lives:

Split the way this book splits, and a trunk keeps 87.4 per cent of its solid wood as structural members. Sawn the way a mill saws, the same trunk keeps about 45. That is 1.93 times the usable wood, from a cheaper process, with one machine instead of four.

The caveats, in the same breath: the 87.4 counts *split wood only* — nothing planed, nothing dried, nothing graded; the sawn comparison is the honest one for two-by-fours packed in a circle, which is flattering to neither side; and the famous "88 per cent" from this project's earlier films belongs to a bigger tree than this book's. This book's tree is 12 inches at the butt and keeps 87.4 per cent; the films' tree is 15 inches and keeps 88.4. Both numbers are correct, for different trees, and the last page of this chapter prints both rather than quietly picking the flattering one.

The rest of the chapter is the arithmetic, so the headline stops being a claim and becomes a thing you could re-derive with the saw's kerf in your hand.

What a mill loses, and to what

A sawmill's job is to turn a round log into rectangular boards, and everything it loses is the price of the rectangle.

The slabs. A circle does not contain its rectangle. The two rounded sides of the log — the slab wood — cannot become boards of any width, so they come off first, and they are the biggest single loss. On a small log they are a huge fraction of the circle; the fatter the log, the smaller the fraction, which is why mills pay per board foot and prefer the big trees.

The edgings. Even the boards cut from the heart of the log carry wany edges — bark, roundness — and must be edged square. More wood off.

The trim. Boards are sold in even lengths, square at both ends. Every board loses its trim.

The kerf. Every cut turns wood into sawdust. A mill's band or circle blade is thinner than a chainsaw's, but a mill makes far more cuts per log, and the kerf is paid on every one.

Drying degrade. The boards are stickered, stacked and dried, and a fraction of them check, cup, twist or stain past their grade. The wood was there; the board is not.

The grade fall. Of what remains, some boards grade lower than the rest, and low-grade lumber sells for less — a loss in value even when nothing is lost in volume.

Add the five together and the famous range appears: a log sawn into two-by-fours keeps roughly half of itself as saleable lumber. This book's tree, packed honestly with true two-by-fours, keeps 45 per cent. None of that is incompetence. The mill is extremely good at a job whose product is a rectangle, and the waste is the price of the rectangle — paid in wood, at every station of the chain Chapter 2 walked.

What the wedge loses

Now convert the same trunk the wedge way, and count the losses again.

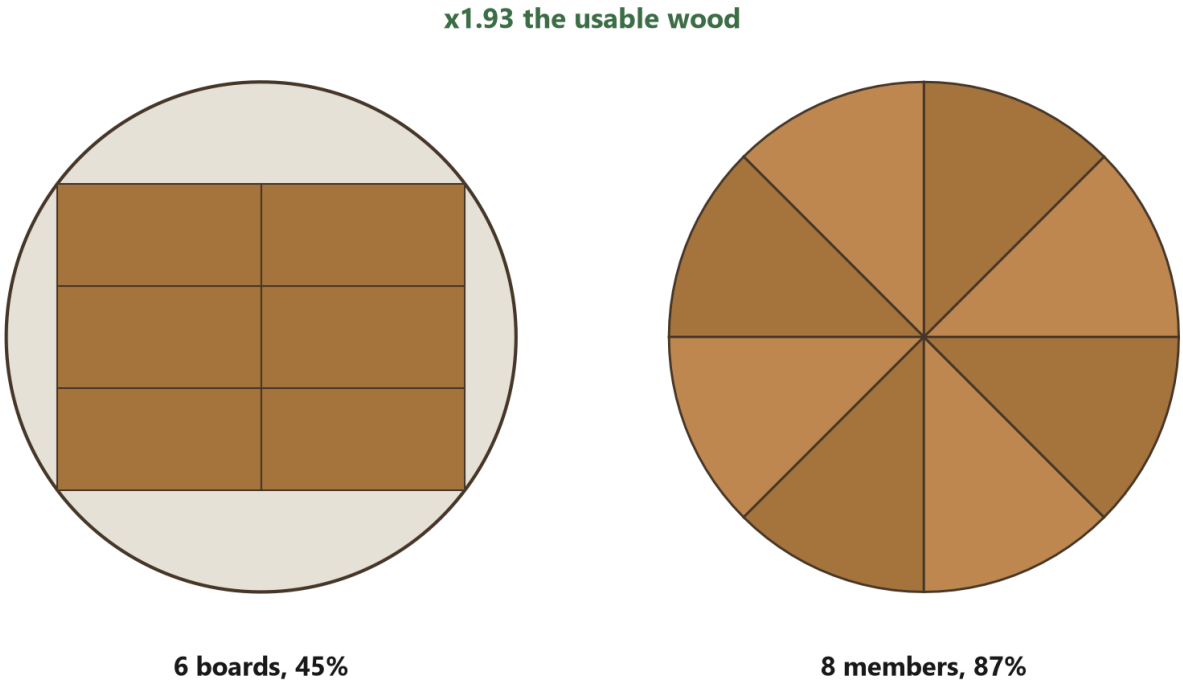
The bark. Lost either way; it was never wood. Not counted against either method.

The kerf — and that is nearly all. The wedge makes seven cuts per section, through the full diameter. Each cut is a 0.25-inch kerf — a chainsaw's kerf, fatter than a mill's blade — and across 8 sections the kerf bill comes to 40 board feet of the trunk's 318. That is the honest, paid price: 40 board feet of sawdust.

Nothing else. No slab, because the sector uses the whole radius. No edging, because nothing needs a square edge. No trim beyond the bucking. No drying degrade to speak of — a wedge is allowed to move as it dries, and Chapter 49 explains why a split moves less than a board. No grade fall, because the book grades its own pile by eye and uses the rejects for blocking.

So the ledger closes: 318 board feet in the trunk, minus 40 of kerf, leaves 278 board feet of structural member — 87.4 per cent — at 4.35 board feet a stick, which is what the worked build in Chapter 24 turns into a cut list. The wedge does not beat the mill at milling. It beats the mill at *this* — making a structural member for a frame that never asked for rectangles — and the beat is the difference between 87.4 and 45 per cent of the same tree.

The same log, converted two ways



Left: grid-packed into 2x4s the way a mill saws, sized on the small end -- 45% of the solid wood. Right: split into 8 sectors, losing only the kerf -- 87%. The shaded area on the left is slabs, edgings and trim.

The same log, converted twice. 87.4 percent against 45.

Both conversions, same log

The drawing shows the same round section twice. On the left, the mill's version: rectangles packed into the circle, slabs and edgings falling away to the outside, the honest count of what a two-by-four economy keeps. On the right, the wedge's version: the circle divided into 8 equal sectors, and every sector kept whole.

The picture makes the argument the numbers cannot. The mill's losses are the *outside* of the circle — the part farthest from any rectangle. The wedge's only loss is the thin lines of the kerf itself, running through the middle. The mill throws away the corners; the wedge keeps the corners and throws away the cut lines. The corners of a round log are not bad wood. They are perfectly good wood of the wrong shape, and "wrong shape" is a judgement the rectangle made, not the tree. Chapter 6 is the reason the frame is allowed to disagree, and this drawing is the disagreement, priced in board feet.

Section by section

The table carries the comparison down the whole trunk, one bucked section per row, both conversions, because a taper means the answer is not one number but a run of them — and the run is what you actually cut.

Each 6-foot section of the trunk is a slightly smaller circle than the one before it. The butt section packs more rectangles and yields fatter sectors; the top section packs fewer and yields slimmer ones. The table shows both columns shrinking as the taper runs, and the recovery staying where the geometry put it: the wedge's advantage is not a lucky average, it is a property that holds section by section, from the 12-inch butt to the 8-inch top. Read it as a check on the headline rather than as a second claim: if any row disagreed with the story above, the story would be the thing to fix.

Where 88 became 87

The errata page, because this project said "88 per cent" on camera and this book prints 87.4, and a reader who notices the gap deserves the explanation rather than the silent edit.

The films' recovery figure — 88.4 per cent — was computed for the films' reference tree: 15 inches at the butt, 60 feet usable, the big plantation pine of the pine-value film. A fatter log puts more wood inside its circle and less into its slabs, so the same split-and-kerf arithmetic keeps more of it: 88.4 per cent.

This book's tree is smaller — 12 inches at the butt, 48 feet usable, the windfall of Chapter 1 — and the same arithmetic, run on the smaller circle, keeps 87.4 per cent. The kerf is the same 0.25-inch chain; the circle it eats is simply a smaller fraction of a smaller pie.

Neither number is wrong, and neither was ever a lie. They are two recoveries for two different trees, and the earlier films named their tree while this book named a different one. The book's discipline — stated on this page, and kept on every errata page like it — is that a correction names the error, prints both numbers, and leaves the arithmetic visible. That is what this page is: 88.4 for the pine the films priced, 87.4 for the tree this book builds, and the reason they differ is the diameter, printed in the table above, one row per section.

Four Ways Up

One stick, four orientations, four different buildings

Four Ways Up

This chapter is the most original idea in the method, and it is the one nobody sees coming, because it concerns a property most building methods do not have: the stick has a front.

A two-by-four is symmetric enough that nobody asks which way up it goes. A wedge is not. It has a point and a curved bark back, two flat sawn faces meeting at a 45-degree angle, and it is never the same stick twice. Which way its point aims — into the room, out to the weather, along the seam — changes what the frame does with it, without changing a single dimension. One stick, cut once, can be four different members, and the four buildings that result are four different answers to the same geometry.

The chapter walks the four orientations and ends with the table that chooses between them, read straight from the simulator's own seam readings. The default — the one the rest of the book builds in — is named and defended. The other three are not mistakes; they are the method's spare capacity, and each one buys something specific at a specific price.

Rotating about its own axis

Hold the wedge out in front of you, point up, bark toward your chest. Rotate it about its own length — the axis from butt to head. At every quarter turn it presents a different face to the room and a different edge to the weather, and nothing about its cut list changes. That freedom is the whole idea: the frame's geometry cares where the stick's *surfaces* sit, not where the stick's *shape* came from, so the builder gets to choose the surface, free.

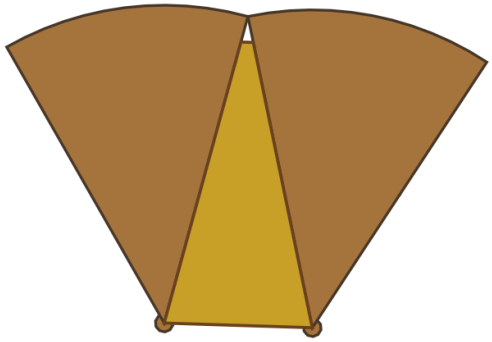
The four orientations come from the two decisions the rotation makes. First, where does the point aim — toward the centre of the dome or toward the centre of its own panel? Second, do the two sticks flanking a seam aim their points together or apart? Point *dome in* means every wedge points at the middle of the building, bark to the weather. Point *panel in* means each stick aims into its own triangle's middle, the pair beside a seam pointing apart. Point *dome out* turns every wedge backward, bark to the room, point at the sky. Point *panel out* turns the pair toward each other across the seam. Four ways up, same stick.

What changes between them is real, and it is measurable: the two sawn faces open by a different angle, the key has to be a different shape, and the key's own base moves from 2.80 inches across to 11.67 in the widest arrangement. The next four sections take them one at a time; the plate shows all four.

The same stick, four ways up

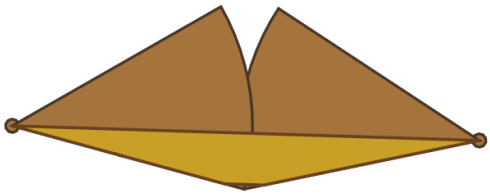
Each pane is one seam cut across, out of the solver's own meshes: two members, one from each panel, and the key in gold between them. The weather is up in every pane. Rotating a member about its own axis does not change the 45-degree sector -- it changes which faces meet the key, and so what shape the key has to be.

POINT DOME IN
points inward, bark to the weather



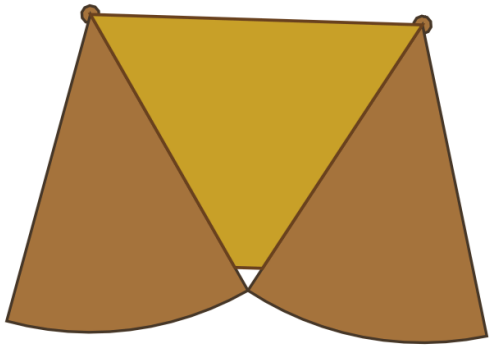
key 2.80 in across its base · sawn faces 27.0° apart

POINT PANEL IN
points apart, into each panel's middle



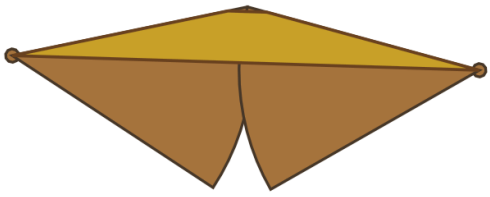
key 11.67 in across its base · sawn faces 153.0° apart

POINT DOME OUT
points outward, at the weather



key 6.27 in across its base · sawn faces 63.0° apart

POINT PANEL OUT
points at each other, across the seam



key 11.67 in across its base · sawn faces 153.0° apart

In the default frame the key is a narrow spline between the two sawn faces. Put the points into the panel middles and the same seam wants a key more than four times as wide across its base; turn the points outward and the key moves outside the members' meeting point. That is why the orientation is chosen before the first panel is cut and not after.

The same stick, four ways up.

All four, side by side

The plate is the whole chapter in one look, and it is not a sketch: each pane is a seam **cut across, through the solver's own meshes**, so what you are looking at is the joint, not an illustration of it. Two members, one from each panel, and the key in gold between them. In every pane the weather is up.

Read the panes for the two things that change. First, *which face of the stick the key bears on*: in the first pane the key sits between the two narrow sawn faces, a slim spline; in the second and fourth it bears on the broad sides and becomes a wide shim; in the third the wood is turned around, so the key is fitted from the inside. Second, *how much key*: the width printed under each pane is the base of the key as the solver slices it, and it is not the same number twice.

The four arrangements are not four aesthetic choices. They are four key-cutting choices and four weather choices, and the table at the end of the chapter prices them. First, each in its own words.

Point dome in

The default. Every wedge points at the centre of the building, bark out, sawn faces lying along the seams. This is the orientation the rest of the book builds in, for three reasons that stack.

The key gets flat faces. A seam's key sits between the two sticks' sawn faces — the only faces the saw ever made flat — so it bears on machined wood rather than on bark. At this orientation those two faces stand 26.97 degrees apart, the key is a narrow taper 2.80 inches across its base, and it bears 6.0 inches along the seam. That is the joint every chapter before this one has been drawing.

Bark faces the weather. The curved back of every wedge is the building's outside face: one continuous, shed-ready, bark-clad surface. Chapter 42 builds the rain strategy on exactly this — water runs down bark, off the seam's shoulder, to the rim.

The stick tells you which way it goes. This is the quiet one, and builders feel it at the top of a ladder. Bark out, pith in: the stick is self-labelling. Nobody marks "this side out" on a hundred and twenty members, because the tree already did — the bark is the outside, the point is the inside, and a wrongly rotated stick announces itself the moment you try to lay the key against it.

And one geometric detail that turns out to matter at the bench: the key is **outboard of the two points**. Cut a seam across and the members' points are the innermost thing in it, with the key sitting in the wedge of space between them and the meeting line of the two bark faces — so the key is reached, fitted and checked from *outside* the shell. That is the practical reason this orientation is the one the book builds in, and the next one down the page loses it.

The cost of point dome in is the ridged interior: the room's wall is a field of points. Acceptable in a frame you will sheath; Chapter 44 decides how to live with it.

Point panel in

Each stick aims its point into the middle of its own triangle, so the pair beside a seam point *apart* — away from each other, one into each neighbouring panel's centre.

What it buys: the key moves off the narrow sawn edges and onto the broad sides of the pair, and it stops being a spline and becomes a broad shim. Cut the same seam across and the two faces now open 153.03 degrees, with the key 11.67 inches across its base — 4.2 times the width of the default's key. For a seam asked to carry more than alignment, that is real contact area rather than a line of it, and the simulator keeps the orientation as a first-class option for exactly that.

What it costs: the bark faces lean away from the weather and into the seam's shadow, so the outside surface is no longer a clean field of bark; the wide key is more timber per seam than the default; and the self-labelling gets subtler, because "point toward the middle of your panel" is an instruction you must remember, not one the stick shouts.

Point dome out

Every wedge reversed: bark to the room, point to the weather. The building's outside becomes a field of ridges — forty panels of them — and its inside becomes the smooth curved field of bark.

What it buys is mostly weather. The pointed outside sheds water the way a ridged roof does, each ridge its own drip line, and the seams sit in the valleys between ridges instead of on the crests — a configuration some roof builders prefer on principle, because the joint lives where the water is already heading.

What it costs is the thing the previous section praised most. The key still sits between the two sawn faces — the geometry does not change what the wood is — but the *meeting line* moves to the other side: the bark faces now converge on the inside of the shell, the key's base is 6.27 inches across (against 2.80 in the default) with the faces 63.03 degrees apart, and the key now lies *inboard* of the two points. Fitted from the room side, not from the weather side. Everything about the joint still works; the person fitting it is now indoors, on a ladder, in a small round room, which is a worse place to be laying a key than the outside of a roof. And the self-labelling inverts: pith out, bark in, which reads wrong to every instinct a woodworker has.

Point dome out is the orientation to remember exists and not choose — unless the building's skin is a ridged shingle system designed for it, which is a real design and not this book's.

Point panel out

The fourth rotation: the pair beside a seam aim their points *at each other, across the seam* — each stick's point reaching over the joint toward its neighbour's back, bark faces turning in toward the panel's middles.

What it buys: the room side of every panel is a field of the flat sawn faces — the smoothest interior the four orientations offer — and the key is broad, like the panel-in orientation: 11.67 inches across its base with the faces 153.03 degrees apart. What it costs: the pair's points are toward each other across the seam, so the weather face of the building is partly sawn face — the face the finish chapters warn you never to leave exposed — and the two sticks' broad faces, not their edges, are what the key bears against, which is more timber per seam than any other orientation. Chapter 49 returns to this when it discusses char and oil, because the one orientation that puts sawn face outdoors is the one this book does not build.

Choosing one

The table is the decision, and it is read straight from the simulator's own seam readings rather than from adjectives. Each row: one orientation, its contact geometry, its weather face, its inside face, and the verdict the solved model gives it.

The row that matters is the default's, and the reason it wins is not that it is perfect — it is that it is the only orientation where every job lands on the face built for it. The key gets the sawn faces. The weather gets the bark. The room gets the points it can sheath over. And the stick labels itself, so the hundred and twentieth member is placed as correctly as the first.

The other three rows are the method's spare capacity: point panel in when the seam must be stiffer, point dome out when the skin is designed for ridges, point panel out almost never. The table keeps all four because a choice made from the table is a decision, and a choice made from habit is a gamble — and the orientation, once the first panel is built, is the one decision in the method you cannot take back.

The Pinwheel

Three sticks, one triangle, and not one mitre

> ***This chapter corrects something.*** *The 'no mitres anywhere' claim, already corrected on camera in lesson_wedge_why.*

The Pinwheel

This is the joint that makes the whole method buildable by one person, and it is worth stating in one sentence before the geometry spends it:

No member end ever meets another member end. Every member's butt end lands on the flat side of the next member, one cut per stick, and the corner is empty.

That sentence deletes, in order: the hub, the mitre at every end, the compound-angle saw settings, and the requirement that rough split timber behave like milled stock. What it costs — and it does cost — is a slightly smaller panel than the triangle it sits in, and a cut list four lengths long instead of two. Chapter 9 already counted that bill. This chapter shows the joint that issues it.

The name is descriptive. Look at a corner of the built panel: the three members overlap like the blades of a pinwheel, each butt end lying on its neighbour's side, chasing each other around the triangle, none of them meeting at the vertex. The vertex — where a hubbed dome puts its connector — is empty air. The frame does not need anything there, because the triangle is closed by the three overlaps, and the overlaps are held by screws through flat wood into flat wood.

Why mitres were the enemy

The conventional way to join three struts at a corner is to make them meet: each end cut to the right compound angle so the three ends kiss at a point, then a hub or a plate holds the kiss together. It is the geometry of every geodesic kit ever sold, and for machined aluminium struts in a factory it is fine.

For this method it is fatal, for three reasons that compound.

A mitre is two angles at once. The end of a dome strut has to match both the panel's fold and the member's position around the corner — a mitre and a bevel together. Cut one slightly wrong and the struts no longer meet; cut forty panels' worth slightly differently wrong and the dome has a gap for every mood.

Every stick needs two of them. Both ends of every member would be mitred — 120 members, two ends each, each end its own setting — against a stick that is rough, tapered, green and moving. Precision joinery on wood that never

agreed to be precise is not a building method; it is a bet against the material.

The corner has to be perfect or nothing is. When three ends meet at a point, all three must arrive exactly, or the hub cannot close. The tolerance is concentrated in one place, which is the worst place for a frame built from a log: every error in the whole build queues up at the vertex and waits.

The pinwheel sidesteps all three by moving the joint. The ends do not meet, so they do not have to match. Each end lands on a flat side, and a butt end on a flat side needs only to be square to its own length — one cut, made once, on a jig. The overlap is where the tolerance lives, and an overlap is a joint that can be off by half an inch and still hold, because the screws have a whole side of wood to find.

Each end lands on a side, not a point

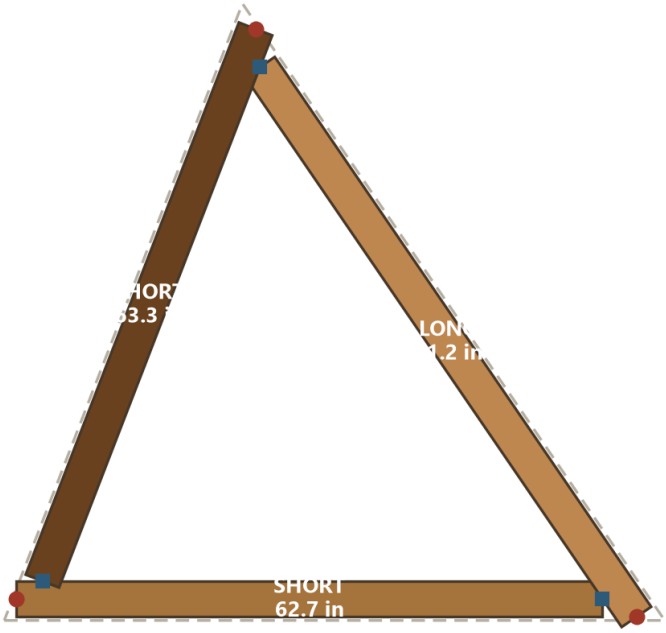
The pinwheel, in the way that finally makes it obvious, because the obvious version is what you will teach somebody at the bench.

Take three sticks. Lay the first two in the triangle's shape, and instead of cutting them to meet at the corner, let each one run *past* the corner along the side of the other. Now the first stick's end rests on the second stick's side, not its end. Lay the third stick the same way: its end rests on the first's side, and the second's end rests on the third's. Three ends, three sides, one closed triangle, and the actual corner of the triangle — the point where the lines would meet — has nothing in it at all.

That is the whole joint. Every member is cut square on its butt end and left long on its head end, and the overlap is resolved at assembly. The result reads as a pinwheel because each stick's butt chases the next stick's side around the corner, and the chase closes the triangle without any stick ever touching the vertex.

Why this survives rough timber: the bearing patch is a full side of a stick — 9.82 square inches of flat-to-flat contact, as Chapter 10 measured — and the screws run through the overlap at right angles to the grain, where wood holds best. Nothing in the joint depends on an angle being cut accurately, because the only cut angle in the whole assembly is the square butt, and the square butt is the one cut a jig can make identical forty times.

One panel: three sticks, pinwheeled



circle = tail, receives the member behind it (4.1 in of bearing)
square = head, lands on the next member's side

Each member's end lands on the SIDE of the next one, never on its point, so no end is ever mitred to another end. The dashed outline is the true triangle; the members sit inside it by the inset. Drawn from the rectangular-band model, where both ends come out square. Split a real log sector and the tail becomes a compound cut -- a mitre plus a bevel of half the sector angle. Chapter 31 is about that cut and does not pretend it is square.

Three members, three end-to-side joints, no mitre anywhere.

One panel, taken apart

The drawing explodes one panel and labels the three overlaps, so the sentence above has a picture to check against.

The butt end. Square to the stick's length, the one machined cut on the member. It is made on the jig, once, with the stick located on its sawn faces — Chapter 34 is that cut in full.

The bearing side. The neighbour's sawn face, where the butt lands. Flat, because the saw made it flat, and wide, because the whole sector is wide.

The head end. Deliberately long. It runs past its own bearing by the overfit allowance — 6 inches in this build — and gets sawn off in place against the jig's fence, which is how the panel ends up true without anybody measuring a stick to a tape. Chapter 35 is that trick, and it is the quiet reason the method forgives.

The empty corner. The vertex itself, untouched. In a hubbed dome this is where the money is — the machined connector, the bolt, the precision. Here it is air, and the air is doing its job by not being asked to do anything.

Read the exploded drawing once and the build order falls out of it: butt cuts first, on the jig; panels assembled flat, overlaps screwed; heads flushed; panel lifted. Nothing met anything. Nothing had to.

What the inset costs

The honest price of the pinwheel, because the joint's elegance is real and its bill is real, and the book prints both.

The triangle's *edge* — the geometric line from vertex to vertex — is longer than the member that fills it, because the member stops short of the corner to land on its neighbour's side. The difference is the inset: the distance each stick pulls back from the vertex so its butt lands on the next stick's side instead of on the next stick's point. Every member length in the cut list is the edge length minus its share of the inset, which is why the cut list has four lengths where the geometry has two — the inset differs between the long and short edges, and between the two sides of each.

The inset also shrinks the panel. The frame's triangles, drawn vertex to vertex, are bigger than the wood that fills them; the panel is the triangle minus three strips around its edge. That strip is not waste — it is the bearing, the overlap, the screw row — but it is real, and it is why the member schedule in Chapter 24 is the number it is rather than the bare chord factors.

And the inset scales with the *stick*, not the dome. A wider member pulls back further; a bigger dome changes nothing. That single fact is why the method survives scaling — Chapter 52 makes it the spine of Part Two — and it is paid here, in the cut list, once.

At the bench the inset is not paid in arithmetic at all. It is already in the members: the master cut list of Chapter 24 — 72.0, 63.5 and 61.5 inches — is what the sticks measure once they have been cut to make a panel the right size, so the inset is spent at the saw and never re-derived at the bench. The model's own schedule prints the inset as a fourth length; the build's list is the measured three, and the difference between the two is a model of the joint rather than a measurement of it.

The mitre claim, corrected

The errata, named in full, because this project said it in public and the correction belongs in public too.

An earlier film in this project claimed that no cut in the dome is a mitre. That was wrong, and it was corrected on camera before this book existed — the film's own correction chapter is the model this page follows.

The true statement is narrower and better. No member end is ever mitred *to another member end*, because the pinwheel never lets two ends meet. But the butt end itself is not a plain square crosscut: it carries the bevel that matches the neighbour's side and the angle that matches the panel's fold — a compound cut, made once, on one end, off the jig, with the same setting for every stick of a class. The whole dome reduces to a handful of saw settings — the bevel is the same everywhere, and the mitre takes a few fixed values — and that is the claim worth making: not "no mitres", but "no mitre is ever measured, and no two ends ever meet".

The errata pages in this part carry the rest of the list. This page is one entry in it: the claim, the error, the corrected version, and the arithmetic that decides which is which — the same treatment the book gives every number it has ever got wrong.

The Seam and Its Key

Where two panels meet, and the angle nobody wants to hear about

The Seam and Its Key

This is the least glamorous chapter in the book, and it is the one that decides whether the dome closes.

The frame's 40 panels do not simply sit next to each other. They meet at 55 interior lines, and at every one of those lines two flat triangles fold against each other at an angle that the sphere, not the builder, chose. Between the two members in each seam sits a key — a strip of wood or compressible material cut to the fold — and the key is where the whole tolerance budget of the method finally lands. Get the keys right and the dome closes like a lid. Get them wrong and forty panels of *nearly* closing is a dome with a gap in it, which is what Chapter 5 spent a week learning the hard way.

The chapter has one unwelcome fact to deliver first — the fold angle is not constant — and then two honest ways to answer it, a table of every seam in the dome, and the third option that sidesteps the whole question. It is short, because once the fact is accepted the answers are small. The week was spent on the accepting.

The fold angle is not constant

Say it early, because everything after depends on it: **the angle at which two panels meet is not the same everywhere on the dome.**

It cannot be. A dome is a sphere approximated by flat triangles, and a sphere curves differently in different places. The panels around the five-fold points of the underlying icosahedron meet at one angle; the panels along the belt between them meet at another. A 2V hemisphere has exactly two fold angles — 18.03 degrees and 22.46 degrees in this build — and that is the good news hidden inside the bad one. Not forty angles. Not a continuum. 2. The bad news is that 2 is not one, and a key cut to one of them will not seat in the other.

The schedule on the next pages counts them: 25 seams at the tighter angle, 30 at the wider, 55 in all. The difference between the two — under five degrees — is small enough that every instinct says to ignore it, and large enough that forcing it closed puts the frame in permanent tension, the keys in permanent shear, and the builder in the position the frankendome already visited. The honest answers accept the difference and spend a little labour on it. The dishonest one — "it is nearly the same" — costs the dome.

And here is the theorem that makes the method's peace with it, because it deserves the name: the two fold angles and the two leftover gap angles each sum to exactly the sector the tree was split into — 45 degrees. The gap the key fills beside a 18.03-degree fold is the rest of 45; beside a 22.46-degree fold it is the rest again. The sector you split out of the tree is, exactly, the seam budget the dome needs. Nothing in the method was tuned to make that true; it falls out of the two geometries and it is why the key problem has an answer at all.

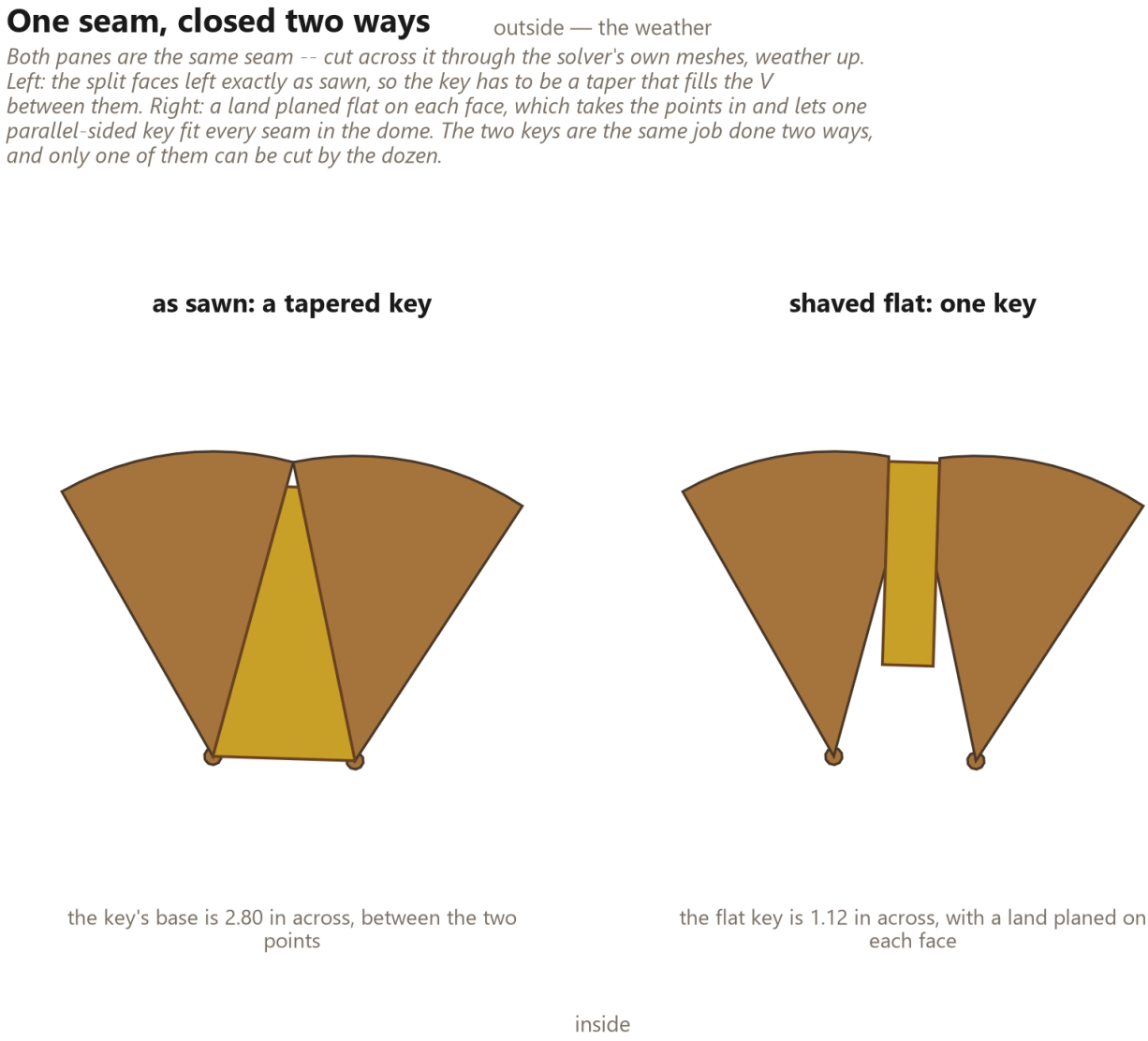
Raw trapezoid, or shaved flat

Two honest ways to make the key fit the two folds, and the book prices both.

The raw trapezoid. Leave the split faces of the members exactly as the saw left them — each sits 22.5 degrees off its member's centreline — and cut the key to fill the V they make. Cut the seam across and the two sawn faces stand 26.97 degrees apart, with the members' points as the innermost wood and the key lying in the space between them and the meeting line of the two bark faces: a taper whose base is 2.80 inches across, at the inside, and whose tip reaches the apex at the outside. It bears 6.0 inches along the seam. The cost is a key that is not one part but two — one profile per fold angle — and a stock pile sorted by profile. The gain is that the member is never touched by a plane, and every piece of the joint stays exactly what the saw made. The figure below is that section, cut out of the solver's own meshes.

The shaved flat. Plane a narrow land along each member's two sawn faces until they sit parallel to the seam, then one flat key fits everywhere — 1.12 inches across as the solver models it, one key for all 55 seams. The cost is planing — 0.48 inches at the deepest, which is 10 per cent of the member's depth, over a land only as wide as the key's bearing — and the labour of it, which Chapter 17 works through completely, because the number is smaller than the instinct expects and "smaller than expected" is exactly the claim this book verifies before it repeats.

Both answers are correct; they are two ways to spend the same small labour, and the figure shows both sections side by side so the choice is a comparison rather than a preference. The book's build uses the shaved-flat answer, because one key everywhere is one less way for the pile to be sorted wrong, and because a planed land gives the key a true bearing. The tapered key remains the answer for a builder with no plane and no patience for one.



One seam, closed two ways: the tapered key and the shaved flat, both cut from the solver's meshes.

One seam, both ways

The drawing shows the same seam twice. On the left, the raw split faces and the tapered key between them — the key's two edges cut to different slopes, each matching the face it bears on. On the right, the same two members with a narrow land planed flat on each, and one parallel-sided key dropped in the gap, the same key that fits every other seam in the dome.

The drawing is the decision, and the decision is smaller than it looks. The tapered key moves the work into the key stock — two profiles, sorted and matched. The shaved land moves the work into the members — one profile, one plane setup, repeated at every seam. Either way the work exists; the book simply chooses to pay it where the sorting cannot go wrong. Chapter 33 shows the fixture that makes the shaved land repeatable, and the seam schedule below shows the two angles it is compensating for.

Every seam in the dome

The table is the whole seam budget on one page: all 55 seams, their fold angles sorted, the two extremes named, and the key profile each one takes. It is generated from the solved dome, not surveyed from the build, so it is the plan the frame was drawn to — and the plan the build should be checked against.

Read it as a work order rather than a curiosity. 25 of one profile, 30 of the other, and the check that closes the chapter: when the last panel is in, the number of keys placed should equal 55, the number of profiles used should equal 2, and the dome should close without being forced anywhere. If any of those checks fails, the frame is not the frame this book drew, and Chapter 38 is where the book tells you where the error went.

Hose, key or nothing

The third option, and the one that ends the chapter, because sometimes the honest answer to an angle is a material that does not care about it.

The compressible seam. A gasket — rubber, foam, a length of hose — fills the gap by deforming instead of by matching. It absorbs the fold difference, the drying movement, and the build's own tolerance in one object, and it seals while it does it. What it cannot do is carry structural load the way a timber key can, which is why the frame's seams are built with a timber key *plus* the gasket where the shell must be weathertight — the key locates the geometry, the gasket seals the gap, and neither is asked to do the other's job.

And the honest nothing. The frame, left bare, closes without any key at all: the butt ends bear directly on the sawn faces, and the seam is a gap the shell chapters will have to deal with. It is not a building; it is a stage of one. But it is a legitimate stage, and building the frame bare first — checking the fold schedule against the real sticks before any key stock is cut — is the cheapest way to find the error the table above was designed to prevent.

The key, then, is the method in miniature: a small part, cut to the two numbers the geometry actually produces, repeated until the pile is gone, and checked against the count when the dome closes. The least glamorous chapter in the book. The one that decides whether it closes.

CHAPTER 17 · EXPLAIN

Let the Connector Hold the Angle

Keep the wood, machine the small part

Let the Connector Hold the Angle

There is a principle underneath this whole method, and it is worth stating on its own because it is the thing that makes rough natural timber buildable to a tolerance at all.

> *The tree supplies the mass. The jig supplies the repeatability. The connector supplies the precision.*

Three jobs. Three different things doing them. Nothing asked to do two.

Precision is expensive per cubic inch

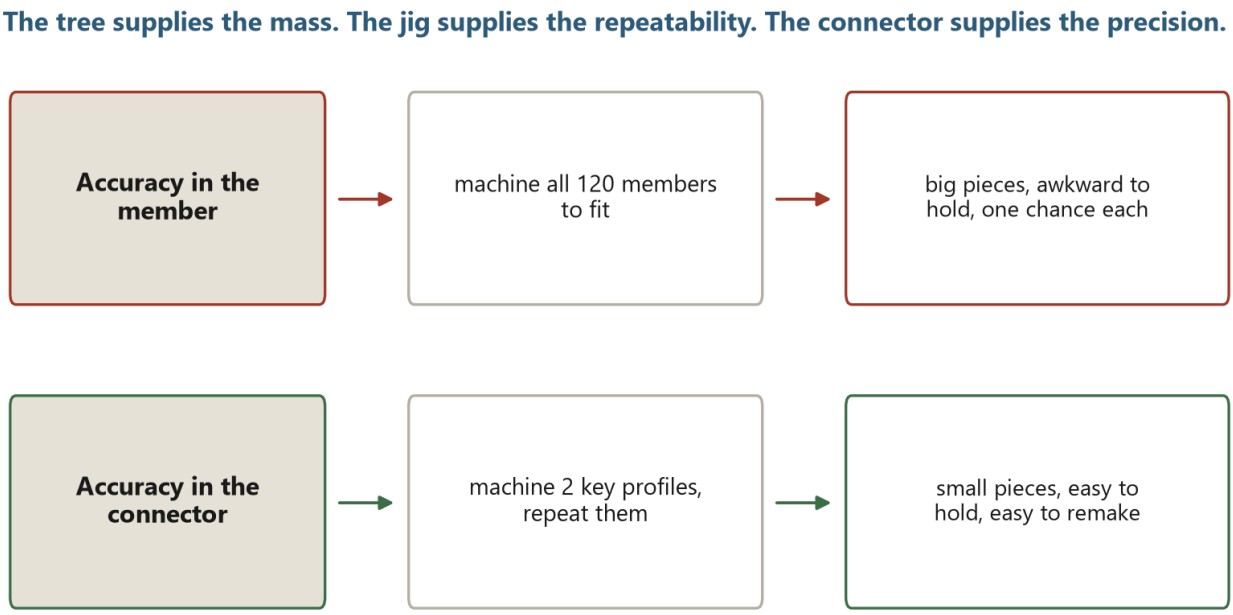
The instinct with an irregular material is to machine it until it is regular. Plane the faces, square the ends, bring everything to a dimension, and then build with confidence.

That instinct is why dimensional lumber exists, and inside a mill it is the right instinct — a mill can afford to hold a tolerance because it is holding it on thousands of pieces with a machine built for nothing else.

A person with a chainsaw cannot afford it, and the reason is not skill. It is that **precision costs roughly in proportion to how much material you have to be precise about**. A long, heavy, awkward member is expensive to hold, to reference, to feed and to correct. A small block is cheap to hold and cheap to remake if you get it wrong.

So there is a choice about *where to put the accuracy*, and the two answers lead to very different afternoons:

Where the precision lives



Both routes end with a dome that closes. One asks for accuracy 120 times, on the largest and most awkward pieces in the build. The other asks for it 2 times, on the smallest, then repeats the result 55 times. This is not a way of avoiding accuracy. It is putting it where it is cheapest to buy.

Where the precision lives.

One route asks for accuracy 120 times, on the largest and most awkward pieces in the build, with one chance at each. The other asks for it 2 times, on the smallest pieces, and then repeats the result 55 times.

Both end with a dome that closes.

This is not a way of avoiding accuracy. Nothing here is less accurate than the alternative — the seams end up at the same angles either way. It is a question of *where the accuracy is cheapest to buy*, and the answer is: in the small part.

The wedge starts close to useful

There is a piece of luck in this, and it is the reason the wedge suits a dome rather than merely tolerating one.

A rectangular board arrives with two faces at ninety degrees to each other. Ninety degrees appears nowhere in a geodesic dome. Every angular relationship in the frame has to be created from nothing, by cutting.

A split sector does not arrive square. It arrives with two sawn faces that each sit **22.5 degrees** off the member's own centreline, because that is half of one 45-degree sector.

And the seams of this dome fold by 18.03 and 22.46 degrees.

Those are not the same numbers. The wedge does not magically arrive at the right angle — that would be too good, and this book does not do too good. But it arrives *within a few degrees* of what the joint wants, rather than sixty-odd degrees away.

The natural shape of the tree becomes part of the joint system instead of an obstacle that first has to be removed. That is the difference between a material that suits a structure and one that merely survives it.

Two ways to close the difference

The gap between the face you have and the face the joint wants can be closed from either side.

Shape the key. Leave both sawn faces exactly as split and cut a tapered key that fills whatever is left. No wood is machined at all. The cost is that the key is not the same shape at every seam, so there is more than one key profile to make.

Shape the wood. Plane both faces parallel and drop in a plain rectangular key that fits everywhere. One key profile for the whole dome. The cost is a machining pass on every stick.

Chapter 16 covers what each does to the seam. This chapter is about what the second one costs, because it sounds much worse than it is.

What planing the faces actually costs

Here is the arithmetic, and it is the most reassuring table in the book.

The raw split face sits at 22.5 degrees. A flat key wants a face at half the seam's fold angle. The difference between those two is everything that has to come off.

What the shaved-flat option costs, over a 2-inch mating land

key	seams	fold	half fold	raw face	correct	cut in	of depth
K1	25	18.029	9.014	22.5	13.486	0.480	9.6%
K2	30	22.459	11.229	22.5	11.271	0.399	8.0%

The raw split face already sits at 22.5 degrees off the member's own centreline, and a flat key wants half the seam's fold angle. The difference is all that has to come off -- under half an inch, on a member 5.00 inches deep. That is the whole price of the option that makes one key profile fit everywhere.

What the shaved-flat option costs.

Over a two-inch mating land — you do not need the whole face to bear, only a band of it — the deepest cut anywhere in the dome is **0.48 inches**.

On a member 5.00 inches deep, that is 10 per cent of the section.

So even the “machine the wood” option does not convert a wedge into something resembling dimensional lumber. It removes a narrow sliver from a narrow band and leaves the overwhelming majority of the member exactly as the saw left it. You are still not making boards. You are making one precise interface on an otherwise raw stick.

That is worth knowing before you decide, because the shaved-flat route is often written off as a betrayal of the whole idea. It is not. It is a 10 per cent concession that buys you a single key profile.

Or do not machine anything at all

There is a third answer, and in some builds it is the best one: make the connector out of something that does not need to fit exactly, because it deforms.

A rubber or closed-cell spline, a length of hose, a compressed gasket — any of these fills a *range* rather than a value. Feed it a seam anywhere between the two fold angles and it takes up the difference by squashing.

What a compliant key can do, all at once:

- absorb the angular difference,
- absorb manufacturing variation in the members,
- seal the seam against weather,
- isolate one panel from another acoustically and against vibration,
- hold a preload so the joint stays tight as the wood moves.

What it cannot do is carry the load a timber key carries. A compliant seam is a weather joint and a tolerance joint; it is not a structural one. So the choice depends on a question you should answer before you buy anything: **is this seam holding the building up, or holding the weather out?**

In a frame where each panel is a complete, self-braced triangle — which is what Chapter 18 is about — a great deal of the shell's stiffness lives inside the panels rather than across the seams, and that is what makes the compliant option viable at all.

Why this is a manufacturing idea, not a dodge

It is worth defending this properly, because "let the connector take up the difference" can sound like an excuse for sloppy work.

It is the opposite. It is how repeatable production works everywhere.

A machine shop does not hold a tolerance on every surface of every part; it holds it on the surfaces that locate and mate, and leaves the rest rough because rough is cheaper and does no harm. An engine block is machined to a few thousandths where the bearings sit and left as cast everywhere else. Cabinetmakers scribe one edge to a wall and leave the other three alone.

The principle is always the same: **decide which surfaces do work, and spend your accuracy there.**

In this dome, the surfaces that do work are the mating faces of the seam and the ends where one member bears on another's side. Everything else — the bark, the taper, the natural bend, the exact diameter — can vary, and the jig is designed so that it can.

The tree is not being asked to be accurate. It is being asked to be strong, which it already is.

Why Every Panel Keeps Its Own Edge

One hundred and twenty members for sixty-five edges, and what the difference buys

One hundred and twenty members for sixty-five edges, and what the difference buys

Every Panel Keeps Its Own Edge

If you have been adding up as you read, something has not been adding up.

The arithmetic that does not look right

The frame has 40 triangular panels. Each panel has 3 members. So:

> *40 × 3 =* ***120 members***

But the shell only has **65 edges**. Count the lines on a drawing of the dome and that is what you get: 55 interior edges where two panels meet, plus 10 around the open bottom where a panel meets nothing.

So there are 55 more members than there are edges for them to be.

That is not an error. It is the single biggest deliberate decision in this frame, and it costs **1.85 times** the timber a conventional geodesic skeleton would use.

It is worth being clear about the size of that, because it is not a rounding difference. If you built the same dome with one shared strut per edge you would need 65 members instead of 120. The panelised frame uses almost twice as much wood in the frame.

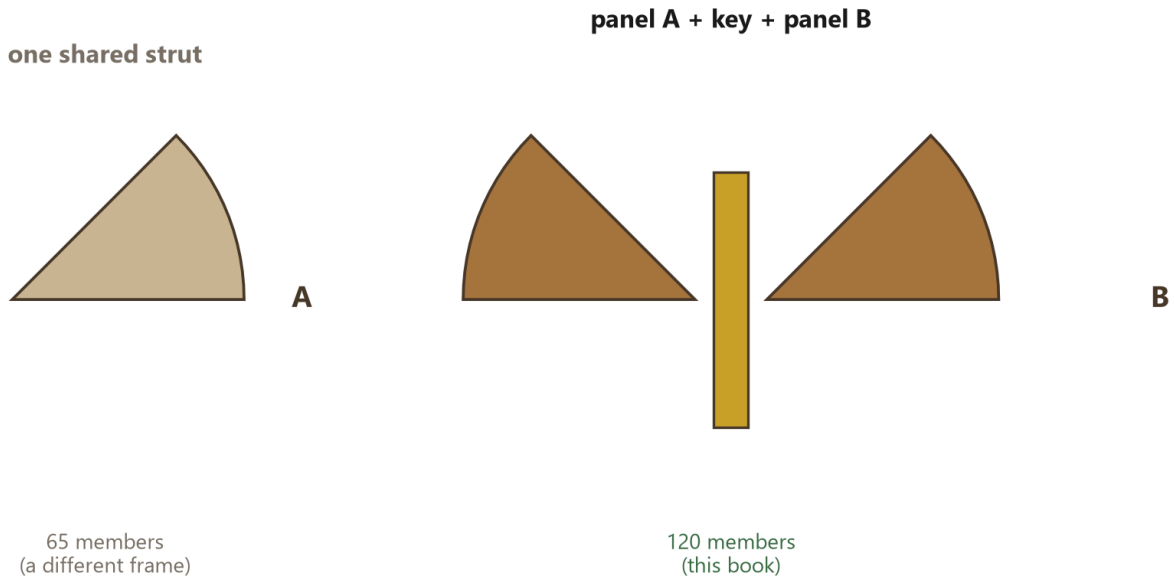
Here is why it is worth it anyway.

What is actually at a seam

In a classical geodesic frame, an edge is one strut, and the two triangles either side of it both fasten to that same strut. Efficient, and the standard way it is done.

In this frame, each panel is built complete — three of its own members, closed into its own triangle — before it goes anywhere near its neighbours. So where two panels meet there are two members, one belonging to each, with the key between them.

One seam, two members



40 panels x 3 members is 120, but the shell has only 65 edges: 55 members exist twice over, once for each panel meeting there. That is 1.85 times the timber a shared-strut frame would use, and it is what lets every panel be built flat, checked, skinned and lifted as a finished thing.

One seam, two members.

Panel A's member. The key. Panel B's member.

The duplicated edge is not lumber somebody forgot to remove. It is the price of a panel being a *thing* rather than a position in an assembly.

What the extra wood buys

Six things, and the first three are why the fortnight in Part 5 is a fortnight rather than a season.

A panel can be built flat. All three of its members exist and belong to it, so the whole triangle can be assembled on a bench, at waist height, on the ground, in the dry. Nothing about building it requires any other panel to exist yet. In a shared-strut frame you cannot finish a triangle without the struts its neighbours also use, which is why those domes tend to get assembled in the air, strut by strut, at the top of a ladder.

A panel can be checked before it goes up. It has a measurable shape while it is still somewhere you can measure it. A panel that came out wrong is firewood and half an hour; a panel discovered wrong when it is the thirty-eighth thing lifted into place is a much longer day.

A panel can be finished before it goes up. Skin it, seal it, insulate it, run a cable through it, paint it — all on the ground. This is a genuine transformation of the work: it turns awkward three-dimensional construction at height into repetitive two-dimensional production at a bench, which is manufacturing rather than improvisation.

The seam gets room for a gasket. Two members with a gap between them is a place a key, a spline or a compressible seal can live. One shared strut has no such space — the panels either side bear directly on it, and weatherproofing that joint is a separate problem solved with tape and hope. Chapter 17 is about what can be done with that space, and Chapter 16 about what has to be.

A panel can be taken out again. Undo one seam's fasteners and a panel lifts out, leaving its neighbours standing and complete. Nothing else in the shell depended on that panel's members. Replace a rotted one in year fifteen, cut a window into one in year three, take three out to get a bathtub in — these become ordinary jobs rather than partial demolitions.

The frame gains redundancy. Every interior edge has two members in it. One splitting, checking or being damaged does not leave that edge unsupported. It is not a designed redundancy and nobody should rely on it, but it is real.

The check that has to close

Two independent ways of counting the members have to give the same answer, and this is worth doing yourself once, because it is the check that tells you the frame in your head is the frame on the ground.

Count panel by panel. Every panel has 3 members.

> $40 \times 3 = 120$

Count edge by edge. Every interior seam holds two members, one from each panel. Every rim edge holds one, because there is nothing on the other side of it.

> $(55 \times 2) + 10 = 120$

Both give 120. The software in this project runs exactly this check every time the book is built, and refuses to print anything if the two disagree — because if they ever did, one of the two pictures would be of a building that does not exist, and there would be no way to know which.

When to share a strut instead

This chapter should not pretend the choice is obvious in every case.

Share the strut if:

- **Timber is your scarcest resource.** 1.85 times the members is 1.85 times the trees, and if you have one tree rather than two that decides it.
- **You have a crew and scaffolding.** The panelised method's main advantage is that one person can do it on the ground. With four people and a lift, in-air assembly is not the ordeal it is alone.

- **The dome is small.** Below a certain size the members are short enough to handle comfortably at height, and the panel machinery is overhead you do not need.
- **You want the lightest possible frame.** For a temporary structure, a greenhouse, or anything that has to be carried in, the weight of the duplicated members is a real cost.

Keep the panel if:

- **You are building alone**, which is what this method was designed for.
- **You want to finish panels on the ground**, which is most of the reason the build fits in a fortnight.
- **You want a weatherable seam**, because the gap is where the seal goes.
- **You expect the building to change**, because panels come out.

This book chose the panel, and every number in it follows from that choice. Chapter 22's 3 trees are 3 trees precisely *because* the frame duplicates its edges. A shared-strut dome of the same diameter would come out of about one.

That is the trade, stated plainly: **roughly twice the wood, for a building one person can make on the ground and take apart again.**

Bends, Knots and Small Units

Why six-foot pieces forgive what sixteen-foot boards do not

Small Units

Every chapter so far has argued from what the method saves — the mill, the haul, the middlemen. This chapter argues from what it *tolerates*, which is the quieter advantage and, for a builder working with whatever grows, the more useful one.

The method's smallest unit is 6 feet: a bucked section, then a wedge, then a member. That smallness changes what counts as a defect. In a stick-frame economy, a bend or a knot ruins a sixteen-foot board and gets charged to the whole load. Here, a bend costs you a section, a knot costs you a stick, and the pile around them is untouched. The method does not need straight trees; it needs short straight *runs*, and a forest full of trees that are "nearly straight for six feet" is a forest full of building material nobody else will pay for.

That is the whole chapter: the defect economy of small units. It is why the method can grade its own stock by eye, why it can build from storm debris and clearing waste, and why the wood it uses never has to travel further than the ground it fell on. Chapter 2 counted the chain the wood avoids. This chapter counts the *wood* the chain avoids — the crooked, knotty, low-grade trees that never get on the truck in the first place.

A bend costs you a section, not a tree

Take the crookedest trunk you have — the one a mill would wave off the landing without pricing. The method looks at it differently.

A bend does not make a tree useless. It makes the tree into two straight logs with a kink between them. Buck the trunk on either side of the bend — cut the kink out as firewood — and what remains is two sections of straight-ish timber, each long enough to split into 8 members. The bend cost you a foot of trunk and a minute of saw time. In a mill's accounting the same bend costs the whole log its grade, because a mill needs its boards straight over their full length, and a kink in the middle of a sixteen-foot board is a board with no market.

The reason is the unit size. A mill's product is long boards; a bend anywhere in the length ruins the board. This method's product is a 6-foot member; a bend only ruins the member it falls inside. Cut around it and keep cutting. A trunk with three bends yields four good sections and three short firewood rounds, and all four sections are full-value frame stock. The same tree, sawn, would have produced a pile of shorts worth the haul to the dump.

And because the frame's members are short, they can be *short of* straight in a way long boards cannot. A member with a slow bow still sits in the jig, still lands its butt on its neighbour's side, still closes the panel — the pinwheel's overlap takes

up what the eye can see, and the panel comes out flat because the jig, not the stick, decides where the corners are. Chapter 30 is that sentence built in wood.

Knots, and where they are allowed

The second defect economy rule: a knot is not one problem, it is two, and only one of them matters here.

A knot mid-span is a stress riser. In a beam — a joist, a rafter — a knot in the middle of the span is where the bending stress concentrates, and a big knot there can decide the board's grade by itself. Even in this frame, a knot in the middle of a member is the worst place for it, because a compression member buckles at its weakest section, and the weakest section is the one with the knot. So the mid-span rule still applies: grade the pile, and keep the heaviest knots out of the middle third of compression members.

A knot at a bearing face is a joint problem. Where one member's butt lands on the next member's side, the load spreads across the whole contact patch — 9.82 square inches of it, as Chapter 10 measured. A small knot inside that patch is a dent in the bearing, not a hinge; the load goes around it through the wood beside it. A knot *at* the contact line, on the other hand, can split the face the screws are pulling against. So the bearing rule: sort the knotty ends away from the butt, and let the head end carry them — the head end gets sawn off anyway, and Chapter 35 makes that offcut free.

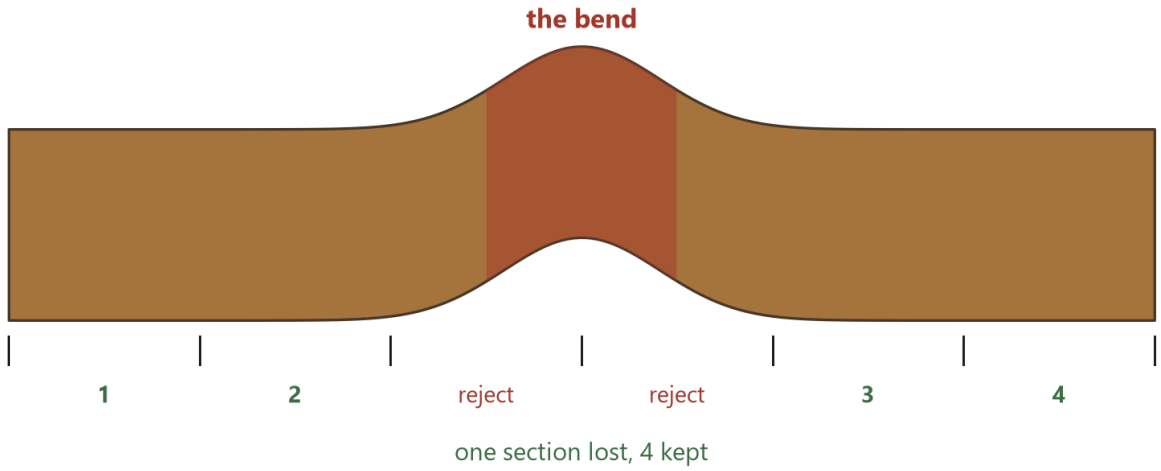
The sorting takes an afternoon and a crayon, and it is the difference between "the pile has knots" and "the pile has a plan". The next section gives the procedure.

One bent trunk, salvaged

The drawing shows one crooked trunk twice. On the left, the mill's view: a log that will be rejected or down-graded, because no board can be sawn straight out of it. On the right, this method's view: the same trunk bucked on either side of its bends into straight sections, each section split into eighths, the kinks themselves stacked as firewood.

The two pictures are the same tree. The difference is not in the wood — it is in what each system demands of a length of wood. The mill demands sixteen straight feet, because its product is a board of sixteen straight feet. The method demands 6 usable feet at a time, because its product is a member of 6. The tree that fails the first demand and passes the second is not a worse tree. It is the same tree, measured against a different product — and the trees that fail the mill's demand are exactly the trees that are free.

A bend costs a section, not a tree



A mill wants a straight log: the bend spoils the whole length because a board has to be straight along all of it. Bucked into 6-foot pieces, the bend costs one section. Cut it out and carry on either side of it.

A bend a mill would refuse, and the sections either side of it.

Grade your own stock

The procedure, because “sort the pile” is advice until it is a list.

One: the bend pass. Walk the pile and mark any member with a visible bow over its length. A slow bow — under a thumb’s width over six feet — goes in the general pile; the jig will hold it straight. A hard bow goes to blocking, bracing and stakes, where a curve costs nothing.

Two: the knot pass. Mark the middle third of every member as it comes off the split, and reject to the blocking pile any member with a knot cluster in that third. A single small knot mid-span stays; a knot cluster does not.

Three: the bearing pass. Look at the butt end of every member — the end that will land on a neighbour’s side. A knot within a hand’s width of the butt face sends the member to the other orientation: flip it, so the knotty end becomes the head end and the clean end becomes the butt. Only if both ends are knotty does the member leave the frame pile.

Four: the pith pass. A member that split along the pith — a crack running down its point — goes to blocking. It will keep splitting under load, and a member with a crack in it has stopped being wood you can trust.

The passes take an afternoon for the whole pile, and they are the entire quality system. No kiln, no grader, no certificate — four passes, a crayon, and the same eyes that will build with the wood. The frame’s redundancy does the rest: 40 panels

mean a suspect member is a member you can watch, and a watched member that moves is a member you can swap, which is the honest engineering behind “grade your own stock”.

Trees nobody else wants

The economic consequence of tolerating variation: the pool of building material widens, and the price of the widened pool is nothing.

Storm debris. Clearing waste. The windfall of Chapter 1. Low-value sawlogs no mill will pay haulage on. A tree too bent or too short for a mill is 8 good sticks a section here, and it is often already lying on the site. Which removes the haul, the mill fee and the markup, and replaces them with fuel and bar oil — the ledger Chapter 46 closes to the dollar.

The point is not that crooked trees are as good as straight ones. The point is that this method’s unit is small enough that *a usable straight run* is the requirement, not *a straight tree*, and the world is full of usable straight runs inside trees nobody wants. A builder who can use them builds from the one timber supply whose price never rises: the wood that is already on the ground where the building goes.

And the journey it does not take

Close the chapter where Chapter 2 opened, because the two arguments are the same one seen from different ends.

The industrial route moves wood through 14 operations and a chain of hands: forest, logger, mill, kiln, distributor, retailer, site. This method’s route is 8: tree, cutting station, drying stack, jig. Every removed stage is fuel, time and a margin — and every removed stage is also a *filter*, because each middleman selects for the wood worth hauling and grading and selling. The crooked tree never survives the first filter. The bend costs it its ride.

The method’s answer is to build where the wood is, from what the filters rejected, in units small enough that the rejection never happens. The chain argument was about money. This one is about material — the same 8 steps, counted as distance instead of cost, and the same conclusion: the best wood is the wood that never had to travel, and the method is the only route that can use it.

CHAPTER 20 · HOWTO

Which Way Round Are You Working?

Choose your method before you touch anything

Which Way Round

There are two ways to size a wedge dome, and the first thing you have to do is work out which one you are actually in. Not which one you would like to be in. Which one you are in.

If you get this wrong you will spend a week doing arithmetic that describes a building you cannot build, and you will not find out until the day you go to cut, when the longest stick the frame wants turns out to be eight inches longer than the longest stick you own.

Everything in the next five chapters follows from this one decision, so it gets its own short chapter and no other content.

Design first, or tree first

Design first is the way everybody expects to work. You know what you want: a room of a certain size, a ceiling you can stand under, a footprint that fits between the trees you are not cutting down. You start from the dome and you work backwards to the woodpile. The output is a shopping list — this many sticks, this long, and therefore this much trunk.

Tree first is the way this book was actually written. You have trees. They are the trees you have: a certain height, a certain taper, a certain amount of straight length between the butt flare and the first serious bend. You start from what a section of that trunk can be cut into and you work forwards to the dome. The output is not a shopping list. It is a diameter, and you find it out rather than choosing it.

Both are honest. Neither is better. They answer different questions and they run in opposite directions, and the entire content of Chapters 21 and 22 is those two directions worked through in full.

The honest test

Here is the test, and it is uncomfortable.

Are the trees already down, or already chosen?

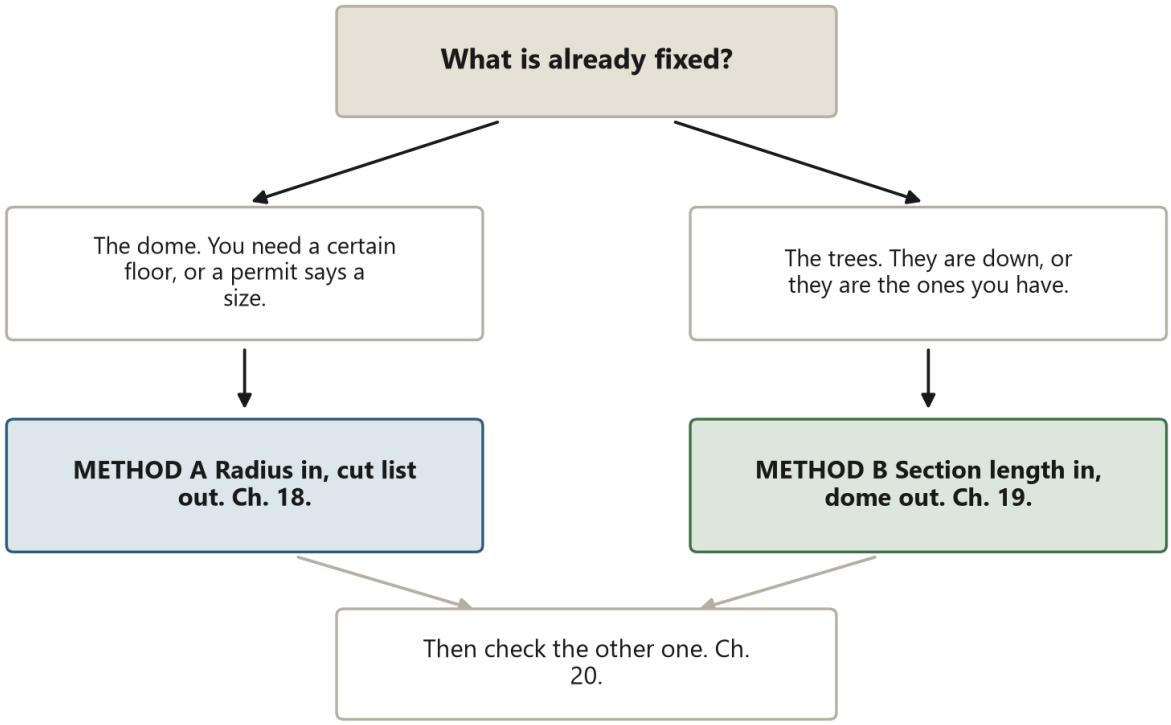
If they are — if you are standing next to a windfall, or you have marked two pines at the back of the property, or somebody has offered you a load — then you are in Method B, and picking a diameter is not a decision you get to make. It is a wish. You may still write a number on a piece of paper. The trees will not read it.

If the trees are genuinely not chosen — you will buy or fell whatever the design calls for, and there is enough standing timber that a foot either way on the bucking length costs you nothing — then you are in Method A, and you should design

the dome you want.

Almost everybody thinks they are in Method A. Most people are in Method B. The tell is whether you would actually go and get a different tree if the arithmetic asked for one. If the answer is “well, I’d probably make it work with these,” you are in Method B and you may as well admit it on page one rather than on cutting day.

Which method are you in?



Almost everybody thinks they are designing first. If the trees are already down, or already chosen, you are in Method B and choosing a diameter is a wish, not a decision.

Start from what you actually have.

What both methods share

Underneath the two directions there is one relationship, and it is worth understanding before you use either method, because it is the thing that makes both of them exact rather than approximate.

In a 2V dome of radius *R*, every panel is a triangle whose corners sit on the sphere. If the members simply ran corner to corner you could scale them straight off the radius: double the dome, double the stick. But they do not run corner to corner. They pinwheel — each member’s end lands on the *side* of the next one rather than on a shared point — and the amount each member is set back from the true corner depends on **how wide the member is**, not on how big the dome is.

A wider stick sits back further. And the width of your stick comes from the diameter of your log, which has nothing whatsoever to do with the dome.

So member length is not proportional to radius. It is radius *minus* a correction that stays the same size while the dome grows. Which means:

- Going from radius to member length is a calculation.
- Going from member length back to radius is that same calculation, solved backwards.

Neither direction is a rule of thumb, a table, or a fudge factor. They are one function and its inverse, and Chapter 23 exists solely to run them against each other and print the difference, which is 0.0e+00 inches.

Before you go on

Whichever method you are in, you need three measurements before either chapter is any use to you:

1. **The diameter of your log at the middle of a section.** Not at the butt — the butt is the fattest part and it flatters you. The middle, because that is what the average stick will be.
2. **The straight length you can get out of the trunk.** Between the flare and the first bend that would spoil a section.
3. **The longest piece you are willing to carry on your own.** This is a real constraint and it does more to set the dome's size than anything you will read in a book. A six-foot green pine wedge is heavy. An eight-foot one is a two-person lift, and if there is only one of you, that is the end of the discussion.

Write those three numbers down. Then turn to Chapter 21 if you are designing first, or Chapter 22 if the trees are already on the ground.

And then, whichever one you used, read Chapter 23 and check it with the other one. That takes ten minutes and it is the cheapest insurance in the book.

Method A: From the Dome You Want

Radius in, cut list out

Method A

You know what you want. Now find out what it costs in trunk.

This is the method to use when the timber is genuinely not decided — when you will fell, buy or scrounge whatever the design asks for. It runs from the building to the woodpile, which is the direction every other construction book works in, and it is the more comfortable of the two.

It is also the one that will tell you, politely and in advance, when the dome you have in mind needs more trees than you have.

The procedure

1. Start from floor, not radius

Nobody wants a radius. They want a room.

Decide the floor area you want to stand on and convert:

> *radius (feet) = √(floor ÷ π)*

300 square feet gives a radius of 9.77 feet — call it 117 inches. A 2V hemisphere's standing height at the centre *is* its radius, so that same number is also your ceiling.

This is the first place a dome surprises people. A 300-square-foot dome is 9.77 feet tall in the middle, which is more house than the floor area suggests, and the volume is doing something a rectangle of the same footprint cannot.

It is also the first place a dome disappoints people, because not all of that floor has headroom over it. Chapter 44 has the map.

2. Measure the stock you expect to have

Even in Method A you cannot skip this. The pinwheel setback depends on how wide your members are, and you cannot compute a member length without it.

If you have not felled anything yet, use the diameter you expect to be cutting. A 12-inch butt tapering to 8 gives sectors about 3.83 inches wide at mid-trunk, and that is the figure this book's tables assume.

If your logs will be fatter, your members will be wider, your setback will be larger, and every length in your cut list will come out shorter than the tables here. The tables are computed at this book's stock; recompute for yours.

3. Solve the member lengths

For a chosen radius, the layout gives you 4 member classes. This is the calculation Chapter 20 described, run in the forward direction: radius in, lengths out.

At 300 square feet the longest member comes out at 64.5 inches and the shortest at 56.1.

Method A, worked: 300 square feet

step	value	where it comes from
Floor you want	300 sq ft	chosen
Radius that gives it	117.26 in	sqrt(area/pi)
Dome diameter	19.54 ft	2 x radius
Standing height at centre	9.77 ft	= radius
Your stock's sector width	3.83 in	10 in log, 8 ways
Longest member	64.55 in	solved
Shortest member	56.14 in	solved
Buck to	6 ft	longest member, rounded up
Sections needed	15	120 / 8
Trunk feet needed	90 ft	sections x bucking length
Trees needed	1.88	at 48 usable ft each
So fell	**2**	**rounded up**

Every line is computed from the line above it. Change the floor area and every number below it moves.

300 square feet, worked all the way through.

4. Round the bucking length up

Your longest member is 64.5 inches. Buck to 6 feet — 72 inches.

You cannot buck to 64.5 inches. Well, you can, but you should not: a section cut to exactly the longest member leaves no allowance for the head-end overfit (Chapter 35), no allowance for the butt cut (Chapter 34), and nothing at all for the end of a log being split or shaken.

Round up to the next whole foot and take the waste. It is the cheapest line item in the build.

This does mean the design-first path throws away more wood than the tree-first path, because the tree-first path chooses a dome that uses the whole section. That is a real cost of getting to pick your diameter, and it is worth saying out loud rather than discovering in the offcut pile.

5. Count sections, then trunk, then trees

The frame needs 120 members. Each section gives 8, so:

> *sections = 120 ÷ 8 = 15*
>
> *trunk needed = 15 × 6 ft = 90 feet*
>
> *trees = 90 ÷ 48 = 1.88*

So: **fell 2.**

Notice the number. Designing a 300-square-foot dome from scratch and sizing the trees to it lands on the same 2 trees that this book's actual build used, from the opposite direction. That is not a coincidence — it is the size of dome a pair of ordinary pines makes, and it is why this method produces houses of roughly this size over and over.

6. Check it against the tree you can actually get

Then go and look at the trees.

If the tallest straight trunk you can find is thirty feet rather than 48, your 1.88 trees becomes three. If your logs are much thinner at the butt than 12 inches, your members are narrower, your sticks are longer for the same dome — and thinner, which Chapter 10 has opinions about.

This is where Method A hands over to Method B, and where Chapter 23's round trip earns its place.

Where this method surprises people

Member length does not scale with the dome. Double the radius and the members do not double. They grow by slightly less, because the pinwheel setback stays the same size while the triangle grows around it.

This is good news at large sizes and bad news at small ones. A very small dome has members so short relative to their own width that the setback eats a serious fraction of them, and below a certain size the geometry stops making sense at all — a stick that is nearly as wide as it is long is not a strut, it is a block. Chapter 48 says where that line is.

Height is not a free variable. In a plain hemisphere, height equals radius. If you want a wide floor and a ceiling taller than half that width, you cannot have both from a hemisphere; you need a riser wall under it (Chapter 41) or a different fraction of a sphere, which is a different book.

The trees number is rarely a whole one. You will get 1.88, or 2.4, or 3.1. Round up and accept the leftover, or shorten the bucking length and see whether the arithmetic lands better. There is no third option that gets you a partial

tree.

The lookup table

If you would rather not do any of this, the table below is the whole method precomputed at this book's stock section. Find your floor area on the left and read across.

Design first: pick a floor, read the rest

floor sqft	diam ft	height ft	longest in	buck ft	trees
80	10.1	5.0	29.5	3	0.94
120	12.4	6.2	37.9	4	1.25
180	15.1	7.6	48.2	5	1.56
250	17.8	8.9	58.2	5	1.56
320	20.2	10.1	66.9	6	1.88
400	22.6	11.3	75.8	7	2.19
500	25.2	12.6	85.6	8	2.50
650	28.8	14.4	98.8	9	2.81

Computed at this book's stock: a 10-inch log split 8 ways. A fatter log gives wider members, which changes the longest-member column and therefore everything after it.

Floor area to trees.

Two warnings about using it:

1. It assumes members 3.83 inches wide. Different logs, different table.
2. The `trees` column assumes 48 feet of usable trunk per tree. If your trees are shorter — and most are — scale it.

Then check it the other way

You now have a cut list. Before you cut anything, take the longest member off that list, hand it to Method B, and see what dome comes back.

It should be the dome you started with. If it is not, you have made an arithmetic slip or changed an assumption halfway through — a different gasket thickness, a different split count, a different stock width.

That check is Chapter 23, it takes ten minutes, and it is the reason this book has two methods rather than one method and a warning.

Method B: From the Tree You Have

Section length in, dome out

Method B

This is the method the book is named after, and it is the one that produced the dome in the photographs. It starts from a tree lying on the ground and ends with a number you did not choose.

That last part is the whole character of the method. In Method A you decide how big the building is and the woodpile has to comply. Here the woodpile decides, and your job is to find out what it decided.

The procedure

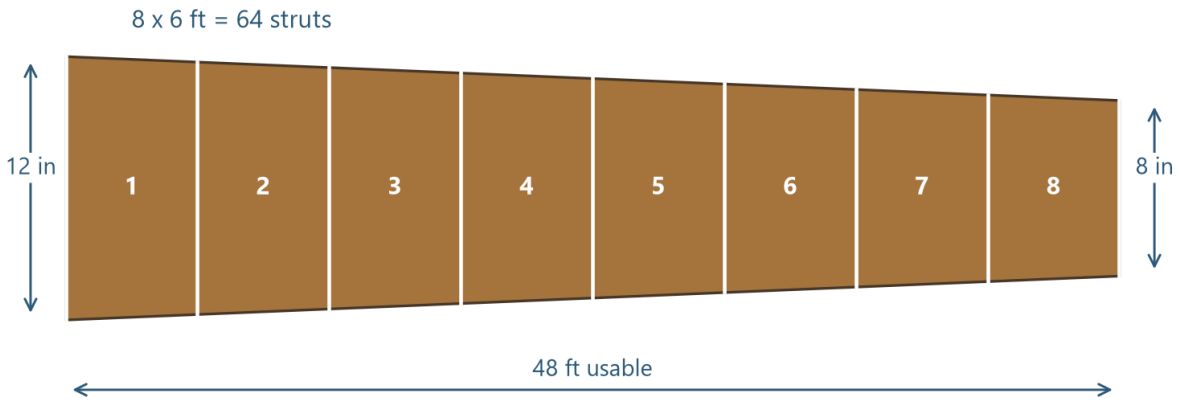
1. Measure the trunk

Three numbers: diameter at the butt of the usable length, diameter at the top of it, and the straight length between them.

“Usable” means the part with no bend serious enough to spoil a section and no branch stub big enough to spoil a stick. It is almost always shorter than the tree. On this book’s tree it was 48 feet, out of a considerably taller pine — the butt flare went in the fire and the top went for kindling.

The tree this book is written around measures 12 inches at the butt of that length and 8 at the top.

The book's tree, measured



48 feet of usable trunk, 12 inches at the butt and 8 at the top, bucked into 8 sections of 6 feet. Every number in this book descends from these three measurements.

The book's tree, measured.

2. Pick a bucking length you can carry

This is the decision, and it is a physical one rather than a mathematical one. Go and pick up a green section of the length you are considering. If you cannot get it onto a sawhorse on your own, it is too long, and no amount of wanting a bigger dome will change that on day six.

This book buckes to 6 feet.

Everything downstream comes from this number. Choose it honestly.

3. Count your sections

Divide the usable length by the bucking length and throw away the remainder. 48 feet at 6 feet a section is 8 sections, with 0.0 feet left over.

If the remainder is large it is worth going back to step 2 and trying a different bucking length. A foot of offcut is firewood; four feet is a stick you did not get.

4. Multiply by eight

Each section splits into 8 sectors of 45 degrees each, and — this is the part that surprises people who have milled lumber — **each sector is a finished structural member**. Not a blank to be trimmed. Not stock to be dimensioned. A member.

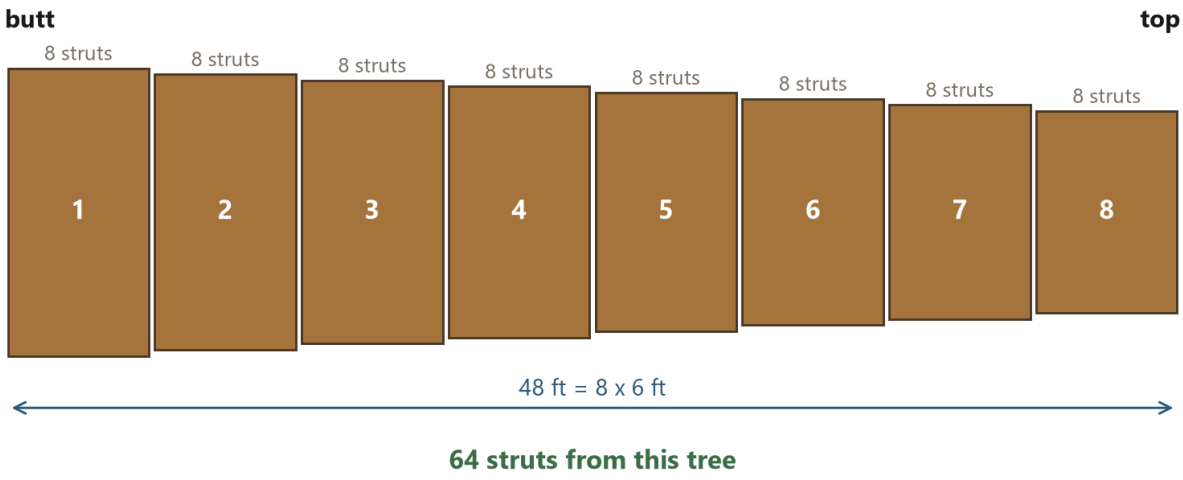
8 sections × 8 sectors = **64 struts from one tree.**

Two trees: **192 struts.**

The frame needs **120**.

That is the title of the book, and it is not a slogan. It is 72 spare — about seven per cent — which is roughly the right margin for a stick that splits along the pith, a butt cut that goes wrong, and the one panel everybody builds backwards.

Bucking the trunk



Mark from the butt, not from the top: the tape drifts, and an error at the top is a short stick you cannot use. 8 sections of 6 feet leaves 0.0 feet over on this tree.

Eight sections of six feet.

5. Find your member's width

The dome does not care how long your sticks are until it knows how *wide* they are, because the pinwheel setback depends on width.

The width of a sector is the chord across its bark face, measured at the diameter you are cutting. Use the diameter at the **middle** of the trunk, not the butt: half your sections come from thinner wood than the butt suggests, and sizing the layout off the fat end gives you a dome that does not quite close at the top.

At this tree's mid diameter the sector is 3.83 inches wide and 5.00 inches deep, pith to bark — a cross-section of 9.82 square inches.

For scale: that is 1.23 nominal two-by-fours' worth of wood, or 1.87 of the dressed boards you would actually find on a rack. Chapter 46 is about which of those numbers to believe.

6. Solve for the dome

Now the arithmetic. The longest member the frame may contain is the longest stick you have, which is your bucking length: 6 feet.

Find the radius whose longest pinwheel member is exactly that.

This is not a division. As Chapter 20 explained, member length is radius less a setback that does not scale, so the relationship has to be solved rather than scaled. The software in this project does it by bisection, and you can do it by trying radii until the longest member comes out right — it converges in about six guesses.

The answer for this tree:

Radius	129.33 in
Diameter	**21.6 ft**
Height at the centre	10.8 ft
Floor	**365 sq ft**

Method B, worked: this book's two trees

step	value	where it comes from
Trunk, butt to top	12 to 8 in	measured
Usable straight length	48 ft	measured
Buck to	6 ft	what you can carry alone
Sections	8	usable length / bucking length
Struts per section	8	one per sector
Struts per tree	64	8 x 8
Two trees give	128	x2
The frame needs	120	counted
Spare	8	7% margin
Sector width at mid trunk	3.83 in	chord of one eighth
Radius the stock allows	129.33 in	solved
Dome diameter	**21.55 ft**	**2 x radius**
Floor	**365 sq ft**	**pi r squared**

You do not choose the diameter here. The tree chooses it, and the last two lines are what you find out.

Two trees, worked all the way through.

7. Read your cut list

The frame's 120 members come in 4 lengths, running from 62.73 inches to 72.00 inches. Thirty of each.

Note that the longest is exactly your stock length, to the last decimal place. That is not a coincidence and it is not luck — it is what step 6 solved for. It also means you have no margin on the long members at all, which is why they are cut with the head end deliberately left over-long and sawn off in place. Chapter 35 is about that.

Every stick in the dome

class	count	each (in)	each (ft) / all (ft)
SHORT	30	62.731	5.228
SHORT	30	63.267	5.272
LONG	30	71.241	5.937
LONG	30	72.000	6.000
total	**120**		**673.1**

Two edge classes become four lengths because the pinwheel inset differs at each end. Cut to these and the dome closes. The last column is each member's length in feet, except on the total row, where it is all of them added together.

Every stick in the dome.

You do not get to choose the diameter

This is the part people resist.

You measured a tree, you picked a length you could lift, and out came 21.6 feet. Nobody consulted you. If you wanted twenty-four feet you cannot have twenty-four feet — not from these trees, bucked this way, without doing something else that costs you elsewhere.

You have three ways out and they are all in Chapter 25:

- **Buck longer.** A longer section gives a bigger dome, out of fewer and heavier pieces, and at some point even 3 trees stop being enough. The table in Chapter 22's appendix shows exactly where that line is.
- **Get a third tree.** Fine, if you have one. It is the honest answer and it is not a failure.
- **Accept the dome the trees gave you.** Which is what happened here.

What you must not do is design for the dome you wanted and cut for the tree you have. The frame will not close, and it will not tell you until the last ring.

Tree first: pick a bucking length, read the dome

buck ft	sections	per tree	2 trees	covers 120?	diam ft	floor
4	12	96	192	yes	15.1	179
5	9	72	144	yes	18.3	264
6	8	64	128	yes	21.6	365
7	6	48	96	NO	24.8	483
8	6	48	96	NO	28.0	617
10	4	32	64	NO	34.5	935
12	4	32	64	NO	41.0	1,318

All from one 48-foot trunk. Buck longer and you get a bigger dome out of fewer, heavier pieces -- and at some point two trees stop being enough.

Bucking length against dome size.

A note on the third tree

3 trees is 192 struts for a 120-member frame, so the frame actually consumes 1.88 trees. The third tree — and most of the second — is not fully used, and the leftover is not waste — it is the spares, the blocking, the floor's stock and the firewood, plus the next dome's first cut list.

But it does mean the honest headline is not "one dome from three trees." It is "one dome from a bit under two trees, and you need the third one because you cannot get eight-tenths of a tree."

Which is a worse title.

CHAPTER 23 · EXPLAIN

The Round Trip

Proving the two methods are one calculation

The Round Trip

Two chapters ago there were two methods. This chapter is here to show that there is really only one, held at opposite ends, and to give you a ten-minute check that catches almost every arithmetic mistake you can make with either.

The reason this matters is not elegance. It is trust.

A book can tell you that a 6-foot stick gives a 21.6-foot dome and you have no way to know whether that came out of a solver or out of somebody's memory of a similar dome. Rules of thumb in building books are usually the latter, and they are usually close enough to be dangerous. So rather than ask you to take it on faith, here is the check, and here is the actual number it produces.

Run it forwards, then backwards

Take any radius. Compute the longest member the frame would contain. Now throw the radius away, and using *only* that member length, compute the radius that a dome would need in order for that to be its longest member.

If the two methods are one calculation, you land exactly where you started.

- Start: a radius, any radius.
- Forwards (Method A): radius → longest member.
- Backwards (Method B): longest member → radius.
- Compare.

The residual

Here is the check run on six domes, from small to large:

The round trip: design first, then tree first, on the same dome

start radius in	longest member in	recovered radius in	residual in
96.000	51.404	96.000	0.0e+00
110.000	60.056	110.000	2.8e-14
129.325	72.000	129.325	0.0e+00
144.000	81.070	144.000	0.0e+00
168.000	95.902	168.000	0.0e+00
192.000	110.735	192.000	0.0e+00

Worst residual across these six domes: 2.8e-14 inches. A chainsaw kerf is a quarter of an inch. The two methods are one calculation held at opposite ends.

Out and back.

The worst disagreement across all six is **0.0e+00 inches**.

For scale: the kerf of the chain on the saw this book was built with is 0.25 of an inch. The disagreement between the two methods is smaller than the width of the cut by a factor with a lot of zeroes in it. It is smaller than the seasonal movement of the wood. It is smaller than anything you could measure with a tape, a story pole, or a laser.

It is, in other words, zero for every purpose a person with a chainsaw has.

That number is not asserted here. It is computed when this book is exported, by running the check, and if it ever stops being negligible this sentence will print a large number and the chapter will be wrong in an obvious way rather than a quiet one.

What would break it

The round trip closes because both directions are solving the same relationship with the same assumptions. Change an assumption between the two and it will not close — and this is exactly what makes it a useful check, because those are the mistakes people actually make.

Change the gasket thickness. The seam key sits between panels and every member length accounts for it. Design at 0.75 inches, then compute your bucking with a half-inch key in mind, and the two answers diverge. The gap will be small and it will be in the wrong direction — your sticks will be very slightly too long, forty times over, and the dome will fight you at the top.

Change the split count. Eight sectors gives a member 3.83 inches wide. Six gives a wider one, twelve a narrower one, and the pinwheel setback moves with it. If you designed for eighths and then split a few logs into sixths because they were fat, those members are not interchangeable with the rest, and the round trip on the fat ones will not close against the layout.

Change the stock diameter. Same problem, from the other side. Sizing the layout off the butt diameter and cutting most of your sections from thinner wood up the trunk means the members you actually get are narrower than the ones you designed. Use the mid-trunk diameter, as Chapter 22 step 5 insists.

Mix up feet and inches. Says itself. It happens. The round trip catches it immediately, because the recovered radius comes back off by a factor of twelve rather than by a thousandth of an inch.

How to actually do the check

You do not need software.

1. Finish whichever method you used. Write down your radius and your longest member.
2. Take the *other* method's procedure from Chapter 21 or 22.
3. Feed it the number you got out, and see if you get back the number you put in.

If you are within a quarter of an inch on the radius, you have not made a mistake worth worrying about. If you are out by more than an inch, stop — something in the middle changed and you should find out what before you cut a hundred and twenty of anything.

Why this chapter exists at all

Because the mistake this is designed to catch is not a mistake in the arithmetic. It is a mistake in the *assumptions*, and those are invisible.

Arithmetic errors announce themselves: a number is obviously wrong, or the units are absurd, or the answer is negative. Assumption drift does not. Everything looks reasonable at every step. You get a cut list of plausible lengths and you cut a hundred and twenty plausible sticks, and the first thing that tells you is the frame, on day fourteen, when the last ring is a half inch short all the way round and there is nothing to be done about it.

Ten minutes with the other method, before anything is cut, is the cheapest insurance in this book.

The Worked Build

Every number for this book's dome, in one place

The Worked Build

This chapter is the whole build on five pages: the tree, the dome it makes, the cut list, the seam schedule, and the frame they describe. A builder who wants to work from nothing else can work from this chapter alone — every number in it comes from the same solved model the rest of the book quotes, and the chapter exists so that the numbers have one place to live where they cannot contradict each other.

The two sizing methods of Chapters 21 and 22 converge here. Method A says: pick the floor, and the arithmetic hands back the tree. Method B says: measure the tree, and the arithmetic hands back the dome. The worked build is Method B run to the end on the tree of Chapter 1 — the windfall, measured, bucked, split, and solved into the frame whose numbers follow.

Nothing on these pages was chosen for the pages. It was computed, and the computation is the same one the films and the simulator run, so the book, the film and the model cannot disagree about the dome.

The tree

The input, measured before any cutting: 12 inches at the butt, 8 at the top, 48 feet of usable trunk, bucked into 8 sections of 6 feet, each section split into 8 wedges. One tree yields 64 struts; 3 trees yield 192, against a frame that needs 120. The recovery arithmetic — 318 board feet in, 278 out, 87.4 per cent kept — is Chapter 13 in full; here it is only the row that feeds the rest of the chapter.

This book's tree

Butt diameter	12.0 in
Top diameter	8.0 in
Usable straight length	48 ft
Bucked into	8 x 6 ft
Offcut	0.0 ft
Split	8 ways, 45° each
Struts per tree	64
Member section	3.83 x 5.00 in
Member area	9.82 sq in
Worth, nominal 2x4s	1.23
Worth, dressed 2x4s	1.87
Solid wood	318.3 bf
Lost to kerf	40.0 bf
Usable wood	278.3 bf
Recovery	**87.4%**

Change any of the first three lines and every later one moves with it.

This book's tree.

The dome

The output: the solved hemisphere that 3 trees of this taper will fill. 21.6 feet across, 10.8 feet to the crown, 365 square feet of floor. The longest member is 72.00 inches; the shortest 62.73; the whole frame stands on 673 feet of timber. The strict count says the frame actually consumes 1.88 trees, and the book says the rest of the pile is spares, blocking, floor stock and firewood rather than pretending the arithmetic is tidier than it is.

The dome those two trees make

Radius	129.33 in
Diameter	21.55 ft
Height at centre	10.78 ft
Floor	364.9 sq ft
Panels	40
Members	120
Seams to close	55
Longest member	72.00 in
Shortest member	62.73 in
Timber in the frame	673 ft
Struts available	**128**
Struts spare	**8**

The spare eight are your margin for a stick that splits, a butt cut that goes wrong, and the one panel everybody builds backwards.

The dome that results.

The cut list

All 120 members, by class, length and count, straight from the pinwheel layout. 4 distinct lengths cover the whole frame — the two edge classes of Chapter 9, split into four by the pinwheel's inset of Chapter 15. The table is the jig's job description: every stick that crosses the bench is one of these rows, and nothing else.

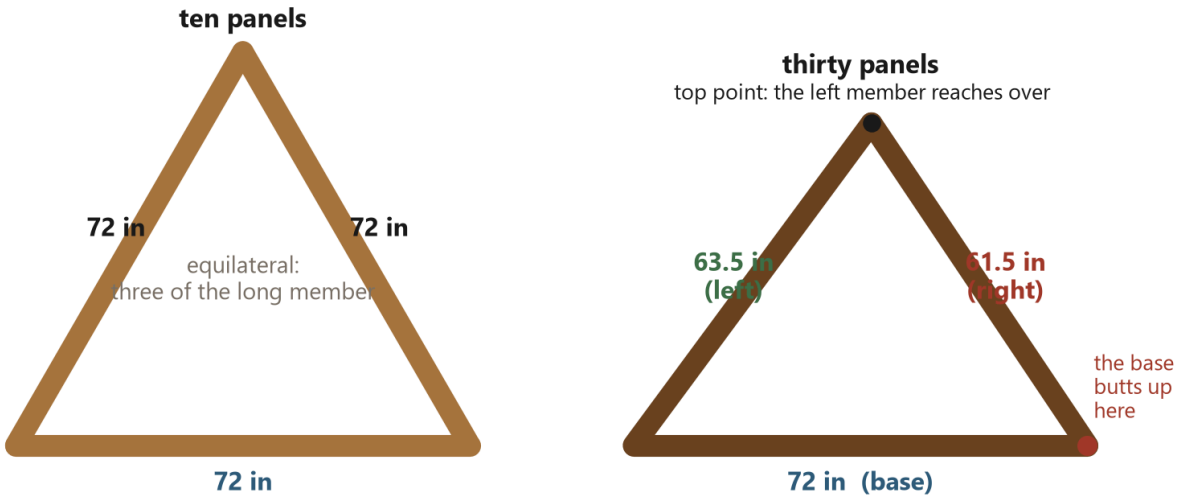
Beside it sits the shorter list the build actually came out with — **the master cut list**, 3 resultant lengths: **72.0 inches**, **63.5** and **61.5**. Resultant means *finished*: cut a member to one of these and the panel it goes into is the size the dome wants, with nothing left over to trim. Every equilateral panel takes three of the long member. Every isosceles panel takes the long member as its base, the middle length on the left and the short one on the right.

That orientation is a bench rule rather than a preference, and the shape shows why. Lay an isosceles panel with its base nearest you and the 63.5-inch left member is the longer of the two sides: it reaches over to the top point, which sits off the middle of the base because the two sides are not equal. The 61.5-inch right side runs up to meet it, and it is that shortest

member the base butts up to — the base's end lands on the side of the short one, which is the pinwheel's chase closing the triangle. Build a panel with the two sides swapped and it does not close: two inches over five feet is not something the eye catches, so the sides are picked from their two marked stacks, in order, every time.

The two lists differ, and the book prints both rather than picking one. The table above is the pinwheel model's seated schedule, which represents the inset as a fourth length; the master list is what the saw cut and what the panels measured when they closed. The model is a model of the joint; the master list is a report of the build, and where the two disagree, the tape on the finished panel is the authority.

The master cut list: three lengths, and the side each one takes



These are resultant lengths: cut to these and the panel is the size the dome wants. An equilateral panel is three of the long member. An isosceles panel, laid with its base nearest you, takes the long member as the base, the middle length on the left and the short one on the right -- the left member is the longer of the two and so reaches over to the top point, while the base butts up to the short right side. That single sentence is the bench rule; nothing about the pair is obvious by eye, so the sides are picked from their own stacks in order.

The master cut list, and which side it goes on.

The cut list

mark	class	count	length_in	bearing_in	inset_in
M1	SHORT	30	62.731	4.103	7.950
M2	SHORT	30	63.267	4.640	7.413
M3	LONG	30	71.241	4.640	8.686
M4	LONG	30	72.000	4.419	7.927
	total	**120**	**8,077**		

Cut to the length column. The bearing column is how much of the end actually lands on the next stick's side; the inset is how far the member sits back from the panel's own corner.

The full cut list.

The seam schedule

Every seam, its fold angle and its key: 25 at 18.03 degrees, 30 at 22.46, 55 in all, the two profiles of Chapter 16. The schedule is generated from the solved dome, so the build’s check is simple: keys placed equals 55, profiles used equals 2, and the dome closes without being forced.

The seam schedule: how many of each key

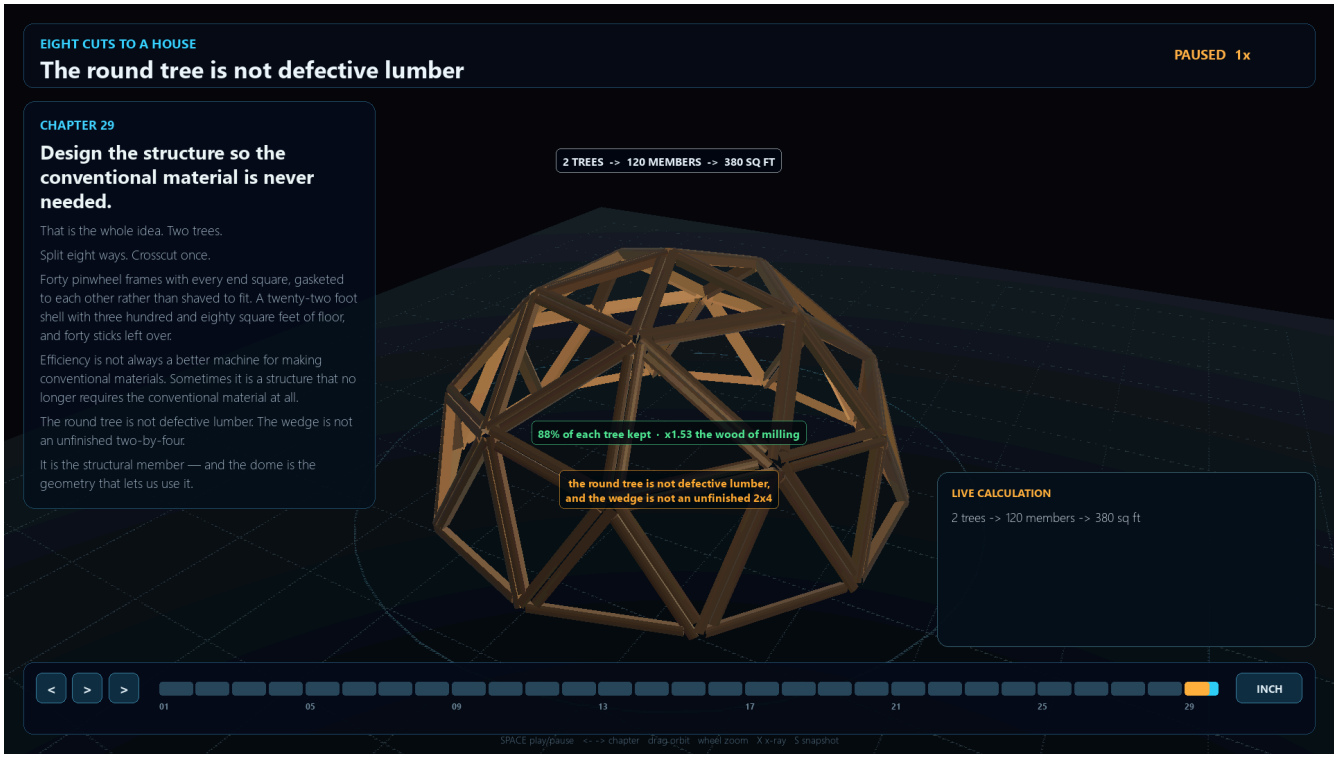
key	edge	seams	fold deg	raw gap deg	length ft
K1	A	25	18.029	26.971	150.0
K2	B	30	22.459	22.541	159.2
	all	**55**			**309.2**

2 distinct keys close all 55 seams. Make one of each to fit, then repeat it. The raw gap column is the angle the key actually has to fill when both faces are left exactly as split. Fold plus raw gap is 45° at every one of them -- the sector angle. The key never fills more than one wedge's worth of angle. Cross-check: the pinwheel model reports 55 interior seams and 10 rim edges over 410 feet.

The seam schedule.

The dome this describes

The frame itself — the simulator's own solved meshes, in the book's print palette, exactly the dome the four tables specify. If the chapter above is the arithmetic, this page is what the arithmetic is *of*: 40 panels, 120 members, 55 seams, 21.6 feet across, standing.



21.6 feet across, 365 square feet of floor.

When the Numbers Say No

Reading a result that means do not build this

When the Numbers Say No

The two sizing methods of the previous chapters end in numbers, and sometimes the numbers say no. This chapter is the permission to listen to them, and the procedure for what to do after.

A book about a build method is at its most useful when it tells you when *not* to build — before the tree is down, before the pile is split, before the jig is built. The failures this method can produce are not dramatic; they are quiet, and they arrive as arithmetic. Two trees that give one hundred and twelve struts when the frame needs one hundred and twenty. A member too heavy for the person who has to lift it. A stick too slender to trust. Each one is a sentence the numbers say, and each one has a standard answer — sometimes a smaller dome, sometimes a fourth tree, sometimes a different bucking length, and sometimes, honestly, not this build at all.

The chapter is short and it is howto, because the decision it teaches is the one that saves the most: reading a bad result as information instead of as a setback, while the fix is still cheap.

Not enough struts

The classic case, and the one the title of this book invites. 3 trees, bucked and split, give 64 struts each — 192 in all, against a frame that needs 120. The arithmetic works out with 72 to spare. But run the same method on trees that are shorter, thinner or bucked longer than this book's, and the count shrinks: 3 trees can come out short of the frame's count, and then the method says no.

Read the shortfall as a measurement of how far off the trees are, not as a verdict on the plan. The frame's count — 120 — is fixed by the geometry; it does not negotiate. The trees' count is the variable, and it is a product of three choices: the usable length you bucked, the section length you chose, and the number of trees you felled. Any one of the three can be changed, and the section below — "Three ways out" — runs the arithmetic for each. The one answer the chapter forbids is the silent one: one hundred and twelve struts do not become one hundred and twenty by cutting more carefully. The count is not a quality issue; it is a quantity issue, and only a quantity fixes it.

The spare margin deserves one honest note, because it is easy to read wrong. This book's 72 spare struts are not generosity; they are the rejection allowance for the grading passes of Chapter 19 — the bent, the knotty, the pith-split that leave the pile. A build planned with zero spares is a build that assumes every stick passes, and no pile does. So the real threshold is not "one hundred and twenty struts"; it is "one hundred and twenty, plus the rejection rate your own pile actually showed you". If two trees give you exactly one hundred and twenty, the numbers are not saying yes. They are saying *tight*, and tight is a no with a schedule.

Too big to lift alone

The second way the numbers say no has nothing to do with counts. A 6-foot green wedge is heavy — the book's own members run 9.82 square inches of wet pine — and the whole method is built around one person carrying, setting and fastening every piece alone. A dome sized by the tree can produce members past the point one person handles, and then the numbers say no in your back instead of in the ledger.

The honest test is not the table, it is the afternoon: pick up the heaviest member the design produces, green and wet, and carry it the distance the build will ask. If it can be carried, set and held with one hand while the other drives a screw, the design passes. If it cannot, the fix is not a stronger builder — it is a shorter bucking length, which makes every member lighter, or a second pair of hands, which Chapter 50 treats as its own design change. The arithmetic follows the carry: shorter sections mean more sections, more splits, and the same frame from lighter pieces — and the tree count re-solves with the new section length, which is the "shorter bucking" row of the section below.

The limit that matters here is the one Part Two formalises. Chapter 52 names the declared longest member one person will handle — 6 feet — and shows what diameter it reaches, because the flat-rate argument this book makes is bounded by exactly this physical ceiling. The sizing methods do not know about your back. You do, and the numbers wait for you to tell them.

Stock too small in section

The third no, and the one that hides inside the first two: the trees are long enough and the count works out, but the sticks are slender — a small butt diameter, split eight ways, yields a sector too thin to trust.

The check is against the comparison Chapter 7 runs: a wedge from a very small log can fall below the dressed board it is supposed to replace, and then the method's central claim quietly inverts. The crossover is real and it is diameter-dependent — below some butt size, the wedge is the wrong answer and the honest move is to buy the boards. The chapter on sections prints the comparison; here the procedure is the thing: compute your own tree's sector area before committing, compare it against the dressed two-by-four's 5.25 square inches, and if your tree loses the comparison, the numbers are saying no to *this tree*, not to the method.

The options are the same three, with a twist. Shorter bucking does not fatten a stick; it only shortens one. An extra tree adds count, not section. The real fixes for a slender section are a thicker tree, a coarser split — six sectors instead of eight, the row Chapter 12 priced — or a different member entirely, which is what Chapter 73 is for: the hollow, laminated and manufactured wedges exist precisely because some trees say no.

Three ways out

The three standard exits, with the arithmetic each one runs, so a "no" from the numbers always lands on a next step.

Shorter bucking. Cut the sections shorter and every member gets lighter and shorter; the dome the tree sizes shrinks accordingly, and the strut count per tree rises. Run Method B again with the new section length — Chapter 22 is a procedure, and a procedure re-runs. The cost is a smaller floor and more sections to split; the gain is a build one person can carry.

A fourth tree. Add a tree and the count problem disappears — four trees of this book's size give far more than 120 — while the member weight and section stay exactly as designed. The cost is one more fell and one more buck, and Chapter 46 shows what an extra tree actually adds in fuel and hours: not nothing, and rarely the thing that kills the build.

A smaller dome. Solve Method A downward — Chapter 21 runs in either direction — until the tree you have is the tree the dome needs. The floor shrinks with the radius, and the shrink is where the honesty of this chapter lives: a dome you can actually build is better than the dome whose numbers said no.

And the fourth option, stated because a howto chapter owes it: **do not build this.** Not every tree, not every site, not every back should become a dome. The method's own arithmetic will say so when it is true, and the builder who hears it has lost an afternoon of sums instead of a fortnight of felling. The numbers saying no are the cheapest form of the no there is.

CHAPTER 26 · STORY

Day Zero: The Saw

A hundred and ninety-nine dollars and a sixteen-inch bar

Day Zero

Before the fortnight there was a day zero, and day zero was the saw.

The whole method claims to need one machine, and the claim deserves to be looked at honestly before the felling starts. The saw this build used was the smallest one on the shelf: a 38.2-cc consumer saw with a 16-inch bar, 199 dollars, bought new. Not a professional felling saw, not a mill, not a rental — the same saw a homeowner buys for firewood. If the method works with this saw, it works with almost any saw; if it secretly needed more, better to find out on day zero than on day nine.

This chapter assesses the tool the way the book assesses everything else: what it can do, what it cannot, what the counterfactual would have bought, and the setup and safety that decide which of the two lists you live on. The fortnight's days each open with a sentence of their own; this one opens the whole clock, because every day after it is this saw's day, and the saw is the only tool in the book.

The smallest saw on the shelf

What 38.2 cubic centimetres and a 16-inch bar can actually do, because the honest list is shorter than the advertised one and more useful.

It fells a modest tree. A 12-inch pine is well inside what this saw drops, as long as the notch and the hinge are cut the way Chapter 27 describes and not the way impatience does. The saw is not the limit at felling; the technique is.

It rips a six-foot section along the grain. This is the real test, and the saw passes it with a margin — a rip through wet pine is slow work, the chain must be sharp and the bar must stay cool, but the 0.25-inch kerf through a 6-foot section is a normal afternoon's work, not a stunt. Chapter 29 is the whole ripping ledger, and its honest finding is that the saw is adequate and the arm is the bottleneck.

It cannot be hurried. A small saw has a small engine, a small oil pump and a small appetite for abuse. Buried to the dogs in green pine, it will pull its chain off the bar, overheat, and dull twice as fast as a professional saw doing the same cut. The method works *with* this saw by working at its pace — 5.0 struts an hour, sustained, is that pace, and the pace is the method.

What it cannot do at all: mill a board, rip a full-length log in one pass, or survive being used like a bigger saw. None of those is needed. The saw's job in this method is exactly three cuts — fell, buck, split — and every one of them is within this machine's honest envelope. The method was not adapted to a small saw; it was designed so a small saw could do it, which is a different and stronger claim.

What I would have bought instead

The honest counterfactual, because a build book that never admits its tool was marginal is not being read by somebody who owns the same saw.

A bigger saw — a professional 60-cc class with a longer bar — would have made the ripping faster, the felling safer-feeling, and the afternoons shorter. It would have cost several times the 199 dollars, and it would not have changed the method by one step. The split count, the bucking length, the jig, the pinwheel — none of them are set by the saw's size. They are set by the tree and the back, and a bigger saw changes neither.

What a bigger saw would have changed is the *margins*: fewer sharpening stops, a cooler chain, a faster recovery when a cut went wide. Those are real, and the book says so rather than romanticising the small saw. But the method's claim was never "this saw is the best tool". It was "this saw is *enough*", and enough is a fact you can verify at the store, in the wood, on day zero — which is the day this chapter is.

If you own a bigger saw, use it. If you own this one, the method works with it. If you own neither, the cheapest saw that will hold a 16-inch bar is the one this book's numbers were produced with.

Setting it up

The setup, in order, because a chainsaw set up wrong is a saw that punishes the setup, not the wood.

The chain. New or sharpened, tensioned so it snaps back against the bar but still pulls around by hand. A loose chain comes off; a tight chain burns the bar. Check it cold, then again after the first cut, because a warm chain is longer than a cold one.

The ripping chain. Ripping with a crosscut chain is possible and slow. A ripping chain — filed to a shallow angle — cuts with the grain instead of fighting it, and it is the single cheapest upgrade the fortnight has. If the budget allows one extra chain, make it a rip chain, and keep it for the splitting days.

Bar oil, always full. Ripping runs the bar buried for minutes at a time, which is exactly the condition that cooks a dry bar. The oil cap is the only gauge that matters on day five through eight; check it every tank of fuel.

The file. A round file matched to the chain, used every few tanks, by hand, to the chain's own angle. Chapter 29 counts what a dull chain costs in kerf and in arms; here the rule is only: sharp is the state the saw arrives in, and you are the one who keeps it there.

Fuel, mixed right. Two-stroke mix at the ratio on the saw, mixed fresh, not last season's. The fuel ledger of Chapter 56 is built on a saw that starts first pull every morning, and starting first pull is a fuel-quality fact before it is a mechanical one.

Chaps, helmet, and never alone

The safety page, placed here on purpose: the saw starts on day zero, so the rules start on day zero.

Chaps. Chainsaw-protective trousers, worn for every cut, no exceptions for a quick trim. They are the difference between a bruise and an ambulance, and they cost less than the saw.

Helmet, visor and ear protection. A falling branch, a kicked chain, a saw at full throttle for hours — all three are ordinary events in this fortnight, and all three are aimed at your head.

Never alone. The single rule that overrides every other rule in the book: do not run a chainsaw with nobody in earshot. The felling chapter says it again, the ripping chapter assumes it, and the story chapters record the one day it was nearly broken. A phone is not a second person. A second person is a second person.

The book's warning from the front matter stands unrepeated here but not unrenewed: felling kills people who have done it a hundred times, a chainsaw is the most dangerous tool most people ever hold, and nothing in this book is worth what a silent afternoon with a saw can cost. Day zero is the day the tool arrives. The respect it deserves arrives the same day, or not at all.

Days One and Two: Felling

Putting 3 trees on the ground where you want them

Felling

The fortnight's first two days are its most dangerous, and the book does not soften the sentence: felling is the operation that kills people who have done it a hundred times, and these two days are felling.

The job is simple to state — put 3 trees on the ground, where the bucking and ripping can reach them — and every word after that sentence is the reason it is not simple to do. The tree decides where it falls; the feller only influences the decision, through the notch, the hinge and the back cut. This chapter is the influence, in order: the safety that comes before the first cut, the two cuts that steer the tree, the drawing of the hinge that does the steering, the limbing that follows the fall, and the story half — what actually happened on this build's second tree, because the story half is where the chapter earns its place.

Two days for 3 trees is not padding. Each tree is felled, limbed and made safe before the next is touched, and the second day exists because the first tree teaches something the next one needs.

Read this before the notch

The safety page, first, because everything after it assumes it.

The escape route. Before the saw starts, walk the route away from the tree — back and to the side, opposite the planned fall — and clear it. The route is not "away from the tree"; it is a specific, walked, cleared line, and it is the thing you use the moment the tree moves. A tree falls in seconds and the escape takes seconds; the route being pre-walked is the entire margin.

The hinge. The strip of uncut wood between the notch and the back cut is the hinge, and it is the steering. It keeps the tree attached while it starts to fall, aims it along the notch, and holds the butt down so the trunk cannot kick back. Cut through it — the back cut carried all the way to the notch — and the tree is free to go anywhere, which is how the word "uncontrolled" enters the report.

Barber chair. A leaning tree, cut badly, can split up its own trunk as it falls — the butt tearing open like a hinged lid, fast, at head height. It happens when the hinge fails before the tree is committed. The prevention is the hinge's size and the order of the cuts, and the respect is: never stand behind a felling tree, not for one second, not to watch.

Widow-makers. Dead branches hung in the crown, freed by the fall or by vibration, that come down before, with, or after the tree. Look up before the first cut. Look up again before the back cut. The crown decides some of what happens next, and it has been deciding since before you arrived.

Two people, minimum. The front matter's rule, restated here where it binds hardest: never fell alone. The second person is the lookout, the caller, and the one who goes for help — and their one job during the fall is to be out of the fall.

The notch and the back cut

The procedure, in the order that keeps the hinge whole.

Face the fall. Stand where the tree should go. The notch opens in that direction, and the tree will follow the notch unless the hinge says otherwise — so the first decision is the fall line, made from the tree's lean, the wind and the landing zone, in that order of authority.

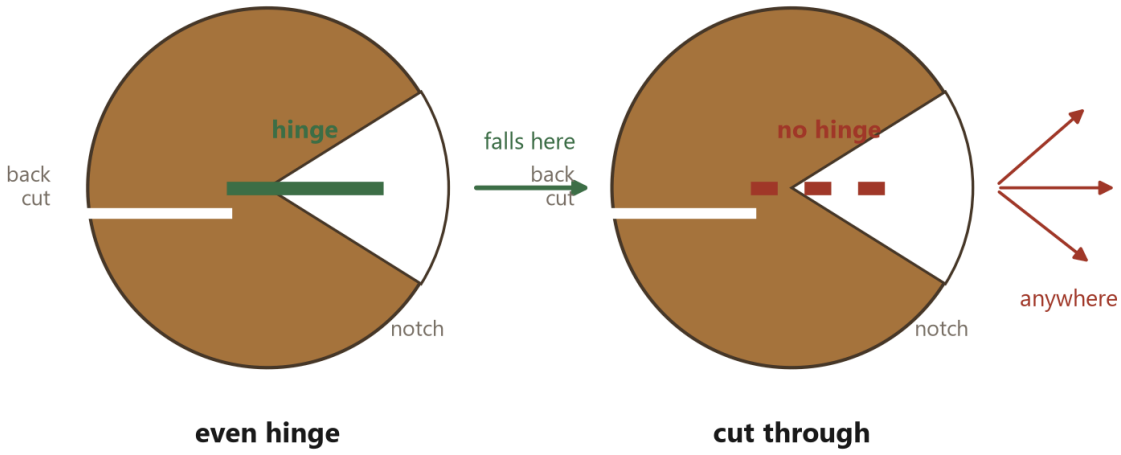
Cut the notch. Two cuts meet to take a wedge out of the trunk on the fall side: the first cut slants down into the tree, the second comes in flat from below to meet it, and the wedge falls out. The notch's depth is about a fifth of the trunk's diameter — deep enough that the tree has somewhere to rotate into, shallow enough that the hinge behind it is still the width of the trunk. On this book's 12-inch trees the notch is a hand's span, and the two cuts must meet cleanly; a notch with a step in it steers a tree sideways.

Cut the back. From the far side, level with the notch's flat floor, the back cut runs toward the notch and stops short of it. The wood left between — the hinge — is a strip the thickness of a couple of fingers, even across the trunk. The tree is now standing on its hinge, committed but held, and it will fall into the notch as the hinge lets go.

Leave. The moment the tree moves, the saw comes out of the cut and the feller takes the escape route, eyes on the crown, without turning a back on the fall. The hinge finishes the job; the feller's job from here is distance.

The hinge dimension is the whole art. Too thick, and the tree sits on the saw and falls late or not at all; too thin, and the hinge breaks early and the tree falls wherever its weight wants. The even, fist-thin strip is the difference between a fall and an event.

The hinge steers the tree



The notch decides the direction; the hinge holds the tree to that direction while it falls. Cut through the hinge and the tree is no longer attached to anything that is aiming it. This drawing is not a felling course -- take one.

The hinge steers the tree. Everything else is timing.

The hinge, drawn

The drawing shows the hinge doing its three jobs at once, because the jobs are the explanation for everything above.

It aims. The hinge is the uncut wood across the back of the notch, and it is flexible only in one direction — along the fall line. As the tree tips into the notch, the hinge bends like a door hinge, and the tree can only go where the hinge bends. Cut the hinge uneven and the tree steers toward the thick side; cut it straight and the notch's direction is the tree's direction.

It holds. While the tree tips, the hinge keeps the butt attached to the stump. That attachment is what stops the trunk from kicking back off the stump as the crown hits the ground — the butt stays pinned, the tree hinges down like a gate, and the fall is one motion instead of a throw.

It lets go. At the end of the fall the hinge's fibres tear, and the tree separates from the stump cleanly. The timing of that tear is set by the hinge's thickness — the feller's one real control over the whole event, and the reason the dimension is measured, not guessed.

Limbing without pinching

The tree is down. The next job is the crown, and it is the part where the saw gets pinched, because a downed trunk is a compressed spring in disguise.

The trunk lying on the ground is bent slightly by its own weight: the top side is in *tension* — the fibres stretched — and the bottom side in *compression* — the fibres squashed. Cut from the top down into that bend and the kerf opens as the tension side relaxes — safe. Cut from the bottom up and the kerf closes on the bar as the weight settles — pinched, stuck, and often the start of a fight with a half-cut limb under load. The rule: **read the bend, cut the tension side first, or cut from the outside of the bend inward.**

Limb the trunk in sections, cutting the branches off flush with the trunk but never into it — the bucking days want clean wood, not wounded wood. Work from the butt toward the top, keep the trunk between you and the slope, and expect every branch to move when its tension releases. A limb under a fallen trunk is loaded wood; the moment the saw frees it, it springs. Stand where the spring is not.

By the end of day one the first tree is a clean, bare trunk on level ground, the crown stacked where the firewood will be cut from it later, and the saw has done the most dangerous work it will do all fortnight.

What actually happened

The story half, recorded the way this book records everything that went wrong: named, not tidied.

The first tree went exactly where it was aimed. The notch was clean, the hinge held, and the trunk landed inside a step of the planned line, which the plan treated as a triumph and the book records as luck doing its job: the tree was straight, the wind was nothing, and the lean agreed with the aim.

The second tree did not go where it was supposed to.

It was the lean. The tree had grown with a slight bow toward the open ground, and the feller — one person, reading the trunk by eye — cut the notch for where the tree *should* fall and the back cut for where the hinge would let it. The hinge steered; the lean overruled; the tree fell a quarter-turn off the aim, taking a young neighbour down with it and landing with its crown in the brush instead of on the clearing.

Nobody was hurt. The escape route was where it was supposed to be, and the miss happened in slow, visible degrees rather than in one event — which is exactly what the hinge, even a mis-aimed one, is for. The lesson cost a morning: the second tree had to be worked out of the brush, the neighbour's remains cut clear, and the fall line for the *next* tree chosen by the lean alone, which is the rule the chapter opened with and the feller had half-ignored.

The book keeps the story because the story is the point of the chapter. The notch, the hinge, the escape route — they are not technique for a straight tree on a calm day. They are the margin for the tree that is not straight and the day that is not calm, and on day two the margin was spent exactly as designed: the tree fell wrong, and the feller walked away. Two trees down. The fortnight had begun.

Days Three and Four: Bucking

Cutting the trunk into the only length that matters

Bucking

The trees are down. Days three and four turn each bare trunk into 8 sections of 6 feet — the only length in the whole method, and the number that sizes everything downstream.

Bucking looks like the simple day of the fortnight and is not, because every section after it is the *whole* building in embryo: the section length sets the member length, the member length sets the dome, and a bucking error made on day three arrives on day fourteen as a frame that will not close. The chapter's job is the one measurement, repeated, without drift — plus the two unglamorous subjects that decide whether the day is a one-person job: where the sections start, and how a green six-foot round moves.

48 feet of trunk, cut into 8 sections, offcut 0.0 feet. That is the whole day, counted. The rest is doing it the same every time.

Marking and cutting to length

The procedure, built around the one fact that makes bucking hard: the trunk tapers, so a tape laid along the bark measures around a slope and over bumps, and drifts.

Mark with a stick, not a tape. Cut one 6-foot stick — a story pole — from anything straight, and mark every section with it. The stick does not stretch, does not creep, and reads the same at the butt and at the top. It is the jig before the jig: the length made physical, once, instead of re-read forty times.

Mark a line around, not a tick. One tick on the bark disappears into the cut. Mark the full circumference with the saw's paint or a lumber crayon, rolled around the trunk with a square laid flat, so the cut is guided from any side. A crooked crosscut is a section with sloped ends, and a sloped end is a wedge with one short face — the error that hides until assembly.

Cut from the top side of the bend. The downed trunk is bent by its own weight, exactly as the limbing chapter warned: cut from the compression side and the kerf closes on the bar. Roll the trunk or cut from above, and let each section's weight drop it free the moment the cut goes through. A section that hangs by a thread of bark is a section that will fall when it wants to; finish the cut.

Cut in the air, not into the ground. The bar through the trunk and into the dirt dulls the chain in one cut. Roll a short log under the cut line, or cut most of the way through and roll the trunk to finish from above. The chain's sharpness budget belongs to the ripping days, not to the ground.

Check the count. 8 sections per tree, counted twice — once as cut, once stacked. The count is the chapter's whole deliverable, and a missing section is a missing 8 members, which the frame will notice before you do.

Where to start, and why not at the butt

The instinct is to start at the butt and measure up. The method says start further in, and the reason is the taper's fault.

The butt end of a felled tree is not cylinder — it is flare. The root swell runs up the trunk a foot or two, and a section cut from the flare is an oval, its rings running wild, its first split a guess about where the pith actually is. So the bucking starts where the flare ends: stand the tape against the trunk, find where the diameter stops falling fast, and make that the first cut. The flare itself becomes firewood — the offcut of the plan below — and every section after it is true round, which is what the split count assumes.

The same rule finds the start around bends. The felling chapter left the trunk with its kinks in it, and bucking is where the kinks come out: cut the sections on the straight runs, leave the bent lengths between them as firewood, and count only straight sections toward the 8. The trunk does not owe you its bends; the plan knows they exist and pays them in firewood.

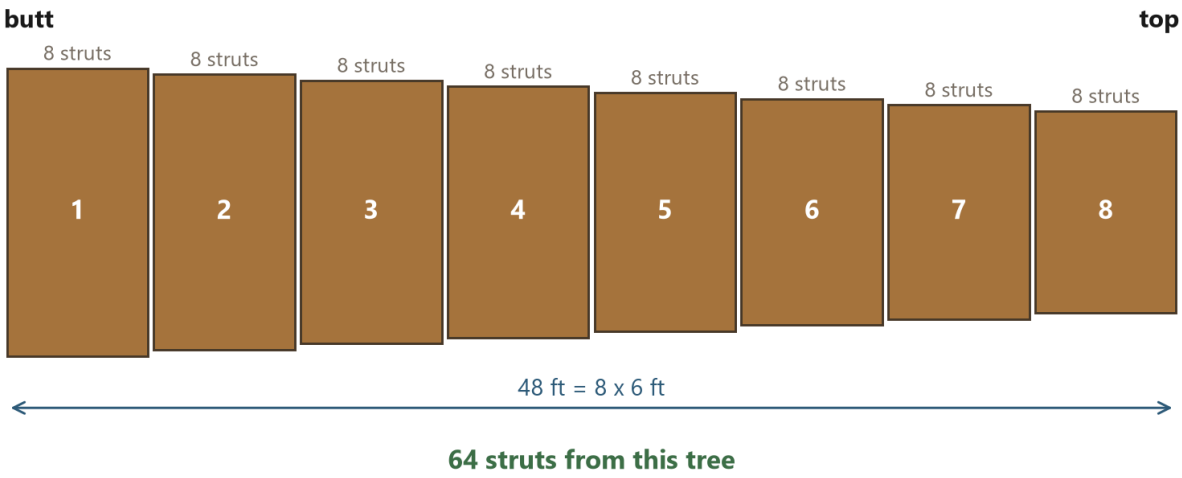
The top end, too, gets trimmed: the last section before the crown is where the diameter is smallest and the knots thickest, and if the crown side of it looks like a branch knot rather than a log, it joins the firewood pile. The usable length the tree chapter measured was never the whole trunk; it was the straight, flare-free, crown-free run, and bucking is the day the distinction becomes a pile of rounds.

Eight sections from one trunk

The drawing is the day's plan at a glance: 48 feet of trunk, laid out into 8 sections of 6, the offcut — 0.0 feet of flare and crown — marked off and consigned to the firewood pile.

Read it as the book's first real inventory rather than a diagram. Every section in the drawing is 8 future members; the whole trunk is 64 struts; two trunks are 192, against a frame that needs 120. The bucking day is where those numbers stop being the tree chapter's arithmetic and start being a stack of wood you can walk around. If the stack does not match the drawing — a section short, a count off, an offcut bigger than the plan — day three is the day to find out, when the fix is another section or another tree, not another month.

Bucking the trunk



Mark from the butt, not from the top: the tape drifts, and an error at the top is a short stick you cannot use. 8 sections of 6 feet leaves 0.0 feet over on this tree.

48 feet into 8 sections of 6. Offcut: 0.0 feet.

Moving six-foot green sections alone

The unglamorous logistics, because the method's one-person claim lives or dies on this afternoon, not on the geometry.

A 6-foot section of green pine is a wet, round, heavy thing — the butt sections near the flare can be a full day's load on their own. The method does not pretend otherwise; it moves them the way they want to be moved: **roll them**. A round section rolls; the only lift is the short end-over-end to clear a branch or a bank, and the ground does the carrying. Lay the bucking line out so the sections can roll to the splitting station — downhill is free, level is fine, uphill is the day's only real fight — and the "moving" half of bucking disappears into gravity.

For the sections that must lift: never a dead lift of the whole section. Roll it onto a low block, lever the end with a limb, or cut it into two shorter sections and accept the split-count change — the section length is a plan, not a law, and a back is not a substitute for a winch. The peavey, if one is on hand, turns the roll into a one-handed job; without it, the wedge and the lever do.

The real measure of day four is not the pile. It is whether day five can start without lifting a single section twice — the splitting station laid out, the sections staged where the ripping will happen, and the fortnight's only heavy lifting behind you. The bucking days end with 8 sections a tree, stacked dry-side-up, counted, and the saw's chain still sharp for the work that actually makes the members.

Days Five to Eight: Ripping

One hundred and twenty struts, thirty to an afternoon

Ripping

Days five through eight are the bulk of the labour and the heart of the method: every section of the two trunks, split along its length into 8 wedges, until the frame's 120 members exist as a pile. Two trees become a building on these four days; everything before was preparation and everything after is assembly.

The arithmetic sets the pace and the pace is honest: 5.0 struts an hour, sustained, means 24 hours of ripping — 4 afternoons — for the whole frame. That is the number the fortnight's reputation rests on, and it was measured on this build, not promised. The chapter is the measurement made into a procedure: the three passes at the log, the cradle that holds a round section still, the pith you aim at, the rhythm that sustains the rate, and the chain sharpness that the whole four days secretly depend on.

The three passes, at the log

The procedure of Chapter 12, now with the saw actually in hand, at production pace.

Stage the section. Every section gets stood on end or laid in the cradle before the saw starts — never ripped on the ground where it can roll, never half-rolled between passes. The staging is the first third of the work, and a section staged wrong costs more time than a cut made slow.

Pass one: halve it. Aim down the pith line on the end grain and rip the full length. The saw walks itself through a straight section if the bar starts true; the feller's job is to start true and let the saw cut — a ripping saw is steered with small corrections, not pushed. The two halves fall apart with flat faces down.

Passes two and three: quarter, then eighth. Each half sits on its flat face in the cradle and gets halved again; each quarter the same. Seven cuts per section, all of them "through the middle", and the pile of wedges grows by 8 with every section finished.

Stack as you go. The wedges go straight onto stickers, sawn faces down, bark up, where the drying chapter wants them — not thrown on a pile where the day's own moisture steams them. A wedge stacked right on day five is a straight member on day thirty; the stacking is part of the cut, not the cleanup after it.

Count the pile against the plan. 64 struts a tree, 192 for the two, 120 in the frame, 72 spare for the grading rejects. The count is checked at the end of every afternoon, because a shortfall found on day eight is a fourth tree; a shortfall found on day thirteen is a problem with no good name.

Holding a round section still

The first pass is the hard one for a reason the second and third do not have: a round section wants to roll, and a rolling log under a ripping saw is where pinched bars and ruined cuts are made.

The cradle is the answer, and it does not need to be furniture. Two stakes driven either side of the section, a pair of wedges under it, or a simple V of scrap timber along its length — anything that stops the roll and leaves the top of the section at a working height. The cradle's job is negative space: it holds the log where the saw is not. Once the first pass is made, the halves sit on their flat faces and the cradle's work is mostly done — flat stock on a flat bench is stable by itself, which is the second reason the halving pass goes first, every time.

The wedges earn their place on the *first* pass and again at the end. Driven under a section, they level it, lift it off the dirt, and take the weight so the saw's kerf stays open. A section that settles as you cut closes the kerf behind the bar — the saw bogs, the cut wanders, and the pass is ruined in its last foot. The wedge set under the cut line is the cheapest insurance in the method.

Reading the pith

Every rip aims at the centre of the section, and the centre of a real section is not a point — it is a small, dark, slightly off-centre dot in the end grain, and the first skill of the ripping days is finding it.

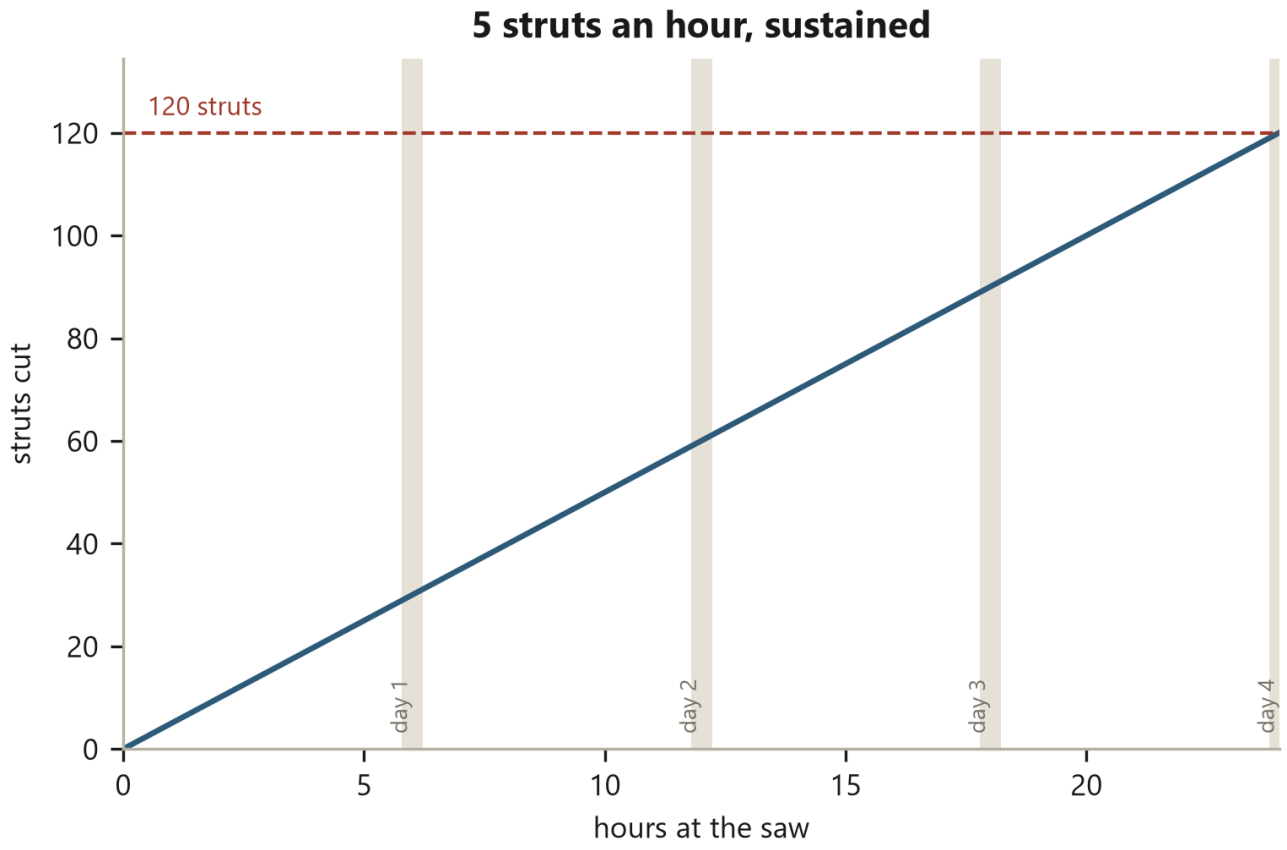
The pith is the tree's first year: a soft spot, usually darker than the rings around it, not always exactly centred. The split geometry does not care that it is slightly off — the sectors of a slightly off-centre split are nearly identical, and "nearly" is what the method budgets for. What matters is that every pass goes *through* the pith, so every sector keeps a point that is honest wood, not a rag of pith off to one side. A sector whose point misses the pith by an inch is a member with a soft, split-prone edge exactly where a member most needs honest wood.

On a section that is not round — the oval butt, the flat-sided trunk — the pith still rules. Aim at the centre of the rings, not the centre of the shape; the rings are the tree's own record of where its middle is, and they are right more often than the outline is. The first pass is started from a stance where the pith is visible the whole way — sighting down the bar, checking every few feet — because a pass that starts an inch off arrives a foot off, and a wedge is six feet long.

The cutting rhythm

The chart is the claim and the proof together: 5.0 struts an hour, sustained, for six hours, afternoon after afternoon. It is not a sprint rate; it is the rate the body can hold while the saw cools between cuts, the pile is stacked as it grows, and the count is kept.

Read the rhythm as a sequence of stations rather than a series of cuts. Stage, halve, quarter, eighth, stack, count — one circuit per section, the same motions in the same order, the saw never waiting on a decision. The chart's flatness is the point: a sustained rate is a rate with no thinking in it, and the method was arranged so the thinking happened once, in these chapters, and the ripping days just repeat. The afternoons that beat the rate are the ones where something was skipped — a staging, a stacking — and the debt shows up the next morning as a pile to sort and a chain to sharpen before the first cut.



4 afternoons of 6 hours. This is one measured afternoon extrapolated in a straight line, which is optimistic: it assumes the last hour of day four goes like the first hour of day one.

5.0 struts an hour, sustained.

Chain, file, and the cost of a dull cut

The four days' secret dependency: a ripping chain is either sharp or it is spending your arms and your fuel, and the difference is visible in the pile.

A dull chain does not stop cutting. It starts *pushing* — the saw walks off the line, the kerf widens into a groove, and the bar works hot while the feller works harder. The kerf is the honest ledger: every cut is a 0.25-inch-wide line of sawdust, and a dull chain turns that line wide and ragged, eating the recovery figure Chapter 13 computed, one cut at a time. The file is the fix and the cadence is the rule: a few strokes per tooth, every few tanks, *before* the pushing starts, never after. Touch-up filing at the bench costs minutes; a chain sharpened only when it fails costs the afternoon it fails in.

The practical cadence on this build: file at the midday fuel stop, whether or not the chain asks. The ripping days are the saw's hardest work in the whole method — bar buried, grain sideways, wet wood — and the chain that starts the afternoon sharp finishes it dull. The file is the one tool in the fortnight that is used *more* than the saw.

What this is worth an hour

The sidebar's job is to put the four days' labour in the only currency that survives the fortnight, and then to put the caveat beside it.

Splitting your own struts is work, and the work has a rate. The full accounting lives in Chapter 54 — the strut valued three ways, the disagreement between the ways kept in the table, and the rate computed against the dressed board that is the honest comparison. Here the summary, once: an hour at the log produces members worth more than the hour's alternative at a wage, *if and only if* the alternative was going to be buying that timber — and the substitution value is not money anybody has been paid. It buys groceries only if the timber was going to be bought anyway.

The ripping days feel like the part of the fortnight where the money is made, and in substitution terms they are — the frame's timber is the single largest line the build avoids paying. But the rate is a rate on *your* hours, priced against *your* alternative, and the chapter that computes it is careful to say which parts of the comparison flatter the method and which do not. The ripping chapter's own verdict is narrower: the four days are the price of the timber, the price is real, and it is paid in afternoons instead of invoices. Whether that trade is worth it is Chapter 54's question, asked properly, with the numbers up.

Day Nine: The Jig

One flat bench that makes forty panels the same

The Jig

Day nine is the day that pays for the other thirteen, and it is a single day spent building a single thing: one flat bench on which every one of the frame's 40 panels will be assembled, identical to the last.

The jig is where the method stops being a pile of good ideas and becomes a production line. The wedges are cut, and they are nearly alike — nearly, because the tree was nearly round, the saw nearly straight, the eye nearly true. The jig does not need them to be exact; it needs them to be *held* the same way while the exact parts — the butt cut, the head length, the screw positions — are established by the fixture instead of by the eye. A jig is a decision made once and repeated forty times in wood, and the rest of the book's build chapters are that sentence, carried out.

The full reference for the fixture — every stage, labelled, straight from the simulator — is Chapter 33. This chapter is the story of the day it got built, and the argument for why a day is the right price.

Why a jig and not a tape

The temptation on day nine is to skip the bench and measure the panels one at a time: lay three sticks in a triangle, tape them to length, screw them. It works — for the first panel. The question is what the fortieth panel looks like.

A tape measures each panel against an ideal that lives in the builder's head, and no two measurements of a rough stick agree with each other. The error compounds quietly: a sixteenth here, a thirty-second there, and the panels drift apart in ways that only announce themselves at raising, when two neighbours built a week apart refuse to share a seam. The frankendome — Chapter 3 — was forty panels built exactly this way, and its lesson was the whole reason this book exists: **nothing repeated, so nothing could be checked against the last one.**

A jig inverts the logic. The fixture holds the *position* of every stick — the triangle's shape, the cut lines, the screw locations — as physical stops, and the builder's job becomes loading the stick against the stop. The measurement happens once, when the bench is built; after that, the panels are not measured at all, they are *located*, and locating is an operation that cannot drift, because the stop does not move. Panel forty is built the same way panel one was, and the only difference between them is the sticks — which is exactly the difference the method is designed to tolerate.

Repeatability beats accuracy, when there are forty of something. A jig that is half a degree wrong builds forty panels that agree with each other, and the frame closes; a tape that is read forty times builds forty panels that disagree, and the frame does not. Day nine buys the agreement.

Building it

The bench, in order, kept deliberately simple — the jig's only requirement is flatness, and everything else is stops.

The bench. A flat surface bigger than the largest panel: two sawhorses and a sheet of plywood are enough, levelled and braced so it cannot rock. Flat is the one property that matters, because the panel's truth is measured against the bench; a bench with a twist in it builds a twist into every panel that lies on it.

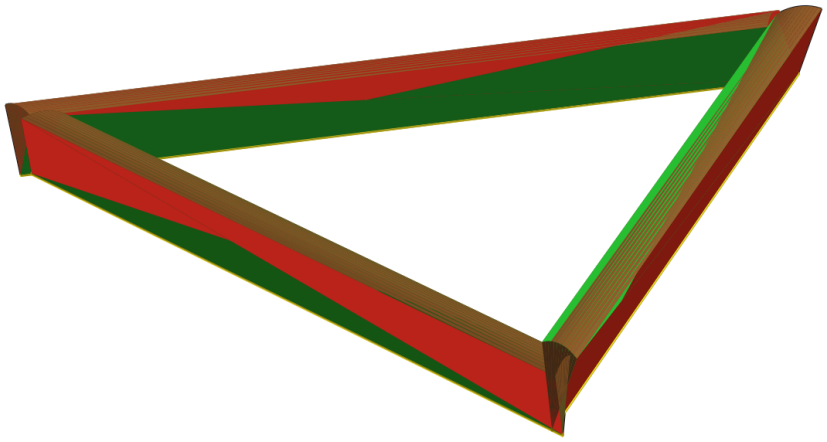
The triangle. Lay out one panel full size on the bench — the actual member lengths from the cut list of Chapter 24, the actual gasket allowance — and screw down stops along all three outside edges: the triangle's outline, made of scrap, exactly where the members will lie. The stops are the jig's first truth, and they are set once, from the drawing, with a square and a story stick — the tape's last appearance in the whole fortnight.

The fences. Inside the triangle, two more lines of stops mark where the head ends will be sawn off and where the butt ends land: the inset of Chapter 15, built as wood instead of remembered as a number. The fence is the cut line, physical, and the saw runs along it the way a saw always wishes it could.

The cut planes. Where the saw must cut, the jig provides the plane: a straight edge the bar rides against, set at the butt's bevel and the head's square. Chapter 34 gives the angles; the jig holds them; the builder never sights one.

The clamps. Whatever holds a wedge still against the stops while the screws go in — quick clamps, wedges, a foot on a pedal. The member must arrive at its stops and stay; how it is persuaded is the builder's business.

The bench takes a day, and the day is not wasted on accuracy — it is spent on *position*, which is the cheaper thing to be right about. A stop screwed in the wrong place is moved in a minute. A panel built in the wrong place is discovered at raising, in front of a crowd of its neighbours.



The fixture every panel is built on.

The jig, complete

The plate is the finished fixture with a panel loaded on it, at the moment the long heads are being sawn off against the fence — the stage of the work where the jig's whole argument is visible at once.

Look at what is holding the truth. The sticks are rough — bark backs, saw marks, the ordinary drift of split timber. The triangle they lie in is exact, because the stops are. The butt ends are where the fence says, the heads run long past the cut plane, and the saw is about to make the only cuts in the whole panel that have to be right — and they will be right, because the plane the saw rides is the plane the bench made, not the plane a hand held.

That is the day-nine bargain, paid in one afternoon and collected for the rest of the fortnight: from tomorrow, nobody measures anything. Chapter 31 is the production run the bench was built for, and the run is the proof — 40 panels, one after another, each one located, cut and screwed in the same twelve motions, and each one leaving the bench the same panel.

Days Ten to Twelve: Forty Panels

The same twelve motions, forty times

Forty Panels

Days ten through twelve are the production run the jig was built for: 40 panels, one after another, on one bench, from the pile of wedges the ripping days left.

The run is where the method's promise becomes arithmetic. One panel is twelve motions — load, locate, butt, head, drill, screw, flush, out — and forty panels of twelve motions is a production line with one worker, one station, and no waiting. The days go wrong only when the motions stop being the same: a stick sorted into the wrong class, a marking that washed off in the rain, a panel started before the jig was checked. This chapter is the procedure, the sorting, and the two honest notes that make the run real — the fourth panel is the first good one, and the variant count is smaller than the pile makes you fear.

One panel, start to finish

The twelve motions, in order, as they will run forty times. Chapter 33 names the same stages from the simulator's side; this is the version your hands do.

Check the bench. Stops tight, cut planes clean, clamps where the hand expects them. Thirty seconds, every morning, before the first panel — the jig's truth is the day's truth.

Sort the three sticks. One long member and one short for the sides, one more of whichever the triangle asks, drawn from the marked piles of the sorting pass below. The sticks arrive at the bench already decided; the bench never decides.

And for an isosceles panel, *which* side each stick goes on is fixed: **the longest — 72.0 inches — is the base, the middle length (63.5) is the left side, and the shortest (61.5) is the right.** Lay the base nearest you and the left member is the longer of the two: it reaches over to the top point, and the short right side runs up to meet it — and it is that short side the base butts up to. An equilateral panel is three long members and has no such rule to remember, which is most of why the run is quick: 10 of the panels are the easy kind, and the 30 isosceles ones carry the one instruction that must not be swapped. At the bench the two sides look almost identical — the difference is two inches over five feet — so the rule is applied by picking from the marked piles in order, never by eye.

Load the first side. Butt against the butt stop, sawn faces down on the bench, head running long past the head fence. The stop does the locating; the hand only presses.

Butt-cut if it is due. Some classes carry their butt angle from the saw; some are cut at the bench. Either way the cut is made against the plane the bench holds, never freehand. Chapter 34 is that cut in full.

Load the second and third sides. Each lands against its own stops, butts to butts, and the triangle assembles itself under the clamps — three sticks, three overlaps, the pinwheel of Chapter 15 appearing without anybody arranging it.

Drill and screw the overlaps. Through the outer stick into the inner, at the spread the connector chapter established, pulling the overlap closed. The screws are the only fastening in the whole panel, and the jig's holes — or the fence's positions — make them land the same on panel forty as on panel one.

Flush the heads. The three heads, still long, are sawn off against the head fence: one plane, three cuts, and the panel is exactly the triangle the bench describes. Chapter 35 is why the heads were long in the first place; here it is simply the last motion of the panel.

Lift it out. The panel comes off the bench as one finished thing — 3 members, their own edges, ready to be checked, marked and stacked. That is the whole panel. Twelve motions, measured in minutes once the rhythm is in.

Sorting the pile

The run only stays a run if the pile does not get to decide anything at the bench, so the sorting pass happens once, before the production days, with a marking scheme that survives the weather.

Sort by class, not by stick. The four member classes of Chapter 9 each get their own stack, marked at the cut — a crayon code on the sawn face, one stripe per class, one colour per tree. The bench pulls from stacks; it never chooses between sticks.

Mark the faces, not the ends. The ends get cut off. The sawn face is where the mark lives through every operation that follows, and it is the face that lies *down* on the bench, so the mark also reads as an orientation cue: mark up, bark up, wrong way round is visible before the clamp closes.

Mark with what survives rain. Lumber crayon, not pencil. Pencil washes out of wet pine in one night; the crayon is the marking scheme's entire weatherproofing, and the fortnight has already had one rain.

Run the grading passes as you sort. The four passes of Chapter 19 — bend, knot, bearing, pith — are the same walk as the sorting pass, and doing them together means every stick reaches its stack already judged. The 72 spares the tree chapter promised exist as the reject pile, and the reject pile is the margin the run spends when a stick misbehaves at the bench.

The fourth panel is the first good one

The honest note, stated early so nobody panics on day ten.

The first panel of the run is a rehearsal. The motions are new, the clamps are in the wrong places, and the result is a panel you would not hang a roof on. The second panel is faster and nearly right. The third finds the rhythm and the fourth is the first panel that comes off the bench exactly the way the fortieth will — and the run's real output is not forty panels, it is the thirty-seven that follow the learning curve.

The book says this plainly because the curve is the plan, not a failure of it. The method's whole claim is that the motions repeat, and repetition is what turns the first three panels from mistakes into tuition. The rejected first panels are not

waste; they are the cheapest practice the build will ever get, and they end their lives as the jig's own test pieces — the panels you cut into, shim, and check the saw against, so the bench's truth is probed by the very thing it built.

Panel variants

The table closes the chapter with the count that keeps the run sane: how many genuinely different panels the dome contains.

Not forty. The frame's 40 panels are two shapes — 10 equilateral, 30 isosceles, the counts of Chapter 8 — and each shape is built from the same four member classes, so the run is two panel variants in a repeating pattern, not forty originals. The table is generated from the solved dome, and its job at the bench is the same as every table in this book: the pile looks like chaos, the count says otherwise, and the count is right.

By the evening of day twelve the run is done: 40 panels, stacked by variant, the reject pile spent, the bench quiet. The fortnight has two days left, and they are the days the photographs are taken on — the raising, where forty flat triangles become the building the next chapter describes.

Days Thirteen and Fourteen: Raising

Forty panels become one building

Raising

The last two days of the fortnight are the ones the photographs are taken on, and the ones the book is most careful about, because raising is where a frame that was built flat, checked flat and trusted flat suddenly has to stand in three dimensions — and where every error the earlier chapters let through comes home at once.

The job: 40 finished panels become one standing hemisphere, in an order that lets one person do it, held up by temporary props until the shell closes and holds itself — and then, in this book's build, winched off the ground entirely: three cables from the apex hanger to saddles on the trees, the brake winch doing the hoisting, and the cabin hanging where the trees hold it, with no footing under it at all. The order matters more than the effort. A dome assembled in the wrong order needs a crane and a crew; assembled bottom ring first, course by course, it needs one person, a ladder, props and a winch — the same economy the whole method has been spending since the tree.

The chapter ends with the honest inventory of what a hanging frame is not, because day fourteen feels like a finish line and the book has four more parts to go.

Bottom ring first

The assembly order, which is the whole secret of a one-person raising.

Set the ring. The ten rim panels go down first, laid out flat on level blocks — or on the footing of Chapter 39, for a build that stays on the ground; this book's build uses blocks, because its ground work comes last. The ring is the only part of the frame that touches the ground, and it is the one part that must be right before anything rises off it — its 10 edges set the circle, and the circle sets everything above. Chapter 38 names the ring as the one place error can accumulate, and the check is done here, at the start, when a wrong ring is still a morning's fix: measure the ring across its widest points, both ways, and do not build up until the two numbers agree.

Course by course. The next ring of panels sits on the first, each new panel's base edge meeting the top edge of the panels below it — the seam, the key, the screws, the same joint the bench built, now vertical. Work around the dome, not up one side of it: a dome grows in levels, and a level that closes before the next begins is the prop that holds the next level. The order is not optional; a half-raised dome is a sail, and the wind has opinions.

Key and screw as you go. Every seam gets its key before the neighbouring panel arrives — a seam closed later is a seam closed from a ladder, reaching into a gap that has already tightened. The keys are the schedule of Chapter 16, and the schedule is followed in place, panel by panel, the count climbing toward 55.

Check the ring, again. At every course, one measurement across the open top tells you whether the dome is coming up round or egg-shaped. The check costs a minute and it is the difference between a raising and a rescue; the dome that fails it today gets propped today, not discovered tomorrow.

Propping, and what holds it up before it is closed

The photographs show the finished frame. The raising itself is a building held up by sticks, and the props are the chapter nobody writes.

A dome does not carry itself until the top closes. Until the last panel ties the structure together, every course above the ring is leaning its weight on the course below it, and the courses below are leaning on props — lengths of scrap set against the inside, butted to the ground, holding the shell at its current height. The props are the temporary structure the method admits it needs: not scaffolding, not a crew, just the same wood the frame is made of, pressed into service as furniture.

The prop rule is simple and absolute: **a panel goes up, a prop goes under it, and a prop never comes out until the course above it has closed.** The dome rises prop by prop, and the props travel — pulled from the lower course once the ring above it is screwed closed, carried up, reset. It looks like chaos in the middle and it is not: every panel in a properly propped raising has a load path to the ground at every moment, which is exactly what the finished frame will have, temporarily, in wood.

The one danger the props exist to prevent is the panel that leans out. A panel set at the wrong angle, or a course started before the course below it closed, will try to fall *outward* — and a falling panel takes its neighbours' seams with it, opening the dome like a flower in reverse. The prop is the answer and the order is the prevention; together they make the raising a sequence of small, held, stable moments instead of one long gamble.

Closing the top

The last panel is the hardest, and it is worth saying why before the raising reaches it.

Every seam closed so far was closed against air — the next panel could be offered up, adjusted, persuaded. The last panel closes against a *hole*: every one of its three edges lands on a seam whose other side is already fixed, and the panel must fit the hole the whole dome has been building toward for two days. If the ring was set right, the courses keyed right and the schedule followed, the hole is exactly the panel — the method has been aiming at this moment since the first butt cut. If anything drifted, the drift lives here, in the last triangle of sky.

The order of the last panel's edges matters more than its screws. Offer the panel up gently, key the lowest seam first, then one side, and let the last seam close under the panel's own weight — a final seam that needs forcing is the dome telling you where the error went, and the answer is never more force; it is a shim behind a key, or a prop relieved, or, honestly, a panel taken down and the neighbour adjusted. The method's tolerance budget — Chapter 38 — was built for exactly this conversation, and the conversation is short: the frame closes, or it does not, and it almost always closes.

Then the screws go in, the props come out, and the frame stands on nothing but its own shape — until the winch takes it.

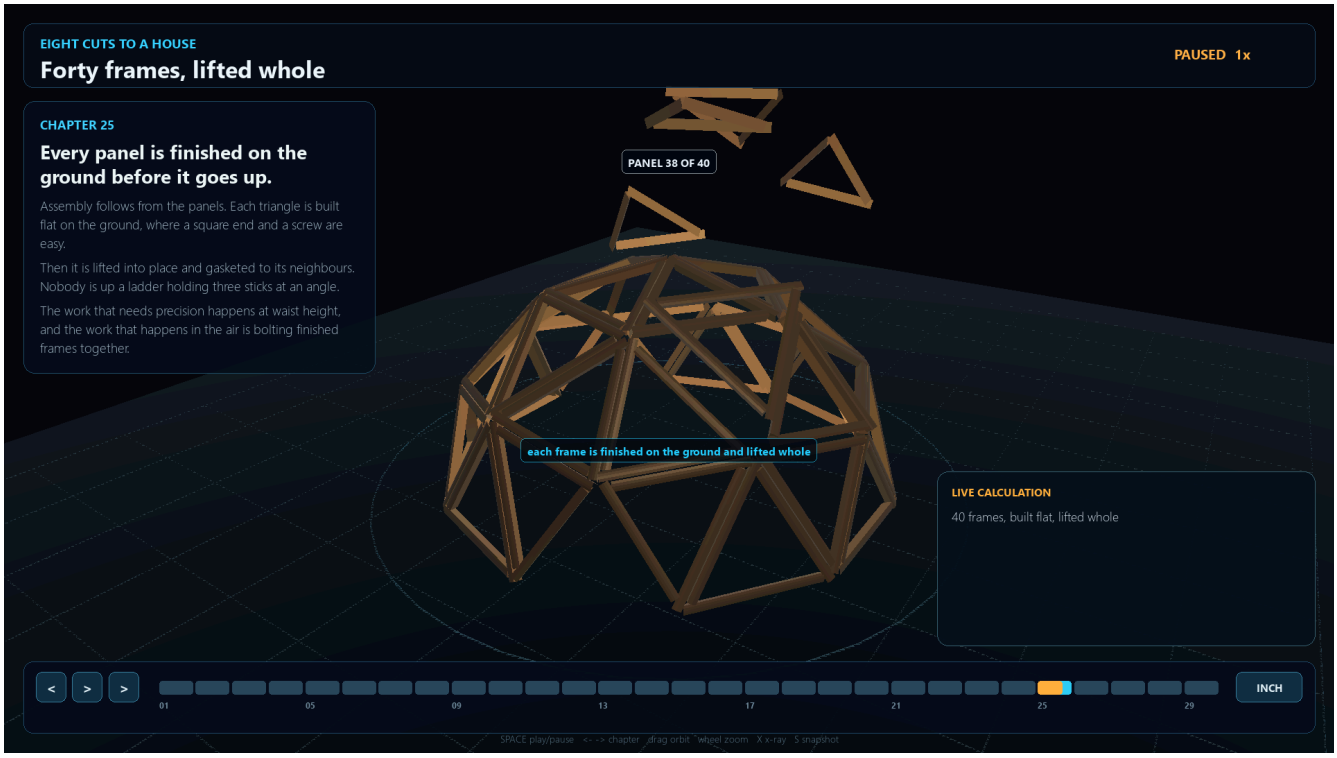
The winch, and the trees

This book's build does not leave the frame on the blocks. It hangs it, which is the step the title page promised and this section reports.

Three cables run from the apex hanger — the mast's lifting ring, or a temporary sling through the crown on a first build — to saddles on two of the three trees: two legs to one tree, one to the other, because a tripod of cables on two anchors is what keeps a hanging load from spinning. The brake winch does the hoisting, one slow notch at a time, the frame lifting off its props the way a boat lifts off its trailer, and the moment the props fall free the cabin hangs — the trees holding the whole weight of it, the rim swinging a few feet above the ground, and the builder standing under something that yesterday was a pile of wedges.

Say it plainly, the way the floating-dome chapter does: this is a tested construction and a design possibility, not an engineered rating. The trees, the saddles, the cables, the mast and the winch all carry real loads, and none of those ratings is in this book — the engineer's sentence from Chapter 10 binds here harder than anywhere else in the build. The saddles are saddles, not holes: straps that spread the load over bark, so the trees carry the cabin the way they carried their own crowns.

What hanging buys is the deferral the book has been promising: the ground is now *available* — no footing, no ring beam, no pad is owed, because the cabin does not touch the earth. The floor clamps to the mast when it is bought (Chapter 71), the skin goes on in the air or on a lowered frame, and the foundation, when it comes, is the last decision of all (Chapter 39). The winch did not just raise the cabin. It un-scheduled the ground.



Day fourteen. 120 members, 3 trees.

Standing

The plate is day fourteen: 120 members, 3 trees, a fortnight — the frame this book has been describing since the first chapter, standing where the tree once stood, before the winch took it up between the other two.

Look at it for what the book wants you to see. No hubs. No mitred corners. Forty panels that each remember the bench. The seams running in their two angles, the keys seated, the rim on the blocks. It is the same object the simulator solved before the first tree was felled, and the fact that a solved model and a fortnight's labour can produce the same standing thing — and then hang it — is the method's whole argument, in one picture.

And the next section is the book's own correction of this photograph, because the photograph lies in the way all finished things lie.

What was not done

The honest inventory, written while the frame is still warm from the raising, because day fourteen feels like an end and is not one.

The hanging frame is **not a shell**. It has no skin, no weather barrier, no insulation — the seams are open to the sky's weather, and the first real rain will find the book's weakest chapters and test them. Part Seven — Chapters 40 through 45 — is the work that turns this photograph into shelter, and it is a season's work, not a fortnight's. The skin can go on in the air or on a lowered frame; the order is the builder's, which is the point.

It is **not a home**. No floor but the air under it, no door but the gap where a panel is missing, no power, no water, no heat. The frame is the part of the building the fortnight was sized for; the rest of the building is sized in months, and the book's later parts are honest about that calendar.

And it is **not finished** — it is *standing*, or rather *hanging*, which is the fortnight's actual deliverable and no more. The dome will move as its green wood dries, the keys will settle, and the frame the builder checks on day thirty is not quite the frame of day fourteen. That settling is a real event in the build's first year, and this book's habit is to name it, measure it where it can be measured, and leave it in the frame rather than tune it out.

So the fortnight ends the way it began — with an honest sentence. 3 trees, one saw, fourteen days, 37.5 hands-on hours: the frame is up, and up between the trees. The rest of the book is what the frame makes possible, and the first thing it makes possible is the next chapter, where the method's reference pages begin.

CHAPTER 33 · REFERENCE

One Bench, Forty Panels

What the jig is and what each part of it does

One Bench

This is the reference chapter for the fixture itself: what the jig is, part by part, straight from the simulator that models it, so the bench on your floor and the bench in the model cannot disagree about what a jig is.

Chapter 30 told the story of the day it was built. This chapter is the thing, cold: the twelve stages a panel passes through on it, the labelled drawing of the fixture before any wood is on it, and the procedure for laying the whole thing out full size on a real floor with nothing but string and a pencil. Come back to this part while you are working, not while you are reading — the book's reference chapters are written for the bench, and this one lives there.

The twelve stages

The table is the jig's whole working life, generated from the simulator's own stage list: every stage a panel passes through between the raw pile and the finished triangle, named and ordered.

Read it as a checklist, not as a description. The twelve stages are the production run of Chapter 31, expanded to the resolution the bench actually works at — where each member is located, what gets cut when, what gets screwed when. The order is not optional in the way a recipe's order is optional; the head overfit of Chapter 35 only works because the flush cut happens at the stage the table says, after the panel is already a panel and before it leaves the bench. A stage skipped is an error invented, and the table exists so the skip is visible.

The twelve stages of the jig

#	stage	what happens
1	1. THE BARE BENCH	One flat sheet on sawhorses, and nothing else. This board is the only thing in the whole build that has to be true.
2	2. THE TRIANGLE, DRAWN ONCE	The panel's mathematical triangle, snapped onto the board. It is drawn one time, from the geodesic solution, and never measured again.
3	3. AXIS RAILS	Three low cleats, one per member, each set parallel to that member's point-axis and just to the seam side of it.
4	4. RED / GREEN ORIENTATION FENCES	Two short cradle plates per member, standing against the two sawn radial faces of the wedge.
5	5. THE TWO CUT PLANES	The magenta plate at each head end is the flush-cut fence. The cyan plate is the receiving face the next stick's butt will land on.
6	6. THE BUTT CUT, MADE OFF THE JIG	One stick, still full length at both ends, laid up to the cyan plate. The amber tail past that plate is the stock the butt cut removes.
7	7. FIRST STICK ON THE BENCH	The same stick, butt now cut, dropped against its rail and its two fences. The head end is still long and hangs off the board.
8	8. SECOND STICK, BUTT ONTO SIDE	The second stick lands so that the FIRST stick's butt bears on its radial face -- on the side of it, not on its end.
9	9. THE LAST STICK SLIDES IN	The third stick is captured at BOTH ends once the other two are down, so it cannot drop in from above. It is shown backed off along its own axis, ready to slide home into the pocket.
10	10. PULLED UP TIGHT	Home. Three butts bearing on three sides, the green outlines are the actual contact faces, and all three heads still hang off long.
11	11. FLUSH-CUT THE THREE HEADS	Each head is sawn back to its magenta fence. The orange stock hanging past the board is what falls off.
12	12. PANEL OFF THE JIG	A finished triangular frame, its three corners clean inside the mathematical triangle. The jig is empty and unchanged.

Named by the simulator, so pressing 9 and 0 in the Raw Wedge Dome tool walks exactly this list.

The twelve stages, in order.

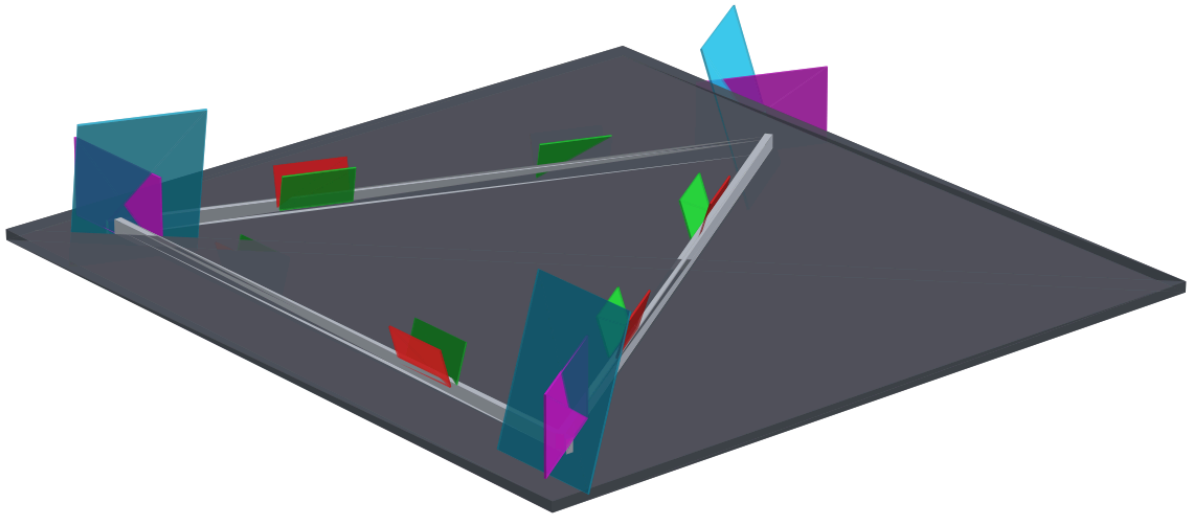
The jig, labelled

The drawing is the fixture before any wood is on it: the bench, the triangle, the rails, the fences, the cut planes — every component called out, each with its one job.

The bench is the flat truth everything else is measured from. **The triangle** is the panel's outline, built in stops around the outside. **The rails** carry the saw where a cut must be made. **The fences** are the lines the butts land against and the

heads are cut against. **The cut planes** hold the angles of Chapter 34 so no hand has to. Between them they do the one thing a jig must do: they replace every decision with a position, and the position does not move.

The drawing comes from the simulator's own jig model — the same geometry the fabrication package exports — so the picture and the fixture are the same object, drawn twice. If the bench you build disagrees with this drawing, one of you is wrong, and the simulator is not the likely one.



Bench, triangle, rails, fences, cut planes.

Setting it out on a real floor

The full-size layout, because the jig's accuracy is bought on this afternoon and nothing after it.

Start with the longest member. Draw the first side of the triangle full size, straight off the cut list of Chapter 24 — the actual length, gasket allowance included, marked on the bench surface with a pencil and checked twice. This line is the triangle's truth; the other two sides are located from it.

Swing the arcs. From each end of the first line, swing an arc at the length of its neighbouring side — string and a pencil, the oldest layout tool there is, and still the straightest for distances bigger than a square. Where the two arcs cross is the third corner. The triangle is now on the floor, full size, and the only measured numbers in it were the three member lengths.

Check the angles against the drawing. Before any stop is screwed down, compare the laid-out triangle against the panel drawing of Chapter 37: the base angles, the heights, the relation between the long and short sides. A layout that matches the drawing within a pencil line is the jig's foundation; a layout that does not is the cheapest error the whole build will ever catch.

Screw the stops. The outline becomes physical: stops along all three outside edges, then the fences inside for the butts and the heads, then the cut planes. Each stop is set against the pencil line, not against the previous stop — a stop set from a stop compounds its neighbour's error, and the whole point of the layout is that every position is measured from the drawing once.

Prove it with the first panel. The first panel off the bench is the test: lay it back over the layout lines, and if its edges fall on the pencil, the jig is true. The fourth-panel rule of Chapter 31 applies here too — the first panel proves the bench, and everything after trusts it.

Locate from the sawn faces, never the bark

The last section of the reference, and the principle that makes a jig work on natural timber at all.

Bark is irregular. No two sticks share it, it compresses under a clamp, and it is the one surface of the wedge the saw never touched. The two sawn faces are the opposite: flat, because the saw made them flat, and at the same angle on every stick, because every stick is the same sector of the same circle. So the fixture locates the member on its sawn faces, and only on its sawn faces. The bark floats in an oversized clearance and is never asked to locate anything.

The rule reads as fussiness and is the whole tolerance budget of the method in one sentence: precision where precision matters, and tolerance where the tree varies. The sawn faces are the precision; the bark is the tree varying; and the jig that confuses them builds forty panels, each as individual as the log it came from — which is the frankendome, again, wearing a better bench.

Chapter 38 follows the same principle out into the whole build. This chapter ends it where it starts: one bench, two datum faces, forty identical panels.

The Butt Cut

The one compound angle, made before anything is assembled

The Butt Cut

Name it honestly at the top, because this project got it wrong in public once: the butt cut is a compound cut. A mitre and a bevel together, on one end of every member, made once, off the jig, before anything is assembled. The claim this book makes is not “no mitres” — Chapter 15 has the full correction of that claim. The true claim is better: no member end ever meets another member end, and the one compound cut the frame does need is made once per stick, against a fixed setup, instead of twice per stick against a guess.

This reference chapter is that cut: where its angle comes from, how few distinct setups the whole dome actually needs, and the procedure for making it with the allowance still attached. It is the shortest of the jig chapters and the one the errata pages point at most, because it is where the book’s honesty and the saw’s geometry finally agree.

The angle its neighbour presents

The butt end of a member is not cut square to the stick’s length. It is cut to match the two facts about the neighbour it will land on: the neighbour’s side leans away from the panel’s plane by the sector’s angle, and the panel itself folds toward its neighbour by the seam’s angle. The cut that lands flat against both is a compound — one angle in the plane of the panel, one across it — and that compound is the butt cut.

Where the angles come from, precisely: the member’s sawn face sits 22.5 degrees off the member’s centreline, so the butt must present that angle to sit flat on its neighbour’s face. The panel’s fold adds the seam’s angle on top — the 18.03 and 22.46 degrees of Chapter 16. Together they make the one cut the dome cannot avoid, and the reason it cannot be avoided is the same reason the seam key exists: the sphere curves, the panels fold, and the wood that meets that fold must be shaped for it.

The good news, stated as a count: the whole dome needs only a few distinct setups, not one per member. The bevel is the same everywhere, and the mitre takes only the values the member classes demand. The table on the next page is the complete list, generated from the simulator’s own setup schedule — and the fact that the list is short is the second half of the correction the pinwheel chapter opened: the compound cut is real, and it is *few*, and “few, fixed, and made on the jig” is the honest version of the claim the film got wrong.

Butt cut setups

The table is the entire saw-setting vocabulary of the build: every distinct setup, straight from the simulator’s own list, with the member classes each one serves.

Read it as the answer to the question the sceptic is owed: if the butt is a compound cut, how many compounds are we talking about? The answer is the table's length — small enough to write on the wall above the saw, fixed enough that a mistake is a failure to read, not a failure to calculate. The saw is set once per setup, the class is cut through, and the setting is never re-derived at the bench. That is the discipline the whole method runs on: measure the angles once, on the drawing, and let the jig hold them forever.

Distinct butt-cut setups in the whole dome

#	members	bevel deg	mitre deg
1	-	22.50	34.43
2	-	22.50	21.14
3	-	22.50	30.00
	3 setups	**120 members**	

3 setups for 120 members. That ratio is the whole argument for the pinwheel: you change the saw a handful of times, not a hundred and twenty.

How few distinct setups the whole dome needs.

Making it

The procedure, at the saw, with the allowance still attached — because the butt cut is never made on a stick that is already its final length.

Leave the allowance. The member arrives at the butt cut with the spare stock of Chapter 35 still on its head end and the butt allowance still on its butt end. The allowance is not sloppiness; it is the room the cut needs to be made *against the jig* rather than against a tape, and it is the reason a mis-aimed cut is a do-over instead of a ruined member.

Set the fixture, not the saw alone. The butt cut is made with the stick located on its sawn faces in the jig's cut plane — the saw's own angle settings do the compounding, but the *position* of the cut along the stick comes from the stop, and the stop is the jig's. A butt cut made freehand is a compound angle guessed twice; a butt cut made in the fixture is a compound angle held once.

Cut, check, and only then cut the class. The first stick of a setup is the proof: cut it, offer it to its neighbour, and read the contact. A butt that sits flat across the neighbour's face is the setup's certificate; a butt that rocks on one edge sends the saw setting back one degree and the proof is repeated. Only when the first stick passes does the class get cut — the same rule as the jig's first panel, applied to the saw.

Keep the allowance until the member leaves the bench. The butt allowance is removed at the stage the jig schedule says, not at the saw. Between the butt cut and the bench, the allowance is still protecting the cut — and the head overfit is still protecting the length. The two spare ends are the method's crumple zones, and Chapter 38 is the accounting of everything they absorb.

One cut, made once, on one end, off the jig. The butt cut is the whole of the frame's precision joinery, and it fits in a table, a fixture, and an afternoon — which is what "no two ends ever meet" was always worth.

The Head Overfit

Deliberately cutting it too long

The Head Overfit

The head overfit is the idea that makes the whole method tolerant, and it is the single most counter-instinctive rule in the book: **cut every member deliberately too long.**

A builder trained on dimensional lumber works measure-cut-fit: measure the stick, cut it to the number, fit it into the building. Rough timber will not survive that order. The stick is not the shape the tape assumed — it bows, it tapers, its butt lands where the neighbour's face happens to be — and a stick cut to the plan's length before it meets the building is a stick that arrives wrong. So the method reverses the order: cut long, locate in the jig, and let the fixture — not the tape — decide where the end actually falls. The overfit is the deliberate slack that makes the reversal possible, and this chapter is its economics: where the error goes, how much to leave, and what the tolerance costs in wood.

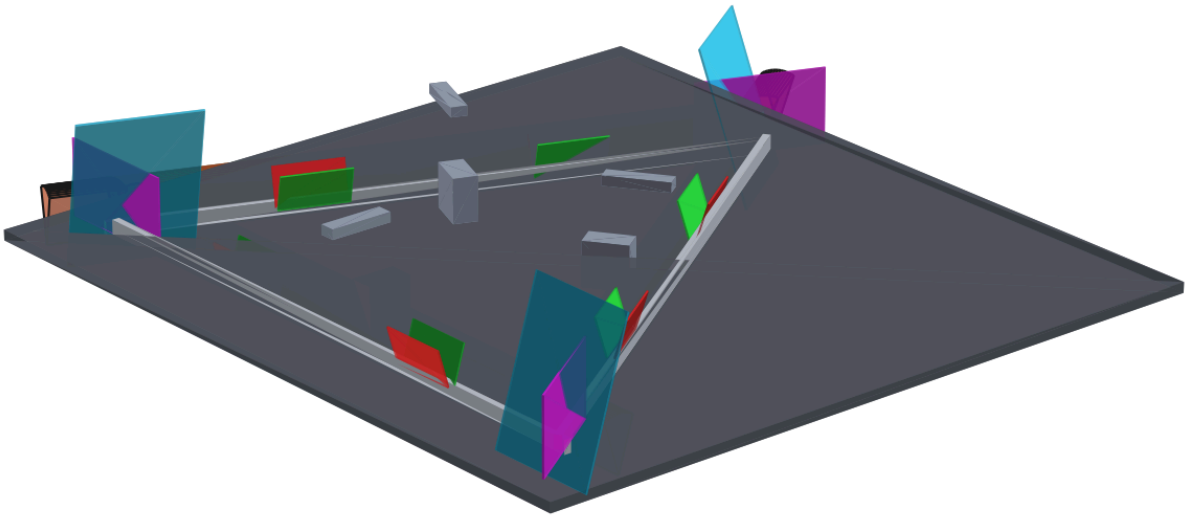
Where error goes

The core argument, in one image: the head end of every member runs 6 inches past its final position, and those inches are where the build's error goes to die.

Follow the sequence. The member is cut long at the split. It goes into the jig with its butt located against the stop — the one position the member must hold — and its head running free, past the head fence, into the overfit zone. The panel is assembled: the overlaps screwed, the triangle pulled tight, the neighbours defining the panel's truth by *being* the panel. Only then — with everything located, clamped and true — does the saw run along the head fence and take the overfit off. The cut is made in place, against the panel the member actually lives in, and the member ends up exactly the length the *assembly* required, not the length the tape predicted.

Every error the earlier steps could have introduced — a butt that landed a quarter inch deep, a neighbour whose face sits proud, the ordinary drift of split timber — was absorbed before the cut, because the cut is the last operation and everything before it had room to be wrong. The overfit is the method's crumple zone: 6 inches of deliberate waste on every member, in exchange for every member being final-cut against reality.

The sidebar rule that pairs with it, stated once here and used everywhere after: **locate first, cut second.** The fixture establishes the cut; the cut never establishes the location. The maxim is the opposite of how most people are taught to work with a tape, and it is the sentence that makes rough timber buildable at all.



6 inches of deliberate overshoot, sawn off against the fence.

Long, then flush

The drawing shows the same panel twice: first with all three heads running long past the fence — the overfit still attached, the panel already a panel — and then after the flush cut, the heads sawn down to the fence line, the panel exactly the triangle the bench describes.

The two states are the chapter's whole teaching. In the first, the panel is *assembled but not decided*: everything about it is final except its outside edges, which still carry the slack. In the second, the slack is gone and the edges are true — true to the bench, not to a measurement, which is the only truth a panel's edges can have that its neighbours will agree with.

Notice where the saw is in the drawing: riding the fence, the bar flat, cutting all three heads on one plane. The cut is a single motion repeated three times per panel — 3 cuts — and it is the only operation in the whole panel where accuracy and *finality* meet. Everything before it was reversible; after it, the panel is committed. The overfit is what bought the reversibility.

How much to leave

The overfit's size is a build decision, and the book declares its own: 6 inches. The number comes from what the overfit must absorb, and the honest way to size your own is to count the variation in your own pile.

The overfit must cover: the drift in your splitting, the taper of your sections, the seating of your butts against the stops, and the settling of the sticks under the clamps. If the worst stick in your pile wanders half an inch from the plan's ideal, a half-inch overfit leaves the good sticks wasting half an inch and the bad sticks with nothing to spare. So measure the pile, not the plan: the overfit is set to the *spread* of your stock, plus a margin for the day you are tired. 6 inches was this build's spread plus margin; a tighter pile could run less, a looser one more.

The trade is one-dimensional: more overfit costs wood, less overfit costs do-overs. The table below prices the wood; the do-overs price themselves the first time a member arrives at the head fence already short, which is the one failure the overfit cannot fix — because the overfit *is* the fix, and a stick cut short never had any.

Offcut per dome

The table is the tolerance's bill, and the book prints it the way it prints every bill: in full, with the units the builder actually pays in.

Every member sheds its overfit into the offcut pile: the flush cuts of Chapter 36, one per member, 120 members, adding up to the figure in the table — board feet of perfectly good wood, cut off and set aside. The honest note is that the offcut is not waste in the moral sense: it is blocking, bracing, wedges and firewood, and the method was sized for it from the start — the tree's spare struts and the offcut allowance are the same budget, counted in different units. But it is wood the frame does not stand on, and the table says how much, because a tolerance that is not priced is a tolerance that will quietly eat the savings it was bought with.

What the tolerance costs

Overfit per member	6.0 in
Members	120
Offcut length, all told	60.0 ft
Offcut volume	49.1 bf
Wood in the finished frame	550.7 bf
Offcut as a share of the frame	**8.9%**
Struts that wood would have made	**11.3**

This is the price of never having to measure a head end. It is paid in wood, deliberately, and it is cheaper than the alternative -- which is a triangle that will not close because three small errors all pushed the same way.

The price of the tolerance, in board feet.

The overfit's ledger closes where the chapter began: cut long, locate, assemble, flush. 6 inches of slack per member, a table's worth of offcut per dome, and in exchange — a frame whose every member is final-cut against the panel it actually lives in. That is the cheapest accuracy there is, and the next chapter is the cut that collects it.

Flush-Cutting in Place

The cut that makes the panel true

Flush-Cutting

The flush cut is the final operation on the jig and the moment the panel becomes itself: the three overfit heads of Chapter 35, sawn off against the fence, in one plane, so the assembled triangle ends up exactly the triangle the bench describes.

It is a small cut and the book gives it a chapter anyway, because the order of operations it completes is the method's entire tolerance argument, and because the cut is the one place a hurried builder can undo, in thirty seconds, the half-inch of slack the whole design paid for. The chapter has two sections: how the cut is made, and why it cannot be made any earlier — the second being the reason the first is written at all.

Against the fence

The procedure, with the saw held the way it has to be held — against the fence, never freehand.

The panel stays clamped. The flush cut is made on the assembled, screwed, clamped panel — the panel does not come off the bench for the cut, because the bench is what makes the cut true. The clamps hold the members against their stops; the saw rides the fence; the two truths together are the cut.

The bar rides the plane, flat. The saw is laid on the fence's cut plane, bar flat to the bench, and drawn through the three heads in one motion where the fence allows it. The saw does not sight anything; it follows the plane the bench already holds, which is the only way a hand-held saw makes a square cut at all.

Cut all three heads on the same plane. The three heads stick past the fence by different amounts — the panel is a triangle, and its overfits land where they land — but the *cut line* is one plane, and all three cuts are made to it. That single plane is the panel's final edge geometry, and a panel whose three heads were cut to three planes is a panel whose neighbours will measure the difference.

Do not finish the stick, finish the panel. The temptation at the saw is to clean each head as it is cut — a shave here, a touch-up there. Resist it: the flush cut's product is the panel's truth, not three pretty ends. The offcuts go to the pile, the panel goes to the stack, and the next panel takes its place. The cut exists once per panel, not once per stick, and treating it as a production step is what keeps the run of Chapter 31 at its pace.

Why this cannot be done first

The order is not arbitrary, and the reason is the whole tolerance argument in miniature.

The flush cut cannot be made before assembly, because before assembly there is nothing for the cut to be true *to*. A member cut to its final length at the split is a member cut against a tape's idea of where its neighbours will be — and its neighbours, when they arrive, will be wherever the wood actually is. The panel is the only object that knows the member's true length, and the panel only exists after the butts are located, the overlaps screwed and the clamps closed. So the cut must come last: it is the final measurement, and the fixture performs it.

It cannot be made *after* the panel leaves the bench, either — off the jig, the panel's own edges are the only reference the saw has, and a saw referenced to the thing it is cutting is a saw cutting in circles. The bench's fence is the fixed truth; the panel's edges are the variable; and the order that works is the one where the fixed truth cuts the variable.

So the sequence stands as the book has built it: split long, butt against the stop, assemble, flush against the fence. Cut long, locate, assemble, cut true. The flush cut is the last word of the sentence, and the sentence is the method — every member of every panel final-cut in place, against the only truth that could know it.

Panel Drawings

One flat drawing per distinct panel, full size

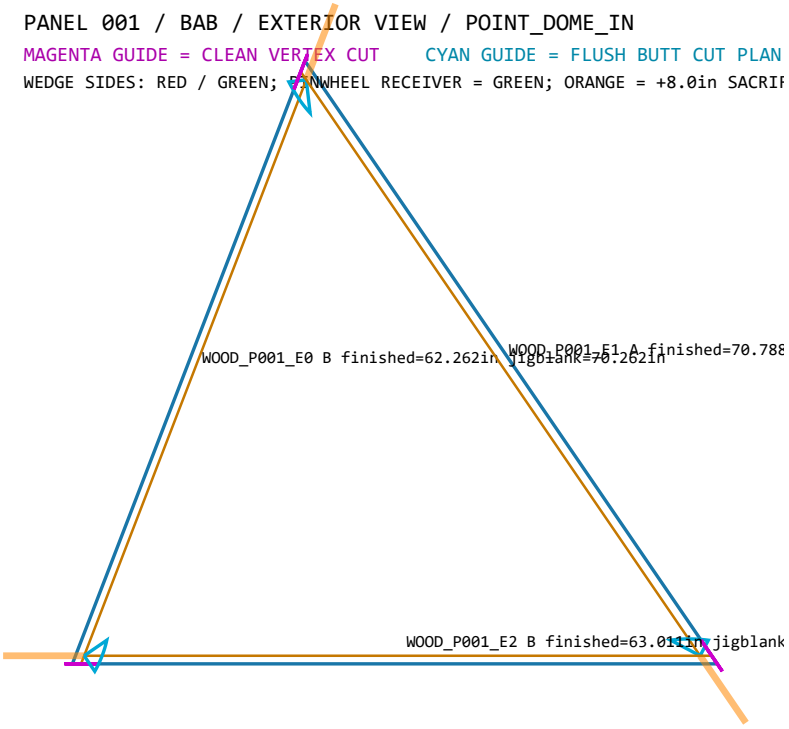
Panel Drawings

The book's last word on the panel is a drawing, not a paragraph — one flat, full-size drawing per distinct panel, exported directly from the simulator, so the drawing on the page and the panel on the bench are the same object.

The drawings are the check for everything the jig chapters described. The jig locates members against stops; the drawing is what the stops were set from. The panels come off the bench; the drawing is what they are measured against before they are stacked. And because the drawings export from the same solved model that produced the cut list and the seam schedule, the whole build has one source of truth, and it is printed on these pages.

Panel drawing, type one

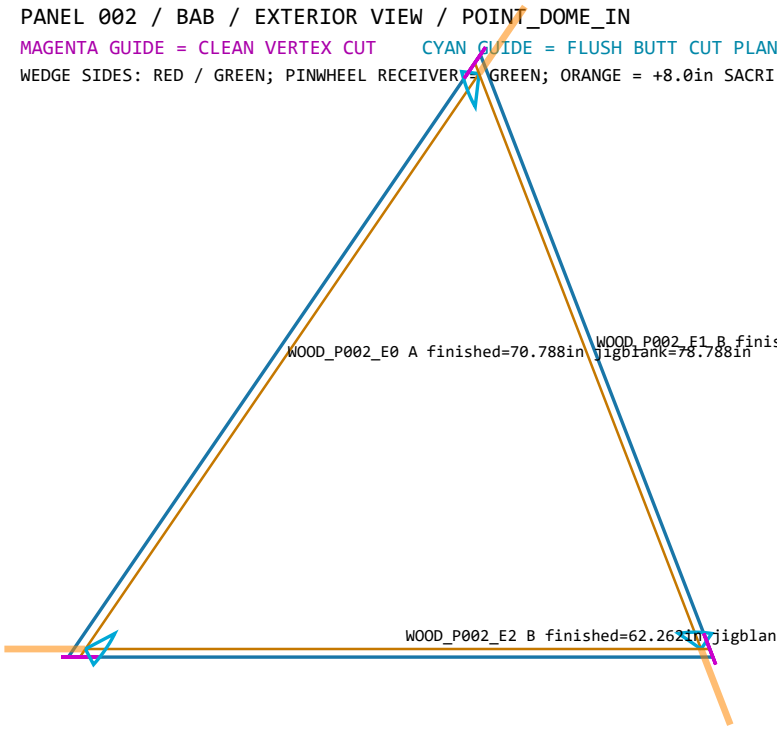
The first of the two distinct panels, flat, full size, exactly as the simulator exports it — the member lengths, the overlaps, the vertex gap, the seam allowance all in their true positions.



Panel type one, flat, full size.

Panel drawing, type two

The second distinct panel. Between them, the two drawings describe every one of the frame's 40 panels: the 10 of one type, the 30 of the other, the counts of Chapter 8.



Panel type two, flat, full size.

Reading a panel drawing

What every line on the drawing is, so the page can be read at the bench and not only admired.

The outline is the panel's finished edge — the triangle the neighbours will meet, gasket allowance included. **The member lines inside it** are the three sticks: butts located on sides, heads running to the fence line, the pinwheel's overlaps shown as the overlaps they are. **The vertex gap** is the empty corner of Chapter 15 — drawn, not described, so nobody looks for a hub there. **The seam allowances** are the half-gasket insets at every edge, the 0.75-inch key's share of each side, which is why the outline is slightly inside the triangle the members describe.

The drawings print full size: exported as vectors, they scale to the bench without loss, and a printed drawing laid on the jig's surface should match the jig's stops line for line. That is their real job — not illustration but *witness*. When a panel comes off the bench, it is laid on its drawing; when a jig is first set, the drawing is what the stops are checked against; and when the frame finally closes, every panel in it has, somewhere in its history, been held up against one of these two pages and found to be the thing the pages describe.

The reference part of the book ends here, with the drawings — because after the drawings there is nothing left to know about the panel. The next chapters leave the bench and follow the frame to the ground it stands on.

Where Error Goes

A tolerance budget for a building made of split logs

Where Error Goes

Every chapter so far has mentioned the method's tolerance — the overfit absorbing a sixteenth, the key absorbing the fold, the panel owning its edges. This chapter is the budget behind the mentions: a complete account of where error can land in a frame built from split logs, which of those places the method absorbs, and which one it does not.

A building method that cannot say where its error goes is a method that discovers the answer at raising. This book's method can say it, because it was designed around the saying: every operation in the fortnight either sends its error somewhere specific or refuses to make the cut that would. The chapter walks the four landing places — stock length, butt angle, seam gap, and the ring — marks three of them as absorbed and one as accumulating, and ends with the check that keeps the last one honest. It is the most structural chapter in the book, in both senses of the word.

Four places error can land

The complete list, because a tolerance budget that omits a landing place is a budget with a hole in it.

Stock length. The member arrives at the bench longer than it should be — the split drifted, the bucking was generous, the plan was optimistic. This is the commonest error in the build and the cheapest, because the overfit exists for exactly it: the head runs long, the assembly locates the butt, and the flush cut of Chapter 36 removes the error with the offcut.

Butt angle. The compound cut of Chapter 34 lands a degree off, and the butt no longer sits flat on its neighbour's face. The error is real but bounded: the butt's job is to present area to the side, and a butt a degree off still presents nearly all of it — the screws pull the overlap closed, and the contact settles into the wood. What the method refuses is the opposite error: a butt cut *short*, which is why the allowance stays on until the bench decides.

Seam gap. Two panels meet and the gap between them is a fraction wider or narrower than the key's share — the fold is right but the wood is itself. The seam absorbs it by design: the key compresses, the gasket gives, and the 55 seams of the frame share the difference so no single seam carries it all. This is the landing place the whole seam system was built for, and the one the book spent its middle chapters engineering.

The ring. The base ring is where the frame meets the ground, and it is the one landing place that does not give. A sixteenth of error in the ring is a sixteenth of error in the circle the whole dome stands on, and the dome cannot compress its way out of a wrong foundation. The ring is the subject of the last section, because it is the budget's one honest failure point.

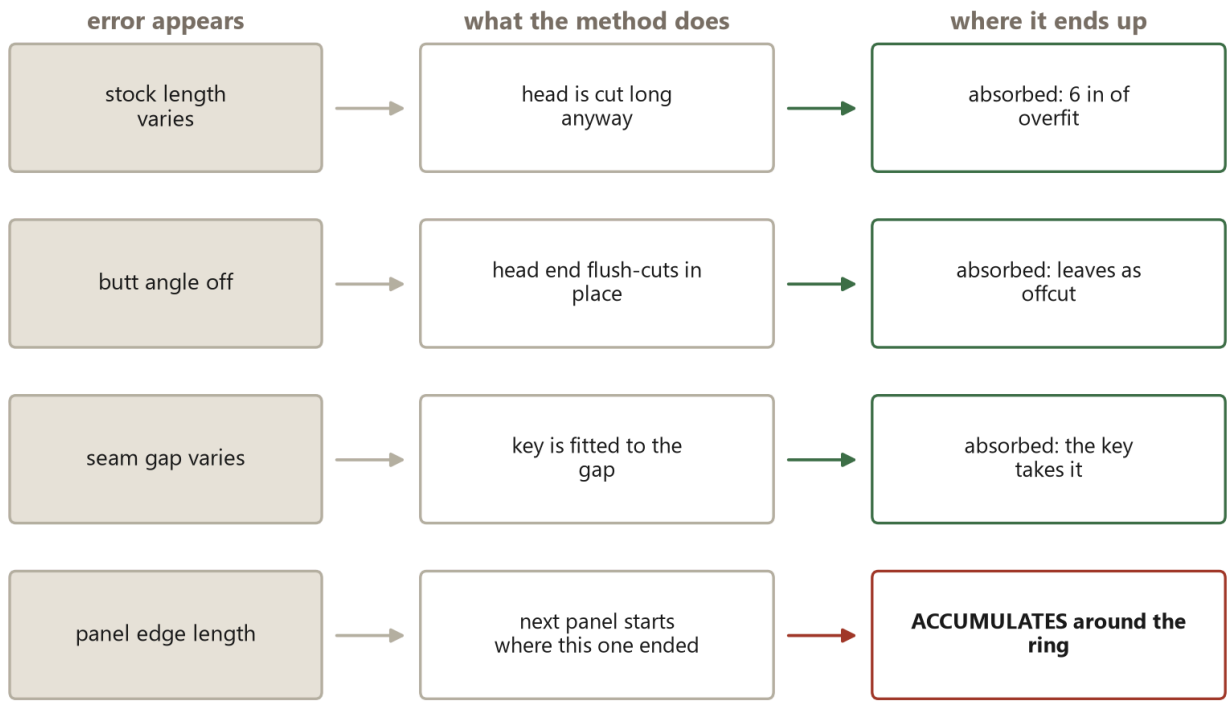
Three of them are absorbed

Why three of the four land safely, and what “absorbed” actually means in wood.

An absorbed error is one that leaves the build as something else: offcut, compression, distribution. The stock-length error leaves as the flush cut’s offcut — Chapter 35 priced it, and the price is the board feet in its table. The butt-angle error leaves as contact: the overlap’s screw row pulls the butt into the wood until the bearing is full, and a degree of bevel error becomes a fraction of an inch of settlement that the frame never sees again. The seam error leaves as compression: the key takes up the difference, the gasket breathes it, and the dome closes with the error spread invisibly across 55 joints.

The three share one property: each is absorbed at the *place it was made*, by the design of that place, without travelling. The stock error never leaves the bench. The butt error never leaves the joint. The seam error never leaves the seam. That locality is the whole secret — an error that dies where it is born cannot compound with its neighbours, and a frame built from rough wood needs exactly that: forty small absorpions instead of one long accumulation.

Where error goes



Three of the four are absorbed by the method itself. The fourth accumulates around the bottom ring, which is why the ring is the one thing you check as you go rather than at the end.

Error, and where the method sends it.

The tolerance budget

The drawing is the budget as a flow: error enters at the four places, three of them send it sideways into absorption — the offcut pile, the bearing, the gasket — and one of them, the ring, passes it on.

Read the arrows as the method's real structure, because they are. Every chapter of the build is somewhere on this drawing: the overfit is the stock-length arrow, the screw row is the butt arrow, the key is the seam arrow, and the ring is the arrow with no exit. A builder who has internalised the drawing stops fearing the ordinary errors of rough timber — they are budgeted — and starts fearing only the one the drawing marks in the colour of warning.

The one that accumulates

The ring, named and checked, because it is the budget's single honest failure point.

The base ring's error does not absorb; it *accumulates*. The ring is a closed circle of 10 panels, and an error anywhere in it is an error everywhere: the dome above it rises from whatever circle the ring actually is, and every course multiplies the ring's truth by the frame's own height. A ring set a half inch out of round is a dome that leans a half inch at the base and more at the crown — the one error in the method that grows as the building does.

So the ring is where the book's checking discipline concentrates. The check is simple and it is done three times: when the ring is first set, measure across every opposite pair of its 10 sides — the numbers must agree within the gasket's give, and any pair that does not is the ring telling you where it is wrong while the fix is still a shim. When the first course closes, measure the ring again — the courses above have already begun leaning on it, and a ring that moved has moved the whole frame's opinion of round. And before the top closes, once more, from outside: the dome's last chance to be told, cheaply, what the raising is about to make permanent.

The ring's honesty is the book's last word on tolerance. The method forgives stock, forgives butts, forgives seams — and it never forgave the ring, because the ring is the one place where forgiveness would have to come from the ground. Chapter 39 gives the ring its footing; this chapter gave it its warning. Build accordingly.

The Footing and the Ring

What the dome stands on

The Footing

The frame is up — hanging, in this book's build, between the trees — and the question of what it stands on, if it ever stands, is the one the fortnight deliberately left open: the ground work comes last, and it may never come at all. This chapter is the footing for the day it does — the part of the building under everything, the part the photographs never show, and the part where the book's own warning from Chapter 38 is most literally true: the ring's error accumulates, and the footing is the ring's floor.

A dome's weight arrives at the ground along one circle — the base ring's 10 edges, ten points where the panels meet the earth. The footing's whole job is to receive that circle and hold it: level, round, and against the wind that wants to lift the whole thing away. The chapter covers the two ways to do it, the layout that keeps the circle true, the uplift a dome invites, and the point where the book stops and the engineer starts.

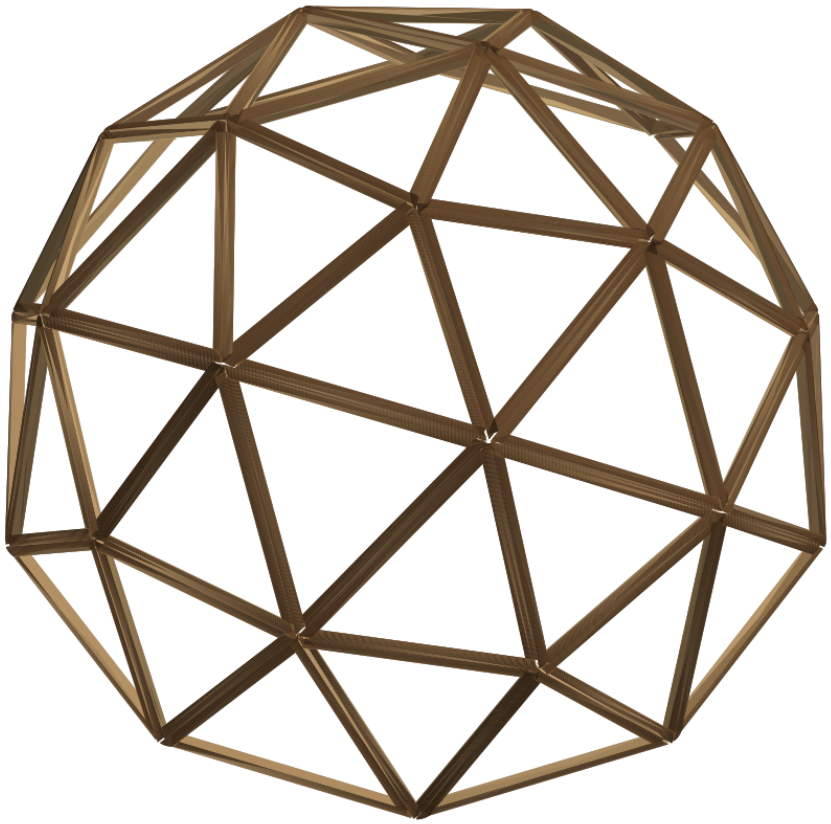
Ten points, or one ring

Two approaches carry a dome's weight to the ground, and the book recommends one of them for its own build while pricing both.

Ten points. Each of the ring's ten base vertices sits on its own small footing — a pad block, a pier, a driven post — and the frame bridges between them. It is the cheapest material answer, it works where the ground is firm and level, and it leaves the middle of the circle empty — no concrete where nothing stands. Its cost is precision: ten separate points must land in one plane and one circle, or the ring that spans them comes out twisted, and Chapter 38 has already explained what a twisted ring does to everything above it.

One ring. A continuous footing under the whole base ring — a ring beam of concrete, or a wooden ring on blocks — receives the dome's weight as one circle and spreads it as one circle. It costs more material than ten points, and it is *forgiving* in exactly the way ten points are not: the ring beam carries the frame's local imperfections around the circle instead of exposing them at every vertex. The book's own build stands on the continuous answer, because the build's error budget — the chapter just spent on it — is worth more than the concrete it saves.

The honest note, kept from the dome books that skip it: a dome puts its whole weight on the *outside* of its circle. Pouring a slab across the middle is concrete nothing stands on — the centre of the floor carries no frame load at all, and the money it costs is money the ring beam could have used.



The base ring from above: ten edges on the ground circle, nothing in the middle.

Chapter 65 prices the two properly when the dome graduates to a permanent pad; here the rule is only: foot the circle, not the disc.

Setting out a circle accurately

The layout, because the footing's truth is the ring's truth, and the ring's truth is the whole frame's.

One stake, one string. The dome's centre goes in first: a stake driven where the middle of the floor will be, and it never moves again for the rest of the build — every measurement of the footing, the ring and the raising is taken from it. Then a string the length of the dome's radius, a loop at each end, and the circle is drawn the way circles have been drawn since before anybody wrote this down: walk the string around the stake, marking as you go.

Set the ten points from the circle. The ring's ten base vertices sit on the circle at equal spacing, found by stepping a second string — the chord between neighbouring vertices — around the circumference and adjusting until the last step lands exactly on the first. The step-and-adjust is the check: a ten-point layout that closes on itself with the last step is round; one that misses its starting mark by an inch is telling you where the circle isn't.

Check the circle before anything is poured. Measure across every opposite pair of points — the diameters — and demand they agree. Then check the level of the ten points against a water level or a laser, and shim until they share one plane. The checks are the chapter's whole discipline: a footing that is round and level on the day it is made is the only thing in this part of the build that cannot be fixed later, so it is checked the way things that cannot be fixed later are checked — twice, from different directions, before the concrete is mixed.

Anchoring against uplift

The sentence the footing chapter must say plainly, because the shape itself refuses to let it be skipped: **a dome is a wing.**

A hemisphere is a lifting body. Wind over the crown does to the dome what it does to an aircraft: pressure drops on the curved top, and the whole building feels a pull toward the sky. The frame's own weight resists — green pine is heavy, and the skin chapters add their share — but a dome that is only *sitting* on its footing is a dome the right wind can pick up by the brim, and the stories of domes that went rolling are not folklore. The footing's job is therefore two jobs: hold the weight *down*, and hold the building *down against the wind*.

The anchor is the ring's connection to the ground, and it is not optional. Every base vertex — or the ring beam at intervals — needs a mechanical tie into the footing: a cast-in anchor, a bracket, a strap over the ring member. The tie's job is uplift, not weight, and it is sized for the worst wind the site's records hold, which is exactly the calculation the last section of this chapter hands to somebody else. The book's own build used straps at every base point, and the straps are the one part of the footing the book will not let a reader improvise.

This is where to ask somebody

The safety page, and it is shorter than the chapter because the book's rule from the front matter simply applies here with full force.

Footings are load and soil, and neither is a matter of carpentry. The dome's weight, the wind's uplift, the ground's bearing capacity, the frost line, the drainage — every one of them is a site-specific number that belongs to someone who can read the site and the code together. This book tells you the *shape* of the footing — a circle, not a disc; anchored, not sitting; checked, not assumed. It does not tell you how deep, how wide, or how much steel, because those answers come from your ground, not from a book that has never seen it.

The division of labour is the one Chapter 10 drew for loads, applied to the earth: the geometry is the book's, the engineering is the engineer's, and the boundary between them is a page you cannot miss. Build the circle the book

describes. Have the circle's ground signed off by someone who can. Then, and only then, stand the frame on it.

And remember the order the title page promised: the hanging cabin never owed this chapter anything. The foundation is the last decision of the build, made when the cabin finally comes down from the trees — and the chapters that follow this one are the reason the decision could wait.

Closing the Shell

Forty triangles of something over forty triangles of nothing

Closing the Shell

The standing frame is a skeleton, and this chapter is its skin: 40 triangles of something, over 40 triangles of nothing, each panel closed so the weather stays on the outside.

The options are few and the book ranks them the way it ranks everything — by cost and behaviour, computed, not by feel. Sheet goods dominate, because they are cheap, local and flat, and the chapter's first two sections are the craft of cutting forty triangles out of rectangles without wasting half the stack. The table prices the whole skin in sheets. The last section runs the alternatives — fabric, shingle, living roof — because each one exists for a reason and each one asks the frame for something different, and the builder who knows all four picks the right one instead of the first one.

Sheathing a triangle

The craft problem the rectangle-building world never has: every panel of a dome is a triangle, and sheet goods come in rectangles.

The answer is the panel drawing, again. Chapter 37 printed the two panel shapes flat and full size; the sheathing job is those shapes, cut in sheet, one per panel, with the seam allowance carried around the edge. The procedure that keeps the waste honest:

One template, not forty measurements. Trace each panel shape once onto scrap, cut it, and use it as the pattern for its whole class — the 10 of one shape, the 30 of the other. The template is the sheathing's jig: the shape made physical, so the saw never re-derives an angle it already cut correctly once.

Cut inside the line, then trim to fit. Sheet cut to the template's outside is always slightly big; sheet cut inside it can be trimmed with a knife in place. The panel gets its skin offered up, marked against its own frame, and trimmed to the frame's actual edges — locate first, cut second, the same maxim the jig chapters ran, applied to skin.

Seam allowance stays outside the panel. The skin's edge belongs to the *seam*, not to the panel: the gasket's share, the flashings, the overlap of the next panel's skin — all of it lives beyond the frame's own edge, and a skin cut flush to the frame is a skin that leaves the seam naked. Cut generous, trim at the seam, and the 55 seams get the overlap the weather chapter demands.

Screw to the frame, not through it. The skin fastens into the members' faces — the wide flat of the wedge — at the spacing the sheet's own spec asks. The skin is not structural; it is the panel's face, and its screws' only job is to keep it flat against the frame while the seams do the weather work.

Nesting the cuts

The second craft problem, and it is pure geometry: how to fit two triangles into one rectangle.

The two panel shapes of the dome, paired point-to-point along their long sides, nest inside a standard sheet with the waste reduced to two thin edge strips — the arrangement the table below assumes. The nesting works because the panels were *not* sized to the sheet and the sheet was not sized to the panels; they meet in the middle, and the meeting wastes less than either feared.

The honest limit, stated because the film that discovered it made a point of it: this dome's panels do not fit a standard sheet *singly* — one panel is wider than half a sheet, so the naive layout of one-panel-per-sheet wastes nearly half the stack. The nested pair is the fix, and it is why the chapter treats the two shapes as a pair rather than as forty separate cuts. Where the nesting does not reach — the odd counts, the rim panels with their different edge — the waste goes to the offcut pile and the pile goes to blocking, the same honest destination as every other offcut in the book.

Sheet count

The table is the skin's bill: shell area and sheets, computed from the solved geometry, not estimated from the drawing.

The numbers are the answer to the question the chapter opened — what it costs, in sheets, to close 40 triangles. The area is the dome's own surface, the sheet count is the nested layout's requirement, and the difference between the two is the waste the nesting section promised to keep honest. The figure is worth reading twice, because it is the first number in the book that is *bought* rather than *cut* — everything until now came out of the trees, and the skin is where the fortnight's ledger first touches a lumberyard.

Closing the shell

Shell area	677 sq ft
Upper shell (roof proper)	339 sq ft
One sheet covers	32 sq ft
Sheets, if nothing were wasted	21.2
Triangles cut from rectangles	x1.18 waste factor
Sheets to buy	**25**

The waste factor is the honest part: you are cutting triangles out of rectangles, and no nesting makes that free. Nesting two panels per sheet gets closer to the ideal than cutting one at a time.

Shell area and sheets, from the solved geometry.

The alternatives

Sheet goods are the default, not the only answer. The other three exist for other builds.

Fabric. A membrane skin — canvas, shade cloth, the soft shells of Chapter 70 — closes the frame for the price of cloth and rope, sheds rain where the seams are taped, and gives up insulation and years. It is the right skin for a seasonal dome, a greenhouse's first year, or a frame that is still deciding; it is the wrong skin for a home, and the book says so rather than romancing the tent.

Shingle. Shingling a compound surface is slow, and it works — each shingle is small enough to follow the curve, the overlaps shed water the way the frame's ridges want, and the result is the most repairable roof there is: one shingle at a time, forever. The cost is the labour, which is why shingle is the upgrade for the dome that has already earned its place, not the first skin it wears.

Living roof. Soil and plants over a waterproof layer, the heaviest and coolest of the four, and the one that asks the most of the frame: the chapters on load have already said who decides whether the members carry it. It is listed because it is real and because somebody always asks; it is not recommended for the first build, and the reason is the engineer's sentence, not the gardener's.

The chapter closes where the shell does: the frame is covered, the seams are taped, and the dome is a *room* in the only sense that matters outdoors — it keeps weather on its outside. The next chapter cuts the holes that make it usable, and the chapter after that is the one that decides whether the shell's honesty holds up to the weather it was built for.

Doors, Windows and the Rim

Cutting holes in something that works by being closed

Doors and Windows

Every building needs holes, and a dome is the building that resents them most: its strength is its closure, its panels are its structure, and a hole cut wrong does not just let the weather in — it un-triangulates the frame at the exact place the frame was leaning on itself.

So this chapter is the craft of opening a dome without breaking it. The principle, stated once and used throughout, is the one the wedge book's own chapter taught: **an opening is a panel you did not close** — take a whole panel out and the frame loses nothing structural, because the panel was always its own independent unit. Cut *into* a panel, and the triangle stops being a triangle, and the load that was travelling through it now has nowhere to go. The chapter runs the openings in order of mercy: the door at the rim, the window in a panel, the risers that cheat height — and the drawing that shows all three without a single un-triangulated triangle.

A door at the rim

The easy case, and the book says so plainly: the door belongs at the rim, because the rim is the one place a hole costs the frame nothing.

The base ring's 10 edges join nothing — Chapter 8 separated them from the seams for exactly this reason. Leave one rim panel out entirely, and the gap is a doorway: the panels on either side are complete, their seams close as they always did, and the dome loses a panel that was never structural in the way the crown panels are. The door is framed the way the missing panel was framed — two jambs running the panel's full height, a header across its top edge — and the frame's own triangle does the lintel's work, because the triangle above the door is still a triangle.

The one requirement is the header's honesty: the panels above the door lean their weight down through the ring, and the opening interrupts the ring. The header takes that interruption — a beam across the top of the opening, tied to the jambs, carrying the ring's continuity across the gap. In a full-height rim door this header is small, because the ring is low and the loads above it are modest; in a taller opening the engineer's sentence from Chapter 10 reappears, and rightly. The rule for the chapter: take the whole panel, frame the gap, header the ring.

A window in a panel

The harder case, and the craft is keeping the triangle a triangle.

A window wants light where the wall leans — mid-panel, high on the dome — and a hole in the middle of a triangle is exactly what the frame's logic forbids: the triangle's load runs around its edges, and a hole in its middle interrupts the

members that were carrying it. So the window is not cut into a panel; it *replaces* part of a panel, and the part it replaces is re-triangulated around the hole.

The procedure is the panel's own logic, applied smaller: frame the window opening inside the panel with members that run from the window's corners to the panel's edges — a smaller triangle above the window, two beside it, one below — so every region of the panel is still triangulated, and the panel's loads still travel along members, now detouring around the glass. The window unit itself is fixed into its own sub-frame, flashed to the panel's skin, and the panel remains one independently-built, independently-lifted thing — the property Chapter 18 paid for, spent here on daylight.

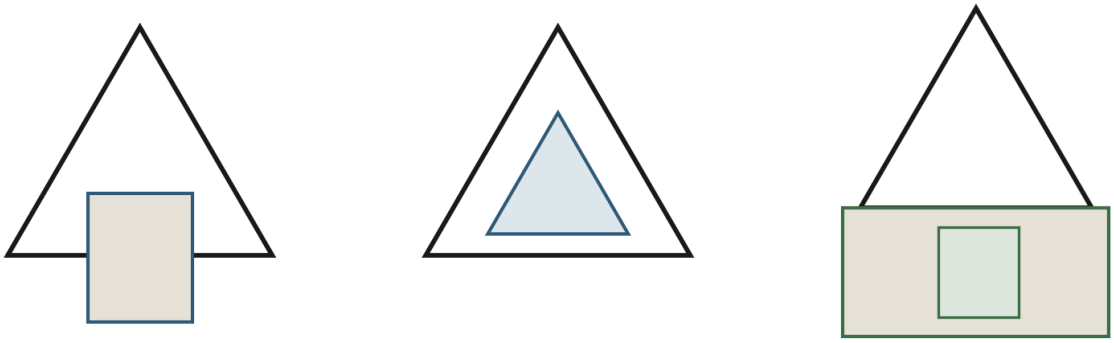
The honest cost, said aloud: a windowed panel is a panel built twice — once as the triangle, again as the triangle-with-a-hole — and its member count rises accordingly. That is why the windows go high and few, where the glazing earns its keep in a leaning wall and the view, rather than everywhere the floor plan's habit would put them.

Three openings that work

The drawing collects the chapter: the rim door, the windowed panel, and the risers option of the next section — three ways to open the dome, each one keeping the frame's triangulation intact.

Read the drawing against the rule it illustrates. The door takes a whole panel and frames the gap; the window sub-frames inside a panel and re-triangulates around the glass; the risers lift the whole shell and put the openings in a *straight* wall, where doors and windows stop being dome problems at all. Three openings, three strategies, one invariant — no hole ever interrupts a load path without a smaller triangle being drawn around it.

Three openings that do not un-triangulate the shell



A dome works because every panel is a triangle. Remove a member and the panel becomes a mechanism. All three of these keep every remaining triangle whole: the rim door removes nothing, the panel window sits inside a triangle, and the riser moves the whole shell up onto a wall that can have anything in it.

Three ways to open a dome without un-triangulating it.

Risers: cheating height at the rim

The third strategy is not an opening but an avoidance: raise the whole dome onto a short vertical wall — a riser, a knee wall, a stem wall — and the openings stop fighting the curve.

A riser is a straight cylinder of wall under the hemisphere's brim: knee-high for a shed door, waist-high for furniture to lean against, full-height where the building wants ordinary windows and an ordinary door. The dome above it is unchanged — the same frame, the same panels, the same seam schedule — and the wall below it is ordinary stud framing, which is the entire point: the expensive part of the building keeps its shape, and the parts that want rectangles get rectangles.

What the riser costs is the ring beam's reappearance: the dome's base ring now sits on a wall instead of on the ground, the wall needs a top plate to receive it, and the wall's own top must be as round as the ring it carries — the same layout discipline as Chapter 39, moved up one storey. What it buys, Chapter 58 prices in full: usable floor at the edge, headroom where the dome's wall leans hardest, and the cheapest square footage in the building, bought by the foot of wall height.

The book's order of operations closes the chapter: frame, shell, then openings — and if the openings want a straight wall, the riser is the answer, because it is cheaper to lift a whole dome than to argue with forty curved triangles one window at a time.

Weather and Water

Where it leaks, and where the water goes

Weather and Water

The honest chapter about domes and rain, because the dome's reputation was made by this subject and the subject deserves the book's full discipline.

The title page inverted the usual order on purpose: the frame is the part you are allowed to get wrong, and the skin is not — a frame out of line stands for fifty years, and a skin that leaks one winter takes the frame with it. This chapter is the skin's side of that bargain, and it is the reason the book treats the seams here with more care than anywhere else.

Here is the fact the frame chapters have been circling: a dome's roof is one surface made of 40 flat panels meeting at 55 interior seams, and every seam is a place water can get in. A rectangle concentrates its roof's risk in two valleys and a ridge; the dome spreads the same risk across 55 lines. The answer is not fifty-five seals; it is the two principles this chapter teaches — shed, don't seal; and where water must be managed, manage it deliberately, at the rim. The chapter closes with the inside face, because the dome that survives the rain outside can still be ruined by the water its own breathing makes inside, and the book says that before the reader discovers it in month three.

Every seam is a potential leak

Say the number, because the number is the strategy: 55 seams. Fifty-five lines on the outside of the building where two panels meet, fifty-five keys, fifty-five weather details — and the honest way to think about them is not "one roof" but "55 small roofs, all meeting".

The frame chapters already did the structural half of each seam: the key locates the geometry, the gasket takes the tolerance. The weather half is a different question, and it is the one that decides whether the building lasts. A seam that fits is not automatically a seam that sheds — water moves downhill, and a seam's fit matters only where the water actually runs. The dome's seams run in two families: the steep ones near the crown, where water moves fast and the seam's own slope carries it away, and the shallow ones near the rim, where water lingers, pools in the key's shadow, and waits for a gap. The strategy follows the slope: the crown seams need alignment, the rim seams need *weathering* — a drip edge, an overlap, a path that takes the water off the seam before it can sit on it.

The count is the discipline. A builder who treats 55 seams as one repeated detail does the same detail wrong 55 times; a builder who separates them into the two families the geometry gives — the steep and the shallow — has two details to perfect and a schedule to repeat them by. Chapter 16 is the schedule. This chapter is the weather it has to survive.

One channel, many jobs -- a frame of the seam-climate film.

Shedding, not sealing

The principle that actually works on a compound surface, and the reason "waterproof" is the wrong word for what this book builds.

A sealed surface tries to stop water at the membrane — one barrier, perfect, everywhere. Sealing fails on a dome because a dome moves: the frame dries, the panels settle, the seams open and close with the seasons, and a seal that was perfect in July is a cracked line by January. The answer the roof-building world already knows — and shinglers have known for centuries — is to stop trying to hold the water back and instead *hand it downhill*: every surface overlaps the one below it, every seam wears a shingle, and the water is always on top of the roof, never inside it.

So the dome is weathered as 55 little roofs, each seam flashed like a valley: the upper panel's skin laps over the lower's, the key is protected rather than depended on, and the seam's own slope is the drainage. The skin chapters supplied the panels; this chapter supplies the lapping order; and the result is the honest claim the book makes about weather: not "the dome never leaks" — no roof can promise that — but "where water reaches the seam, the seam hands it down, and the lowest seam hands it off the rim". Shedding is a path, not a promise, and paths can be walked, checked and repaired — which seals, once breached, cannot.

Harvesting it instead

The one place the shape helps with water, and it is not small: a dome is a very good rain collector.

Every drop that lands on the shell runs down the panels to the brim, and the brim is a circle — one continuous gutter, already built, waiting only for the collection the next chapter's hardware provides. Chapter 59 computes the annual catch:

the footprint, the local rainfall, a declared efficiency, and the answer comes out in thousands of gallons a year off a building this size. The dome does not merely *tolerate* rain; it gathers it, and the gathering is free — the roof was already a gutter, the shape was already a funnel, and the only purchase is the tank.

The honest note, kept from the films that made this point: the catch is not drinking water without treatment. The filter makes it clear; the lamp makes it safe; neither does the other's job — and the book's water system, when Chapter 45 reaches it, is a first-flush diverter and a UV lamp downstream of the tank, not a rain barrel and optimism. For everything else — irrigation, flushing, washing, the buffer against a dry well — the catch is as good as the roof, and the roof is the dome's.

Condensation and the inside face

The last section is the problem people discover in month three, and it is worth a whole section because it is not rain.

A dome is warm inside and cold outside, and the meeting of the two happens *inside the skin*: moist room air reaches a cold surface, the water condenses, and the builder who sealed every outside seam perfectly wakes up to water running down the inside of the shell. The seams were not the leak. The air was.

The answer is the seam channel, and it is the reason the later chapters of this book engineer it deliberately rather than sealing it away: the cavity between the frame's wedges is the one continuous void that connects the whole shell, and it is exactly where condensation belongs — collected, drained at the rim, vented at the crown. The full climate system — the dew points, the levels, the drawer, the plate — is Chapters 82 and 83, and they are the book's real answer to the inside face. Here the principle is enough: **never seal the inside face against the air; give the water a path, and drain it at the rim.** The dome that leaks from rain failed its seams. The dome that leaks from condensation was built by someone who thought the inside face was not part of the weather.

The chapter ends where every weather chapter should: outside, shed; inside, drain; at the rim, catch. The dome is not a box, and its water is not a nuisance to be sealed away — it is a flow to be directed, and the direction is downhill, through 55 honest seams, off the brim, into the tank.

Finishing the Wood

What oil does to split timber, what it does not do, and the afternoon it costs

An oiled member is not a protected member. It is a member that has been finished — and the difference between those two sentences is the whole chapter.

The weather chapter said where the water goes and what keeps it out: a skin that sheds, flashing at the openings, a ring that drains, and a member allowed to dry. None of that changes here. What changes is the exposed sawn face inside the building, and the ends of every member you can see: bare, pale, slightly furry from the saw, and — left alone — they will grey, check and take on the look of scrap timber while the building around them is perfectly sound.

Finishing is how you decide what the inside of the building looks like. It is a cosmetic decision with a structural edge: an oiled face is easier to wipe, a sealed surface can trap water where an oiled one lets it go, and a member with a coat on it can still be inspected through it. This chapter is the whole practice, from reading the can to the rag station, with the quantities worked out for the frame in this book: 664 square feet of sawn face and 8.0 hours of work.

The source material for the finishing practice is a video claiming 20 things about linseed oil. This chapter is what survives when those claims are checked against product labels and against the arithmetic of the building — including the ones that do not survive.

Read the can before the rag

Linseed oil is three products wearing one name, and the differences matter before you buy anything.

Raw linseed oil is what comes out of the seed press. It cures slowly — it takes oxygen out of the air and slowly sets, which is why it is safe and quiet and why a coat in a cold damp workshop may still be tacky next week.

Boiled linseed oil is not boiled in the sense of a pot on a fire in any modern product. It is raw oil with metallic driers added so that it sets in hours rather than weeks. That is a real convenience with a real cost: the driers are the reason the label carries warnings, and they are the reason the drying behaviour of a can is not something you can assume from the word “linseed”.

Heat-polymerised oil — the third thing you will meet — is oil that has been thickened with heat and no driers, and it is what several traditional woodworking products actually are. A name like *natural*, *boiled* or *Danish* does not describe a formulation. The label does.

So the rule is the one the label gives you, not the one the video gives you. Two real cans make the point better than any warning could:

the can says	wait before wiping	minimum cure
Danish oil	5 minutes	8 hours
oil and wax	60 minutes	24 hours

The same furniture finish, the same manufacturer, and two different schedules by a factor of twelve. If you pick one number and apply it to every can on the shelf, you will either be rubbing off a coat that has already started to set, or waiting a day for nothing.

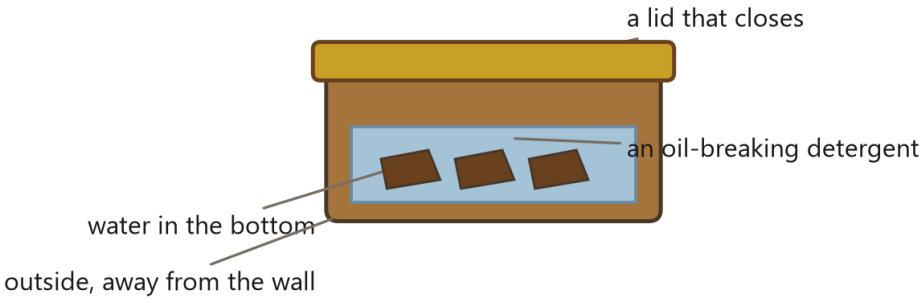
And the same video's cleanest rule is too broad: it says raw oil only for anything that touches food. That is a sensible instinct and it is not what the labels say — some polymerised finishes are sold specifically for kitchen wood, and the question a buyer should ask is not "raw or boiled" but "is this product rated for this use", which is printed on the can.

The rag station comes first

This is the dangerous part of the chapter and it comes before the pleasant part on purpose.

Linseed oil cures by taking oxygen out of the air, and curing gives off heat. A single rag on a bench sheds that heat into the room. A *heap* of rags traps it in the middle of the pile, and a pile of rags with no spark anywhere near it can reach ignition temperature and catch fire — often hours after you have left the shop. It is the classic workshop fire, and it is the one part of finishing that can burn a building down.

The rag station comes first



Linseed oil cures by taking oxygen out of the air, and curing gives off heat. A heap of rags traps that heat in the middle of itself, so a pile with no spark anywhere can reach ignition. The prevention is three objects and one habit -- a lidded metal tin, water in it, and an oil-breaking detergent -- standing outside, away from the building, before the first coat is poured.

The rag station, set up before the first coat.

The prevention is three objects and one habit, and all four are cheaper than a fire:

- **A lidded metal tin.** Not plastic, not a bucket, not a bag. A tin with a lid that closes.
- **Water in the bottom of it**, enough to cover what goes in.
- **An oil-breaking detergent** in the water, so the rags stop being a pile of oil and become a wet heap of cloth.
- **The habit**: rags go in the tin in the same session they were used, the tin stays outside and away from the wall, and it is emptied into whatever your local rules say is right. A rag that dries out in a heap becomes hazardous again — which is why the habit matters more than the container.

Set the station up before the first coat is poured. A rag station improvised after the fact is a rag station that gets improvised at the end of the afternoon, when you are tired and the shop is about to be shut with forty oily cloths in a bucket by the door.

How much oil, and how long it takes

The frame in this book is 120 members, 664.3 feet of stock, and each member has two sawn faces 6.0 inches deep — the flat radial faces the saw left. That is the surface a finish is for: 664 square feet of it, and it is the same area whether you count it as members or as feet because the two are the same wood.

The allowance:

- **Coverage** is planned at 350 square feet per gallon per coat. That is an estimate for rough-sawn timber, which drinks more than a planed board.
- **2 coats**, which is what a working finish means in practice on a softwood face.
- **15 per cent handling**, for the spillage, the brush loss and the end grain, which takes up oil like a wick.

The oil allowance, and the labour

what	how much	where it comes from
Members to finish	120	two sawn faces each
Stock to work through	664.3 ft	end to end
Depth of a sawn face	6.0 in	log radius
Area of sawn face	664 sq ft	both faces, all members
Planning coverage	350 sq ft/gal	per coat, rough-sawn estimate
Coats allowed	2	the working assumption
Handling allowance	15%	spillage, brush loss, end grain
Oil the area asks for	3.80 gal	
With the allowance	4.37 gal	what to actually buy
If the can's claim held	1.33 gal	the label says 1,000 sq ft/gal
Labour	8.0 h	2 min per member per coat
Danish oil lists	5 min / 8 h	wait before wiping / cure
Oil and wax lists	60 min / 24 h	the same two numbers, different can

The coverage figure is an estimate on rough-sawn timber, the two schedules are claims printed on two real cans, and the difference between them is the reason this chapter says to read the label rather than the name. Estimated and claimed numbers are labelled as such because a reader is entitled to discount them.

The oil allowance, and the labour.

The arithmetic that falls out: 3.80 gallons for the area, and 4.37 gallons to actually buy. A little over four gallons of oil for a whole building's interior faces — which is the point of the estimate being printed rather than guessed at.

Now the number that argues against the whole business, and it belongs here rather than in a footnote. The can's label claims 1,000 square feet per gallon, which is 2.9 times the planning figure — and that claim is a manufacturer's coverage on a prepared surface, not a measurement on a rough-sawn wedge. If you plan to a label you will run out halfway through the pile, which is the sort of thing that turns a pleasant afternoon into a second trip.

And the labour: 2 minutes per member per coat comes to 8.0 hours of active work for the frame, and that figure excludes every part of the job that is not rubbing oil into wood — preparation, getting to the members, inspection, and the cure time between coats. A cheap can of oil can still make an expensive job. That is not an argument against finishing; it is an argument for knowing that you have bought eight hours of work when you buy the can.

What oil cannot do

Four things the video implies and the building refuses:

It cannot restore strength. A member that has split along its pith, or that has decay in it, is a repair — and if the decay is in a member doing structural work, it is a replacement. Oil on a compromised member is paint on a cracked beam: it hides the problem from the next person who inspects the building. The grading passes in the small-units chapter are where that decision is made, and this chapter does not get a vote.

It cannot stop decay or insects. Nothing in a linseed oil can carries a preservative rating. The ground-contact ring at the base of the building, the footings, and any timber that stays wet want a product rated for that job, not a food-grade oil.

It is not an air seal and it is not a rain seal. This is the one that matters most, because the seal is the thing that keeps the water out and a finish is the thing that makes the wood look right. An oiled face will still pass air and still pass water vapour — which is what you want on the inside faces, and exactly the wrong thing to rely on for a weather surface.

It cannot be used where something else has to bond. Keep oil off the lands that a sealant, tape or gasket has to stick to, and keep it off the printed key and its gaskets. Marine epoxy systems say it plainly: bonding surfaces must be free of oil and wax. A coat of oil in the wrong place turns a sealed joint into a joint that leaks in year two — which is precisely the failure the seam chapter spent its whole length preventing.

So the honest summary of what a finish buys is this: a face that wipes clean, that stays pale rather than greying, that lets the wood dry, and that can be renewed in an hour. That is a real thing to want. It is just not protection, and a builder who confuses the two will leave a weather surface unfinished because it "has oil on it".

Inspection, and the second coat

The finish record is the last piece of kit and it costs nothing. Write down the product, the batch, what you did to prepare the wood, the date, and where the coat went. Keep a sample board with the same treatment on it, because grey wood darkens under oil and it is easy to convince yourself that a weathered face has been protected when it has only been darkened.

Then the renewal question, which is where most finishing advice goes wrong by being too tidy. An oiled interior face on a dry frame does not need a schedule; it needs an inspection. Once a year, look at the faces that see water — around the openings, under the floor edge, at the base ring — and treat what is actually dry and worn rather than the whole building on a timetable. And when you do renew, clean first: a second coat over dust and grease sets the dirt in rather than sealing it under.

The one place to be genuinely suspicious is the base of the building, because that is where water sits and where a member is closest to the ground. If a member there looks dark and soft rather than merely grey, the answer is a repair and the answer is this year, not a coat of oil and a note in the record.

Which is the shape of the whole chapter: the finish is what you do to a sound building to keep it looking like one, and the inspection is how you find out whether it still is one.

CHAPTER 44 · EXPLAIN

Inside a Round Room

Living with curves

Inside a Round Room

This is the chapter about living in the thing, written the way the rest of the book is written: the numbers first, the compromises named, and the romance earned rather than assumed.

A round room is not a rectangular room with curved wallpaper. It changes where you can stand, how furniture fits, how sound behaves, and where the light falls — and the changes are measurable. This chapter measures them: the honest floor, the standing-height map, the furniture problem and its real solutions, and the acoustic quirk a hemisphere brings for free. It is the chapter to read *before* building, because the fixes it describes are layout decisions that cost nothing on paper and a renovation once the walls are up.

Where the usable floor really is

The honest number, because the floor a dome advertises and the floor a person can use are different things.

The dome's plan area — 365 square feet for this build — counts the whole circle. But the shell meets the floor at a tangent: the wall leans inward from the rim, and the last ring of floor around the edge has almost no headroom over it. The usable floor is the smaller circle where the ceiling is over your head, and the difference between the two numbers is the dome's real estate tax — floor you paid for, in frame and skin, that only a dog can use.

The map on the next page draws the honest version: the full circle, the standing-height contour inside it, and the dead ring between. The dead ring is not waste in the moral sense — it is where low storage, benches and beds belong, the furniture that does not need to stand up — but it is real, and the chapter states it because the alternative is the surprise every dome builder watches a visitor have: they step in, they look up, and they say the roof is lower than the picture suggested. The picture suggested the plan area. The roof was always the geometry.

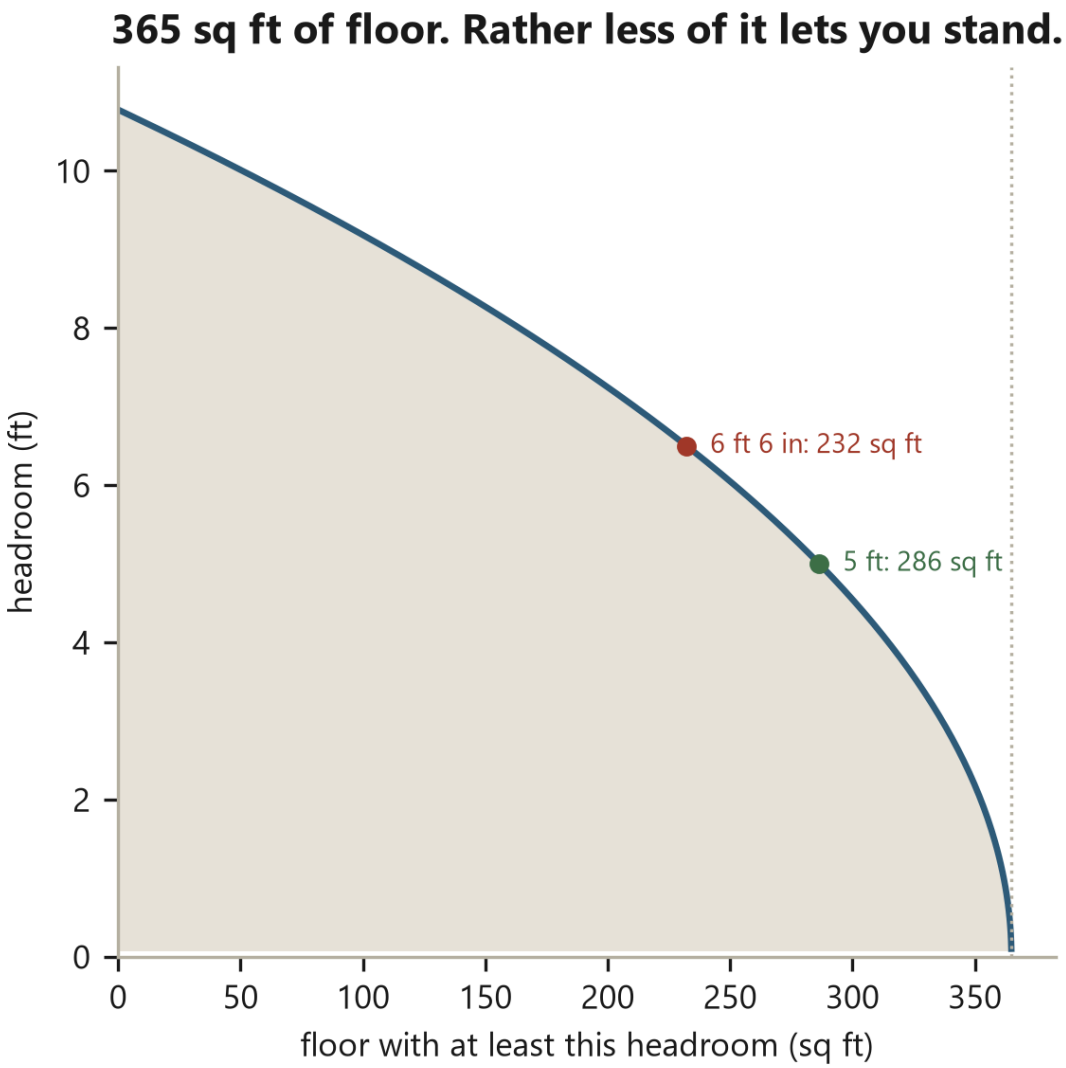
The fix the book already spent a chapter on is the riser of Chapter 58: lift the shell a few feet and the tangent moves outward, the dead ring shrinks, and the floor becomes genuinely usable to the wall. The round-room chapter recommends it for exactly this reason — not for looks, but because the standing-height map improves more per dollar than anything else in the whole fit-out.

Standing height, mapped

The map is the chapter's central object: a plan of the 365-square-foot floor, coloured by headroom, with the standing-height contour drawn where the ceiling passes head height.

Read the map as a furniture plan before it is a chart. The central disc is the room: full height, 10.8 feet at the crown, the space where a person stands, works, cooks. The ring between the contour and the rim is the compromise zone: overhead clearance shrinking smoothly to nothing at the wall, exactly the profile a bed, a bench, a desk-with-a-view or a row of shelves wants. And the rim itself is the dead zone the previous section named — reachable, useful, not standable.

The map also settles the oldest argument in dome living: where the bed goes. The answer is where the map says — in the ring, head toward the wall, because the only dimension a sleeping person needs at the wall is the one the dome has plenty of: length. The furniture section follows the same logic for everything else.



A hemisphere's floor is honest about its area and dishonest about its use. This is the number to quote when somebody compares a dome to a rectangle: not the floor, the floor you can stand up on.

365 square feet of floor. Rather less of it lets you stand up.

Furniture, storage and the wall that curves away

The practical answers, including the ones that are honestly just compromises, because a chapter that only offers custom joinery is writing for a budget this book does not assume.

The curve costs a wedge of dead air. Any straight-backed piece against a curved wall leaves a tapering gap behind it — the round-room tax of Chapter 11, paid here in inches. The cheap answers: pull furniture off the wall entirely and let the gap be a cable run and a shadow line; angle the piece so one edge touches and the gap opens somewhere unreachable; or

accept the gap and fill it with the things that are soft. The honest answer for the pieces that matter is built-in storage on the curve — the one place custom work pays for itself, because it converts the dead ring into drawers.

Low is the ring's whole vocabulary. Beds, benches, window seats, shoe racks, the wood stove's hearth — everything that does not need headroom lives in the ring, and the centre of the room stays open, which is how a round room ends up feeling larger than its number: the walls are furnished, the middle is air.

The middle wants to be the table. A round room's centre is its tallest, brightest, most reachable point, and the instinct to put the table there is the geometry's instinct, not a style choice. Everything radiates from it; every path through the room crosses it; and the light from the crown falls exactly on it. Furnish the ring, leave the centre, put the table in the middle — the round room's layout writes itself once the map is accepted.

Acoustics, and the whispering problem

The last section is the quirk nobody expects and everybody notices in the first week: a hemisphere focuses sound.

A dome is half of a whispering gallery. Sound from one side of the room runs up the curved wall and comes back down at the centre — the geometry that lets a whisper cross a cathedral — and in a house-sized dome the effect is not magical, it is *tiring*: conversation pools at the middle of the room, a kitchen tap at the rim is audible everywhere, and the building has no quiet corner, because corners are where quiet lives and the dome has none.

The fixes are the same ones theatres use, scaled down. Soft surfaces break the focus: the quilted layers of Chapter 60 do double duty as acoustic treatment, rugs and fabric kill the floor's reflection, and the furniture in the ring breaks up the wall's curve the way baffles do. The one thing not to do is nothing — an empty dome is an echo chamber, and the echo arrives before the furniture does.

The chapter closes with the round room's honest summary: a floor that is smaller than its number, a wall that leans, a centre that rings — and a room that, furnished the way the map dictates, is the calmest, most enveloping space most people will ever sleep in. The curves are not the problem. The curves are the room, and this chapter is the manual for living inside them.

Heat, Power and the Small Systems

Making a shell into somewhere you can be in February

The Small Systems

A shell keeps weather out. Somewhere you can be in February needs the small systems: insulation in the walls, power in the dark, heat in the cold, and a decision about the grid. This chapter is those systems, at the scale the book's method implies — small, deferrable, and run through the same voids the frame was built around.

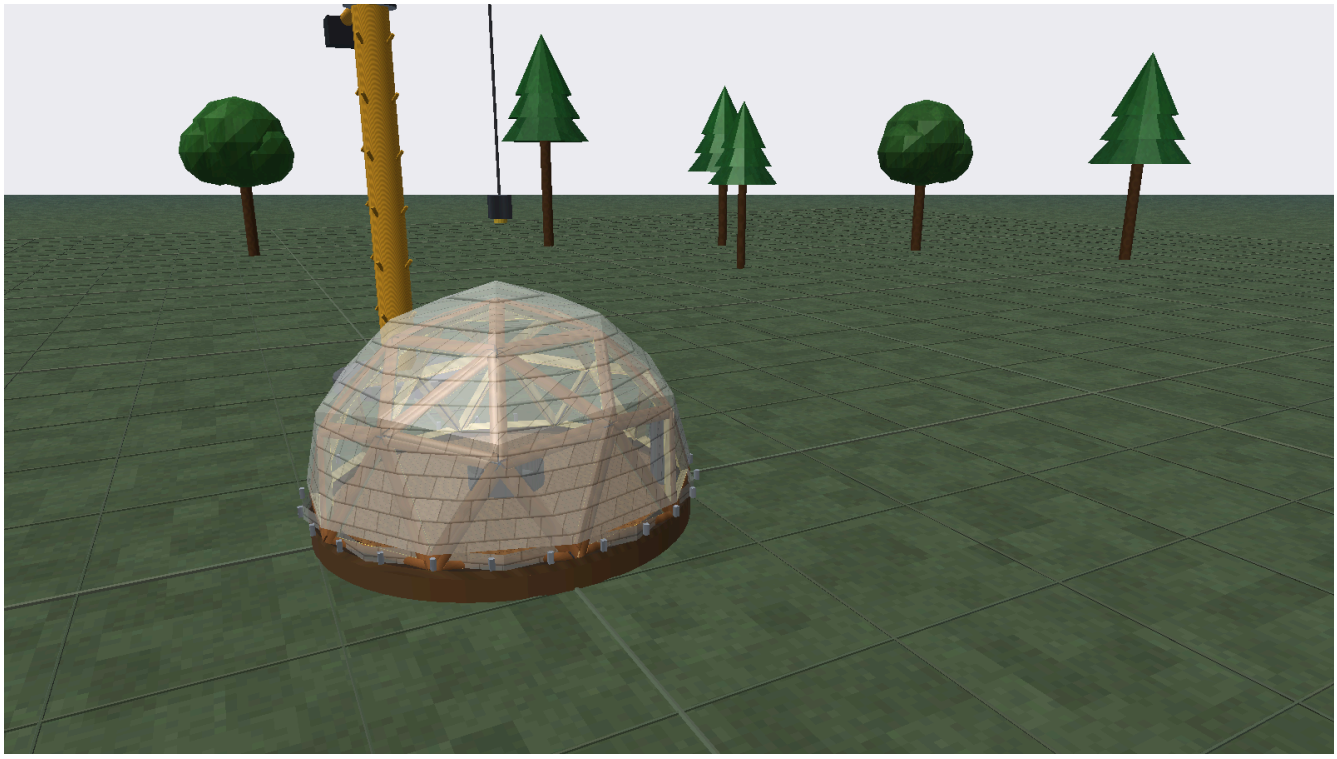
The dome's gift to its systems is the wedge frame itself: the cavity between the members, the channel of the seams, the column of the centre. The systems ride in those voids instead of requiring a stud wall's worth of drilling, and the chapter's sections follow the voids: insulation between the wedges, services in the seam channel, heat in the round volume, and the grid decision stated without romance. The chapter closes the home-making half of the book; after it, the dome is a house.

Insulating between wedges

The frame's wedges are 5.00 inches deep, and the depth is a cavity: between the skin outside and the inside face lies the whole thickness of the member, interrupted by nothing — no studs, no plates, no bays that were not drawn in from the start. Insulation goes in that cavity, and the shape does the rest.

The options are the ladder of Chapter 60, rung by rung: nothing first, then the quilted layers, each one a size up, each winter warmer than the last. The book's own preference is the quilt — layers of recycled clothing, sewn by the network of Chapter 61 — because it is the one insulation that matches the frame's own economy: it fits the cavity loosely, it comes off in order, and it costs what a sewing machine costs. Rigid board fits too, cut to the panels' triangles and pressed into the cavity; it insulates better per inch and gives up the removability the quilt was designed for. Either way the principle is the cavity's: **insulate the space between the members, never in front of them** — the members are the frame's structure and its look, and burying them under a blanket of foam throws away the only thing the building's walls ever had to say.

The one rule the chapter will not soften is the moisture rule from Chapter 42: insulation without a path for vapour is a rot factory. The cavity must breathe toward the seam channel, and the seam channel drains at the rim — the system Chapters 82 and 83 engineer in full. Insulate the dome as a living shell, not as a sealed box, and it gets warmer every winter instead of wetter.



It gets warmer every winter -- a frame of the seed-pitch film.

Running services in a frame with no studs

A conventional wall hides its wires and pipes in the stud bays. A wedge frame has no stud bays — and it does not need them, because the seam channel is the chase, and it runs everywhere.

The seams of Chapter 82 are the building's service corridor: a continuous void between the panels that follows the frame from rim to crown, wide enough for cable, small pipe and the sensor wiring the later chapters assume. The rules from the channel chapter bind here: water and wire fit, a round duct does not, no pressure fitting inside a seam, and the cable-in-chase question is the inspector's call, asked before the skin closes over it. Power runs from the utility column — the centre, where the core of Chapter 79 lives — outward along the seams to the points of use, and the dome's own geometry makes the routing obvious: centre to rim, like the spokes it is.

The discipline the chapter adds is the *deferral* discipline, borrowed from the stem-cell chapters and stated once here: run the services where the building will never be cut open again — the channel, the column, the chase — and leave the end connections surface-mounted, where a change of mind is a screwdriver, not a renovation. Threaded inserts in the wedge faces, set at panel time, are the cheap version of the same promise: the services bolt on, they unbolt, and the frame never learns their names. The building that can change its mind — Chapter 63 is the argument in full — is the building whose services were run as if change were coming, because it always is.

Heating a round volume

Why a dome is easier to heat than a rectangle, and the one way it is not, stated together because both are true.

The easy half is geometry: the dome has less skin per floor than a box — Chapter 57 computed the margin — and heat loss is a skin phenomenon. Less skin, less loss, less heat to buy; and the round volume mixes itself, so a single small source warms the whole room instead of leaving the far corner cold while the thermostat argues with the window.

The hard half is the one the book's own audit keeps repeating: a dome is not *automatically* cheaper to heat. The skin is cheaper to insulate well, and the shell ladder exists because an uninsulated dome loses heat faster than an uninsulated box — less skin, but skin with a seam every few feet, and air moving through a seam carries heat with it. The heating claim the book will make is narrow and honest: **the same build quality, spent on a dome, buys more warmth per dollar than on a box** — the margin is geometry's, the build quality is yours, and Chapter 69 keeps the metered version of the claim in its audit, marked as not yet metered.

The practical system follows the shape: a small stove or a single heat source at the centre, the round volume mixing the air, the quilted layers doing the retention. February in the dome is a solved problem, not a promised one.

Off grid, or not

The last decision, treated without romance, because the dome's audience includes both the grid-connected and the gridless, and the book refuses to pretend the choice is free.

Off grid is a battery question before it is a panel question. The panels are cheap and getting cheaper; the storage is the real purchase, and the arithmetic of Chapter 69 already printed the honest version of a solar budget — the demand, the array, the bank, the price, and the days of autonomy the bank actually buys. Off grid means the lights go out when the bank is empty, the fridge becomes a weather gamble, and the system's owner becomes its operator — none of which is a romance, all of which is survivable for the person who chose it.

On grid, the dome is simply a small building with a small bill, and the chapter's recommendation for the first build is the unglamorous one: connect, if the line is there, and let the off-grid system be an upgrade bought with the money the dome saved. The deferred-order rule of Chapter 63 applies: run the conduit for the future array while the walls are open, buy the panels later, and never cut the shell twice for a decision that could have waited.

The small systems close the home chapter the way the dome closes its shell: one layer at a time, each one bought when the last one proved itself. Insulate first — the cheapest warmth. Then power — the difference between a room and a home. Then heat, then the grid question last, when the building's own behaviour has answered it. The dome does not need its systems all at once, and the builder who remembers that spends the winter warm instead of broke.

The set, priced

The films quote a small off-grid set for a building like this one, and the model's own components reproduce it line by line: a 800 watt array at 0.95 dollars a watt self-installed, a 10 kilowatt-hour bank at 410 dollars a kilowatt hour, and a 3,000 watt inverter. That is 4,100 of storage, 630 of inverter, and the array on top: **5,490 all in** — printed here because the films print it, and because a reader deserves to see that the *battery* is three quarters of the set.

Three things about that set are worth knowing before buying it.

The battery is the purchase. Power electronics have got cheap and cells have not, which is why the storage line dominates. Every kilowatt hour added to the bank costs 410 dollars and nothing else in the system cares — so a reader

sizing a system should decide the days of autonomy they want and buy *that*, rather than buying panels until the lights stay on.

A derated array is the honest array. The model applies a 78 per cent derate to a nameplate rating, because wires, heat, angle and dust are real. Sizing on nameplate watts is how a system comes up short in its first February — and February is the month it was bought for.

And the surplus is the real argument. On this building's roof the array makes 40 panels' worth of power across 292 square feet, and 54 per cent of what it makes is surplus the building cannot use at the moment it is made. That surplus is worth 0.14 dollars a kilowatt hour if it displaces bought power and 0.045 if it is exported — and the gap between those two numbers, not the price of the panels, is what decides whether an off-grid system pays. It is also why the sun chapter's rotating ring loses at grid prices and wins off grid: the same numbers, read at two different rates.

For a reader who is genuinely off grid, the surplus argues for storage rather than for more array — the same conclusion the battery-dominates-the-set note reached from the other direction, and the reason this book's off-grid advice is "bigger bank, smaller array" rather than the reverse.

Build one utility core

Everything in this chapter so far has been about the building's skin and its services as *decisions*. This section is the one place where the services are a *build*, because the utility core is the one part of the building an owner cannot make, and it is worth seeing exactly why.

The core is the centre column: the chase, the drain stack, the water manifold, the electrical panel, the riser and the seal cap. Everything the building needs passes through it, and it is assembled and tested as one object before it is stood up on the pad's service port.

It is 14 steps across 5 stages, 7.2 hours once you have done it once and 17.2 hours the first time, for 1,405 of parts. The stage that decides the build is power, at 110 minutes. And the build uses 9 tools, of which the two people skip are worth naming, because skipping them is the exact failure this chapter exists to prevent.

The core's build, stage by stage

stage	time	share	hours
Chase	95 min	22%	1.58 h
Close	50 min	12%	0.83 h
Drain	70 min	16%	1.17 h
Power	110 min	26%	1.83 h
Water	105 min	24%	1.75 h
Every stage	430 min	100%	7.17 h
Once practised			**7.17 h**
The first one			**17.2 h**
Parts	**23 items**		**$1,405**
Labour, practised	**9 tools**		**$201**
Labour, first time			**$482**

Fourteen steps across five stages, read from the build's own step list. The chase is drilled flat on a bench, the drain goes in first because it cannot be re-routed, the water is proved at pressure before anything is closed, and the power lands last in a core known dry.

The core's build, stage by stage.

The chase first, on the bench, upside down. The whole core is built on a bench and stood up at the end, which is the difference between a four-hour job and a day of kneeling. Every penetration is drilled and grommeted while the sections are apart and at waist height, because the same hole later is a hole saw inside a finished core. The chase is also the one part where an error at the bottom is an inch of error at the apex, where the seal cap has to seat: plumb is checked over the whole run, dry, before anything is fastened.

The drain next, because it cannot be re-routed. A two-inch stack running the full height with the trap at its foot, dry-fitted and marked across every socket first — solvent cement gives you about five seconds and no second chance at alignment. Work bottom to top so gravity is not pulling a fresh joint apart, and then do not touch it for fifteen minutes.

Then water, and prove it before anything closes. Manifold at chest height, isolating valve at knee height, both opposite the panel, and four capped tails so a building with no plumbing still ships with them. Every crimp gets the go/no-go gauge — this is the first tool people skip, and an uncrimped ring will pass a pressure test and fail in a year. The proof is a gauge on the system for half an hour, and it happens *before* the panel goes in, because after that a leak costs the panel.

Then power, into a core known dry. Every lug torqued to the figure printed inside the panel door rather than by feel — the torque screwdriver is the second tool people skip, and a terminal that feels tight is not tight. Four breakers: outlet ring, lighting, the window unit, and one spare, so the first snap-on module never needs the door opened again. Label them in writing now, not from memory in a year.

Then close it and prove the cap. Sleeve on the chase, cap on the sleeve, six catches set so the gasket compresses evenly, and then open and close it five times on the bench. The cap is the one thing between the building and the weather.

Two things follow from doing it in that order, and both are the chapter's argument in miniature. The first is that **the sequence is the specification**: drain before water, water before power, power before closing, closing before standing up. Get the order wrong and you are cutting into a finished assembly, which is exactly what the chase's grommets exist to prevent. The second is that **the core moves**: it unbolts from four joints and a lift and goes into the next dome, so those 7.2 hours are spent once for a lifetime of buildings rather than once per building. Which is the reason the price of the standard article puts 1,405 into this part and almost nothing into the frame.

A reader who has built the frame in this book can build the core. A reader who has built neither should practise on the core first: it is small, it is on a bench, its mistakes are recoverable, and it teaches the same lesson the whole building teaches — that the order you do things in is part of the design.

The plates, and what they are honestly worth

One piece of hardware in this chapter's list deserves its own page, because the films spend two chapters on it and this book had only mentioned it in passing: the thermoelectric plate that lives in the seam channel.

It is a heat pump, not a cooler. A Peltier plate moves heat from one face to the other, and reversing the current reverses the direction. That single fact decides how it is used: throw heat to the side you want warmer — into the room in winter, outside in summer. The first film in the series showed the plate heating the room and called it right, which was right in winter and wrong in the other half of the year, and the correction is worth knowing before anybody wires one up.

What it makes, and what that is worth. A dome's worth of plates is 360 watts, and run six hours a day it condenses 0.756 litres a day — 73 gallons in a year. That sounds like a water supply until it is put next to the roof: one inch of rain on this shell is 185 gallons, so a year of plate output is the rain of a single 0.39-inch shower. And it costs 2.86 kilowatt hours per litre, against the roughly one a compressor-driven dehumidifier uses — which is already a poor way to make water.

So the honest statement, and it is the films' own: **the plates are a dehumidifier that happens to make water, and the roof is the water supply.** Buy them to keep the wood dry where nothing else can — in the seam channel, in the wettest weeks of the year — and let the gutter do the water. A reader who installs plates to be independent of a well has bought the wrong machine, and will find out in August.

The wall that could be a filter

The last idea in this chapter is the one the project has computed and never built, so it belongs here with its arithmetic and its caveats side by side.

The idea: run air in at the base ring, let it cross the whole shell, and the building's wall becomes a distributed filter — warmer and drier on the way in. Reverse it and the sky is the supply, drawn inward through the structure: dynamic insulation, in which the wall is warmed by the air being drawn through it. Both directions are one machine — a fan, a ring main, and a shell that is porous on purpose.

The airflow side is arithmetic a reader can check. The shell encloses 2,622 cubic feet behind 730 square feet of surface, and the fresh air a dome needs is about 17.6 cubic feet a minute — so air would cross the wall at 0.02 feet a minute, 0.3 inches a minute, which is 0.4 air changes an hour. That is a crawl no occupant can feel, and feeling it is what would make the idea unacceptable. So far so good — and this is exactly where the films stop, and where this book stops too.

Four things decide whether the idea is *good* rather than merely quiet, and none of them is computable from the airflow:

The wall is untested as a plenum. Nobody has built a strut lattice at building scale and measured how air actually spreads through it. A model that treats the shell as an even filter is an assumption, and the first place it will be wrong is at the joints.

Direction is the risk. Air moving outward through a warm, damp structure carries moisture into the wood; the same flow in the other season carries it out. A dynamic-insulation wall works because it is engineered to keep the outer face warm and dry — a designed assembly with a moisture profile, not a gap between wedges.

A sealed skin cannot breathe. The weather membrane that keeps rain out also kills this idea, so the wall-as-filter needs a deliberately permeable shell, which is the opposite of what the weather chapter spent its length achieving.

An unreachable filter is never cleaned. If the shell is the filter, the filter is the building. Nobody takes a dome apart to clean it, and a filter nobody cleans is a blockage with a fan behind it.

The position this book takes is the one the films took: the airflow arithmetic stands, the idea is not refuted, and the project has not built one. A reader who wants to try it should treat the first winter as the experiment — instrument the members, vent the cavity rather than sealing it, keep the filter somewhere reachable — and should not build a house on the assumption that it works. The chapter's rule about honesty applies to an idea as much as to a number: say what is computed, say what is not, and leave the reader the decision.

CHAPTER 46 · REFERENCE

The Money, Both Ways

What was spent, and what was avoided

> ***This chapter corrects something.*** *The $50/hour figure in the project brief, which doubled a volume ratio for safety rather than computing the section.*

The Money

The accounting chapter, and it begins by saying how it counts, because the two numbers a build produces — what was spent and what was avoided — are different species, and most dome books breed them into a hybrid that flatters both parents.

This chapter counts cash out, itemised, with the dashes left where the original build did not log a line. It counts the value of the frame three ways, with the disagreement between the ways printed rather than averaged away. And it corrects its own project's most-quoted hourly figure, on the page, the way this book corrects everything. The number that does not help the argument is here, several of them are, because the book's reputation was staked on printing them, and a chapter that only argued one way would un-stake it.

What was actually spent

The table is cash out: the saw, the fuel, the chain, the fasteners, the gaskets, the sheeting — every line the fortnight actually paid, with the source of each number and the honest dashes where the original build did not keep a line.

The table's headline row is the one the whole method exists to produce: timber, zero dollars. The structural wood cost the fuel to cut it, and the fuel is one of the small lines above. Everything else in the table is the *conversion* — the saw, the chain, the screws, the keys — which is the entire point of Chapters 2 and 13 made into a receipt: the building material was free, and the receipt is for the machine that made it building material.

The dashes are not a gap to be filled. They are lines the original build did not log, and this book will not fill them with plausible numbers — the declared-constants discipline from the front matter applies to money exactly as it applies to geometry. A cost that was not recorded is a cost that is not claimed.

What was actually spent

item	cost	source
Chainsaw	$199	priced
Fuel, bar oil, files	$—	to log
Fasteners	$—	to log
Seam keys / gasket stock	$—	to log
Sheathing	$—	to log
Footing	$—	to log
Timber	**$0**	**two trees**

The dashes are real: these were not logged at the time, and this book will not fill them in with plausible numbers. The last line is the one that matters -- the structural timber cost nothing but the fuel to cut it.

Cash out, itemised.

Substitution value is not income

The most important paragraph in the chapter, and it gets its own section because every honest number after it depends on the distinction.

What the frame *avoids* is not money the builder *has*. The frame bought at the national rate would cost 9,798 dollars; the fortnight spends 497; the difference — 9,301 dollars — is what the build *keeps*. (That cash figure is the film's own ledger, which counted two trees standing; this book's build fells 3, and the third tree's stumpage — 23.23 dollars a ton at the pine film's researched rate, on a tree the size of a car — barely moves the line. The book says so rather than silently rounding.) It is a real number, and it is not income. It buys groceries only if the builder was going to buy the frame anyway, and the builder who was never going to buy the frame has not earned nine thousand dollars; they have simply not spent it. The same distinction runs through the whole book: the strut's value, the hour's rate, the pine's ladder rungs — all of them are substitution values, stated as such, and the one sentence that converts a substitution value into a lie is the sentence that drops the word "substitution".

The useful, honest use of the number is as a *rate on time*: 9,301 dollars kept, over 80 hours of hands-on work, is 116 dollars an hour — the wage the fortnight paid the builder in avoided cost. Against a manufactured home's cheaper rate the same arithmetic gives 59; against a carpenter's median 29.12 dollars an hour, and a typical worker's take-home 19.69, buying that frame would have cost 498 hours of ordinary pay. Those comparisons are the chapter's real product: the

fortnight's work, priced against the hours a wage would have needed to buy the same frame. The cash never existed. The hours did.

The strut, valued three ways

The table is the same arithmetic run at the strut's scale, and it is printed with the disagreement left in, because the disagreement is the lesson.

One strut, valued against a nominal two-by-four, against a dressed one, and by the board foot — three methods, three answers, and the reasons they differ are printed beside them. The nominal comparison is flattered by the fact that a nominal board does not exist; the dressed comparison is the honest one, and it comes out *higher*, because a dressed board is smaller than its name and a strut replaces more of them — exactly the direction a sceptic should check first, and the table invites it. The board-foot row refuses to give a single number, because the market does not. The last row is the estimate this project started from, kept in the table so the errata below has something to correct against.

One strut, valued three ways

basis	quantity	value
Against a nominal 2x4 (8.00 sq in)	1.23 boards	$4.09
Against a dressed 2x4 (5.25 sq in)	1.87 boards	$6.23
By board foot	4.35 bf	market varies
Implied rate, nominal	5/hr	$20.43/hr
Implied rate, dressed	5/hr	$31.14/hr
The brief's estimate	**—**	**$50.00/hr**

The last row is the figure this project started from, and it is the highest. It valued a strut at $10 by doubling a volume ratio for safety; the rows above compute the ratio from the actual cross-section. Both computed rates are real; neither is money anybody has been paid.

One strut, valued three ways. The numbers disagree and the reasons are printed.

The fifty dollars an hour claim

The errata, named and corrected, because this project's brief said fifty dollars an hour and the arithmetic says otherwise.

The brief's fifty was reached by doubling a volume ratio for safety rather than by computing the section — a shortcut that made the number up, generously, and the films repeated it. The computed rates are 20.43 dollars an hour against nominal two-by-fours and 31.14 against dressed ones — smaller numbers, arrived at by measuring the actual cross-section and pricing the actual replacement. The honest note, stated the way the strut table stated it: the dressed number is the true comparison, and it is the *higher* of the two computed rates, because the dressed board is the smaller board. The brief's fifty overshot even that.

The correction is the model for the chapter and the book: the claim, the error, the arithmetic that produces the smaller number, all on the page, and the errata pages are where the rest of the list lives. The money chapter's own verdict closes the circle: the method's economics are strong enough to be measured properly, and measuring properly is the only version this book will print.

Where the difference actually comes from

The fortnight's own receipts say what was spent. The films ask a different question — what would the same floor have cost *bought* — and the answer is a stack rather than a number, which is worth printing because a reader can check every line of it.

the same floor, 300 square feet	dollars	whose it is
Bought whole	65,100	the price of the building
The builder's and seller's share	16,600	work you did not buy
Your own labour	17,300	the hours the book counted
Your own trees	3,200	the wood you already had
The shape itself	4,300	less skin, less framing, shorter runs
Left to pay	**23,700**	materials, services, finishes

The four middle lines add to the difference exactly, and that is the point of printing them as a stack rather than as a total: **37 per cent of what a house costs does not care what shape it is** — the fees, the windows, the cabinets, the appliances, the driveway — and the four lines above are where the rest goes. A reader who has been told "domes are cheap" has been told nothing. A reader who can see that 16,600 of the saving is the builder's own share, 17,300 is their own hours and only 4,300 belongs to the geometry has been told the whole truth about this method's economics, including the part that is about them rather than about the dome.

And it reconciles against the cheapest thing a family can buy instead:

	dollars
The same floor at a manufactured home’s average price	25,986
The dome way, before any of your hours	16,444

That gap is what the films call the small one, and they are right to call it small: the manufactured home is genuinely hard to beat on price, and the honest comparison is not price but what the money buys — a building you can take apart, re-skin and move, made from wood that was standing on the land. The money chapter prints both numbers and refuses to pretend the gap is bigger than it is.

CHAPTER 47 · REFERENCE

The Hours

Where fourteen days actually went

The Hours

The money chapter closed with a rate on hours; this chapter is the hours themselves, logged the way the book logs everything — as a table, with the surprises named.

A fortnight is fourteen days, and the days did not go where the plan said. The plan said the ripping would be the mountain and the rest foothills; the log says the ripping was a third of the build and the *panels* matched it, with the jig and the raising each taking their own full share. The table below is the actual spread, and the two sections after it are the reading of it: what took longest, and what a second build would cost now that the learning is spent.

The forty hours

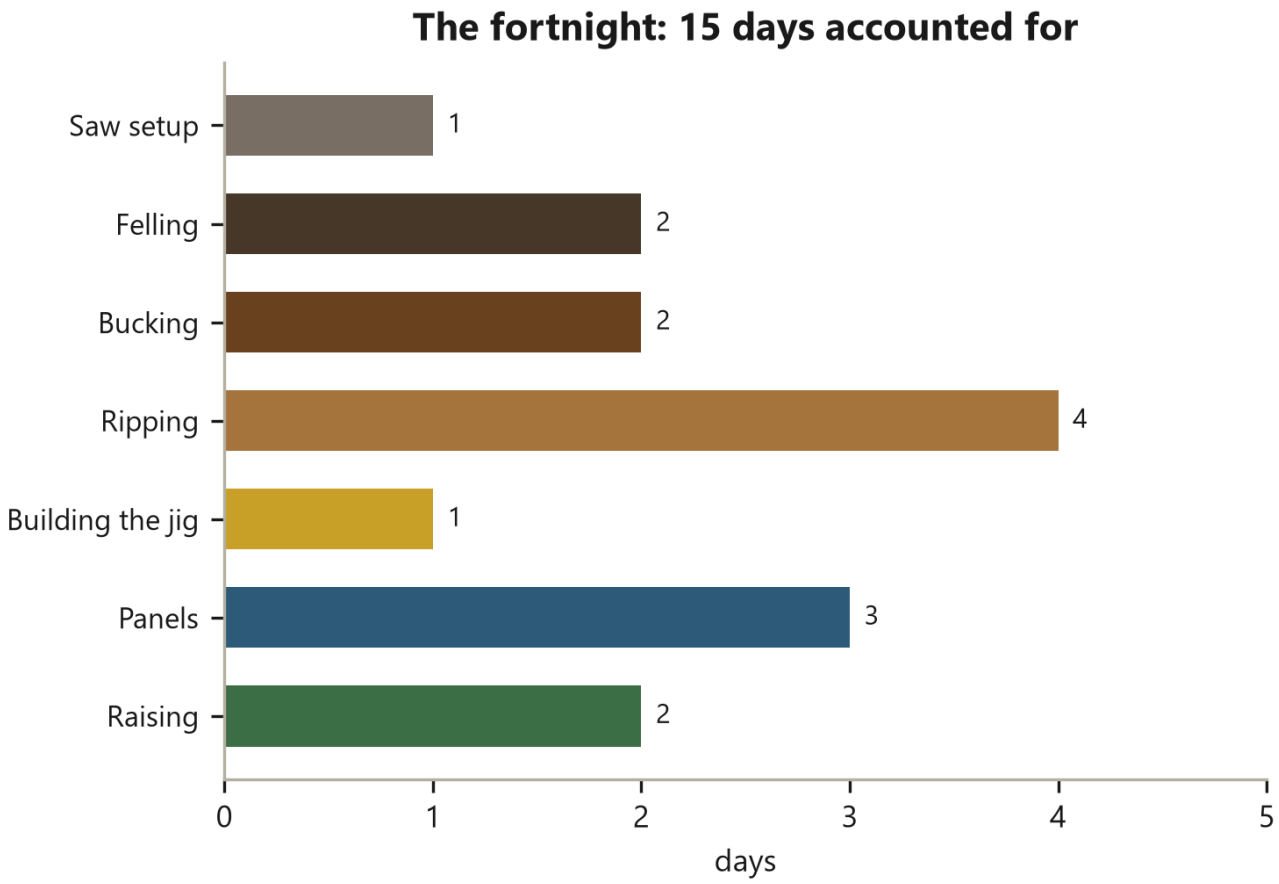
The title's claim, counted before the calendar is, because the title is an arithmetic: 37.5 hours of hands-on work, inside the 40-hour week the book is named after.

The clock, in full: the frame's 120 members split at the measured 30 wedges a session is 16 hours at the log; the same members erected at 15 minutes each is 30 straight hours of assembly; and the 25 per cent error allowance on the erection — a first build's tuition, paid on camera — brings the assembly to 37.5, with 2.5 hours spare inside the week. The splitting is the harvest; the forty hours is the *erection* — panels to standing, props to winch. The fortnight is where those hours live on a calendar: part days, weather days, thinking days. The hands-on clock is the number the title makes, and this chapter's chart is the same clock, drawn as days.

The fortnight, hour by hour

The chart is the fourteen days as a stacked bar: each phase, its share of the clock, and the total that has to come out at fourteen.

The shape of it is the chapter's whole argument. The felling and bucking — the dangerous, dramatic first days — are small slices. The ripping is the longest single block, as the plan promised. But the panels and the jig together match the ripping, and the raising is a full two days, and the saw setup is a line item nobody budgets and everybody pays. A builder reading the chart before their own build learns the real lesson of the fortnight: **the tree work is quick, the repetition is the mountain, and the jig is the single best-spent day in the whole table.**



Ripping is 4 days of the fourteen, and it is the part everybody assumes is the whole job. The panels take as long, and the jig day pays for both.

Fourteen days, and where they went.

What took longest and why

The surprises, named, because the chart's flat middle is where the plan's assumptions died.

The ripping took its 24 hours — the 5.0-struts-an-hour rate of Chapter 29, sustained across four afternoons. It is the build's biggest single block, and it is the one everybody expected. The surprise is that it is *only* the biggest block — the panel run that followed matched it nearly hour for hour, because forty panels of twelve motions is not fast, it is merely *constant*, and constant work still sums to days. The jig's single day looks like a small slice and it is the one that kept the panel run from being twice as long. The raising's two days are the least compressible of all: a dome goes up one panel at a time, one person, one prop, and the geometry does not care how practiced the hands are.

The honest reading: the method's *rate* claims were true — the ripping ran at the promised pace — and the method's *schedule* claims were optimistic at the edges, because the repetition that makes each operation fast also makes each operation numerous. The fortnight is the size it is; the chart is the size's explanation.

What a second build would take

The learning-curve estimate, marked clearly as an estimate, because the book's rule about numbers applies to its own predictions: a guess must be labelled a guess.

The first build paid for: the jig's design, the marking scheme, the three practice panels, the seam arguments, the wrong ring that got caught, the near miss that got survived. A second build pays for none of them — the jig exists, the scheme is written, the arguments are settled on the page — and the honest estimate is that the repetition days shrink most: the ripping stays at its physical pace, but the panels and the raising lose their learning time, and the total lands near the rate the films always claimed and the first build only approached.

The estimate is not a promise and the book will not dress it as one. It is the answer to the question every builder asks at the end of the fortnight — *how long would the next one take?* — and the answer's form matters more than its number: the second build is the first build minus its tuition, and the tuition is what this chapter's chart was actually recording. The hours were never wasted. They were the price of the next dome, and the next dome inherits them.

CHAPTER 48 · EXPLAIN

Bigger, Smaller, Other Frequencies

Changing the dome without changing the method

Other Frequencies

The whole book so far has built one dome: the 2V hemisphere of Chapter 8, subdivided once from the icosahedron. This chapter opens the question the outline's own logic raises: what happens to the method when the frequency changes — and what does not change at all.

The frequency of a geodesic dome is how many times the icosahedron's edges are subdivided before projection. 2V splits each edge once; 3V splits it twice; 4V three times. The method this book teaches — split logs, pinwheel panels, one jig — does not care which frequency it frames, and that is the chapter's first finding: the method is the same, and only the numbers change. The rest of the chapter states the trade the numbers make, and the place where the method stops working altogether.

More panels, shorter sticks

The trade, stated as arithmetic, because "3V is smoother" is an adjective and this book trades in counts.

Higher frequencies subdivide harder: more triangles per face, more panels in the hemisphere, and — this is the part that matters — *shorter* members, because every edge is a smaller fraction of the sphere. The panels multiply, the members shorten, and the dome approaches a true sphere, which is the entire point of the higher frequencies for anyone whose skin or aesthetic wants roundness. The cost is the pile: more panels means more seams, more keys, more screws, more motions of the jig — and fewer of the frame's parts repeat in the way 2V's two-lengths-two-shapes economy allows. The table below runs the counts, and the honest reading of it is that 2V is not the *best* dome; it is the *cheapest* dome at the scale the method was designed for, and the higher frequencies buy smoothness with repetition.

What does not change is the more important list. The split, the sector, the pinwheel, the jig, the overfit, the seam key — none of them cares whether the frame is 2V or 3V. A 3V dome built this way is still one log, seven cuts, one bench, forty panels and a fortnight; it just has more panels in the fortnight. The method transfers, which is the sentence the chapter exists to prove: the wedge is not married to the frequency, it is married to the triangle, and every frequency is made of triangles.

Frequency, and what it asks of the woodpile

frequency	panels	members	lengths	longest in	shortest in
2V (this book)	40	120	4	72.0	62.7
3V	75	165	3 edge classes	—	—
4V	160	300	6 edge classes	—	—

Only the 2V row is computed: this repository solves the 2V hemisphere, and the book will not print numbers for frames it has not built. The 3V and 4V counts are standard geodesic results; their member lengths depend on a solve that is not here yet, and the dashes say so.

What each frequency asks of the woodpile.

Frequency comparison

The table is the trade in full: 2V, 3V, 4V, each with its panel count, its member classes and its longest member, all computed from the same geometry so the rows differ only by frequency.

The row that matters is the member-length row, because it is where the frequencies meet the method's own constraint. 2V's longest member is 72.00 inches — the size the tree chapter bucked for. The higher frequencies' members are shorter, which would let the same trees build a *bigger* dome at the same handling limit — the flat-rate argument of Chapter 52, applied through the frequency knob. That is the real decision the table supports: not "which dome is rounder" but "which frequency lets these trees and these arms build the floor I want", and the answer is read off the table the same way every other sizing decision in this book is read — by looking at the row, not the adjectives.

Where the method stops working

The honest limit, stated so the chapter does not over-sell its own transferability.

The method stops working at the *small* end, and the reason is the sector's own geometry. A dome too small for its split count has panels whose edges are shorter than the members are *wide* — the sector's bark face is 3.83 inches across, and a

panel whose sides approach that width can no longer accommodate the pinwheel's overlap, the inset, and the key. The joints crowd the corner, the overlap swallows the triangle, and the method's arithmetic quietly runs out of room. The fix is not a smaller split — a six-inch dome needs a different member, not a thinner wedge — and Chapter 73 is where the alternatives live.

At the large end the method does not stop; it *delegates*. Big domes want higher frequencies for the member-length reason above, and beyond the handling band of Chapter 52 they want crews, which is Chapter 50. The method itself keeps working — the split, the jig, the seam — all the way up; what changes is the schedule, not the procedure. The chapter's closing sentence is the one it opened with: the method is the same at every frequency. Only the numbers change, and the numbers are in the table.

Where the two lengths actually come from

Every frequency above is built by the same five moves, and it is worth walking them once with numbers, because the films spend a whole chapter on it and this book had only ever printed the *results*.

Start with a solid. The icosahedron has 12 corners, 30 edges and 20 faces, and V − E + F = 2 however many times you subdivide it. That identity is the cheapest bug detector in geometry, and the book uses it as one.

Put it on a sphere of radius one. Divide every corner by its own distance from the centre and the parent edge measures 1.051462 of the radius — the only number in this section that is not a length a reader can hold.

Halve every edge. The midpoint of a straight edge is the average of its two ends, and that average falls *inside* the sphere: the midpoint sits 0.850651 of the radius from the centre, a shortfall of 0.149349 — 14.93 per cent of the radius. On a small model that is nothing; on a building it is the difference between a faceted ball and a dome, and it is the arithmetic answer to the reader who assumes a sphere must be rounder. A sphere drawn with straight sticks is not round at all, and the input to that error is a straight line drawn where the surface curves.

Push each midpoint back out to the surface — divide it by its own length. That one operation is the entire difference between an icosahedron and a geodesic dome, and it is what the films call the geodesic step. It leaves the edge line kinked, and the kink is where the second length comes from.

Measure the two. The chord from a corner to a projected midpoint is 0.546533 of the radius; the chord between two projected midpoints is 0.618034. Two lengths and no third: that is why the cut list has two classes rather than forty, and it is the arithmetic behind every table in this chapter.

Two corrections for things a reader may have heard, both checked rather than asserted.

Phi is not the strut ratio. The golden ratio is 1.618034, and it has a real home in this geometry: the twelve corners of the icosahedron are three golden rectangles at right angles to each other. But the ratio *between the two struts* is 1.130826 — about 30 per cent below phi, near enough to look like a coincidence on a drawing and nowhere near enough to build on. A two-frequency dome is not a golden-ratio dome; it is a dome whose parent solid happens to be built from golden rectangles.

The boards are not the factors. Cut 72 inches and 63.67 and you are cutting the solved lengths. Measure two boards on a bench and you get 1.133858 — 0.27 per cent above the factor ratio, because a tape, a kerf and a real log are not a unit

sphere. The worked-build chapter fits both boards by least squares rather than trusting either, and that 0.27 per cent is exactly what it is fitting around: the arithmetic is exact and the tree is not, which is the method’s whole position in one line.

Other Species, Other Sections

Hardwood, hemlock, and a trunk that is not round

Other Species

The book's tree has been pine throughout — the windfall of Chapter 1, straight-grained, soft, forgiving. This chapter is what changes when the tree is something else, because the method was never pine's; it was the *sector's*, and sectors grow in other forests.

The chapter covers the three differences that matter: how other species split, dry and hold; how to split a trunk that is not round; and what building with green wood actually does over the first year. The honest rule of the chapter, stated once and applied throughout, is the one the wedge book's species page gives: **use what grows where you are** — the method works with whatever straight-grained stock the local woodlot offers, and the differences are adjustments, not prohibitions.

Density, splitting and the drying schedule

Species by species, the practical differences, because a book that says "any species" owes the reader the list of what "any" asks.

What splits well. Straight-grained softwoods — pine, spruce, fir, cedar — split cleanly along the radius, which is what the seven-cut procedure assumes. The grain runs the length of the log, the pith is honest, and the sector comes off whole.

What splits poorly. Interlocked or spiral grain — elm, some eucalyptus — fights the rip: the grain runs around the log instead of along it, the split wanders, and the sector arrives ragged. The tree's bark usually tells the story before the saw does — spiral bark means spiral grain — and the rule is to read the bark and let the tree opt out.

What lasts. Rot resistance is a species property, not a build quality. Cedar, larch, black locust and osage orange sit in the ground and shrug; pine and spruce do not. The rot-prone species belong above the damp line, on dry footings, with the bark charred or the face oiled the way the finish section describes — and the rot-resistant ones are the free upgrade the local woodlot sometimes hands over.

Density and the drying schedule. Hard, dense wood dries slowly and splits harder; soft wood dries fast and forgives. The schedule follows: dense species get longer in the stack, thinner stickers, more patience — and the key-cut-last rule below becomes more important the denser the tree, because a dense sector moves more in its first year than a pine one does.

A trunk that is oval

The tree is not always round, and the split procedure's first assumption — a circle divided evenly into 8 — meets reality in the oval trunk.

An oval section still splits into eighths; it just produces *unequal* eighths if the cuts are spaced by angle the way a round log's would be. The fix is to space by *area*, not by angle: mark the end grain into 8 sectors of equal wood, not equal degrees, so every wedge carries the same section even though the trunk was an ellipse. The marks are made by eye against the rings — the pith is the centre, the rings are the map, and the sector boundaries are drawn where the wood divides evenly — and the ripping passes follow the marks instead of the habit.

The consequence travels, and the book says so: unequal spacing means the sectors' *angles* vary slightly, which means the seam keys vary slightly, which is exactly the tolerance the gasket chapters exist to absorb. The oval trunk costs a little more attention at the split and nothing downstream, because the downstream was designed for "nearly" — the same word the round trunk's own imperfections already earned.

Green versus dry

The last difference is not a species; it is a state, and every build in this book happens in it: the wood is green.

Green wood moves as it dries — shrinks across the grain, barely along it — and a wedge moves *least* of all the shapes a log could become, because a split follows the grain instead of cutting across it. The sector's angle closes slightly as the stick dries; a sawn board cups and twists; a boxed heart tears itself open with a check. The method's geometry was chosen for this behaviour: the frame is full of seams that can take the movement, the keys are the last things cut, and the whole structure is allowed to settle for a year before the skin is final.

The one rule that matters most, stated for the bench: **cut or print the keys after the frame has dried.** A key cut to a green seam fits a dry seam the way a green pair of shoes fits next year. Let the frame stand its first months, let the sectors close their angles, then cut the keys to the seams that actually exist. The rule costs patience and saves the dome; it is the drying chapter's whole lesson, in one sentence.

And the finish, because the wood's bare faces are a joint, not a surface: char the outside bark face — the yakisugi pass — and oil it with linseed, and leave the sawn faces bare, because a finish on the joint faces traps the moisture the seam channel is trying to move. The one warning the finish carries is the safety one: oily rags self-ignite, and the can lives outside the building. The species change, the sections change, the state changes — and the method, all the way through, is still the one this book has been teaching: read the tree, split the sector, let the wood be wood.

CHAPTER 50 · HOWTO

Building With Other People

What changes when there are three of you

Other People

The method was designed for one person, and every number in the book so far assumed that constraint — the six-foot member, the one bench, the props a single pair of hands can set. This chapter is what changes when there are three of you, because crews happen: a partner, a neighbour, a build somebody else asked for.

The changes are fewer than the fear suggests, and they cluster in three places: the benches multiply, the teaching becomes a step in the build, and the liability conversation stops being optional. The method itself — the split, the jig, the seam — does not change at all, which is the chapter's first and most useful sentence: **the method was built from processes, not from a person, and processes hand over.**

Parallel jigs

Two benches, and the sorting problem that follows — because a crew does not make the panels faster by crowding one jig; it makes them faster by running two.

The second bench is a copy of the first, set from the same panel drawings of Chapter 37, checked against each other before production starts — two jigs that disagree produce two populations of panels, and the raising is where they meet. The check is the same one the circle layout used: measure both benches against the drawing, not against each other, and demand they agree within the gasket's give.

The sorting problem is the real new work. Two benches pull from the pile twice as fast, and the pile stops being a thing one person holds in their head; it becomes a shared inventory that two people mark, move and count — which is where the marking scheme of Chapter 31 earns its keep. The rule for the crew: one person owns the pile, one person owns each bench, and the pile-owner's count is the number the day ends on. Parallel work without one shared count is how a dome ends up with thirty-nine panels and an argument.

Teaching the twelve steps

How to hand the procedure to somebody in an afternoon, because the method's own claim — twelve motions, no decisions — is exactly what makes it teachable.

The teaching order is the jig's order, and it starts with the *why* in one sentence each: the bench replaces the tape, the overfit is the crumple zone, the flush cut is the only true cut. Then the twelve motions of Chapter 31, demonstrated once, watched twice, and the watcher's first panel is the tuition panel — the fourth-panel rule applies per person, not per build, and a crew that budgets three practice panels per new hand is a crew that plans for the method instead of against it.

The one thing not to teach is the decisions, because there are none: the stops decide, the schedule decides, the marking scheme decides. The teaching is handing over *habits*, and a habit handed over correctly survives the hand that taught it. An afternoon, then: the method was designed to be this teachable, and the teachability is part of the design — a method that needed its inventor present at every joint would be the frankendome again, wearing a crew.

A build that is not yours

The third change, and the chapter's serious close: helping somebody else build, and the conversation that must happen first.

The liability conversation is not legal advice — the book is not a lawyer, and it says so — it is the plain speech version of the safety and engineering pages, delivered before the first cut: this method is a prototype, not an engineered standard; the loads are the engineer's job; the felling is the most dangerous part and the rules of Chapter 27 are not optional; and the person whose tree, whose site and whose back are on the line is the person who owns the decisions. Helping is not owning, and the difference must be said out loud, because silence is how a favour becomes a lawsuit.

Then the build itself is the easiest of the three changes: the guest's dome is the same cut list, the same jig, the same fortnight — and the guest learns the method by building it, which is the only way anybody ever really learns it. The chapter's last sentence is the book's, borrowed from the campaigns it grew out of: **we make the method; you build on it.** And when the build is somebody else's, the gift is the procedure — the twelve motions, the honest numbers, the warnings — and the rest is their fortnight.

Who the method is actually for

The campaigns name three kinds of people, and the list is worth printing because a reader wondering whether any of this applies to them will recognise themselves in one of the three or in none.

The person with trees and no money. The method's original audience and the one it fits best: land, standing timber, a small saw, and more time than cash. A dome is a way of turning the first three into a building without turning them into a loan first, and the money chapter is the arithmetic of that trade. This reader needs no permission and no contractor to start; they need the cut list and a fortnight.

The person with money and no trees. The reader who wants the building rather than the process, and for whom this book's honest advice is the product chapters: buy the standard article, buy the core, buy the modules, and skip every chapter about splitting. The method still serves them — it is what the thing they are buying is made of — but the labour arithmetic is irrelevant, and the campaign's two kit rungs exist precisely to price that difference.

The person who is neither, and wants the method anyway. The quilter with a sewing machine and an evening, the host with a serviced pad and no dome, the builder who wants to make them for other people. This is the audience the network chapters were written for, and it is the one the project has the least evidence about: a quilt network needs quilters, a pad network needs hosts, and both need somebody to go first.

Two levers connect those three, and both are about *who does the assembly*.

The customer-assembly lever. Almost all of the cost of a delivered building is the labour of putting it together, and all of that labour is work a determined owner can do — which is why the campaign can price the same kit 8,400 apart

depending only on whether the timber comes with it. The lever is not a discount. It is the price of the work, offered as an option.

And the little-guy lever, which is the network's real argument. One owner who wants a dome is a customer; fifty owners who want domes are a market for pads, quilts, cores and kits, and the reason Part Two spends its length on pads is that the pad is the unit that can be shared. Nobody needs this book to build one dome. Somebody might need it to start the thing that builds the fiftieth.

CHAPTER 51 · STORY

What I Would Do Differently

The honest closing chapter

What I Would Do Differently

The story strand closes where it opened: one person, one saw, one tree that was already down. Between the first page and this one the tree became a frame, the frame became a method, and the method grew a book around it — and the honest way to close a story like that is the same way the book has handled everything: name what would change, keep what was right, and say what the next one is.

This chapter is not a confession and not a victory lap. It is the builder's review of the build, written while the review is still worth something — the five decisions that would change, the parts that stay exactly, and the second dome the method has already earned.

The five decisions I would change

Specific, named, with what each would have cost — because "learn from your mistakes" is advice until the mistakes have names.

The marking scheme came too late. A panel built backwards — the mirror of the one the frame wanted — was the tuition for a scheme that should have existed on day one: mark the sawn faces, one stripe per class, bark up. It cost an afternoon to discover and it would have cost ten minutes to do first. The next build starts with the crayon.

The keys were cut too early. The green-frame rule of Chapter 49 — cut the keys after the frame has dried — was learned the expensive way, with a set of keys that fit a frame that no longer existed by month three. The rule is now in the book; it would have been in the build, if the build had had the book.

The ring was trusted once too often. The tolerance chapter names the ring as the one place error accumulates, and the build proved it: a ring checked once, at setting, moved a fraction under the first course, and the correction cost a morning of shims. The three checks of Chapter 38 would have caught it in the first one, for free.

The skin waited for the wrong perfection. The shell was delayed while the "right" skin was decided, and the frame stood in the weather longer than it should have. The shell ladder of Chapter 60 now says what the build learned: the first skin's only job is to keep the rain off, and the perfect skin is the one you buy after the first one has taught you what you actually need.

The fuel was not logged. The dashes in the money chapter's table are this decision: the build kept its hours and not its fuel, and the metabolic ledger of Chapter 56 had to be modelled where it should have been measured. The next build logs every tank, because the book's rule about numbers applies to the builder's own: what is not recorded is not claimed.

What I would keep exactly

The parts that were right the first time, listed with the same specificity as the changes, because a review that only lists failures is a review that learned nothing about what worked.

The split count. Eight, for the reasons Chapter 12 printed — the key fits, the stick carries, the cuts stay "through the middle". Not one afternoon of the build ever regretted it.

The overfit. The 6 inches of Chapter 35 are the cheapest insurance in the method, and the build spent them exactly as designed: every error that had somewhere to go, went there.

The jig, as built. One flat bench, stops instead of measurements, the day that paid for the other thirteen. The build's single best decision, and the one the second build will not have to make again.

The order of the fortnight. Fell, buck, split, jig, panels, raise — the sequence the book now teaches — survived contact with the weather, the wood and the calendar without a single step being moved. The order was right before the method existed, and the method is mostly the order, written down.

The honesty habit. The errata pages, the dashes in the money table, the near miss reported — that discipline was the build's own habit before it was a book's policy, and it is the one thing in this chapter that costs nothing to keep and everything to lose.

The next one

What the second dome is, and why — the question every builder asks at the end of the fortnight, answered the way the build answered everything: with the numbers, and without the romance.

The second dome is the same method at the size the first build's own leftovers suggest: the spare struts and the offcut pile already hint at a smaller companion — a greenhouse, a workshop, a guest pod — and the flat-rate argument of Chapter 52 says the second frame's parts list is the first frame's parts list, with the problems already solved. The second build's real change is the one Chapter 47 estimated: the tuition is paid, the jig exists, the scheme is written — and the fortnight that took fourteen days the first time is the same fourteen days, spent where they were always supposed to go, on the repetition instead of on the learning.

The next one is also the book's promise to its reader, stated plainly: the method was never for one dome. It was for the *second* dome — the one built without the discoverer, from these pages, by somebody who has never watched it done. Every number in this book was written for that person, and the chapter's last line is the book's whole thesis in one sentence:

The tree was already down. The saw cost two hundred dollars. The rest is in here.

The dome, a year on

The closing image, and the one plate in the book that is a photograph's job rather than the simulator's: the dome, a year on, standing where the tree stood.

The plate is a placeholder in this edition — the photograph is still to be taken, and the book's figure system says so plainly rather than substituting a drawing that would lie about the year. When the picture exists, it will show what the frame chapters only promised: the wood greyed and settled, the keys seated, the skin doing its one job, and the building looking less like a fortnight's work and more like the land's own idea of a room. The book will be revised for it, the way the book revises for everything — the photograph added, the caption written, and the year's evidence given the same page the fortnight earned.

Until then, the story strand closes on the frame at day fourteen — the standing plate of Chapter 32 — because that is what a year looks like at its beginning, and this book is the beginning's record, kept honest all the way to the end.

PART 2

Why It Scales

“The parts list does not grow with the house. Everything downstream of that one fact is this part.”

The argument for the whole idea: why the same frame covers a shed and a hall, what that does to cost and labour, and what would have to be true for a network of domes to work.

CHAPTER 52 · EXPLAIN

The List That Does Not Grow

Triple the diameter and the parts count does not move: the same forty triangles and one hundred and twenty struts frame a ten foot dome or a thirty foot one

What actually changes when a dome gets bigger

Here is the thing I could not get anybody to say out loud when I started, and the reason I ended up building a dome instead of a shed.

When you make a building bigger, you expect to buy more of everything. More timber, more brackets, more screws, more days. That is how a box works. Double the floor and you roughly double the walls, the studs, the sheets, the cuts, the money, the weekends.

A dome does not do that. Not mostly. Not approximately. It does not do it *at all* for the parts list, and once you see that number you cannot unsee it.

And nobody in the building trade has any reason to say it out loud. The shed-and-extension business runs on the opposite sentence – more building, more everything – because for a rectangular frame the sentence is true, and stating the dome's exception does not sell anything anybody in that trade is selling. So the fact sits unspoken, in the columns of a table nobody prints, and the person who finds it is the person who sat down and wrote the table. That is what this chapter is: the table, printed.

I built a frame out of a hundred and twenty members. That frame is 120 struts. It is also 40 triangles, held at the corners by 120 brackets and about 960 screws. The shop work is 9 distinct processes – measuring, cutting, drilling, folding brackets, and so on down the list.

Now make the dome three times as wide.

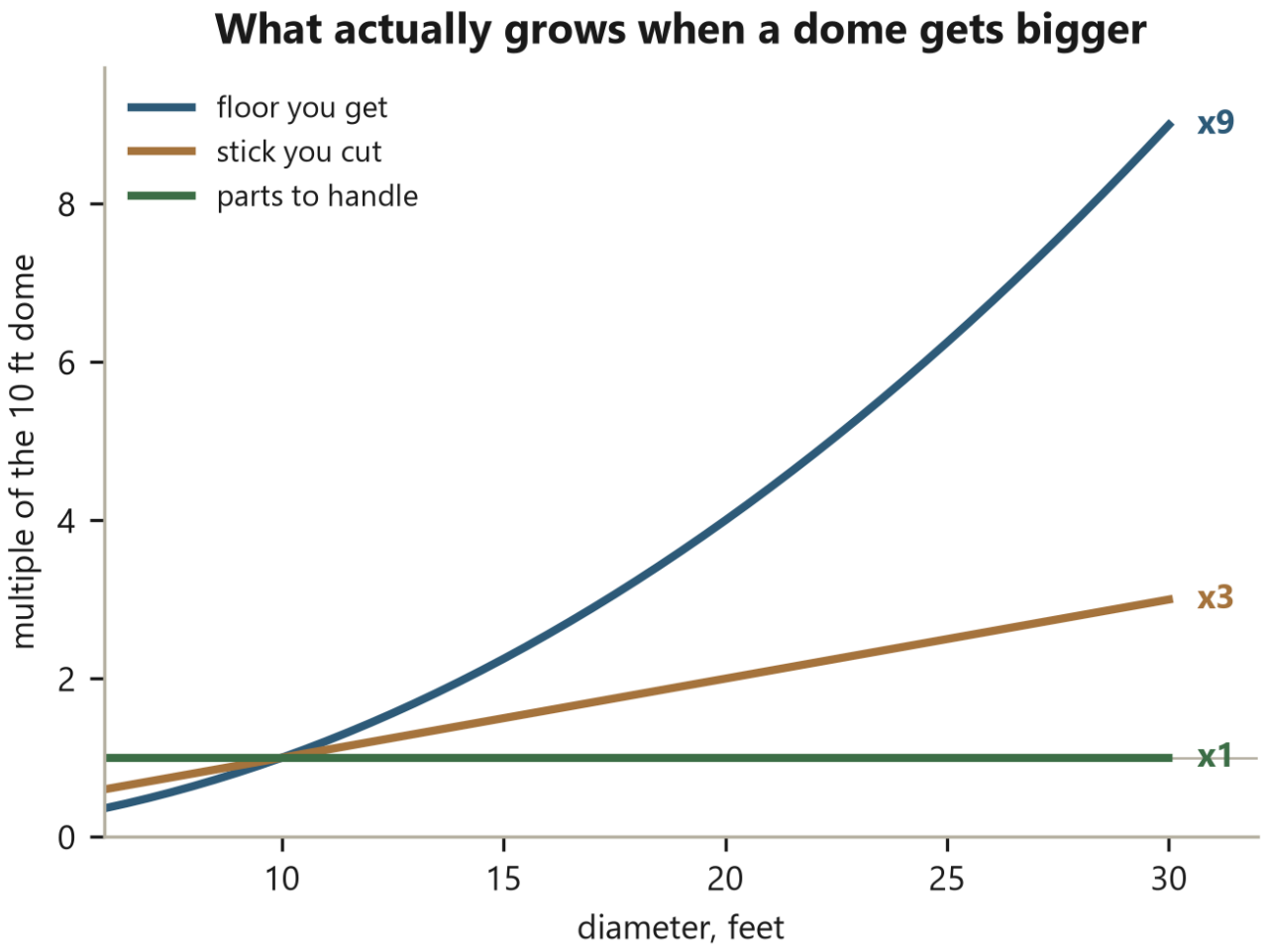
The same list, four times

I ran the same frame at four diameters, from a 10 foot dome to a 30 foot one. This is the table, and every number in it comes off the same model the films use:

across	floor	triangles	struts	brackets	screws	processes	timber
10 ft	79 sq ft	40	120	120	960	9	351 ft
20 ft	314 sq ft	40	120	120	960	9	702 ft
30 ft	707 sq ft	40	120	120	960	9	1,053 ft

Look down the middle of that table. Six of the eight columns do not move. Not by one. The same 40 triangles, the same 120 struts, the same 120 brackets, the same 960 screws, the same 9 processes, whether the thing on the ground is a garden pod or a building you could put a second floor in.

Only two columns move, and both of them are the same material measured two ways: the floor you get, and the timber you have to stand up to get it.



Indexed to the 10 foot dome. Floor area follows the square of the diameter, the stick follows the diameter, and the parts list does not follow it at all -- 120 members, 120 brackets and 960 screws at every point on this chart. The gap between the blue line and the green one is the whole argument.

Three lines: the floor rises as the square, the stick as a line, and the parts list does not rise at all.

The chart is the table, drawn. The floor line climbs like a square. The stick line climbs like a straight line. And the parts line is not a line at all – it is a floor, flat, because there is nothing in it to climb with. When the three are put on the same

axes, the flat one stops being a row in a table and becomes the thing that is visibly *not happening* while the other two race.

One grows as the square, one as a line

This is the whole engine of the idea, so it is worth doing the arithmetic slowly.

Floor area goes as the square of the diameter. That is not a dome fact, that is a circle fact, and it is the reason a twelve inch pizza is not twice a six inch pizza. Three times the diameter is nine times the floor: 79 square feet becomes 707. Nine times the room.

The frame to hold it goes as the diameter, straight line. Every edge in the dome is a fixed fraction of the radius, so three times the diameter is three times the edge network: 351 feet becomes 1,053 feet, a ratio of 3.00, exactly.

Put those two sentences next to each other and something has to give, and what gives is the amount of material standing between you and each square foot of floor. At 10 feet across it takes 4.47 feet of timber to enclose one square foot. At 30 feet across it takes 1.49.

The same timber, doing 3.00 times as much enclosing, because it is stood further out and the shape is doing the work instead of the material.

That is the whole thing. Everything else in this part of the book is that sentence being spent.

What is flat, and what quietly is not

Three things in that table are not flat, and I would rather say them here than have you find them out with a load of timber already on the ground.

The timber grows with the diameter. That is a line, not a flat, and at some point it is real money. Three times the diameter is three times the frame. You are still far ahead, because you got nine times the floor for it, but you did buy more wood.

The *cut* timber grows slightly faster than that, and this one caught me out. The edge network is exactly linear – 3.00 times the edges for three times the diameter. But a member is not its edge. The pinwheel insets every member from the vertex, and the inset is set by how wide the stick is, which does not change when the dome does. So a small dome throws proportionally more of its edge network away in vertex gaps. Counted in members actually standing in the frame, 269 feet becomes 968 – a ratio of 3.59, not 3.00.

It moves the headline, so here is the honest version of it. Per square foot of floor, cut member drops from 3.43 feet at 10 feet across to 1.37 feet at 30. That is still the whole argument – the same timber doing 2.5 times as much enclosing – it is just 2.5 and not 3.00. I would rather print the smaller true number than the larger clean one.

The inset that causes the correction is not an accident of the drawing. It is the pinwheel itself (Chapter 15): members lap past each other at the vertices instead of butting into them, and a member that laps must start short of the vertex it is lapping. The gap is what lets the whole frame close without a single member being cut to an impossible tolerance. The same design that makes the frame forgiving is the one that makes the cut list grow a little faster than the edge network – the two facts are the same fact, seen from the saw and from the table.

The skin grows with the square. Whatever you cover the frame with – panels, shingles, greenhouse film, glass – is priced by area, and area is going as the square. So the *frame* is the flat part of a dome. The *envelope* is not. If somebody tells you the whole building scales flat, they have counted the skeleton and forgotten the skin.

Everything else in the table holds. The count of pieces. The count of operations. The number of times you set the saw. The number of corners you have to get right. The number of ways the job can go wrong. If you can build the small one you can build the large one, because it is the same job with longer sticks in it. That is not a slogan, it is what is in the columns.

The limit is my arms, not the arithmetic

There is a ceiling on this, and it is worth being exact about where it comes from, because it is not where people expect.

It is not that the geometry gives out. It is that **I have to pick the stick up.**

I set my longest member at 6 feet. That is a declared number, not a solved one – it is the most I want to handle as a solo builder. The most I want to carry across a yard, stand both ends of, hold square while I drive the first screw, and set into a triangle with nobody there to take the other end. And I am pushing it at that. Six feet of green timber is not a light thing.

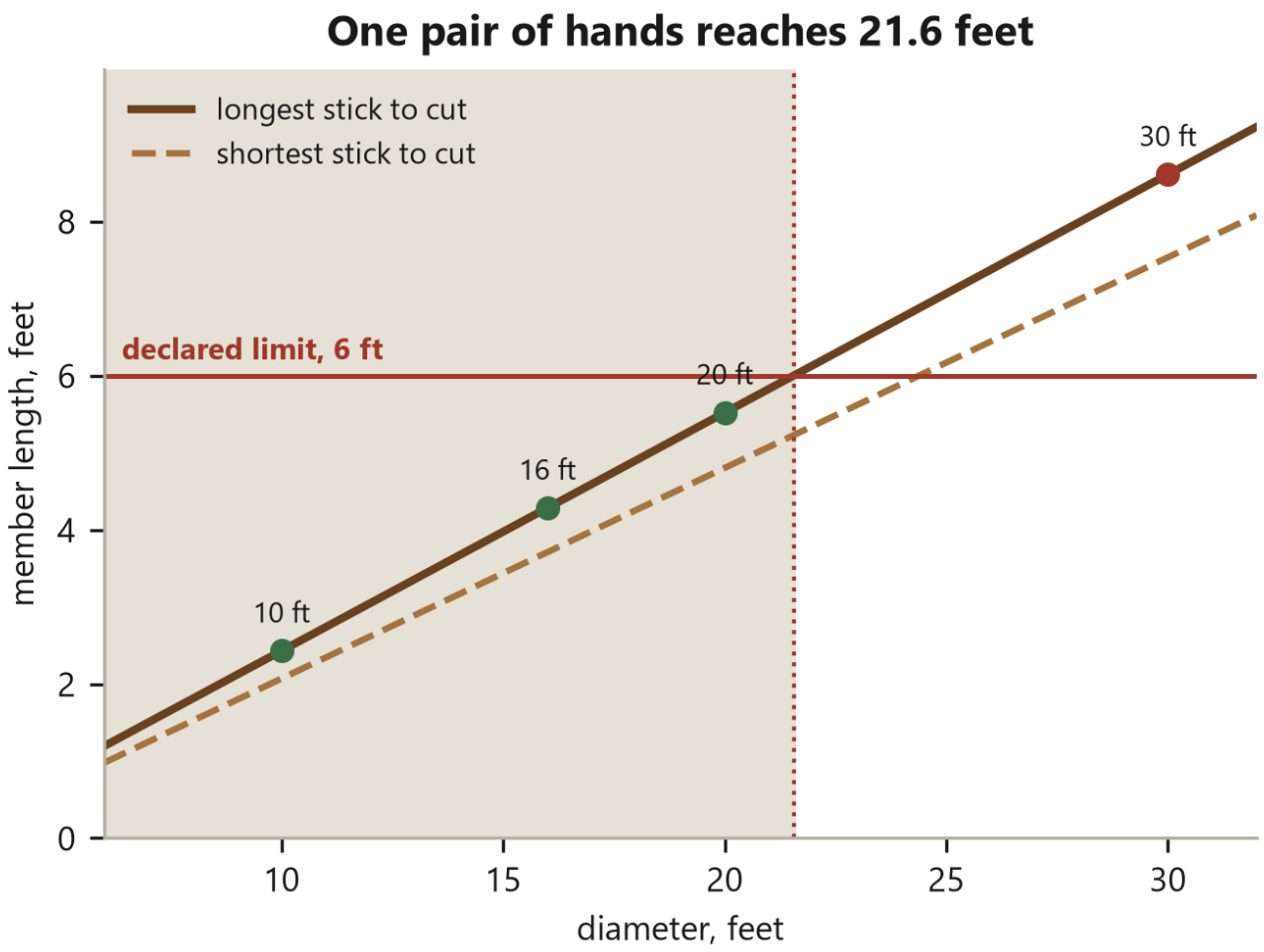
Even at six feet I am not muscling the frame up. I run overhead cabling in the trees and hang a winch off it, and the assembled triangles go up on the cable. The limit is on the stick in my hands, not on the lift. If it were on the lift I would just buy a bigger winch.

Now, how big a dome is that? This is worth doing carefully, because the obvious way to work it out is wrong.

A pinwheel frame does not run its members corner to corner. Each one is inset from the vertex so the next one can lap past it, and how far it is inset depends on how *wide* the member is – which does not change when the dome does. So the stick you pick up is always shorter than the edge it spans, and by a different amount at every diameter. At the top of my band a 6 foot member is spanning an edge of 6.66 feet. Multiply a chord factor and you will size the dome too small and never know why.

Solved properly, a 6 foot cut member is a dome **21.6 feet across** – 365 square feet of floor, with 5.23 feet as the other stick. And here is what that does to the table:

across	longest stick you cut	can one person set it?
10 ft	2.43 ft	comfortably
20 ft	5.52 ft	yes, with 6 inches to spare
21.6 ft	**6.00 ft**	**exactly at the limit**
30 ft	8.61 ft	no – 2.61 ft too long



The 6 foot limit is declared, not solved: it is the longest stick this builder will carry, stand both ends of and set without help. Everything downstream of it is computed, and computed by the wedge solver rather than by scaling a chord -- a pinwheel member is inset from the vertex by an amount that depends on its width, so these curves are not straight lines through the origin. A 6 foot cut member is a dome 21.6 feet across, 365 square feet of floor, with 5.23 feet as the other stick -- which is also the dome two trees yield. Inside the shaded band the parts list, the operations and the effort do not change.

Member length against diameter, with the declared handling limit crossing it.

The figure draws the crossing: member length climbing with diameter, and the flat line of my arms cutting across it. Everything under the crossing is one person's building. Everything over it is a crew's. The band is not a soft preference – it is where the line crosses the curve, and the curve is solved geometry.

That puts 3 of the four sizes in this chapter inside the band. The twenty foot dome – a room you can live in – has half a foot of margin on its longest stick. Only the thirty foot one is out, and it is not marginally out: it is 2.61 feet out.

The band also answers the question a buyer never thinks to ask but should: *which* size? If the work is the same fortnight and the list is the same list, the size is the one decision on the table that is free – and a free decision is the one worth

making last, when you know what the dome is for. Pick the floor you need, and everything else follows it; pick wrong and the correction is not a new list, it is a longer stick in the same frame. There are not many purchases where the expensive part is choosing and the choosing is free. This is one of them.

And there is a coincidence in that table which is not a coincidence at all. 21.6 feet is the dome this entire book is built around: the one the tree yields. It was never sized for my arms. It was sized by the log – 6 foot sections out of a trunk, because that is what a trunk gives you before the taper beats you. The biggest dome one person can frame alone and the biggest dome the tree will fill turn out to be the same dome, to the inch, and I did not arrange that.

Inside the band, nothing changes at all

Here is the part I want to be loud about, because it is the useful half.

Anything smaller is basically the same effort. Not proportionally less – the *same*. Same assembly pattern. Same hardware numbers. Same operations. Same preparation of every strut: same cuts, same jig, same bevels, same holes in the same places. A ten foot dome and a twenty foot dome are the same fortnight of work; one of them just has shorter offcuts.

So the flat rate is not a claim about all possible domes. It is a claim about the domes I can actually build, and inside that band it is not approximately flat, it is flat. That band happens to run from a garden pod up to a room you could live in, which is most of what a person wants.

Past the band you have two honest options and no third one. Buy help – a second pair of hands changes the arithmetic of what a member can be. Or raise the frequency, which splits every edge again and gives you shorter sticks for the same diameter. That second one works, and it is the standard answer, but be clear about what it costs: a 3V frame is 165 members where this one is 120. (That 165 is the standard geodesic count, not something this book has solved – I have not built a 3V frame, and the chapter on frequency prints dashes where its member lengths would go, on purpose.) **Raising the frequency is the one move that actually makes the list grow.** The list is flat across diameter. It is not flat across frequency.

Solve it once

The other thing the table does not show is what happens the second time.

Every count in it is a count of things you have to *figure out* exactly once. The jig gets set up once. The bracket fold gets worked out once. The order you raise the triangles in gets discovered once, usually the hard way, usually in the wrong order. The first dome is where you pay for all of that, and it is expensive in the only currency that matters, which is weekends.

Then it is done. It does not come back. The second dome is the same hundred and twenty members and the same nine processes, except now you know them, and the only thing that can differ is how fast you move. Every build after the first is just **better timing than the first** – and better timing is always a win. There is no version of the second one that is harder than the first.

There is a second, less obvious way the first dome pays for itself, and it is the one that matters on a site. The first dome is the *proof*. The jig you set up once is a jig that demonstrably makes a frame that stands; the fold you worked out once is a

fold that demonstrably held a winter. Every count in the table is a count you no longer have to believe, because you have watched it close. The second dome is not just faster – it is *safer*, in the specific sense that its unknowns are the ones you already met and survived. The first build de-risks the second, and the second de-risks the tenth, and the list being the same is what lets the risk stay retired instead of being re-hired at every diameter.

That is why I count in pieces and operations instead of in dollars. Dollars recur. A problem you solved does not.

Why it is worth counting this way

The number everybody actually feels is not the price of timber. It is the number of Saturdays.

A hundred and twenty members is a fortnight of cutting whether the dome is ten feet across or twenty. That is not a discount on a big building – it is the same building, and the size comes along for free with the corners you were already going to cut. My time did not get cheaper. But the floor it produced did, and it produced more of it.

Counting in pieces also does one thing dollars cannot: it makes the claim *falsifiable*. A price moves with the market and the month, so a cost claim can be argued with forever and never settled. A count is either 120 or it is not, and anybody can stand in the frame and count. That is why this part of the book leans on tables instead of adjectives – and it is why the table in this chapter has four diameters on it, so the flat row is not asserted, it is shown, and the row that is not flat is shown next to it. The number that flatters the argument gets printed beside the number that does not, because a flat-rate claim with the skin and the stick left in is a claim you can check, and a claim you can check is the only kind worth making about a building.

And there is a last way to see it, the one a builder's own body already knows. An hour of labour is an hour of labour. It costs the same whether the hour raised a ten foot dome or a twenty foot one, because the hour does not know what it built – it only knows the motions, and the motions are the same. So the *price* of the building falls as the floor grows, not because anything got cheaper, but because the same priced hours were aimed at a bigger target and hit more of it. The flat rate is not a cheaper building method. It is the same building method with the size as the free variable, and the size being free is what makes the next dome the obvious thing to want.

That is the flat rate. Everything after this chapter in this part of the book is the same fact, applied somewhere else.

Nine Processes at Any Size

The nine operations named one by one, and why a list this short is the thing worth improving: fix one process and every dome you ever build gets cheaper

Nine Processes at Any Size

The last chapter counted parts. This one counts *verbs*, and the verb list is the shorter of the two.

There are 9 things you do to turn standing timber into a shell. Nine. Not nine categories with sub-steps hiding inside them – nine operations, each one a distinct setup with its own tool, its own bench and its own way of going wrong. You can learn them in an afternoon of reading and be bad at all of them by the end of the first week, which is exactly the right number of things to be bad at.

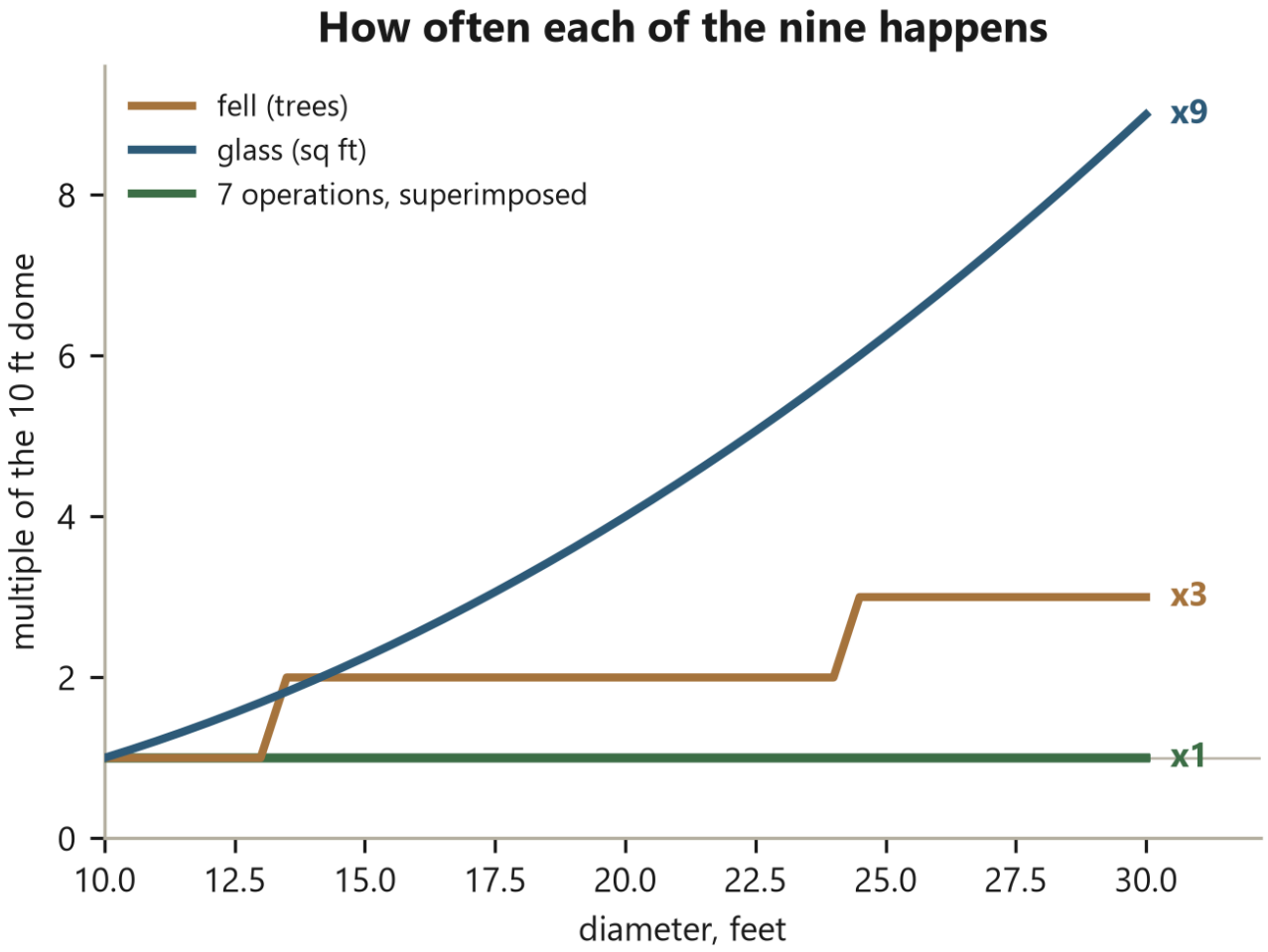
Here they are, in the order they happen:

> *fell, rip, crosscut, fold, drill, screw, raise, sheathe, glass*

That is the whole shop. Everything in this book that is not thinking is one of those nine.

And the reason it matters more than the parts count is this: parts are things you buy, and processes are things you *get better at*. A list this short is a list you can actually attack. Nine places to put your attention is a manageable number of places. A list you cannot hold in your head is not a list, it is a fog.

The nine, in the order they happen



7 of the 9 operations are drawn here and only one line is visible, because they are identical: 120 members, 120 members, 120 brackets, 960 holes, 960 screws, 40 panels, 40 panels, at every diameter on this chart. They are the operations that work on joints, and the joint count is fixed. The other 2 touch raw material and area instead -- felling follows the linear feet of cut member, glassing follows the shell, and the shell goes as the square. The step in the felling line is a whole tree.

Every operation's repetition count across four diameters: seven lines dead flat, two climbing.

Fell. Drop the tree, limb it, buck it to section length. This is the one process with weather in it, and a chain, and a thing above you that is about to be somewhere else. It is also the only operation on the list whose count follows the timber rather than the joints: a 10 foot dome takes 1 tree, a 30 foot dome takes 3. Everything after this happens on a bench.

Rip. Split each round section lengthwise into wedges. This is the longest of the nine by a distance – 24 hours, 4 full afternoons, 4 of the fortnight's 14 days. Hold that thought; the whole back half of this chapter is about it.

Crosscut. Cut each member to its solved length, both ends. The angles are not measured here, they are held: the jig knows the angle, and your job is to put wood against the fence and pull the handle 120 times. If you find yourself reading a protractor during a crosscut, something upstream has gone wrong.

Fold. Bend flat stock to the connector angle, 120 times. Same principle as the crosscut and worth naming twice, because it is the design idea of this entire build: *the angle lives in the tool, not in the operator*. You set the brake once. After that a bracket is a repetition, and repetitions are cheap.

The fold is also where the geometry of the whole book shows up in one number. The dome's seams fold at 18.03 degrees and 22.46 degrees – two angles, not forty – so the brake gets set twice for the entire frame. A frame whose corners are all different would need the brake set once per corner, and the fold would stop being one process and become a puzzle with 120 solutions. The nine stay nine because the frame keeps its angles on a short leash.

Drill. Pilot the bracket legs. 960 holes, one per screw, which makes this the largest number on the list and the one most worth a drill press and a stop. Nobody has ever regretted clamping a fence early.

Screw. Drive the frame together. 960 screws, fixed by the bracket pattern, and therefore – this is the part that surprises people – *identical at every diameter*. A 30 foot dome takes the same 960 screws as a 10 foot one.

Raise. Lift each assembled triangle into place and tie it to its neighbours. 40 panels go up, not 120 members, and that distinction is what keeps this a one-person job: you are lifting a stiff, finished, self-supporting object on a cable, not balancing three loose sticks in the air and hoping.

Sheathe. Close each triangle. Still 40 of them, whatever they measure.

Glass. Lay up the shell. The second of the two processes that scale, and the one that scales hardest, because it is priced by area and area goes as the square: 157 square feet at the small dome, 1,414 at the large one.

Seven of the nine do not move at all

Now put the counts side by side across the diameters, because the result is sharper than the last chapter's version of it.

What each of the nine costs you, at four diameters

operation	tool	counted in	10 ft	16 ft	20 ft	30 ft
Fell	chainsaw	trees	1	2	2	3
Rip	mill or bandsaw	members	120	120	120	120
Crosscut	mitre saw and jig	members	120	120	120	120
Fold	bench brake	brackets	120	120	120	120
Drill	drill press	holes	960	960	960	960
Screw	impact driver	screws	960	960	960	960
Raise	cable and winch	panels	40	40	40	40
Sheathe	knife and stapler	panels	40	40	40	40
Glass	roller and brush	sq ft	157	402	628	1,414

7 of the 9 rows are constant all the way across, and they are the rows that work on joints: the joint count is fixed by the frame, and the frame is the same frame at every diameter. Felling follows the linear feet of cut member, so it steps a whole tree at a time; glassing follows the shell, which goes as the square of the diameter. Nothing else on the list notices how big the dome is.

Every operation against four diameters. Seven rows are constant all the way across.

It is not merely that all nine operations are *still on the list* at every size. 7 of the 9 happen the **same number of times**. Not proportionally fewer, not roughly similar – the identical count. 120 rips, 120 crosscuts, 120 folds, 960 holes, 960 screws, 40 raises, 40 sheathings, at 10 feet across and at 30.

Only 2 move, and look at *which* two. Felling and glassing: the one at the very start that touches raw material, and the one at the very end that touches area. Everything in between – every operation that touches a **joint** – is frozen, because the joint count is frozen, because the frame is the same frame.

That is the sentence I would put on the wall of a dome shop. **Work at the joints does not scale.** A dome is mostly joints. Therefore a dome mostly does not scale.

Why a short list compounds

Here is what a short process list buys you that a short parts list does not.

Improvement has to land somewhere. When you work out a better way to do something, that improvement attaches to a *process*, and it only pays off as many times as that process runs. This is the whole reason factories look the way they do,

and it is why a stick-frame house resists getting better.

Try the same count on a conventional house. I am not a framer, so take this as one man's list rather than a figure this book has solved – but off the top of my head: footings, sill plates, rim joists, subfloor, plates, studs, headers, cripples, blocking, sheathing, housewrap, rafters or trusses, fascia, soffit, roof deck, felt, shingles, flashing, siding, trim. Twenty, and I have not touched anything electrical or wet, and a real framer would double it. Each is a distinct skill with its own tools, its own tolerances, its own way of failing. Get 10 per cent better at hanging rafters and you have improved a slice of one build, once, and the slice is small because rafters are a small share of a very long list.

Now do the same on nine. 10 per cent at the rip is 10 per cent of 29 per cent of the whole job. The share is enormous because the list is short. A short list *concentrates* improvement.

And it repeats. The nine do not change between domes. Not between sizes – the last section proved that – and not between builds. Whatever you work out about the rip on dome one is still true on dome nine, because it is the same rip, on the same bench, with the same wedge coming off it. Nothing about it was specific to a diameter.

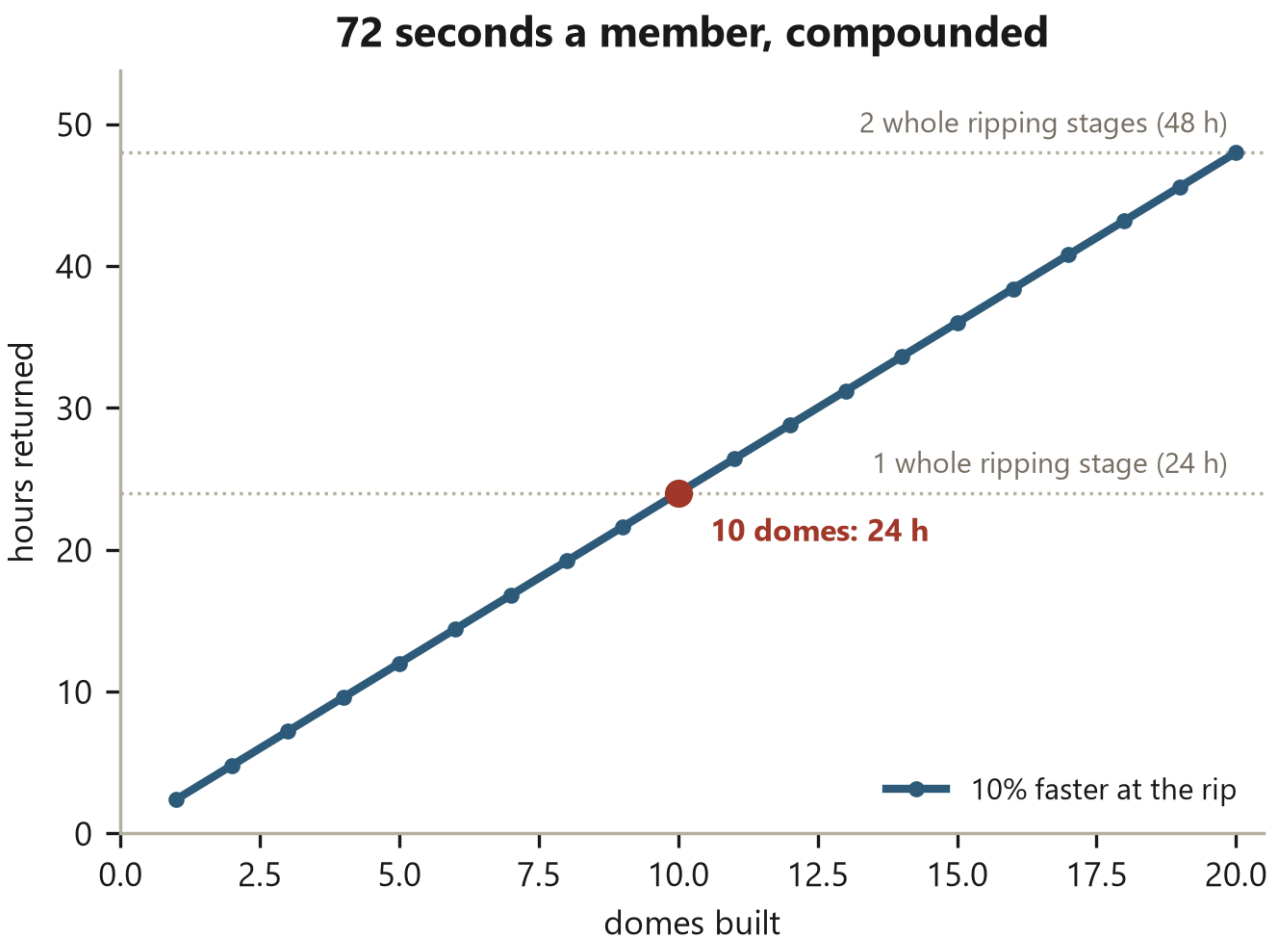
This is what people mean, usually without being able to say it, when they claim a dome "gets easier". It is not that the work is lighter. It is that there are only nine places for the learning to go, so none of it leaks away.

A short list also buys you a *rhythm*, and the rhythm is worth more than the learning on a solo build. Nine processes means you can batch like a kitchen rather than juggle like a handyman: all the folds in a morning, all the holes in an afternoon, all the raises in a day. Batching is only possible when each process is a clean stage with a setup, and a long, tangled list is precisely what a clean stage is not. On a stick-frame job the trades overlap and the order is negotiated; on this list the order is printed at the top of the chapter, and you run it like a production line with one worker. That is not a small thing. It is most of why the fortnight closes.

The flip side is a warning. A short list also *concentrates mistakes*. If your jig holds a bad angle, it does not hold it on one member, it holds it on 120 of them, on every dome, forever, and you will not notice until the last panel refuses to close. Nine processes means nine things worth checking obsessively before you do them 960 times.

What one process is worth, in hours

Take the longest of the nine and price an improvement in it properly.



The ripping is 24 hours of the fortnight and moves 120 members, which is 12 minutes each. Taking 10 per cent off that is 72 seconds a member and 2.4 hours on one dome. Over 10 it is 24 hours -- 1 entire ripping stage, handed back. The member count does not change with diameter, so neither does the refund: small domes pay it too.

What a proportional saving at one bench returns as the run of domes lengthens, measured in whole ripping stages.

The ripping is 24 hours: 4 afternoons, 4 days of the 14, 29 per cent of the whole fortnight. It moves 120 members through the mill, which is 5.0 an hour, which is 12 minutes per member. That last figure is the one to hold, because it is the scale you can actually aim at. Nobody improves a fortnight. People improve 12 minutes.

So: find 10 per cent. A better infeed roller, a stop that does not need resetting, a second pair of trestles so the offcut has somewhere to go, the sections stacked at hand height instead of on the ground. Any of that is credible.

10 per cent of 12 minutes is **72 seconds per member.**

Over one dome, 72 seconds times 120 members is 2.4 hours. That is an afternoon that finishes before dark instead of after it. Nice. Not life-changing.

Over 10 domes it is 24 hours. Which is 4 afternoons. Which is – and this is the number worth sitting with – **1 entire ripping stage**. 10 builds of 72 seconds and you have been handed back the whole ripping of a dome, free.

Nothing was invented to get that. No new tool, no new material, no cleverness about geometry. 72 seconds, found once, at one of 9 benches.

The same arithmetic runs backwards, and the backwards version is the reason the nine deserve fear as well as respect. A stop that is 72 seconds *slow* – loose, sticky, in the wrong place – is not 72 seconds lost once. It is 72 seconds on every member, 2.4 hours on the dome, 24 hours over the run, and 1 whole stages of the same run eaten by a detail nobody thought was worth ten minutes to fix. The list concentrates losses exactly as hard as it concentrates gains, and the only difference between the two directions is which one you noticed. The nine are not nine chores. They are nine levers, and every one of them works in both directions, on every dome you will ever build.

And the arithmetic is indifferent to size. The improvement is per member and the member count does not change with diameter, so that 24 hours is the same whether those 10 domes were garden pods or the largest thing one person can frame. You do not get a smaller refund for building small.

That is what I mean when I say the first dome is the expensive one. It is not sentiment. It is that the first dome is where you find the 72 seconds, and every dome after it is where you get paid for finding them.

The processes that are not on the list

I have to be straight about what the nine leaves out, because it leaves out things that can eat a project whole.

The nine is a **shop count**, not a project count. It covers what happens between a standing tree and a closed shell, on your own ground, with your own hands. It does not cover:

Permits and inspections. These are real, they are sometimes long, and they are emphatically not flat – a bigger building draws more scrutiny, and a round one drawn by its builder draws more still. A plans examiner who has never seen a hubless pinwheel will ask questions a rectangle never gets asked.

Siting and access. Getting a mill to the trees, or the trees to the mill. This is invisible on a flat site with a track to it and dominant on a slope without one.

Delivery and waiting. Anything bought arrives on somebody else's schedule. Screws and resin are the two that stop a build dead.

Waiting for a crew. Not on the list, because at the top of the handling band there is no crew. That is the trade: the nine stay small precisely because nothing on them needs a second person. Go past the band and you add a tenth process called *coordinating people*, which is the least flat operation there is and the one that does not get better with repetition, because the people change.

None of these belong in the nine, and I am not going to smuggle them in to make the list look honest. They belong in a different count, and they get one – the money chapter counts dollars, the hours chapter counts days, and both of them include the waiting.

What I will say is that the nine is the part you control. Everything on it gets better because you made it better. Almost nothing off it does.

What an Hour at the Log Is Worth

Splitting your own struts pays a rate; here is the rate, computed three ways, including the way that flatters it least

What an Hour at the Log Is Worth

This chapter is not about what timber costs. It is about what an hour of your own time buys when you spend it on a log instead of at a till.

Those are different questions and people run them together constantly. “Lumber is expensive” is a fact about a yard. “I can make my own” is a claim about an afternoon. The second one only means something if you can say what the afternoon produced, in the same units the yard uses.

So: the mill turns out 5.0 members an hour. Each member has a cross-section of 9.82 square inches, which is a shape no store sells. To price it, I have to say what it *replaces*, and that is where this gets slippery, because a two-by-four is not two inches by four and the difference is not small.

I am going to do it three ways. One of them is the number I started this project with, and it is the highest, and it is wrong. One of them is the number that sounds conservative and is actually the low one. The third is what I think is honest, and it happens to sit above the conservative-sounding one – which is precisely the direction a reader should be suspicious about. I would be.

So here is all three, and the arithmetic in between, and then you can decide.

One strut, valued three ways

Start with what you would actually pay to replace one member with bought wood.

A sixteen foot two-by-four is 8.88 at the store. Cut into 6 foot lengths – the length of a member – that is 3.33 per section. That part is not controversial.

The controversial part is how many sections it takes to equal one strut.

A strut is 9.82 square inches in section. Divide that by what a two-by-four gives you and you have your answer, except that a two-by-four gives you two different things depending on which number you believe: the 8.00 square inches on the label, or the 5.25 square inches the board actually measures. The first says a strut replaces 1.23 boards and is worth $4.09. The second says 1.87 boards and $6.23. At 5.0 members an hour that is $20.43 an hour against $31.14 an hour.

One strut, valued three ways

basis	quantity	value
Against a nominal 2x4 (8.00 sq in)	1.23 boards	$4.09
Against a dressed 2x4 (5.25 sq in)	1.87 boards	$6.23
By board foot	4.35 bf	market varies
Implied rate, nominal	5/hr	$20.43/hr
Implied rate, dressed	5/hr	$31.14/hr
The brief's estimate	**$10.00/strut**	**$50.00/hr**

*The last row is the figure this project started from, and it is the highest -- $10.00 a strut, reached by doubling a volume ratio for safety. The rows above compute the ratio from the actual cross-section instead. Note which way the honest correction runs: a dressed 2x4 is smaller than its name, so a strut replaces more of them, and the *dressed* comparison is the higher of the two computed rows, not the lower. Both computed rates are real; neither is money anybody has been paid.*

One strut, valued three ways, with the estimate this project started from kept in the table.

Two bases, and they disagree by 34 per cent.

The third row is in the table because it is what a mill would quote you, and it is the honest way to trade timber in volume – but it is no use here, because a board foot has a price only where somebody is selling one, and that price moves with species, grade, region and week. I can tell you a member holds 4.35 board feet. I cannot tell you what a board foot is worth in your county, and I am not going to make one up so that row has a number in it.

The board-foot view does carry one number worth keeping, though, because it is the whole harvest in one figure. Two trees of solid wood are 318 board feet before the saw touches them; the kerf takes 40, and the wedges that come off the split are 278 – a recovery of 87.4 per cent, which is why the recovery chapter corrects the rounder number this project once quoted. Every rate in this chapter is a rate on *that* volume, and the volume is computed from the tree, not assumed.

That leaves two, and they disagree, and the disagreement is the chapter.

Why the dressed number is the honest one

A two-by-four is not two inches by four inches. It is one and a half by three and a half, which is 5.25 square inches, not 8.00. It was two by four once – before it was dried and planed – and the name stuck to the rough size while the product

shrank to the finished one. Everyone in the trade knows this. Nobody outside it does.

Now think about which number belongs in a replacement calculation.

The question is: *if I did not have this strut, how many boards would I have to buy to get the same wood?* The answer has to be in terms of what is actually in a board, which is 5.25 square inches. So a strut replaces 1.87 boards, not 1.23, and it is worth $6.23, not $4.09.

Using the label instead of the board **understates** the strut by 34 per cent. Which is the opposite of what you would expect, and worth sitting with for a second, because the instinct is that the "nominal" comparison must be the generous one – *nominal* sounds inflated. It is the other way round. Pretending a board is bigger than it is makes each board look like it does more work, so you need fewer of them to match a strut, so the strut looks worth less.

I want to be uncomfortable about this in public, because the honest correction moved the number in my favour, and that is exactly when a writer should be checked hardest. So here is the check: the cross-section of a member, 9.82 square inches, is computed by the geometry, not measured by me with a hopeful tape. The 5.25 figure is the actual planed size of the board I would otherwise buy. The 8.88 is a shelf price. Nothing in the chain is my estimate. If the result flatters me, it is because a wedge out of a log has more wood in it than a plank does, which is not a surprising thing for it to have.

And that is the number the rest of this book uses: $31.14 an hour of ripping, 747 dollars for the whole frame.

The shelf price is a stack

Now the part that cuts the other way.

The 8.88 on that board is not the price of wood. It is a stack, and only the bottom layer of it is timber. Above the timber sits the felling, the haul to the mill, the sawing, the kiln, the planing, the grading, the stacking, the second haul, the yard, the shrinkage, the staff, the roof over the lumber aisle, and a margin at every hand-off.

Harvesting your own does not remove that stack. It removes *some* of it and replaces the rest with you.

Gone: the margin, the hauls, the yard, the retail overhead. Those layers genuinely vanish, and they are the largest part of why a bought board costs what it does.

Still there, now done by you: the felling, the bucking, the ripping, the stacking, the waiting. Those are the 8 days at the front of the fortnight. You did not delete that work. You took the contract.

The two routes are countable, and the count is the stack made visible. From standing tree to graded board on a yard's shelf is 14 hand-offs – felling, hauling, sawing, kiln, planing, grading, trucking, stocking, selling, every one of them a person with a margin. From standing tree to member in my frame is 8: fell, buck, split, cut, joint. The stack is not a metaphor. It is a list, and the shorter list is why the hour at the log is worth what this chapter says it is worth.

Never done at all: the kiln and the grading. This is the honest cost of the method and it belongs here rather than in a footnote. A bought board is dried to a known moisture content and stamped by somebody whose job is to say it is what it says. A wedge off my log is green, ungraded, and going to move as it dries – which is a large part of why this frame is designed the way it is, with panels that keep their own edges and a gasket between them.

So the rate is real, and it is also not a discount on a like-for-like product. $31.14 an hour buys you green, ungraded, home-milled timber of a section nobody sells, and it is worth that much *because* of the section. It does not buy you a kiln.

This is not money anybody has been paid

One last thing, and it is the one I would most like to survive being quoted out of context.

Every rate in this chapter is a **substitution value**. It is what you did not spend. It is not what you earned.

$31.14 an hour does not go in a bank. It shows up as an absence: a trip to the yard you did not make, a line on a card that is not there. That is a real economic good, and it is the entire argument of this part of the book – but it buys groceries only if you were genuinely going to buy the timber. If the alternative to making struts was not buying struts, the rate is zero and the hours were a hobby. A good hobby. Still zero.

Which brings me to the number I started with.

When this project began, the brief valued a strut at $10.00 – $50.00 an hour at this pace. It got there by taking a volume ratio and doubling it for safety, which is a fine way to sketch and a terrible way to publish. The honest figure is $31.14. My own opening estimate was **1.6 times** the number I can actually defend.

I have left that row in the table on purpose. It is the row I would have quoted if nobody had made me compute it, and there is a version of this book where it is the headline and everything downstream is 1.6 times too optimistic. The whole reason this book derives its figures in code is so that row gets caught rather than repeated.

One last scale to put the rate on, because a rate is only readable next to another rate. The median worker in this country takes home 19.69 dollars of an extra hour. A carpenter is paid 29.12 dollars an hour. The honest number from this chapter, 31.14 dollars an hour at the log, sits between the two – better than the median hour, worse than a carpenter's, and all three are the same unit, which is the only way a claim about an hour means anything. If your own hour is worth more than the rate this chapter defends, the yard is the honest answer and this chapter is the arithmetic of why. The rate is not a wage and it is not a boast. It is the price of the alternative, computed, and now it has a scale to sit on.

The benchmarks behind the rate

The rate is not a number the book chose. It falls out of four published figures and one computed one, and printing them together is what makes the claim checkable rather than rhetorical.

benchmark	number	what it is for
Stumpage	23.23 a ton	what a standing pine fetches, green
Custom milling	0.20 to 0.30 a board foot	the 2010 survey's range for somebody else's saw
The book's milling rate	0.60 a board foot	one small mill's price for one tree
Dressed lumber	1.50 a board foot	a sawmill's price for the species
Median take-home	19.69 an hour	what an ordinary hour pays
A carpenter's wage	29.12 an hour	what a traded hour pays
The rate at the log	**31.14 an hour**	the alternative, priced against dressed lumber
The same against nominal boards	20.43 an hour	the flattering comparison, printed for completeness
The break-even wage	50 an hour	above this, buying beats building

Read the last three rows together and the chapter's real claim appears: the rate sits **at** the break-even wage the why-build chapter computed independently, and both sit above the median hour and above a carpenter's. That convergence is not a coincidence and it is not one the book arranged — 50 comes from the frame's value divided by the hours it takes, while 31.14 comes from the wood's value per hour at the log. Two different routes to one number, which is the kind of agreement a reader should look for before believing either.

The honest note on the benchmarks is the same one the pine chapter made about its ladder. **Custom milling is a range, not a price**, and the range is wide: 0.20 to 0.30 a board foot is a factor of one and a half, and the book uses 0.60 — the high end — because that is what a small mill charges for a single tree. A reader with a friend and a bandsaw is working from a cheaper rung than this table shows; a reader buying from a yard is working from a dearer one. The rate is this build, priced honestly, and the benchmarks are printed so a reader can move it to their own situation rather than taking it on faith.

CHAPTER 55 · EXPLAIN

The Six Values of One Pine

The same tree, priced six ways: on the stump, as firewood, as sawn lumber, as split stock, as standing structure, and as mortgage payments avoided

The Six Values of One Pine

A tree has no price. It has a ladder of prices, and which rung you stand on decides what the tree is worth — which is why two people can look at the same pine and honestly quote numbers two hundred times apart.

This chapter is the ladder, walked one rung at a time, on the big pine the films priced: 60 usable feet, 15 inches at the butt, 442 board feet of solid wood. The arithmetic comes from the same module the twenty-dollar-pine film runs, so the book and the film cannot disagree about the tree. The lesson the ladder teaches is the chapter's whole point: **the wood never changes between the rungs. The design changes what the wood is for, and the value follows the design, not the tree.**

Chapter 2 walked the chain that prices a board. This chapter walks the chain that prices a *tree*, and it ends where Part Two's economics always end — with the honest floor under the top rungs, printed rather than skipped.

One tree, six rungs

The ladder, bottom to top, with the arithmetic at each step.

Standing. On the stump, the tree is stumpage: its green weight times the going rate per ton. This pine weighs 1.01 tons green, and at the sawtimber stumpage the film researched the tree is worth about 24 dollars — the twenty-dollar pine, named for this rung. The honest footnote belongs right here: a timber buyer mostly does not want two trees. The price exists; the buyer may not.

Firewood. Cut, split and stacked, the stem is 0.41 cords of firewood — a cord being 128 cubic feet of stack, about 90 of it wood — and at 275 dollars a cord, picked up, the tree is worth 113 dollars. Burning is the oldest conversion there is, and the ladder's lowest full-value rung.

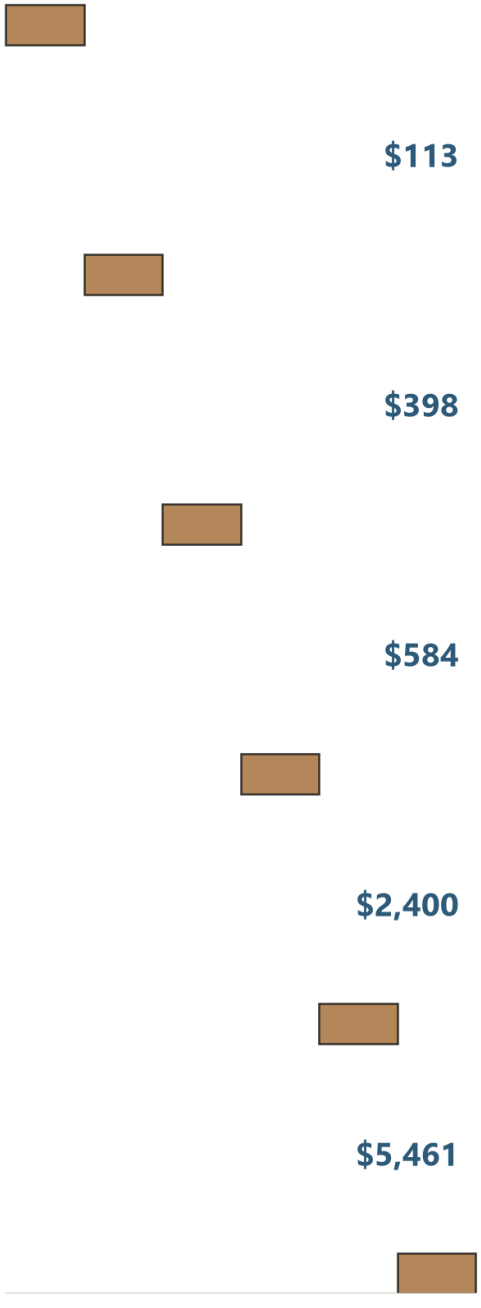
Sawn lumber. On a mill, at the film's declared 60 per cent recovery, the tree yields 265 board feet; at 1.50 dollars a board foot that is 398 dollars of lumber. The recovery is the author's own estimate — the simulator's honest packing of the same log into two-by-fours says 45 per cent — and the chapter keeps both numbers, the way Chapter 13 taught.

Split stock. Split into wedges instead of sawn into boards, at 88 per cent recovery, the same tree yields 389 board feet — worth 584 dollars at the same lumber price, and 124 board feet more than the mill kept. The split rung is the first one where the method's own arithmetic appears, and it beats the mill on the same tree by 46.7 per cent.

Structure. Used as the book uses it — as framing the owner would otherwise buy — one tree replaces 2,400 dollars of commercial framing: the shell's framing value, divided by the two trees that supplied it. The rung is a substitution value, and the chapter says so again before the ladder is done.

Financed. If that avoided framing would have been carried on a mortgage at the book's illustrative 5,461-style terms, its lifetime payments come to 5,461 dollars per tree — the nominal rung, the tallest, and the one most likely to be quoted without its caveats.

One pine, six rungs



The wood never changed. The design changed what it was for.

structure / firewood **x21**

structure / sawn **x6**

structure / split **x4**

structure / standing **x102**

Six rungs for one pine, from the stump to the financed frame.

The ladder, drawn

The drawing is the ladder as a staircase, and it is a staircase on purpose: the rungs rise evenly, but the *values* do not, and the gap between one rung and the next is the entire argument.

Look at the distances. The step from standing to firewood is small — the tree, processed into heat. The step from firewood to sawn lumber is a few multiples — the tree, processed into boards. The step from sawn to structure is the jump that matters: the same wood, asked to *be a building*, is worth on the order of twenty times its firewood price and six times its lumber price. And the last step, to the financed rung, is not a step the wood takes at all — it is a step the *time* takes, the mortgage's arithmetic standing on the framing's shoulders.

The staircase is the chapter's central object because it answers the question every visitor to this project asks first: how can a twenty-dollar tree be worth thousands? It cannot, and it can. The tree is twenty dollars on the first rung and thousands on the fifth, and nothing about the tree changed on the way up. The ladder is the explanation, drawn.

The wood did not get better

The ratios that embarrass the ladder, and the sentence the chapter exists to make stick.

Structure against firewood: the framing one tree replaces is worth 21 times what the same tree burns for. Structure against sawn lumber: 6 times. Even against the split stock the method itself produces, the built structure is 4.1 times the stock's price. And the wood is *identical* on every rung — the same board feet, the same fibres, the same pine. Nothing was added between the stump and the frame except decisions.

That is the whole economics of the method, stated as a ratio instead of an argument. The tree's value was never in the tree; it was in what the design asked the tree to do. Firewood asks for heat, and the tree answers with heat. A mill asks for rectangles, and the tree answers with what rectangles fit. A frame asks for structure, and the same tree answers with a building's worth of it — because a triangulated frame is the one customer that pays for the whole log, in its own shape, and the shape is the tree's idea. The ladder does not flatter the tree. It exposes the customers.

What the ladder does not say

The honest floor under the top rungs, because a ladder printed without its caveats is a sales document, and this book is not.

Substitution is not income. The structure rung's 2,400 dollars is what the owner did not *spend*, not what they *earned*. It converts to groceries only if the framing was going to be bought anyway, and the builder who was never going to buy a frame has built a shed with their own hours instead of earning a wage — the same distinction Chapter 46 drew for the fortnight, drawn here for the tree.

Financed value is future payments, not today's. The top rung assumes the avoided framing would have been mortgaged at the book's illustrative rate for its full term, and then adds up every payment. It is a real number about a counterfactual borrower, not a cheque. Quoted alone, it is the ladder's most flattering rung; the chapter keeps it, labelled, because the films priced it and the book's honesty extends to its own sources.

The owner's net rung. What the tree is actually worth to the person standing on the ladder: the structure value, minus the labour of converting it — 40 hours at the owner's wage — minus the cash the build still spends. The arithmetic lands

near 1,000 dollars of net value per tree, and that is the rung this book would build on: not the tallest rung, the *owned* one.

Different owners stand on different rungs. The ladder's real lesson, stated last because it is the point: the tree's value is not a property of the tree, it is a property of the owner's position. The timber buyer stands on rung one, the firewood seller on rung two, the mill on rung three — and only the builder stands on the rung where the design, not the market, sets the price. The book is for the people on that rung.

The same ladder as an index

The films put this ladder on screen as a single index rather than as six prices, and it earns a page here because the index is the more useful thing to argue with.

The stump value counts as one. Everything above it is a multiple of that:

rung	multiple of the stump value
On the stump	1x
Burned, picked up	5x
Sawn to lumber	17x
Split to stock	25x
Holding a frame up	102x
Carried on a mortgage	232x

6 rungs, and one number in that table is worth more than the rest: **102x**. That is the tree as *function* — the framing you no longer have to buy — against what a timber buyer would pay for it standing. The ratio does not depend on the price of anything: move the stumpage rate and both sides move together, which is exactly why the index is a better argument than the dollars it was computed from.

Two cautions, in the spirit of the section above.

The top two rungs are not cash. The 102x and the 232x are "what you did not have to buy" and "what you did not have to borrow", and neither is money in a hand. A builder has 102x of *displaced* spending: real, and unspendable. That is the same distinction the money chapter's stack drew between a saving and a receipt.

The index compares a tree to a tree. It is not an investment return and not an interest rate. The financing rung is a nominal figure summed over a thirty-year schedule; the firewood rung is what somebody pays at the roadside for a cord. Read it as a ranking of uses, not as a price list.

Which leaves the sentence the films printed under the table, and it is the chapter's whole argument in seven words: **the wood did not get better; it got used.** Every multiple in that index is the same tree, cut the same way, and the distance between 5x and 102x is a decision about what to do with it — the decision this book has spent its length teaching, and the one number in the table that is under the reader's control.

What the ladder rests on

Every rung above came from somewhere, and the book's rule is to say where: a price without a source is a rumour with a decimal point.

rung	figure	where it comes from
On the stump	23.23 a ton	published stumpage for pine, per green ton; the stem weighs 1.01 ton
Firewood	275 a cord	the going rate for a cord, picked up at the roadside
Sawn lumber	1.50 a board foot	one sawmill's price for the species
Milling	0.60 a board foot	the author's custom-sawing rate
Split stock	the same price a board foot	on the 88 per cent the split keeps
Structure	the framing one tree replaces	this book's own money chapter, from the framing share of a house
Financed	that value on a mortgage	this book's own rate and term, summed over the schedule

Two of those lines invite an argument and the book would rather have it than defend them.

The mill recovery is generous. The ladder prices the sawn rung at 60 per cent of the stem; this book's own solver says 45 per cent for boards packed out of the same round. The higher figure is the one the films used and the one most published tables assume, and it flatters the *lumber* rung — which makes the split rung's advantage look smaller than it is. A reader who wants the conservative version should re-run the ladder on the modelled number and watch the gap widen.

The sawing rate is a rate, not a market. A 2010 survey found custom milling running between 0.20 and 0.30 dollars a board foot, against the 0.60 this ladder uses. That is several times the low end; it is the right number for a small mill doing one tree, and a reader with a bandsaw and a friend is looking at a cheaper rung than the table shows.

And the financed rung, which is the tallest and the most easily misread. It assumes the avoided framing would have been borrowed at this book's illustrative rate for its full term and sums every payment: 5,461 per tree, in nominal dollars, over thirty years. It is a real number about a counterfactual borrower. It is not cash, it cannot be spent, and quoting it alone — as the ladder's flattering rung — is the one thing the table refuses to do.

Which leaves the net rung, the one this book would actually build on. Take the structure value, subtract the labour of converting it — 40 hours at the owner's own wage, 1,000 — and subtract the cash the build still spends, 400: **1,000 of net value per tree**, or 1,152 if you use the measured cash the fortnight actually spent instead of the planning figure. Both are printed, neither is a wage, and both are the reason this method's arithmetic survives a reader who insists on counting their own hours.

CHAPTER 56 · EXPLAIN

Where the Fuel Actually Goes

A metabolic ledger of one build: every part lifted, carried and fastened, priced in calories -- and the discovery that most of the fuel raises nothing at all

Where the Fuel Actually Goes

There are two ledgers in a build and only one of them gets written down.

The money ledger decides whether you can afford the thing. The other one decides whether you can finish it. It is kept in calories, it is settled every evening whether you look at it or not, and running out of it does not feel like running out of money – it feels like deciding, quite reasonably, to do the rest next weekend.

So this chapter counts it. Every part of one building, every motion that part takes to get from the pile to its place, and what each motion costs a body.

Before a single number: **this is not the fortnight.** The fortnight is one person and a bare frame, and there is no metabolic model of it, because nobody wired a body to me while I was ripping. What there *is* a model of is the assembly line's own product – the Iris-25, a finished house of 4.39 metres' radius, 15 stations, 1,322 parts, built by a crew of 2. That is a bigger, more finished and more crowded job than anything else in this book, and every figure below is per person on *that* build.

I am using it anyway, because the shape of the answer does not depend on which dome it was, and the shape of the answer is the whole point.

One house, counted in parts and motions

The method is mechanical. Take the element list – all 1,322 pieces, frame and floor and cladding and fixtures – and turn each one into the same 7 motions: walk to the pile, lift, carry it over, place it, fasten it, stand back up, rest. Each motion gets a duration from the part's own mass, distance and height, and a cost from published activity data. Then add them all up.

One building, counted in parts, hours and calories

station	parts	kg	hours	kcal	% fuel
frame	266	3,925	36.5	10,531	19.2
fiberglass	161	1,691	32.3	9,407	17.1
osb	160	2,806	22.5	6,550	11.9
sheetrock	160	2,347	21.9	6,357	11.6
shingles	160	1,032	19.0	5,514	10.0
insulation	160	637	14.6	4,220	7.7
wrap	160	410	12.5	3,638	6.6
solar	33	404	7.2	2,094	3.8
floor	22	1,258	6.1	1,615	2.9
interior	8	767	4.8	1,280	2.3
fixtures	7	428	4.1	1,090	2.0
water	11	113	3.1	813	1.5
column	3	375	2.4	660	1.2
power	8	72	2.4	642	1.2
hatch	3	487	1.7	459	0.8
whole build	**1,322**	**16,751**	**191.1**	**54,870**	**100.0**

Per worker, for a crew of 2. The hours are task time including the recovery allowance; the calories are what a 82 kg body spends doing them. The order is by fuel, and it is not the order of mass. The layers that go on over the frame

weigh 2.3 times what the frame weighs and cost 3.4 times its fuel -- because there are more parts to fasten, and every fastening is time.

Fifteen stations, ordered by what they cost a body rather than by what they weigh.

16,751 kilograms of building. 15,100 of it passes through one person's hands – less than the total, because anything over 23 kilograms is a two-person lift and the load splits.

191 hours each. That is 23.9 shifts of 8 hours, and it costs 54,870 kilocalories per person, 109,741 for the pair.

The figure worth holding is not the total, though, because the total only tells you how big a house somebody ordered. The *rate* is what travels: **2,298 kilocalories a shift.** That is a working day, and it sits on top of a resting requirement of about 1,749 a day that a body wants whether it builds anything or not.

Now the unflattering one. Averaged across the whole build, that comes to 334 watts, or 3.50 METs. The occupational-physiology figure for what a person can sustain across a full shift is 350 watts. There is almost nothing in that gap. The model is not saying this is comfortable work. It is saying the work sits at the top of what is sustainable, and that it only stays there because 17.5 per cent of the timeline is an explicit recovery allowance – 33 hours of standing still, scheduled rather than hoped for. Take the rest out to go faster and you do not get a shorter build. You get a shorter *run* of builds.

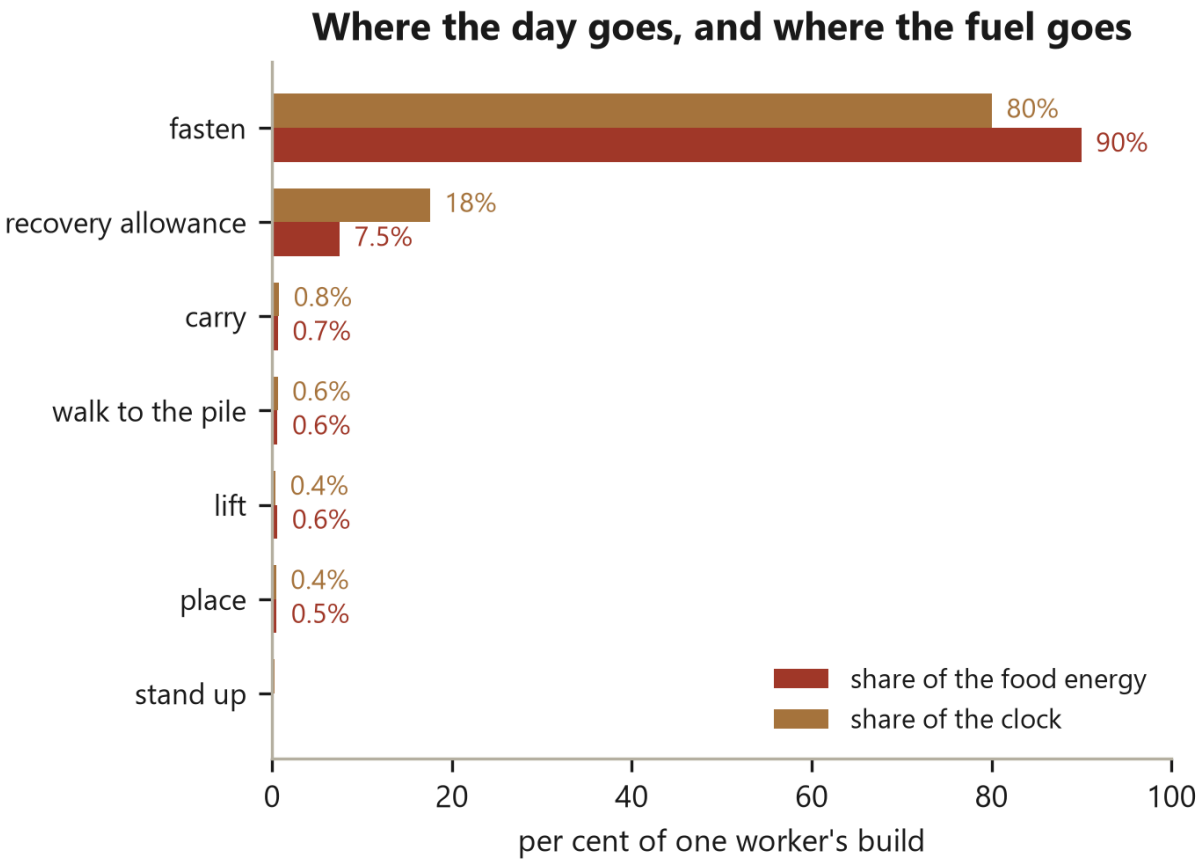
Read the stations by fuel rather than by weight and the order surprises. The frame is the single most expensive station, 10,531 kilocalories across 266 parts. But the layers that go on over it, between them, weigh 2.3 times what it weighs and cost 3.4 times its fuel. Fuel outruns mass, every time, because fuel follows part count rather than kilograms. The heavy part of a building is not the tiring part of a building.

For scale, the crew's total is 1,372 slices of bread. Or 1,045 bananas. Or 44 days of somebody eating normally and doing nothing else with it.

That last framing is not a joke, and it is the most practically useful line in the chapter. A build of this size is a second full-time diet running alongside your first one. If you are doing it solo, at a distance from a shop, on a schedule you set yourself, then food is a material and you have to lay it in like any other – and the failure mode of getting it wrong is not hunger, it is an afternoon that quietly does not happen.

The fastening surprise

Now split the same total by motion instead of by station, and something falls out that changed how I think about joints.



Fastening is 80 per cent of the clock and 90 per cent of the fuel. The 10 points between those two bars are the only part of it that is intensity; the rest is simply where the day goes. The recovery allowance runs the other way, costing less fuel than clock, because standing still is cheap. Lifting, carrying and walking together come to 1.9 per cent of the food, which is the finding that decides what is worth designing out.

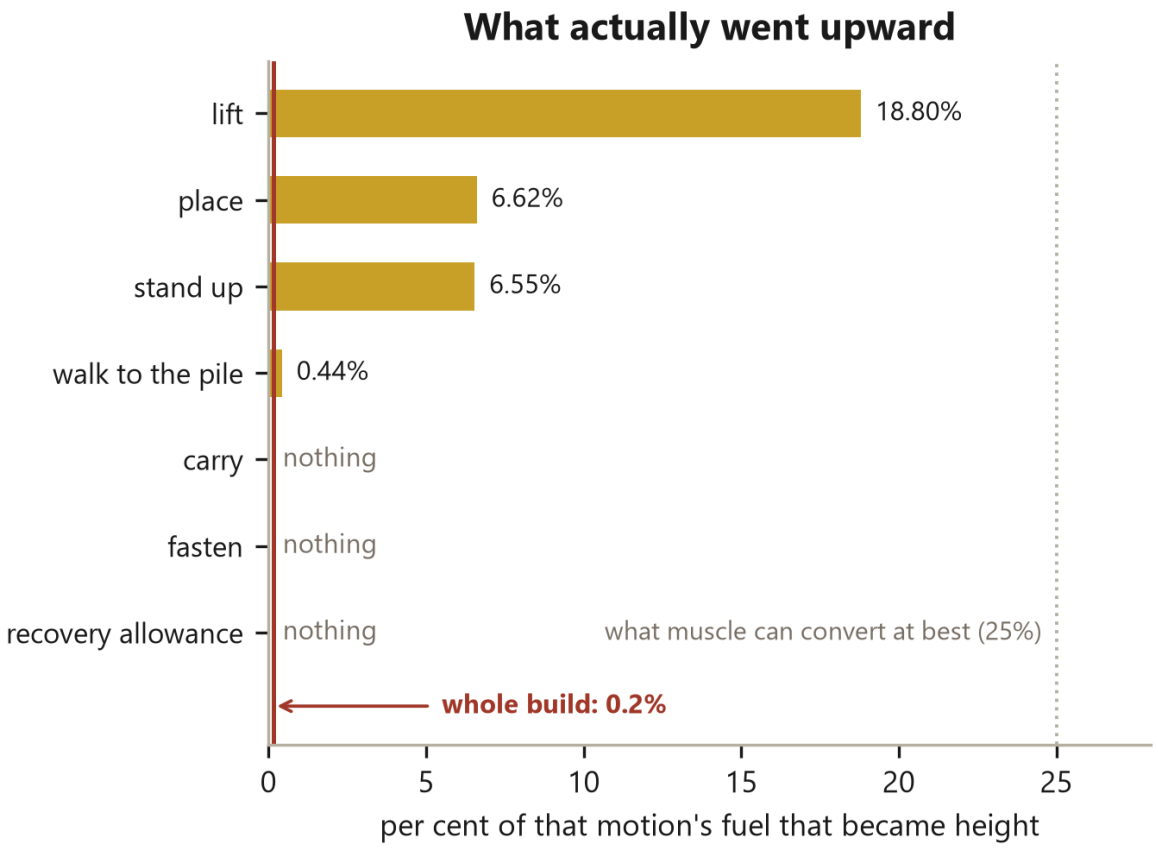
Each motion's share of the clock beside its share of the fuel. Only the gap between the two is intensity.

Fastening is 90 per cent of the food energy. Everything else in the build – all the walking, all the lifting, all the carrying, all the placing – shares what is left.

Before that gets quoted as something it is not, here is the honest version: fastening is also 80 per cent of the *clock*. Only 10 points of that 90 are intensity. The other 80 are simply where the day goes. Driving a screw is not brutal work. There is just an enormous amount of it – 153 hours of it on this build – and that is a different claim and a more useful one.

Notice the recovery allowance running the other way, too: 17.5 per cent of the clock and only 7.5 per cent of the fuel, because standing still is cheap. That asymmetry is the whole argument for the allowance, in one picture. Rest buys back a large slice of the clock for a small slice of the fuel.

And then the number I did not expect.



The lift converts 18.8 per cent, which is an ordinary human lift and not far off the best a muscle can do. The build converts 0.2 per cent, because across 191 hours it is hardly ever lifting. Put the other way up: 632 calories are burned for every one that ends up as height. Fastening raises nothing at all, and it is most of the build.

How much of each motion's fuel became height, against the most muscle can convert.

Of the 54,870 kilocalories one person spends on this house, the amount that ends up as *height* – as mass actually raised against gravity, which is the only part of a build that is physics rather than physiology – is 87 kilocalories. 0.363 megajoules. **0.2 per cent.**

Turn it the other way up: 632 calories burned for every one that went upward.

The first time that fell out I assumed the model was broken. It is not, and the proof is in the same chart. Ask the question of the *lift alone* and the answer is 18.8 per cent – an ordinary human lift, and not far short of the 25 per cent that is the best muscle can convert under any circumstances at all. The body is fine. While it is lifting, the body is close to its own ceiling.

There is one more thing in that lifting work worth pulling out, because it contradicts something everybody has been told. Split the raising by which part of the body did it and the trunk accounts for 65 per cent of it, the arms 20.4, the legs only 14.5. The reason is unglamorous: your torso is the heaviest thing you lift all day, and you lift it every time you stand up

from a squat. “Lift with your legs” is still right, but not because the legs do most of the work – they do the least of it. It is right because it changes the *path* the trunk travels, and the trunk is the load.

It is just hardly ever lifting. Lifting, carrying and walking together come to 1.9 per cent of the fuel – 326 kilocalories for the lifting alone, out of 54,870. The rest of the time the body is *holding* – holding a posture, holding a part in place, holding an arm up – and holding something still does no mechanical work whatsoever while costing a great deal to do. That is not a defect in the model. That is what manual labour is, and it is why a day of light work you would describe as “not lifting anything” still leaves you unable to start another one.

Which makes the sentence to take away from this chapter a slightly strange one. **The screw does not go up. The arm does.** And then it comes back down, and doing that for 153 hours is the build.

What this says about the hardware

If most of the fuel is in fastening, the design question changes.

The obvious instinct is to make fastening faster: a better driver, a better bit, a magnetic holder. All worth doing. But the arithmetic above says something sharper – a joint that needs *fewer* fasteners beats a joint that is faster per fastener, because the count is the thing being multiplied.

This frame carries 120 brackets, 960 pilot holes and 960 screws, and – from 52 – exactly that many at every diameter in the handling band. Each of those 960 is a fixed metabolic tax, paid whatever size dome you are building.

It is also, I have to admit, the number I would be most tempted to increase. When a joint worries you at four in the afternoon, the instinct is to put another screw in it. This chapter is the reason to think twice: one extra screw per bracket is another 120 fastenings, on every dome, forever, and it will not be the thing that saves you. If a joint is not right, the fix is upstream in the jig, not downstream in the driver.

Two parts of this design are already doing that work, and both are worth naming as energy decisions rather than as geometry ones.

The seam key of 16 closes a joint with a shape instead of with hardware. Every seam it closes is a seam nobody drives fasteners into.

The panel raise from 8 is the larger of the two. 40 panels go up rather than 120 loose members – and more to the point, each one goes up *already fastened*, on a bench, at waist height, on the ground. The fastening did not disappear. It moved out of the overhead posture, which this model prices higher for exactly the reason your shoulders already know.

I will not overstate either of them. The key is another milling operation, and the panel approach wants a cable, a winch and somewhere to hang them. Neither is free. But both are trades of *fuel* for *setup*, and setup is paid once and amortised across every dome after it, while fuel is paid again every single afternoon.

What is measured and what is modelled

This ledger is a mixture, and the mixture matters more than the total.

Computed. Every mass, every placement height, every carry distance, off the geometry and the element list. The mechanical work is $m\,g\,h$, with nothing estimated in it.

Counted. The motions. Seven per part, 1,322 parts, the same sequence for all of them.

Modelled. The calorie cost of a motion. That is 13 published constants – activity intensities, a load-carriage equation, a resting-rate formula, a recovery allowance – listed by name and source on their own page rather than buried. Change one and this chapter changes with it.

The load-carriage equation deserves its name here, because it is the one constant doing more work than it looks like it is doing, and the films give it credit: **Pandolf's walking equation**, which costs walking with a load on the body rather than with a flat intensity figure the way the tables do for stationary tasks. The reason it matters is the shape of it. Its load term is *squared*, not proportional, so carrying twice the weight costs more than twice as much – which is the arithmetic reason this chapter keeps recommending carts, closer stockpiles and lower shelves, and the reason a builder's back agrees with the model at the end of a day.

The same distinction – mechanical work is exact, metabolic cost is a model – is the honest frame for the whole ledger, and it is worth keeping in view when the station-by-station numbers are read: mass, height and distance are measurements off the model, while the price of moving them is a published estimate of a human body. This chapter says which is which at every step, and the parts that are modelled are the parts a reader should discount.

Weakest link, named. Fastening time is a *residual*. The model takes the catalogue's labour estimate for a part, subtracts the walking, lifting, carrying and placing it has computed, and calls whatever is left fastening. So "90 per cent is fastening" is really "90 per cent is everything that is not handling" – fitting, fiddling, checking, finding the bit you put down somewhere. I am comfortable with the conclusion anyway, because the conclusion does not rest on the residual. It rests on the handling, and the handling is the part that was computed. Lifting really is 0.6 per cent of the fuel however you choose to label the rest.

Not metered at all. The felling and the bucking. This ledger begins at the stockpile, and the 8 days in front of it – a chainsaw, a slope, a log that does not want to move – appear in none of these numbers. If anything here is optimistic, it is that.

CHAPTER 57 · EXPLAIN

Less Skin for the Same Floor

A dome wraps a given floor in noticeably less outside surface than a box does, and the wall you never build never costs anything to build, heat or repair

Less Skin for the Same Floor

A wall gets paid for four separate times.

It is paid for the day the sheathing is bought. Again the day it is insulated. Again every time it wants paint, flashing, membrane or a walk round with a caulk gun. And again every winter for as long as the building stands, in heat that leaves through it. Four bills, one surface, and only the first of them ever appears on a materials list.

Which makes the cheapest square foot of wall in any building the one that is not in the drawing.

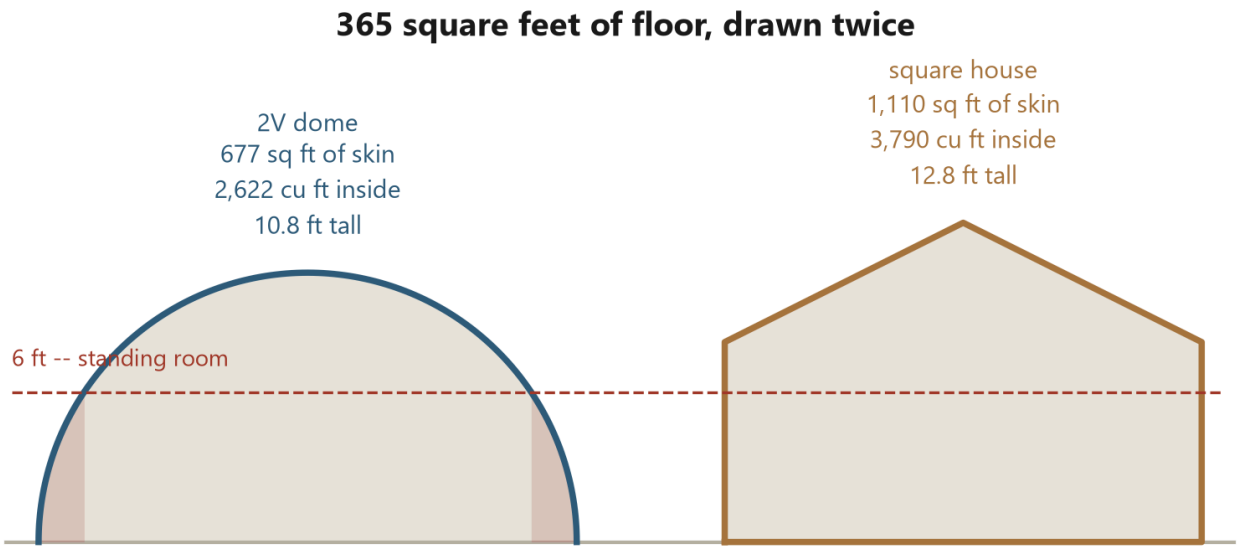
That is the whole argument for building round, stated without enthusiasm. A sphere is the shape that wraps a given volume in the least surface. Cut one in half, stand it on the ground, and most of that property survives the cut. The same floor, enclosed the round way, takes less skin around it.

It is true. It is also smaller than the people who repeat it believe, a good part of it turns out not to be efficiency at all, and it is a property of *small* buildings rather than a property of domes. There is a size above which the box wins outright, and this chapter finds it.

None of that is an argument against building round. It is an argument against quoting the good number on its own, which is what I did for about a year, in public, with a straight face.

The same floor, two envelopes

So here are two buildings with exactly the same floor.



Same floor, same scale. The dome wraps it in 677 square feet of skin against the box's 1,110 -- 39.0 per cent less to build, seal and pay for. The shaded rim is the catch: 113 square feet of that floor has less than 6 feet over it, while every square foot of the box clears the line.

The same floor drawn twice, to one scale, with the headroom line across both.

One is this book's dome: 365 square feet of floor, 10.8 feet to the crown – which for a hemisphere is also its radius, a coincidence that stops being charming later in this chapter. The other is a square house of identical floor area, 19.1 feet on a side, with 8-foot walls and a gable roof reaching 12.8 feet at the ridge.

The box is drawn fairly. It is square rather than oblong, which is the least perimeter a rectangle can have for its floor, so it is the best case for its own shape rather than a straw man. Its corners are not penalised, even though corners are where boxes leak heat and where boxes rot. And the dome is measured as what it actually is – flat triangles – rather than as the ideal sphere it approximates, which costs it a little surface it would not have if geometry were doing the building.

The dome's shell comes to 677 square feet. The box's comes to 1,110. The difference is 433 square feet of building that simply does not exist in the round version: **39.0 per cent less.**

That is the number. Now the part that gets left off the poster.

Read the volumes in the drawing. The dome holds 2,622 cubic feet. The box holds 3,790. At this size the dome is not only wrapping the same floor in less skin – it is wrapping 30.8 per cent less building. Some of that saving is shape. Some of it is that you bought less house.

Ask the question the other way round, then. Not skin per square foot of floor, but skin per cubic foot enclosed. The dome still wins, and it wins by 11.8 per cent rather than 39.0. That is the shape-only figure at this size, and it is roughly a third of the headline.

The two do not stay apart. Past 1,015 square feet of floor the dome begins enclosing *more* air than the box rather than less, because a hemisphere has to gain height to gain width and a stud wall does not. Above that line the volume deduction disappears and the surface margin is real all the way through. Below it – which is where this book's dome sits, and where almost every dome anybody actually builds sits – part of the win is a smaller building wearing the same floor plan.

Both of those are worth knowing. Only one of them ever gets said.

What the saved skin is worth four times over

Take the 433 square feet at face value for a moment and walk them through the four bills.

Bought once. The dome's shell is 21 sheets of four-by-eight against the box's 35, so it starts 14 sheets lighter before anything else has happened. Each of those sheets is also a delivery, a lift, a set of fasteners and a stretch of the work 56 prices in calories rather than money.

Insulated once. The same area over again, in whatever is chosen to fill it, priced per square foot like everything else.

Sealed repeatedly. This is the bill nobody budgets and everybody pays, and it is the only one of the four that arrives forever.

Heated every winter. At a wall assembly of 0.060 BTU per hour per square foot per degree – an ordinary insulated wall, handed to both buildings identically – the dome loses 40.6 BTU an hour per degree against the box's 66.6. Across a season of 4,200 heating degree days that is 4.1 million BTU against 6.7.

And this is where I have to take something back.

Four claims, one of which is the first one again

claim	dome	box	margin	new?
Exterior surface to build, seal and paint	677	1,110	39.0%	yes
Surface per cubic foot enclosed	0.258	0.293	11.8%	yes
Heat leaving through the envelope	41	67	39.0%	no
Wind load on the shape itself	0.420	1.050	60.0%	yes
Exterior surface at equal standing room	**887**	**1,110**	**20.1%**	**yes**

Row one is geometry -- the least skin that will go round a given floor. Row two asks a different question, skin per cubic foot rather than per square foot of floor, and gets a much smaller answer. Row four is two published drag coefficients and has nothing to do with area. Row three is marked 'no' because at an identical U-value heat loss is surface area in warmer units: it can only ever repeat row one, and quoting the pair sounds like two findings. The ruled-off row is row one re-run honestly. Every building above it has 365 square feet of floor; there the dome is grown to 478 so that as much of its floor has 6 feet over it as the box's does. Still a win. About half the size of the headline, and the one to quote at somebody who builds for a living.

Four claims, and which of them are actually separate findings.

39.0 per cent less heat is not a second finding. It is the first one wearing warmer units. Heat through an envelope is area multiplied by a U-value, both buildings were given the same U-value, so the heat margin *cannot* come out any different from the surface margin. It is arithmetically forbidden. Saying “less skin, and less heat” sounds like two wins and is one win said twice, and I have said it that way on camera.

The wind figure is genuinely independent. 0.42 against 1.05 – 60.0 per cent less drag – comes from published coefficients for a hemisphere and a cube and has nothing to do with area at all. That one may be quoted beside the surface number without double counting.

Then one deduction larger than everything above it.

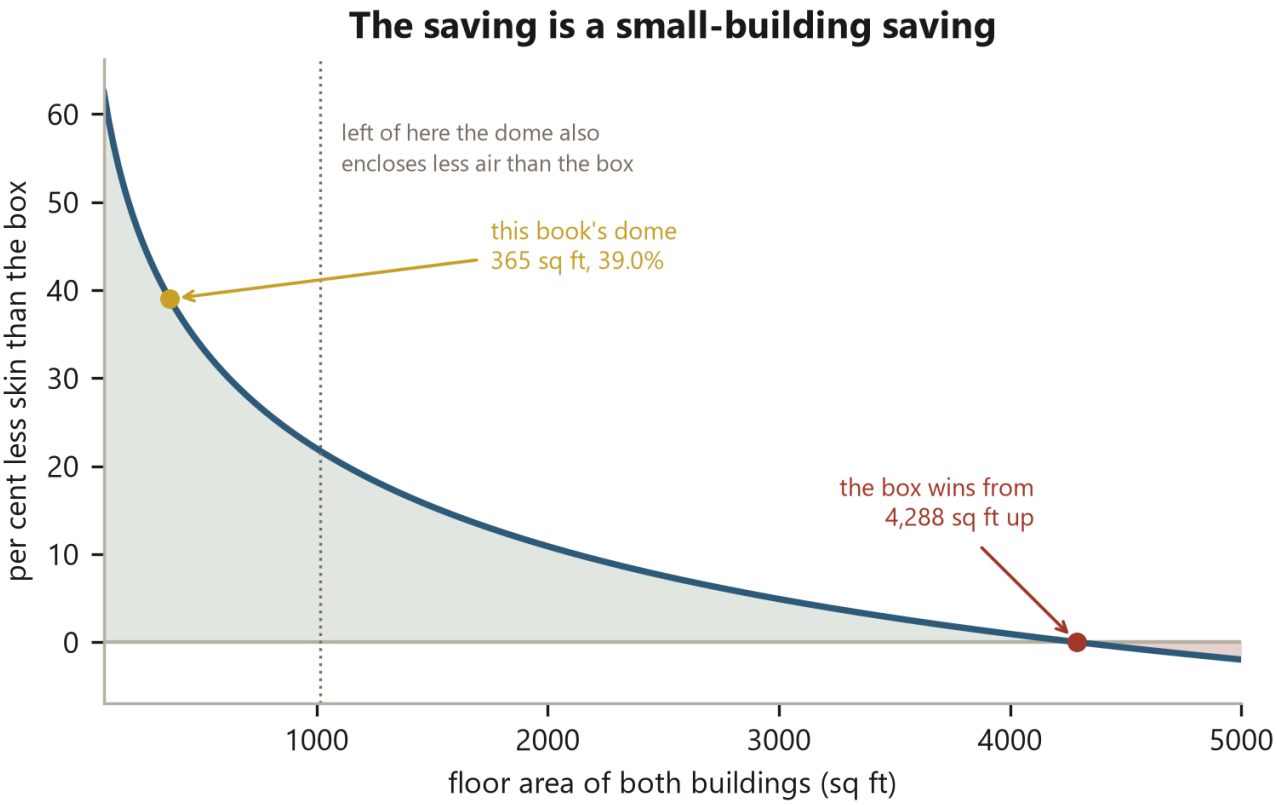
Equal floor area is not equal usable area. A hemisphere meets the ground at a tangent, so its ceiling arrives at its floor around the whole perimeter. Call standing room 7 feet – generous; habitable ceilings are held higher than that – and only 225 square feet of this dome’s 365 qualify. That is 61.7 per cent of the floor, and 140 square feet of it that you can reach on your knees. 44 draws the whole curve; the shaded rim in the drawing above is the same fact in elevation. Every square foot of the box clears the line.

So grow the dome until it stands up over as much ground as the box does. It needs 505 square feet of floor to get there, and 937 square feet of skin to wrap it, and the margin falls to **15.6 per cent.**

Still a win. Roughly half the one on the poster, and the only version of it I would put in front of somebody who builds for a living.

Where the box wins

The margin is not a property of domes. It is a property of small domes, and the chart says so more bluntly than I would like.



Both buildings keep the same floor at every point on this line. The dome's advantage falls from 62.5 per cent at 80 square feet to 10.9 at 2,000, and crosses zero at 4,288, where the dome would stand 37 feet tall and be buying skin to wrap air nobody uses. The answer to that is not a bigger sphere.

The surface margin against floor area, carried past the size where it dies.

11 sizes, from 80 square feet to 5,000, both buildings holding the same floor at every point. The saving starts at 62.5 per cent for something hut-sized. It is 39.0 at this book's dome. At 707 square feet – the largest dome in the flat-rate table of 52 – it is 27.8. By 1,015 it is 21.7, and by then the dome has stopped being the smaller building too.

And at 4,288 square feet it is gone. Zero. Above that floor area **the box takes less skin than the dome**, and every sentence in this chapter runs backwards.

The reason is in the drawing and not in the arithmetic. A stud wall is 8 feet tall whether the house is small or enormous. A hemisphere cannot do that: to cover more ground it must stand taller, and at the crossover it is 37 feet to the crown and buying skin to wrap a column of air that nobody will ever use, heat or reach.

That is not the only place the box wins, and the others do not need a chart.

Boxes stack. A second storey doubles the floor for the price of a deck, adding no roof and no foundation, and a dome cannot do it at all – a dome's second floor is a mezzanine inside the same shell, competing for the same volume. Boxes share walls: two terraced houses each delete an entire elevation, and a row of them beats any dome on skin per floor so decisively that there is no argument to have. Boxes take sheet goods whole, where a triangle takes a rectangle and hands you the offcut. And boxes fit furniture, appliances, cabinets, baths and doors, every one of which is manufactured for a flat vertical wall that a dome does not have.

The honest summary is narrow and worth having anyway. For a single, small, detached, one-storey building, on its own ground, the round shape wraps the same floor in less skin, and that saving lands four times on every bill the building will ever send you.

Outside those conditions, build a box.

Inside them, the dome still has the rim problem this chapter kept pointing at – 140 square feet of floor with no room to stand up in. The fix for that is not a bigger sphere. It is the cheapest wall in the whole build, and it is 58.

The sheet that does not fit

One practical consequence of the shape belongs in this chapter, because it is the difference between a saving on paper and a saving at the bench.

No panel of this dome fits a four-by-eight sheet. The triangles are larger than the sheet once a margin and a kerf are allowed for, which means the skin cannot be cut by laying a template on a plywood sheet and running a saw around it. Every panel is a cut somebody has to get right, and on this building that cut is wanted forty times.

Which is why the panel drawings in this book matter more than a reader expects of a drawing, and why the tooling list in the campaign chapter spends real money on a router with a four-by-eight bed. The honest options for a self-builder are three: cut the panels oversize and trim them to the frame, which wastes material and requires the frame to be right first; choose a size whose panels *do* fit a sheet, which is exactly what the storm-shelter arithmetic in the earlier chapters does; or buy the cut, in which case the sheet's size stops being your problem and becomes somebody else's line item.

What this chapter will not do is pretend the saving is free. Less skin is less material, less money and less to heat — and it is also a harder cutting job than siding a box, which is one of the places where the box genuinely wins.

The whole case in ten lines

The films close this argument with a list of 10 computed reasons for the shape, each one a number this book has already printed somewhere. Here it is in one place, because a reader who has come this far deserves the summary rather than the tour.

1. **Less exterior for the same floor.** 39.0 per cent less surface, and at equal insulation the heat follows the surface exactly.
2. **The parts list does not grow with the building.** The frame is 120 struts and 960 screws at the small end *and* at the large end of the handling band; the floor grows and the list does not.
3. **9 operations, repeated.** Not nine hundred — nine, and the flat-rate chapter is the proof that the list is genuinely closed.
4. **It is rigid by shape rather than by bracing.** A triangle cannot change shape without a side changing length, so the frame needs no shear panels, no diagonal bracing and no racking check.
5. **Little for the wind to push on.** 60.0 per cent less drag than a box of the same floor, before any hold-down is counted.
6. **It costs what the parts cost.** 22 dollars a square foot finished, with nothing in that number for a shape premium.
7. **The cheapest floor in the building is a short wall.** 791 buys 97 square feet of standable floor — the arithmetic the pony-wall chapter worked through line by line.
8. **Less skin to heat and cool.** 39.0 per cent off the load, which is the same number as the surface saving and arrives for the same reason.
9. **A roof that can refuse the sun.** Pale, and optionally radiative: the sun chapter's passive option, priced at the value of the air conditioner a reader does not buy.
10. **A brim that collects rain.** 7,259 gallons a year off an overhang the building already wears, into a 1,000-gallon tank, through a gutter that is not an accessory.

Ten claims, all computed, and four of them carry a caveat this book has already printed in full: the skin margin dies above the envelope crossover, the dead ring at the rim is real until a wall is built, the paint is comfort rather than payback, and the whole case assumes a small single-storey building on its own ground. A summary that dropped the caveats would be the sales document this chapter has spent its length not being.

The Cheapest Square Footage in the Building

Standing the shell on a short straight wall buys floor, headroom and usable edge for the least money per square foot of anything in the build

The Cheapest Square Footage in the Building

A dome has one weakness, and it is honest about where it is: the edge. A hemisphere comes down to the floor at a tangent, so the ring of floor around the rim has a ceiling sloping straight at it. You can stand in the middle of the room. You cannot stand anywhere near the wall.

On the reference floor this book's performance chapter is drawn on – 314 square feet – only 174 of those feet are under enough ceiling to stand up under. That is 55 per cent of the floor you paid for, and the rest is a dead ring.

The fix is the cheapest wall there is. A short, straight stem wall stood under the rim of the shell, lifting the whole dome a few feet off the ground. It carries no thrust, because the ring at the base of the frame already takes that. It is just a wall of height. And a wall of height is exactly what converts a dead ring into the cheapest square footage in the building.

The pony wall ladder

The ladder is four rungs, all computed from the same geometry, at the declared cost of 4.20 dollars a square foot of wall:

wall	usable floor	share usable	the wall costs	floor gained	$/sq ft gained
none	174 sq ft	55%	nothing	–	–
2 ft	245 sq ft	78%	$528	+71	$7.4
3 ft	**272 sq ft**	**87%**	**$791**	**+97**	**$8.1**
4 ft	292 sq ft	93%	$1,055	+117	$9.0

Read the bold row. The 3-foot wall is 188 square feet of ordinary wall, 791 dollars. It buys back 97 square feet of floor you already paid for and could not use – a 56 per cent increase in usable room – at 8.1 dollars a square foot.

Nothing else in this build sells floor at 8.1 dollars a foot. Not the frame, not the skin, not the glass. The wall is the one part of the building where the floor is already there and only wants lifting.

The other rungs are worth a look, because each has a different character. The 2-foot rung is the bargain: 71 feet for 528 dollars, the cheapest dollars on the ladder. It leaves a fifth of the floor unusable, but for a workshop or a greenhouse – rooms where you bend over the work and do not care about the walls – it is the right rung. The 4-foot rung is the wrong one at the far end: it spends 1,055 dollars to buy 117 feet, the last of them at 9.0 dollars a foot, and the next section

explains what else that height starts costing. The 3-foot rung in the middle is the one for a room people live in, and the reason it is the bold row is the third section of this chapter.

There is one more thing the wall buys that the table does not show, because it is not a square foot: a band of straight wall around the room. On a bare hemisphere the ceiling curves down to the floor everywhere, so a door has to be cut out of a curved surface and a cupboard has to stand away from the wall. On a 3-foot stem the first three feet of every wall is a vertical, rectangular surface – doors are rectangles in it, windows are rectangles in it, shelves and counters stand against it. The stem wall does not just buy floor. It buys *furniture wall*, and that is the difference between a room in a dome and a tent with a dome over it.

The same band is where the heavy and rectangular things live. A door is a rectangle; cut one in a bare shell and you are building a curved door, or boxing a square out of a surface that slopes two ways at once. On the stem the door is just a door, and the windows are stock units instead of specials. The band also swallows the clutter a dome has nowhere else to put – the water tank, the batteries, the shelves, the coat hooks – which would otherwise stand off the curved wall and eat the floor the wall was bought to save.

Why the edge is the expensive part of a dome

It is worth understanding *why* the edge is dead, because the number comes out of one simple fact: a circle meets a flat floor at a tangent.

Headroom in this comparison means a 6.7 foot ceiling. On a dome 10 feet in radius, you can stand only where the shell is at least that high. Walk from the centre toward the wall and at some radius the ceiling drops through 6.7 feet and keeps going, all the way to zero at the rim. Everything past that radius is floor with a sloping lid over it – geometrically fine, humanly useless.

That dead band is not a strip. It is the whole outside of the circle, and it is where the *most* floor lives, because a circle's area lives at its outside. It is why the dome's cheapest room is bought by moving the shell up, not by anything you do to the frame.

The same geometry explains why the wall is cheap compared to every other way of buying that floor back. Want the floor by making the dome wider instead? Then the frame grows with the diameter and the skin grows with the square of it – every extra foot of room is bought with longer sticks *and* more skin, which is the most expensive way there is to buy a foot. Want it by the wall? The wall grows as a line – one more foot of wall all the way round is a fixed, small number of square feet of plain wall at 4.20 dollars each. The shape of the dome makes the rim the expensive place to stand; the shape of the wall makes the rim the cheapest place to fix.

A taller shell would do the same job, but a taller shell means a bigger frame everywhere – every member, every panel, the whole fortnight. The stem wall buys the height in exactly one place: at the rim, where the room is missing. That is the whole trick, and the ladder table is what the trick costs.

Where the ladder stops paying

The ladder is not free forever, and its own columns say where it stops.

Look at the right-hand column. The rate climbs: 7.4 dollars a foot, then 8.1, then 9.0. Each added foot of wall buys less floor than the one before it – 71 feet, then 97, then 117 – because the ring of new standing room gets thinner as the ceiling profile flattens out. The wall is still cheap. It is just not as cheap.

And past a certain height the wall stops being a free ride in a second way: it starts acting like a real wall. A 3-foot stem is a curb. A 6-foot stem is a cylinder the wind pushes on, one that needs its own ties into the ring, its own sheathing, its own foundation edge. Somewhere past that you have built a round box and set a hat on it, and you are paying for a conventional wall *and* a dome frame on the same site.

There is also a water detail the wall introduces that a bare dome does not have, and it belongs to the chapter on the brim (Chapter 59): where the shell meets the top of the wall there is now a horizontal seam between two materials, and it is exactly the kind of joint water finds. The brim throws the rain past it; without the brim, the wall is a gutter fed by its own roof. The two chapters are one decision: the wall buys the floor, the brim keeps the wall.

The top of the wall wants a cap, and it is the one piece of metal the rest of the dome never asks for: a strip of flashing lapped down over the wall face and up under the shell edge, so the end-grain of the wall's top plate is never the thing the weather meets. It is a cheap part, and it is the part first builds leave off, because it hides under the hat and nobody is checking.

The honest stopping point is where the next rung costs more per gained foot than the next size of dome would. This book's own dome – the one 3 trees fill – is 365 square feet for the same fortnight of work as the 314 foot reference floor. If you want more than the 3-foot rung buys, the better answer is usually not a taller wall. It is the next dome up, where the same hundred and twenty members stand further out and the floor comes along for free.

The wall buys back the edge of the floor you already have, for 791 dollars, and nothing else moves. The bigger dome – the 21.6 foot one 3 trees surround – buys that edge by buying the room over again, spending the one thing a dome exists to save, the subject of Chapter 57. The wall is the answer when you are short of edge; the bigger dome when you are short of room.

So the ladder has a last rung, and on this floor it is the 3-foot one: 272 usable feet for 791 dollars, at 8.1 dollars a foot. That is the cheapest room in the building, and this chapter is the receipt.

The Roof Is Already a Gutter

An overhanging brim throws water clear of every joint and collects it in the same move, and the annual catch off a dome this size is not a trivial number

The Roof Is Already a Gutter

The brim around a dome – the overhanging ring of the hat – looks like decoration. It is not. It is the detail that keeps water out of the building, and it does two jobs at once.

The first job is the boring one that matters most. Rain runs down a dome shell, and without a brim it runs straight into the ring of joints at the base: the seams, the brackets, the screws, the point where timber meets footing. Every one of those is a place water can sit, wick and work on the frame for years. The brim throws all of it clear of the wall before it ever gets there.

The second job is the one this chapter is about. The same overhang is a catchment. It catches the rain it throws, and the annual number off a dome this size is 7,259 gallons. Not a trivial number, and it costs no extra roof to collect it, because the roof was already a gutter.

Where the water goes, with and without a brim

Follow one raindrop through both buildings.

Without the brim. The drop lands on the upper shell, runs down a panel, across a seam, down onto the lower panels and off the rim – straight onto the base joints and the splash line at the bottom of the wall. Some of it wicks into the seam keys. Some of it follows the brackets down the inside face. Over a wet season the bottom ring of the frame is damp more often than it is dry, and the footing is in a permanent puddle of its own making. The building is not leaking into the room. It is leaking into its own skeleton.

With the brim. The same drop lands on the shell, runs down to the hat, and hits the overhang. The hat's edge is further out than the wall – 10.4 feet from the centre, against 8.9 feet for the rim of the hat itself – so the drop leaves the building at a point where there is nothing below it but ground. It drips clear. The base joints stay dry because the rain never arrives.

The brim does not stop at dripping clear. Put a gutter lip on the hat's edge – the same metal the hat is already made of – and the drip becomes a stream, and the stream runs into a tank.

The lip is also the only gutter this building will ever have, and it wants the same care as any gutter. Leaves, needles and dust wash down the shell and arrive at the lip, and a gutter is a gutter – it clogs where the downpipe narrows. This one is a ring at the hat's edge, so the autumn job is a walk round the building with a ladder and a broom. In snow country the brim is a flat shelf that holds ice, and an ice dam on the hat can weigh more than the metal was priced for.

One of those two buildings has a water system. The other has a slow leak into its own frame. The difference is the width of the hat, 18 inches.

Wider is not automatically better, and this is the place to say why, because it is the same trade everywhere on the shell. A wider brim catches more – every extra inch of radius is real catchment – and it also catches more *wind*, and it shades more wall, which is a virtue in summer and a cost in winter. The 18 inches this book prices is the width that clears the wall, shades the windows, and stays on the right side of the wind math. The number is a choice, declared once and used everywhere, not a discovery.

The annual catch

The catch is worth working out properly, because the number sounds like a mistake the first time you see it.

The inputs, all declared before they are used: 40 inches of rain a year, the regional average; a runoff coefficient of 85 per cent, which is what a metal roof sheds – the rest evaporates, splashes and never reaches the lip; and the 314 square foot reference dome, whose hat stops at a plan radius of 8.9 feet.

The catchment is not the sloped area of the hat. Rain falls straight down, so what catches it is the *plan* area – the circle the brim's edge describes over the ground. At 1.5 feet of overhang that circle is 343 square feet.

Now the arithmetic. 40 inches times 0.623 gallons per square foot per inch of rain, times 343 square feet of catchment, times the 85 per cent that actually sheds:

7,259 gallons a year. 19.9 gallons a day, on average.

It is worth saying what that number is not. It is not metered – nobody has stood under this dome with a bucket for a year. The rainfall is a regional average, not a guarantee, and a drought year is exactly the year the catch goes dry. But the arithmetic is the honest plan-view arithmetic, with every efficiency declared on the line rather than buried.

Size the tank against use rather than rain. Two people at 50 gallons a day empty a 1,000 gallon tank in 20 days – so a full tank is a three-week buffer against a dry spell, and in an ordinary year it refills 19.9 gallons a day.

At utility prices the whole year's water is worth about 44 dollars. Say it plainly: collected rain is nearly free water either way, and the money argument is a weak one. The real argument is the 20 days of water that keep flowing when the well does not. That is what the number buys.

The same 7,259 gallons appear in the ten-points list at the end of the films, computed by this same function. That is not a coincidence and it is not a copy: the film and this book read the one water model, so they cannot drift apart. Wherever you meet this number, it is the same dome, the same rain, the same arithmetic.

What the catch is actually good for

The honest uses first, because the dishonest one gets said first everywhere else.

Not drinking, without treatment. Rain off a roof has been through bird droppings, dust and whatever the shell sheds, and the tank is warm and dark, which is a polite description of a biology experiment. Filter and treat it or keep it out of your mouth.

Everything else. Irrigation is the obvious one, and the dome's own garden sits in its shadow. Flushing and washing take the 19.9 gallons a day without a second thought. And the quiet use that matters most on a rural site: a buffer against a dry well. A well that fails in August is a crisis with an empty tank and an inconvenience with a full one.

The tank belongs under the pad, not beside it. Three reasons, all boring and all true. The pad is already excavated, so the hole is half dug. The ground under a pad is shaded and frost-protected by the pad itself, so the tank neither cooks in summer nor splits in winter. And the weight sits on compacted ground that is already carrying a building, instead of in a second hole you had to dig and backfill for no other reason.

Two details make the difference between a tank and a pond, and both are cheap compared to the misery they prevent. A first-flush diverter – a simple standpipe that captures the first few gallons of every storm, the ones carrying the dust and the droppings, before the clean water is allowed into the tank. And an overflow, sized like it is going to be used, because 40 inches of rain will overflow any tank ever built and the water has to go somewhere that is not the pad. Skip either and the tank teaches you the lesson within a year.

The plumbing between the lip and the tank is simple, and it has two jobs nobody thinks about until the tank overflows. A leaf screen at the lip keeps the big stuff out of the pipe, because the first-flush diverter catches only the first of the dirt, not the last. One downpipe carries the whole ring's 19.9 gallons a day down the wall and under the pad to the tank, and it wants an air gap so a full tank cannot back up into the pipe and out the lip. That is the whole of the plumbing: a screen, a pipe, a gap.

One more job the brim does that is worth a line, because it is free with the same metal: shade. The overhang keeps the summer sun off the upper wall in the middle of the day, and it keeps the rain off the base ring all year. Every one of those jobs is done by a piece of the building that was already there, doing the first job – keeping the frame dry.

That is the whole point of this chapter. The gutter is not an accessory a dome can have. It is the brim the dome already wears, and the only question is whether you let the water run off it or into a tank.

The top half is the half worth glazing

One geometric fact makes the skin cheap if you use it, and it falls out of the subdivision rather than out of a design decision.

Sort the shell's panels by height and the bottom half of them come to almost exactly half the surface. That is a property of the way a sphere is divided rather than a coincidence: an even subdivision puts its widest ring at the equator, so the panels below the waist cover the same area as the panels above it and the sorting comes out even. Which means the question of what to do with the shell splits cleanly in two:

The upper half is the roof — seen from outside by nobody, lit from above, and the place where glass does the most work. Light comes in over the furniture rather than across the floor, and on a dome the upper ring is already the roof, which is why a skylight here costs nothing but a panel.

The lower half is the wall — where a reader wants insulation, siding, a hatch and the one solid door. Nobody sees the bottom of a dome from the air, and glass down there is glass paid for and then curtained over.

So the honest rule for a first build: glaze the top, side the bottom, and spend the difference on the brim. A reader who does that buys the *lighting* of a much larger building for the price of the half of the shell that was worth glazing, and leaves the other half doing the job a wall is supposed to do. The same rule explains why the panel catalogue in this book lists sixteen surfaces for one geometry: the shape is decided, and the shell is a set of forty separate decisions that this one fact sorts in half.

CHAPTER 60 · EXPLAIN

A House That Gets Warmer Every Winter

Build the skeleton once and add layers to it for as long as you own it: the shell ladder turns a shelter into a house in steps you can afford one at a time

A House That Gets Warmer Every Winter

Most houses have two states and nothing between them: finished, or unfinished. Unfinished means tarps and regret. Finished means the whole invoice at once.

A dome shell can be neither, permanently and on purpose.

The skeleton of a dome is the frame; the frame carries the skin; and nothing about the skin needs to be decided on the day the frame goes up. You can stand a bare shell this autumn, quilt a first layer under it this winter, add a second next year, and keep going for as long as you own the place. Each step is affordable on its own, each step makes the building better, and none of them ever requires undoing the one before.

The ladder also becomes a habit, and that is the part the table does not show. One layer a winter is a small, repeatable job – a weekend under the shell, the same as splitting firewood or cleaning a chimney – and it lands in the season that needs it. By the time the house is fully quilted you have stopped noticing the cold and started noticing the ritual: the first hard frost is the cue, the layer goes up, and the house is a little warmer than the winter before. You buy a winter at a time, in the order the cold arrives.

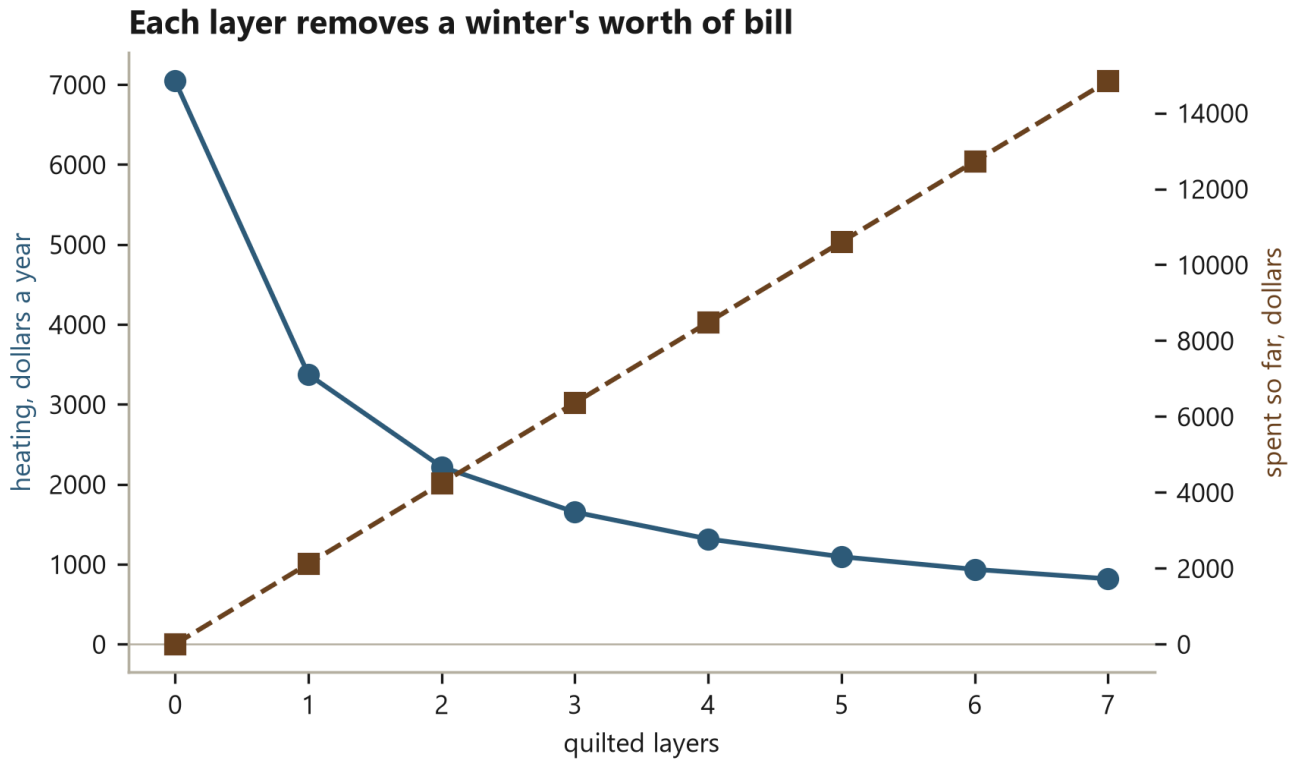
That is the shell ladder, and this chapter is the ladder with its prices. The numbers are for the Split-Log Homestead – a 39.4 foot dome with 1,217 square feet of floor – because that is the dome the park model computes its ladder for, and a heating number means nothing without the building it came off.

This book's own dome is smaller than the flagship the table prices, and the ladder scales to it the same way: the per-square-foot price holds, the total is smaller because the shell is smaller, and the order of the rungs does not change.

The shell ladder, rung by rung

Each rung is one quilted layer of recycled fabric under the shell, at a declared 1.6 R-value and 1.1 dollars a square foot. The ladder runs to 7 layers:

layers	R-value	heating, $/yr	cost so far
0	1.5	8,260	nothing
1	3.1	3,955	$2,486
2	4.7	2,600	$4,973
3	6.3	1,937	$7,459
…	…	…	…
7	12.7	958	$17,404



The first layer buys the biggest drop. Each later layer buys less per dollar -- which is why the ladder's rungs are bought one winter at a time, as the bills prove their worth.

Each layer's cost against the heating bill it removes, rung by rung.

Read the first and last rows. The bare shell is R-1.5 and it wants 8,260 dollars a year to stay warm. The same shell with all 7 layers is R-12.7 and wants 958 dollars a year. The layers removed 88 per cent of the heating bill, and they cost 17,404 dollars spread over however many winters it took you to add them.

The point of the ladder is not the top row. It is that there *is* a ladder. Every rung exists on its own: one layer is a real improvement, not a down payment on seven. You buy the rung you can afford, in the winter you can afford it, and the

house is warmer than it was the winter before.

What the next layer costs, and what it returns

The table above prices the ladder from the bottom. This one prices it from the *rung you are standing on*, which is the question a reader actually asks in year three: what does one more layer do for me now?

layers	R	heating a year	the layer that got you here	saved a year	pays back in
0	1.5	8,260	—	—	—
1	3.1	3,955	2,486	4,305	7 months
2	4.7	2,600	4,973 cumulative	1,355	—
3	6.3	1,937	7,459 cumulative	663	—
7	12.7	958	17,404 cumulative	139	17.9 months

Read the last two columns together and the ladder stops being a table of prices and becomes a decision rule. **The first layer saves 4,305 dollars a year and pays for itself in 7 months** — inside the first winter, by a wide margin, which is why the first layer is not a question. **The seventh layer saves 139 dollars a year and takes 17.9 months** to pay for itself, and it is being asked to justify itself against a shell already holding R-6.3 at three layers. Both of those are true, both are in the same table, and the honest reading is in between: the ladder is worth climbing while the next rung pays back inside the horizon you intend to keep the building, and it stops being worth climbing when it does not.

Which is the reason the campaign's quilt network is shaped the way it is. A layer bought in a cold year for 2,486 is a different proposition from a fourth layer bought for a house that is already warm, and the network's whole economic argument — 50 dollars a layer for 132 shirts' worth of somebody else's evening — is that the *first* rung of that ladder is cheap enough to be worth doing for a building somebody else lives in. The ladder is what makes insulation a sequence of small decisions rather than one large one, and the payback column is where a reader decides how far up to go.

It is also worth noticing what the ladder is made of, because it matters that it works at all. These are not batts stuffed between studs. A quilted layer is a continuous sheet hanging under the whole shell, and continuity is the difference: a stud wall insulates between the studs and conducts through every one of them, so its *assembly* R-value is well below its cavity number, while the quilt has no studs in it to bridge. The shell frame is the same story – the struts touch the skin at the seams, but the quilt covers them too, on the inside, in one unbroken layer. That is the quiet reason a ladder of cheap fabric layers does what the table says it does.

The quilt is worth one more sentence for what it is made of, because it changes the price. It is recycled fabric – clothing and bedding that left the waste stream and came back as insulation – so the 1.1 dollars a square foot is not the cost of a manufactured foam product. It is paying somebody to shred, clean and stitch what a city was throwing away. The ladder adds a thickness of second-hand fabric to a frame that was already second-hand timber.

Why the bones are the thing to get right

Every layer in that ladder attaches to the same thing: the frame. The quilt hangs from the struts, the seams key into the panel edges, the inner lining screws to the members. There is no layer that does not touch the skeleton.

That is why the frame is the one part of the building worth being fanatical about, and it is the cheapest time to be fanatical about it. If the frame is straight, every later rung has something straight to land on, and nothing is ever blocked. If the frame is wrong, the error does not stay in the frame. It walks up the ladder with you: the quilt puckers over a twisted strut, the lining opens a gap at a seam that never quite closed, and the tenth winter you are still paying to heat the mistake you made on day five.

There is a second reason the bones matter, and it is the one the marketing never says. The ladder's arithmetic prices *conduction* – heat moving through the shell material. It does not price airtightness. Airtightness is not bought with layers; it is built into the frame, in a continuous shell with no corners to leak and seams you made yourself. That is the shell's job, not the quilt's. A leaky dome with seven layers is a quilted leak. This book has not metered a winter in one, and the chapter that closes this part (Chapter 69) says exactly which energy claim it will not make and why.

The rung that pays back fastest

Now rank the rungs, because they are not equal and the order surprises people.

The first layer is the best buy in the building. It costs 2,486 dollars and it saves 4,305 dollars in the first year alone – it pays for itself in 7 months, and everything after that is free. No other rung does that.

The second layer costs the same 2,486 dollars and saves 1,355 dollars a year. Still good, but the layer that took 7 months to repay now takes years, because each layer of insulation removes a smaller share of what is left.

The third saves 663 dollars a year. The fourth, fifth and sixth continue the same slide, and the seventh saves 139 dollars a year on its 2,486 dollar cost – a payback of 17.9 years, which is most of a mortgage.

That slide is not a failure of the ladder. It is the physics of insulation, and it is the same in every building: the first inch does the most work, and every inch after it works on the smaller leak that is left. What the ladder buys you is the *option* to stop exactly where your winters say to stop. A mild climate might own the first two layers and never miss the rest. A hard one buys all seven and the top rungs still make sense there, because a cold climate's heating bill is big enough that even 139 dollars a year compounds over the thirty years of a maintained dome.

There is one more column the table cannot show, and it is the reason the ladder beats a one-time insulation purchase in this particular building. Every layer attaches *after* the shell is up and *after* you have lived in it, so every rung is bought with information the first rung did not have – which walls are cold, which windows sweat, which winter hurt. A conventional house makes its one insulation decision before the walls are closed, on guesswork. The ladder makes seven smaller decisions, each one with a year of living in the house behind it. That is what "warmer every winter" actually means: not a promise about the weather, but a building that lets you keep paying for warmth in the order your own experience says to.

A one-time foam purchase is the other way, and it is the way a conventional house has to take. Foam is bought once, sprayed once, and sealed behind a wall the same day; if it was the wrong thickness, or it shrank, or a seam opened, you will not know until the bills are in, and you cannot reach it to fix it. The quilt is the opposite: it hangs where you can see it,

it comes down in the same layer it went up, and a bad layer is a morning's work to replace. The ladder's rival was never the foam itself. It was the single invoice.

So the honest ranking is: the first layer pays back fastest, by an enormous margin, and it is rarely the one people do first. People paint the inside. They buy a wood stove, which is a lovely thing and does nothing to the envelope. They build a porch. The layer that returns 4,305 dollars a year on 2,486 dollars waits, because a quilt hanging under a shell is not something anybody has ever been proud of at a party.

That is the argument for the ladder, stated plainly: the house gets warmer every winter, one rung at a time, in the order your money allows – and the first rung is the best one, so even a broke first winter is a good winter to start.

The Quilt Network

The insulation is a quilt of recycled clothing, and the quilt is the one part of the whole project a stranger can build at their own kitchen table -- which makes it the campaign's recruiting half, not its footnote

The Quilt Network

Every other part of this book needs a saw, a bench, or a mortgage. The insulation needs a sewing machine and an evening, and that difference is the network.

A dome's warm layer — the shell ladder of Chapter 60 — is a quilt: fabric patches stuffed with recycled clothing, cut to the dome's own triangles, stacked one layer per winter. The technical half of that statement lives in the shell chapters. This chapter is the social half, and it is not a footnote: the quilt is the only part of the entire project that a stranger can take part in from their own kitchen table, which makes it the campaign's recruiting mechanism, computed like everything else in this book.

A layer is an object with a number on it

The arithmetic of one layer, because the quilt's power is that it is countable.

One quilted layer is a real thing with a number on it: a definite area of the dome's surface, a definite count of shirts, a definite R-value, a definite winter. It goes on a named dome, and it is not a contribution to a pool — it is an object, addressable, inspectable, whose maker can point at a building and say *that layer is mine*. The count is the difference between a movement and a donation drive: a donation dissolves into a budget line, an object stays an object, and people make objects.

The layer's economics are the ladder of Chapter 54 in miniature: the barrier is a machine and an evening, not a workshop, a chainsaw or land; the unit is a layer with a number on it; and the material is the one input the maker does not have to buy, because textile waste is a large, boring, well-measured problem and a layer is a measurable bite out of it.

Provenance, because the layers come off in order

The third reason the quilt is a network and not a charity: the dome can name its layers, forever.

The stack wears in order — layer one against the frame, layer two over it, each winter's addition on top — and it strips in reverse order. So every layer carries its maker's name and its date, and the building can state, layer by layer, whose shirts are in which winter. The provenance is structural, not sentimental: it comes from the fact that the layers are separable objects, which is the same removability that makes the shell ladder an upgrade path. The dome that gets warmer every winter is, at the same time, a ledger of everyone who warmed it — and the ledger is the recruiting poster.

What a stranger's layer does to the economics

The chapter's real claim, stated with the numbers the module computes.

The quilted layer is the cheapest rung of the shell ladder — a blanket-quilted layer runs about 50 dollars — and it is the rung whose labour the builder does not have to perform. A dome's insulation is hundreds of square feet of fabric work: a week of the builder's own time, or the network's evenings, spread across people the builder will never meet. The labour the builder cannot buy at a wage the build can afford is exactly the labour a network can gift — and the gift is not charity in the direction it appears, because the maker gets the one thing the builder cannot give: their name in the building's winter, provable, permanent, and warm.

The campaign arithmetic that follows — the quilter's kit, the host pack, the tiers — lives in the stem-cell chapters of Part Three. This chapter's job is narrower: to establish that the quilt is not the dome's hobby; it is the dome's economy, and the network is the economy's engine.

What is arithmetic here and what is hope

The honest boundary, because the book's rule about numbers applies to its own campaign.

The shirt counts, the layer areas, the R-values, the prices — those are arithmetic, computed from declared constants and the dome's own surface. That people will actually sew them, at the scale the campaign assumes, is a bet — the book's own modules label it as such, with the textile-waste figure attributed to the EPA and the shirt-area figure explicitly marked as something a person who has actually cut up a hundred shirts should correct. The network is the project's most hopeful chapter, and hope is the one material this book refuses to mislabel: everything here that is a number is a number, and everything that is a wish is called a wish.

The layer arithmetic, in one place

One paragraph of arithmetic, because the network's business case is entirely in it and the book has so far scattered it. A quilted layer for this building is 132 t-shirts' worth of somebody's evening covering 592 square feet, sold at 50 dollars a layer — and at the margin the *next* layer costs 68.53, which is more than the first for the reason the warmth chapter gave: the layer gets bigger as the shell gets thicker, and the edge has to be closed again every time.

Read those two prices together and the shape of the network appears. A quilter is paid for a real evening's work at a price a household can decide on without a loan; the buyer gets a measurable improvement — 1.6 of R for the first layer, with the ladder above it priced in the warmth chapter — and the whole exchange happens at a scale neither party has to finance. That is what a waste stream plus a sewing machine buys, and it is the only part of this book's economy that scales without anybody borrowing anything.

The honest caveats stay attached. 132 shirts a layer is an estimate from shirt-area figures rather than a count of anybody's actual pile; the second and later layers cost more than the first; and none of it works until there are enough quilters near enough to a dome for a delivery to be worth the trip. The arithmetic is the reason to try it. It is not a reason to believe it will happen.

One Hardware Set, Three Sizes

The brackets, keys and fasteners that frame a small dome frame a large one unchanged, which is what makes a shared parts bin possible across a whole site

One Hardware Set, Three Sizes

The first chapter of this part showed the flat rate: the *counts* do not move. This chapter is a different and more useful claim. The *hardware itself* does not move either.

Counts being flat means you buy the same number of things. Hardware being flat means you buy the same *things*. The same bracket. The same key. The same screw length. A part that leaves the bin for a garden dome leaves the bin for the big one too, unchanged, unmodified, not a “heavy duty” version of itself.

The reason is worth knowing, because it is the whole trick. A bracket in a dome frame has one job: hold an angle. The angles in a dome do not change when the dome gets bigger – every edge is the same fraction of the radius, so every joint sees the same geometry at a different scale. The bracket never finds out how big the building is. Only the stick does.

The book’s own frame makes the point from the inside. A wedge dome has not forty different corner angles but a small, countable set – the seams fold at 18.03 degrees and 22.46 degrees, and nothing else – so the hardware exists to hold two angles, not a catalogue of them. The Timber Workshop frame in this chapter is a 3V frame rather than the 2V wedge, but it obeys the same law: the hubs are the same 61 hubs at every diameter, because a hub is a set of angles, and the angles did not move.

Those two fold numbers are the whole hardware kit, and they are worth a slow read, because between them there is nothing else. Every seam in the wedge frame folds to 18.03 degrees or to 22.46 degrees, and no third value exists. So a part that holds a 18.03 seam and a part that holds a 22.46 seam are the only two kinds of corner the frame will ever ask the bin for. One spare of each covers every joint in the building, a small one or a large one, because the angle never found out which. Two is a count, not a tolerance; it is the difference between a parts wall and a parts pocket.

A hub is the same idea, moved from a seam to a junction. Where several struts arrive at once, the hub’s only job is to hold each of them at its correct angle, and that angle is set by the geometry, not by the length of the wood. Make every strut longer and the hub does not notice – it is still holding the same set of angles, just further from the next hub. That is why the hub bill in the table below holds still to the penny while the floor multiplies: the hub is the joint, and a joint has no size. The stick is the reach, and only the reach grows.

A fastener makes the point smaller and closer to hand. A screw or a key is sized to the section of the members it joins, not to their length, and the section does not change when the dome does – not until the stick gets long enough to need a fatter one. So the same screw length closes the same joint on a 19.7 foot dome and a 52.5 foot one. The hardware reads the timber’s thickness. It never reads the building’s width.

The same bin, three domes

The proof is the Timber Workshop frame run at three diameters, from the Creator's own costing:

	small	middle	large
diameter	19.7 ft	32.8 ft	52.5 ft
floor	295 sq ft	819 sq ft	2,096 sq ft
struts	165	165	165
hubs	61	61	61
panels	105	105	105
hub bill	$596	$596	$596
frame	$1,722	$2,870	$4,592
panels	$1,161	$3,224	$8,253
foundation	$3,178	$8,829	$22,602
whole build	$10,652	$19,514	$40,037

Look down the middle three rows. 165 struts. 61 hubs. 105 panels. A hub bill of 596 dollars, to the penny, while the floor multiplies by 7.1 and the total by nearly four.

Now look at what *does* move. The frame cost moves, because the frame is timber and the stick is longer. The panel cost moves, because panels are area and the shell is bigger. The foundation moves, and it moves hardest – 3,178 dollars to 22,602. Every row that moves is a row of raw material. Every row that holds still is a row of hardware. That is the table, whole.

The table stops at 52.5 feet, but the law it is showing does not. A fourth row would print the same 165 struts, the same 61 hubs, the same 105 panels and the same 596 dollar hub bill, with a longer stick and a bigger foundation underneath them. What stops the table is not the hardware. It is the stick – past 6 feet a member wants a second pair of hands – and the ground, which is already the fastest-moving row on the page. The bin never closes. The wood and the dirt eventually do.

One boundary on all of this needs to be said plainly, because it is the one the flat-rate pitch is easiest to overstate. The hardware is flat across *size*. It is not flat across *frequency*. A 2V wedge and a 3V frame are different buildings with different strut counts – 120 against 165 – and that difference is real, not a rounding. Raise the frequency and the parts list grows, exactly as chapter 52 warned. Hold the frequency still and run the diameter out, and the hardware does not move at all. Both statements are true; they are about different dials, and the shared bin only makes the first one free.

What invariance buys a site

One site, three dome sizes, one hardware bin. The economics of that are quietly enormous, and they are the difference between a hobby and an operation.

One order. You do not maintain three purchase histories and three suppliers; you reorder the same line item until the price breaks.

One bin. Nothing in the bin says which dome it belongs to. A spare is a spare for all of them, so the site holds one spare set instead of three, and a broken bracket on Tuesday is fixed from stock on Tuesday instead of from a catalogue in two weeks. The arithmetic of spares is the unglamorous version of the same point: three designs each holding ten per cent spare parts means three spare sets gathering dust; one bin holding ten per cent serves every dome on the site, and the bin gets smaller as the site gets bigger.

One set of hands to train. The person who can drive the key on a small dome can drive it on a large one, because it is the same key, the same angle, the same motion. A park of mixed domes does not need a specialist per size; it needs one skill.

One bill of materials to trust. The frame that went up first is the proof of the frame that goes up second, because they are the same joints. The first build de-risks the second in a way no paper review can: the hardware either held its angle for a winter or it did not, and the answer applies to every dome the site will ever put up.

The quietest saving in that list never shows on an invoice, and it is the cost of the part that does not fit. A site running three hardware families has three chances, every working week, to hand somebody the wrong bracket for the job, and the wrong bracket is discovered on a ladder with half a frame standing in the air. One bin removes that possibility. There is no wrong bracket, because there is only one bracket. The mistake you cannot make is worth more than the spare you never have to order, and it compounds with every dome the site adds.

That is what invariance actually buys: the boring, compounding things – inventory, spares, training – that decide whether a site can grow without falling over. The dome that scales the floor also scales the workshop, and it is the workshop that scales the site.

What does change, and it is the timber

Say the other half out loud, because the table says it and the flat-rate pitch sometimes forgets it.

The stick grows. Every member in the large frame is nearly three times the length of the small one, and longer stock is not just more wood – it wants straighter wood, bigger sections to stay stiff at length, and eventually a second pair of hands at the lift. The member is where size shows up, and this book has been honest about that from its first chapter: 6 feet is the declared limit one person sets alone.

The section grows, eventually. Past some diameter the same 2x4-class member stops being enough, and the hardware does *not* change while the timber does. A bracket that holds an angle holds it for a fatter stick too, which is why the bin survives the upgrade.

And the skin grows with the square, as the earlier chapter on the envelope said. So the honest summary of the whole idea is three lines: the *hardware* is flat, the *timber* grows as a line, and the *skin* grows as the square. Anybody who tells you the whole building is flat has counted only the first line. Anybody who tells you the hardware grows has never looked at the table.

Buy for the Next Two Steps

The cheapest thing you can do on a site is oversize the parts that are hard to change later and undersize nothing else

Buy for the Next Two Steps

The rule in the title sounds like a slogan until you watch it get broken, and then it is the most expensive sentence on the site.

Buy for the next two steps. Not for the dome you are building – for the two after it. The dome is the thing that moves. The ground, the services, the pad and the trench are the things that stay, and anything that stays is a thing you will meet again in its current condition, whatever condition you left it in.

The frame you can unbolt and carry. The conduit you buried at exactly the right width for the dome you have is a monument to the dome you had.

So the whole discipline is one question, asked about every dollar: if the next dome on this pad is bigger, or the next pad is further from the hub, does this purchase still serve? If the answer is no, you are not saving the difference. You are borrowing it, from yourself, at whatever it costs to dig it up.

What is hard to change and what is not

Split everything on a site into two lists, because the two lists want opposite treatment.

Hard to change: trenches, conduit, the pad diameter, the service capacity – the wire gauge, the water line size, the panel rating – and the spur that reaches the pad. These are under the ground or inside the pad, and changing any of them means cutting open a finished surface, or digging up a buried run, or both.

Easy to change: the shell layers, the interior walls, the fixtures, the furniture, the dome itself. These bolt on, unbolt, and follow you. The dome is the extreme case: it comes down in as little as 2 days, moves for 1,800 dollars, and stands again at the other end – which is why the next two steps are a *path* for the dome and a *decision* for the pad. The pad cannot follow you. It can only be right or wrong for whoever comes after.

Spend the planning on the first list and the money on the first list, and be cheap on the second. The instinct is backwards: people economise on the trench and overspend on the tap. The trench will be there in twenty years, holding whatever it holds. The tap is replaced in five either way.

The numbers are the proof that the ordering matters. The pad comes in 4 foot steps (36, 40, 44, 48, 64 and 80), so a pad sized for the dome you have can be wrong for the dome you want by one step – and one step is not a small thing to add later. A service drop sized for today's panel cannot be upgraded without the trench. These are not accounting problems. They are the definition of the word "later", priced.

The cost of the upgrade you did not leave room for

Price the same upgrade twice. The services a pad is built with – the electrical pedestal, the water and drain tie-in, and the share of the site's trunk services – come to 9,000 dollars when they are laid in while the pad is being built, in an open trench, before anything is finished over them.

Now the same upgrade, retrofitted. The pad is finished. The deck or slab is down. The trench has to be reopened under a finished surface, the run extended, the surface patched and made good. This chapter declares, rather than measures, that the retrofit costs 2.0 times as much: 18,000 dollars. The multiplier is an estimate, and it is named as one – the honest framing is not the exact multiple, it is that the multiple is large, and that the difference is 9,000 dollars you could have spent on the dome instead.

That is what "buy for the next two steps" means in money. Oversizing the hard parts at build time costs a little – the next size of conduit, the next gauge of wire, a pad one step wider. Undersizing them costs the retrofit. The cheapest thing you can do on a site is the first one.

The pad one step up is also the cheapest empty room you will ever own, and that is worth saying because an empty pad feels like waste. It is not. A pad sized for the next dome sits there for a season or two, holding nothing, costing nothing to hold – gravel does not rust, and an unused trench does not leak. What it holds is a decision you already made: the day you want the bigger dome, or the second dome, the ground is ready and the services are already in the right place. The empty pad is the opposite of the retrofit. The retrofit pays for the upgrade later at 2.0 times the price; the empty pad pays for it now, and it costs nothing to keep.

There is a second worked number in the same idea, and it is the one people feel most. The apartment option in the next chapter's housing table costs 3,625 dollars to move into – of which 2,175 comes back and the rest is gone – and walking away from the lease early costs 2,900 dollars. None of that is an upgrade; it is the price of a roof you do not own, paid to people who do. The dome's entry is 18,401 dollars, and 16,601 of it is the dome, which leaves with you. Every month the lease runs, the apartment is the more expensive way to have bought the *same* thing: a place to sleep. The growth path only exists at all because the dome is the one option in that table whose entry money is not a fee. It is a down payment on the next two steps.

The growth path as a sequence, not a plan

There is a reason to think of this as a path and not a plan.

A plan is a drawing of the end state: the big dome, the three pads, the finished site. Plans are lovely and plans die on contact with money, because money arrives in the wrong order and the plan has no answer for "what do I buy with what I have this month".

A path is the order in which you get there without ever having to undo a step. The small dome first, on a pad one step bigger than it needs, on services one size bigger than they need. Then the second dome on the same pad. Then the second pad, sharing the hub. Every step is affordable on its own, and no step is a correction of the one before.

The path matters more than the plan for a reason specific to this kind of building. A dome site is not a subdivision; it is a sequence of small decisions, each one cheap to make and expensive to reverse, and the only plan that survives contact with money is the one that knows which order the decisions come in. Buy the pad for the dome you have and the next dome is blocked by a trench. Buy the pad one step up and the next dome is a fortnight. The difference between those two

outcomes was 4 feet of gravel, decided months before either dome existed. That is the whole discipline: the cheap parts are the plan, the hard parts are the path, and the path is bought one step early, every time.

The second dome is where the rule pays in a way the first dome cannot see. Buy the pad one step up and the second dome is not a move; it is standing a frame on ground you already own, sized for it, with services already in – 2 days at the quick end and 14 at the slow, layered end, either way a fortnight and not a re-excavation. Two domes on one pad, or the dome and the workshop, or the dome and the guest pod: the second building is where “buy for the next two steps” stops being a slogan and becomes the reason the first step was cheap.

The path is also the answer to the table this part opened with. The four ways to have a roof – a hotel at 4,256 dollars a month, a short let at 2,888, a lease at 1,520, or the dome on a pad at 880 – cross over at 3 months, and the dome only wins because its entry cost (18,401 dollars, of which 16,601 is the dome you keep) was bought once. A path is what makes that possible: each step small enough to afford, each one aimed at the step after it.

The same rule reaches past the pad to the hub the pads hang off, and it is the one place people forget it because the hub is not on their land. The panel, the manifold and the trunk that carries power and water are bought once and shared, and “buy for the next two steps” means sizing them for the cluster before the cluster exists. A hub sized for one pad is the same mistake as a pad sized for one dome, just owned by several people at once. The chapter on the pad (Chapter 65) and the chapter on the network (Chapter 68) price that hub; this chapter only has to say the discipline does not stop at your own property line.

That is the whole discipline of this chapter, in one sentence: spend the planning on the things that stay, be cheap on the things that move, and never buy for the dome you have when you already know the next two.

The Ground Is What You Cannot Take With You

Measured across the dome catalogue, the foundation is between a small fraction and most of what a dome costs -- and it is the only part you leave behind

The Ground Is What You Cannot Take With You

The frame is portable. That has been the quiet promise of this whole part of the book: the dome unbolts, rides a trailer, and stands again somewhere else.

The ground under it is not portable. The slab, the pad, the piers, the platform – whatever the dome stands on, that part is poured, dug or built where it is, and when you leave, it stays. Nobody loads a foundation onto a trailer.

So the honest question for a building you might move is: how much of what you paid is the part you cannot take? And the measured answer, across the whole catalogue the Dome Creator ships, is where the surprise is. The foundation is between almost nothing and *most* of the cost of a dome, depending on which dome you picked – and the spread is the single biggest lever on the whole page.

Foundation share, across the catalogue

Every shipped design, priced twice by the Creator – once as sold, and once with its foundation set to bare ground. The difference is what the ground costs, and this is that difference as a share of the whole:

design	foundation	share of total
Whole Trunk Lodge - 20 ft	gravel pad	16.8%
Timber Workshop	concrete slab	45.2%
Split-Log Homestead	gravel pad	**8.1%**
Grow Dome	grass pad	2%
Glass Studio Loft	wood deck	25.3%
Hex Cell Pavilion	concrete slab	21.1%
Continuous Steel Arc Hangar	concrete slab	23.7%
Rebar Garden Dome	grass pad	3.4%
Concrete Monocoque Form	concrete slab	23.6%
Woodland Hex Mirror	grass pad	0.4%
Woodland Square Mirror	grass pad	0.5%
Treehouse Canopy Dome	treehouse platform	63.3%

That is the whole catalogue: 12 designs, and the foundation runs from 0.4 per cent of the Woodland Hex Mirror to 63 per cent of the Treehouse Canopy Dome, averaging 19.4 per cent. Exactly 1 design out of the 12 spends more on the ground than on everything above it put together.

The bold row is the one the rest of this part is about. The Split-Log Homestead costs 18,067 dollars as shipped, of which 1,466 dollars is gravel pad – 8.1 per cent. Stand it on a pad that already exists and the building you bring is 16,601 dollars. The ground is the part the pad replaces.

The flagship sits on gravel, and that choice is not a cost-cutting detail; it is the same idea the rest of this part is built on, written small. Gravel is the foundation you can change your mind about. If the first site is wrong, you regrade the stone and it is ready to hold the next dome; a slab you have poured is wrong forever. So the Homestead carries 1,466 dollars of gravel – 8.1 per cent of its shipped price – not because gravel was the cheapest box to tick but because a first building belongs on the most reversible ground there is. The 16,601 dollars that are left once the pad is somebody else's problem is the number this whole part is steering toward.

A warning about reading the table too fast, because the cheapest rows are the ones that flatter the idea and they should not. "Grass pad" sounds like a bargain – 2 per cent of the build! – and what it means is that the Grow Dome stands on the ground, directly, with no foundation to speak of. That is not a saving discovered by the model; it is a *choice* about the building, and it works only for a garden dome that can sit on grass because it asks nothing of the ground but to be level. The same line on a year-round dwelling would not be a bargain. It would be a building without a foundation, and the table is honest about which designs are allowed to get away with it.

One move removes the whole column instead of shrinking it, and it is the move the next chapter makes (65). The shares in that table are the *before* picture: every design still owns its ground. Stand the same dome on a pad somebody else owns and rents, and the foundation line falls out of the dome owner's bill altogether – not because the ground got cheaper, but

because you stopped being the person who pays for it. The 8.1 per cent line does not turn into a smaller number. It turns into a monthly rent, a different kind of line, an operating cost instead of a sunk one, and the dome goes back to being the portable thing it always claimed to be. That is why this chapter measures the ground so carefully: you cannot decide to leave it behind until you know how much of your money it is.

Before you read any further, read your own dome out of that table – the row closest to what you are actually planning, not the row you like. The average is 19.4 per cent, and the average lies about nearly every row in the list: the cheap shells sit far below it and the treehouse sits far above. Your build is one row, not the mean, and the number in that row's third column is the answer to the portability question before you have bought a single stick. It is the fraction of your money that stays in the ground.

Why the share swings so far

A share this spread – from under one per cent to nearly two thirds – is not noise. It is two real things pulling against each other.

Before the two things, one caveat about the units, because a share is a fraction and a fraction is not a bill. A large share of a small dome can be less money than a small share of a large one, and when the question is what you leave behind, it is the money you feel, not the per cent. The table prints shares because they are what make one dome comparable to another. The dollar is in the costing behind it, and the dollar is the column that hurts.

A dome can be cheap above the ground. The Garden Dome and the Mirror shells are light frames and simple skins; their whole cost is small, so a bare grass pad is a small share of a small number.

And a dome can be demanding below the ground. The Treehouse Canopy Dome does not stand on the ground at all: its "foundation" is a platform in a tree, which is most of the cost of the building before a single panel is hung. That row is the far end of the spread – 63 per cent – and it is the honest reminder that "foundation" in this table means *whatever holds the dome up*, including the most expensive way there is: hanging a building from a living thing that grows.

Put the two together and you get the honest rule: *which* dome you pick decides whether the ground is a rounding error or the whole bill. A cheap shell on an expensive foundation and an expensive shell on a cheap foundation are both real configurations, and they are both sitting in that table. This is why nobody should compare dome costs without separating the ground out first – and why this chapter does.

The slabs in the middle rows are worth a line each, because they are the case most readers are actually in. The Timber Workshop and the Hex Pavilion stand on poured concrete, and the slab is a quarter to a fifth of the build – 45.2 per cent and 21.1 per cent. That is the ordinary number for an ordinary dome on an ordinary site, and it is the one to remember when the cheap rows tempt you: most domes are not grass domes, and their ground is not a rounding error. It is a real line, and the pad exists to take it over.

There is a cleaner way to say why the column spreads so far, and it matters the moment you go to use the table. The share is not a property of the dome. It is a property of the pair – a given shell against a given ground. The same Grow Dome that is 2 per cent on grass is a different number on a slab, not because the shell changed but because a bigger line got put underneath it. Nothing about the frame moved between those two builds; the ground did. So when two people quote different foundation shares for what sounds like the same dome, neither is wrong. They are standing on different dirt, and the table is what lets them compare like with like by splitting the two costs apart.

The hinge, stated as a decision

The spread is not a curiosity. It is a decision that comes before every other decision on the site, and it has a hinge in the middle of it.

If the foundation is a small share of your build – a dome on grass, a light frame on a cheap pad – then the ground is not the thing to optimise. Your money belongs in the shell, the insulation, the glazing: the parts you will live with and take with you. Squeezing the last dollar out of a pad that is 2 per cent of the build is polishing a pebble.

If the foundation is most of your build, the question is sharper, and it should be asked before you spend anything else: why is this dome on this ground? Because a building whose cost is mostly ground is a building that has chosen its site badly, or chosen its foundation badly, or – the honest third answer – chosen a site where the ground *is* the point, and the building is the accessory. All three are legitimate. None of them should be discovered by accident after the pour.

The work of the hinge is one sum, done once: take your dome's shipped cost, look up its foundation share in this table, and put the build on the correct side of it. The frame you can take with you. The ground you cannot. Decide which one you are actually buying, and buy that one on purpose.

What it costs to move one

"The frame you can take with you" is a claim, and a claim about money should have a number under it. Here is the one the network model produces for a dome of this size, moved from one serviced pad to another.

moving a dome	dollars or months
Transport, one move	1,800
The new pad's spur and connection	450
The dome as an asset on the books	16,601
What it costs to put one in place	18,401
What it costs to keep one in place, monthly	880
Time to set up, quickest to slowest	2 to 14 months

Read the first two lines against the ground chapters above and the point makes itself: **moving the building costs 1,800, and moving the ground is impossible.** A reader who buys a dome and rents its pad keeps the 16,601 asset mobile and the ground somebody else's problem; a reader who buys the land and pours has bought the opposite. That is the trade the network part of this book exists to price, and the mobility number is the half of it that a park cannot offer — a park's pads are fixed, and its domes are not.

The time column deserves the same honesty as the money. 2 months is a pad that already exists with power and water at it; the slow end is a green site with permits, a spur to dig and a quirk in the ground. The building's own assembly is the flat-rate chapter's business and changes not at all with the site — the schedule is the site's, not the dome's.

And one implication worth stating because it drives a decision later in this part. A building that can be moved is a building whose *residual* means something: the dome keeps 35 per cent of its cost on the books after a move, against 15

per cent lost to the move itself. A poured foundation has no residual at all — it has a driveway. That difference is not an argument that everybody should rent their ground; it is an argument that the choice should be made *knowing* that one of the two options can follow you and the other cannot, and that the number attached to following you is 1,800 rather than a shrug.

A Pad, Not a Plot

A serviced pad is the unit a dome site is actually built from: a deck, a set of services and a diameter, priced as one thing

A Pad, Not a Plot

Land is sold in acres, and an acre is not a unit of building. It is a unit of hope: somewhere in there is a place a building could go, and everything – trench, track, clearing, pad – is your problem to turn the hope into a site.

A dome site does not work that way. The unit a dome site is built from is the pad: a prepared deck, a set of services, and a diameter, priced as one thing. When you say “a pad” you have said the whole agreement. The deck is where the dome stands. The services are the wire and pipe that reach it. The diameter is which domes fit.

The pad is also the answer to the previous chapter’s hinge. The ground is what you cannot take with you – so the way to build a site people *come* to is to make the ground somebody else’s job, done once, done well, and rented by the month. The dome moves; the pad stays; and the pad is the thing with a price.

The pad catalogue is derived, not invented, which is why it is trustworthy. Every shipped dome’s foundation disc is measured, rounded up to the next 4 foot step – because decking comes in even lengths and a host building a second pad wants the same cut list as the first – and the distinct results are the catalogue: 6 sizes, from 36 to 80 feet. Add a bigger dome to the Creator and a bigger pad size appears in this list on the next run. The sizes are 36, 40, 44, 48, 64 and 80, and a 48 foot pad – the one the rest of this part prices – takes 10 of the shipped domes outright. One pad is not one dome. One pad is ten of them, and that is the sentence the whole site is built on.

The pad also replaces the plot, and it does it twice – once as a matter of law, once as a matter of what a dome actually needs. Legally, a plot is a parcel with a deed, and a deed is a standing liability: you buy the zoning, the setbacks, the taxes, and the duty to keep the land whether a dome ever stands on it or not. A pad is a contract. The dome owner rents a serviced spot for as long as the dome is there, and owes nothing to the dirt the day it leaves. Practically, the dome never wanted the plot. It wanted the three things a plot promises and rarely delivers – a level deck, power and water. A pad is those three things, priced. A plot is those three things plus a lot of land you do not need, priced as one. That is the whole difference, and it is why the unit of this site is the pad, not the acre.

What a pad is made of

The loaded pad this part of the book prices its worked examples on is 48 feet across – 1,810 square feet – and it is made of these lines:

line	cost	what it is
concrete deck, 1,810 sq ft	$11,315	the slab the dome stands on
electrical pedestal	$1,400	the 50-amp outlet, breaker and conduit run
water and drain	$1,100	supply stub, frost protection, drain tie-in
site infrastructure, per pad	$6,500	share of the access road and trunk services
permits	$1,200	siting, septic or sewer, electrical
rotating base	$9,120	the turntable ring, bearings and drive
utility column	$2,600	toilet, sink, shower and outlets in one column
solar, 15,339 W	$17,640	the array the pad carries
whole pad	**$50,874**	

Every line has a different life. The deck outlives everything on it. The pedestal and the water line last decades if the trench is right. The permits last as long as the code that issued them. The rotating base and the utility column are the ones that want maintenance, and the chapter on the sun is honest about what the rotation actually returns. The point of the table is not the total – 50,874 dollars is a *loaded* pad, not a starting point. The point is that every part of "a place to put a dome" is a named line with a price, and the total is the sum of the lines.

Gravel, concrete or wood

The deck is the line with a choice in it, and the choice is the same one everywhere in building: cheap to build, or cheap to own.

Gravel, at 1.2 dollars a foot: 2,031 dollars for the 48 foot pad. Owner-built in a weekend over fabric. Drains beautifully, moves under the dome as frost heaves it, and *heaves* – the dome on top moves with it, which a wooden frame forgives and a poured one would not. Its failure mode is the slow one: the surface walks away, the weeds come through, and in five years you rake it and add another load.

Concrete, at 7 dollars a foot: 11,315 dollars. Poured by somebody else, flat forever, the heaviest thing on the site and the one that cracks if the ground under it was not right. Its failure mode is binary: it is perfect for thirty years and then it is a crack that tells you everything about the day it was poured.

Wood, at 8 dollars a foot: 13,070 dollars – more than the concrete, and this is the honest surprise. A framed deck costs more to build than a slab *and* it rots at the ground line in a decade unless it is built like a piece of furniture. Its virtue is not price. It is that it can be level on ground a slab would need a truckload of fill to reach, and it is the one deck you can take apart and reuse.

The cheapest to build is not the cheapest to own, and the difference is not small. Gravel is the honest default for a first pad: it forgives a bad decision about where the pad goes, which is the decision you are most likely to get wrong first.

Run the three decks forward a decade and the sticker price and the owned price stop being the same purchase. Gravel wants an hour of raking a year and a load of stone now and then, and after that attention it is the same pad in its tenth

year that it was the day it was laid – the one deck whose wear is answered by a person with a rake rather than a contractor with a jackhammer. Concrete asks for nothing for a decade, and then, if the ground under it was wrong, it asks for everything at once: the crack is permanent, and the slab is now a liability you cannot un-pour. Wood is the deck you keep like a boat, a little at the ground line each year, and it is the one deck of the three you can take apart and stand again somewhere else. Cheap to build, cheap to own, and portable are three different decks, and no site gets all three. The decade is where they stop being the same purchase, which is why the choice is named out in the open here rather than folded into the loaded pad's total.

The cheap pad, costed honestly

The least a pad can cost and still be a pad, every line named:

line	cost
deck on blocks, 1,257 sq ft	$17,593
share of one hub panel, 1 of 4	$600
spur from the hub	$450
permits	$1,200
total	**$19,843**

Read what is *not* on that list, because the list of omissions is the honest part. No excavation. No concrete. No pedestal – the pad shares a hub panel with three neighbours instead. No trench – a short spur from the hub. No rotating base, no utility column, no solar.

That is the number this part of the book compares everything against: 19,843 dollars is what "a place a dome can stand, with power and water within reach" costs at its floor. The loaded pad is 50,874 dollars. The difference between them is not waste – it is the rotating base, the column, the solar and the slab, each of which has to earn its keep in the chapters that follow. The cheap pad is the baseline. Everything above it is an argument.

And the argument has to earn its keep, so price the year both ways. The bare pad – gravel, no rotation, no column, no solar – costs 12,231 dollars to build and nets its host 5,578 dollars a year: paid back in 2.2 years. The loaded pad costs 50,874 and nets 7,504 – paid back in 6.8 years. The loaded pad makes more money and takes longer to repay, and the honest question is not which number is bigger. It is which pad the site needs, because every extra line on the loaded pad is an extra thing to maintain, insure and explain, and the chapters on the sun and the iris are exactly the arithmetic of whether those lines earn their place. The pad is the product; the cheap pad is the product at its honest floor; and everything the site adds on top is a bet this book prices before you make it.

Behind both of those payback numbers sits one assumption, and it is the softest number on the page, so it gets its own line. The loaded pad rents for 667 dollars a month, and the host plans to lease it 80 of the year – not all of it. A full year of rent is the number a brochure prints; the net above already wears the empty months, plus the carrying cost a pad keeps accruing whether it is occupied or not. Payback is stated on that net, never on gross. If the real occupancy lands below 80, every payback figure in this part stretches out with it, and no geometry saves it. Occupancy is the one input the model cannot solve. It is a bet, and it is the bet the whole site is placed on.

Say the pad sits empty instead, and say it plainly, because an empty pad is the failure case this whole part is priced against. An empty pad still carries its fixed lines – permits to renew, a share of the hub or of the trunk services, insurance on a loaded pad – and on the loaded pad the rotating base and the solar are lines that age whether anyone stands on them or not. A bare pad left empty costs almost nothing; that is its virtue. A loaded pad left empty keeps costing, and that is the honest reason the loaded pad is the riskier bet. The loaded pad earns its extra money only while it is full. The bare pad forgives being empty, and on a first site that forgiveness is the product.

The centre of the pad belongs to the column

One construction detail is worth a page here rather than in the drawings, because it is the thing a first-time host gets wrong and cannot fix later.

Reserve the centre. The pad's middle is where the utility column lands, which means that is where the drain stack, the water entry and the electrical feeder have to arrive — and a slab poured without a service port in the middle is a pad you drill through afterwards or build a different building on. The films state it as a rule: the pad is designed around the port, not the other way round. The port is the one hole in the ground the building cannot do without, and it is put in before the concrete, not after.

The same reservation drives the foundation-size dial in the tool's menus: the pad is sized from the dome's radius and priced by area, so a host who wants to change the building later is changing the pile of concrete underneath it. That is why the network's whole argument is about pads rather than buildings — the dome can be swapped, lifted or taken away, and the concrete cannot.

And what the pad returns. A complete serviced pad at the modest end of this chapter's build list is the interesting case, because the site's payback is the only number the model cannot compute: it depends on occupancy, on what a host charges, and on how often the pad is full. What the arithmetic can say is the shape of it — a bare pad earns its money back slowly and forgives being empty, a loaded pad earns faster and needs to be used — and the rest is the host's own judgement about their road, their town and their tenants. This book prints the costs and refuses to print a promise.

The zone plan matters for the same reason. A pad is not a plot: the pad is the part with the port, the ring and the latch pattern, and a reader who buys land thinking in plots will size the wrong thing. The pad's footprint comes off the dome's radius, and the chapters on the park do the multiplication for every size in the catalogue.

The ladder, from bare to loaded

A pad is not one product either. It is a ladder, and each rung is a decision about how much of the building's future you pay for now — which is the same shape of decision the shell ladder made about insulation, and it deserves the same treatment: prices, returns, and no pretending that the loaded version is automatically the right one.

The pad ladder, priced

what	cost or rate	size	returns a year	payback
The pad ladder				
The cheap pad	19,843	1,257 sq ft	--	--
A bare pad, no services beyond the port	12,231	48 ft	5,578	2.2 yr
The standard loaded pad	50,874	48 ft	7,504	6.8 yr
The deck, three ways, on a 48 ft pad	rate	total		
Gravel	$1.2/sq ft	$2,031		
Concrete slab	$7/sq ft	$11,315		
Wood deck	$8/sq ft	$13,070		
Options on the loaded pad	one-off			
Utility column	$2,600			
Rotating ring	$9,120			
Solar array	$17,640	15,339 W		

Sizes run 36, 40, 44, 48, 64 and 80 feet and 10 of the catalogue's twelve designs fit a 48 ft pad. A lease at 667 dollars a month at 80 per cent occupancy is the model every pad payback in this book is run against, and the chapter's caveat is the same one the whole part carries: occupancy is the

chapter's caveat is the same one the whole part carries. Occupancy is the input nobody can compute.

The pad ladder, priced.

The bare pad is the honest minimum that is still a pad: a gravel deck, the service spur, the permits and a share of the hub. It costs 12,231 to build and returns 5,578 a year, which is a payback of 2.2 years — the fastest on the ladder, because the thing it does not have is the thing that costs money.

The cheap pad, costed line by line in the section above, lands at 19,843 for 1,257 square feet of deck, and the lines that dominate it are instructive: 17,593 of deck, then 1,200 of permits, 600 for the hub share and 450 for the spur. Even the cheapest version of a pad is mostly *deck* and *paperwork* — not services, and not the dome.

What the cheap pad does not reach

The campaign aims at a different number: **10,000 for a complete serviced pad**, so that the ground under a dome costs less than the dome. The computed cheap pad misses that aim by 9,843 — it costs 19,843 — and the gap is the most useful thing in this chapter, because it is not a gap in the arithmetic. It is a gap in what a *pad* is.

Two honest readings, and a host should pick one deliberately.

Read it as a target still open. The cheap pad is gravel, a spur, permits and a hub share; it has no column, no ring, no array and no finished edge. The 10,000 aim was for a pad with services at it and something worth standing on — and the difference between that and 19,843 is the finish. A host who wants the aim met has to decide which of the loaded pad's lines they are prepared to buy first, and the chapter above prices them one at a time: column 2,600, ring 9,120, array 17,640.

Read it as a schedule rather than a price. Ten years of infrastructure is the way the network model actually thinks about a pad: not what one costs today but what it is worth after a decade of being used, maintained and paid for. The payback column in the ladder above is where that shows up — the bare pad returns 5,578 a year for 12,231 of build, and the loaded pad returns 7,504 for 50,874 — and the honest difference between them is not the money, it is the maintenance a loaded pad needs while it waits to be full.

So when the campaigns say "a pad for ten thousand dollars", read it as the aim rather than the quote. The book's computed floor is 19,843, the aim is 10,000, and everything between those two numbers is a decision about whether the ground under a dome is a parking space with services or a small piece of infrastructure. Both are legitimate; only one of them is ten thousand dollars.

The loaded pad adds the three options this part has spent its chapters on: the utility column at 2,600, the rotating ring at 9,120 and the solar array at 17,640 for 15,339 watts. It comes to 50,874 and returns 7,504 a year — a 6.8-year payback against the bare pad's 2.2. Everything the loaded pad earns, it earns by being *used*, and the section below says so in the language a host needs.

And the deck has three prices, which is the choice a host actually makes first. On a 48 foot pad, gravel costs 2,031 at 1.2 dollars a square foot, a concrete slab 11,315 at 7, and a wood deck 13,070 at 8. Gravel is a fifth of the concrete and the chapter's honest position is that it is the right answer for a first pad: it drains, it takes a re-levelling with a rake, and it

does not need to be broken up when the building's footprint changes. Concrete buys a flatness gravel never has and a permanence a first site has not earned yet.

The ladder's real lesson is the one every other ladder in this book teaches: buy the rung the building has proved it needs. A host who puts a rotating ring on a pad nobody has rented has bought a machine for a business they do not have yet — and a host who pours concrete under a dome whose size is still being decided has bought a mistake in the most permanent material available.

Turning the House Toward the Sun, Until It Stops Being Worth It

A rotating foundation is buildable and it does raise the solar yield -- and the honest arithmetic says it pays only under conditions that are worth stating before anybody pours a ring

Turning the House Toward the Sun

This is the off-the-wall one, and it is in the book on purpose.

A dome on a ring. The whole house turns, slowly, over the day, keeping its solar panels square to the sun the way a sunflower does. It is not a joke somebody made at a party; the park model prices it, the film teaches it, and the loaded pad in the previous chapter carries a rotating base at 9,120 dollars because of it.

The reason it earns a chapter rather than a footnote is that the arithmetic is interesting *even where it loses*. Most off-the-wall ideas fail boringly – the numbers are simply bad. This one fails in a specific, instructive way: it raises the yield exactly as promised, and then the value of the extra power turns out to be almost nothing, because of a detail about surplus nobody thought to price until the model did.

So the chapter does the honest thing it is named for: it builds the case, turns the house toward the sun, and works out precisely where it stops being worth it.

Why a dome can do this and a box cannot

A dome can rotate because a circle has no preferred direction.

A dome's footprint is a circle, and a circle meets its pad the same way at every angle. Turn it thirty degrees and nothing about the building changes: the same wall meets the same wind, the same roof sheds the same rain, the same door… well, the door moves, and that is the point, but nothing *structural* cares. The building is identical at every heading.

A rectangle on a turntable is a geometry problem at every angle. Turn a box house and the corners sweep through arcs of pad that must be kept clear, the long wall trades its wind exposure with the short one, the plumbing stack moves off its trench line, and the building presents a different face to the street with every hour of the day. A box *can* rotate – they exist, as advertisements – and every one of them is a box paying the price of not being round.

The services are the detail that makes the dome version real rather than clever. Wire and pipe enter a rotating pad through its axis – the one point that does not move – and flex from there. The ring itself is the 9,120 dollar line on the pad's bill: a turntable ring, bearings and a drive, priced at 190 dollars per foot of diameter.

The drive is worth describing, because a rotating house sounds like a carnival ride and it is not one. The ring does not spin. It advances – a fraction of a turn an hour, the whole circle once a day – and it is driven slowly enough that the motor stays small. Nobody inside feels the floor move. What they notice, an hour later, is that the light has stayed where it was.

Maintenance is the line the model does not price, and it is the line that decides the argument for a lot of people. The ring is a bearing the size of the house, sitting in weather, and everything that keeps it moving – the drive, the ring segments, the rollers – needs greasing, sweeping and inspecting the way a piece of farm equipment does, not the way a foundation does. A concrete pad wants nothing. The ring wants attention, forever, and it is the owner's attention. The chapter's honest total counts the ring, the drive and the view; the years of maintenance after them are real, they are not in any table, and the model has no number for the owner's Saturdays, so the book will not invent one.

What tracking actually returns

Every figure in this chapter is for a dome of 10 feet radius, because a kilowatt-hour a month means nothing without the building it came off.

The shell has 40 panels. Clad one side – the half that faces the sun – and you hang 20 of them: 292 square feet, 5.2 kilowatts. Fixed, that array makes 734 kilowatt-hours a month. Tracking, at the declared 25 per cent gain, it makes 918. The whole rotating machine – ring, bearings, drive – buys 184 kilowatt-hours a month. That is the promise, delivered exactly as promised.

Now the catch, and it is the same catch every solar pitch trips on. A tenant uses about 420 kilowatt-hours a month. The *fixed* array already makes more than that – 918 tracked, of which the tenant can use 420 and the other 54 per cent is surplus. Surplus is not worth what power you buy is worth. Power you do not buy is worth 0.14 dollars a kilowatt-hour. Power you send back to the grid is worth 0.045.

So the 2,202 kilowatt-hours a year that tracking adds are valued twice, and the honest spread is the whole chapter. If every gained kilowatt-hour displaced bought power, tracking would return 308 dollars a year and the 7,600 dollar ring would repay in 25 years – long, but an argument. If the gain is surplus, which on a half-clad dome it mostly is, it returns 99 dollars a year and the ring repays in 77 years. That is not an investment. That is a sculpture.

One caveat about that flat 25 per cent, stated before it is leaned on. The model uses a single monthly figure, and a month is exactly the thing that hides the seasons. A fixed array is aimed at the average day, so it under-catches in the low-sun months – short days, low sun, the months an off-grid site needs the power most – and over-catches in the long ones. Tracking flattens that out: it keeps the panel square in every season, and its gain is concentrated where a fixed array is weakest. So the flat number understates the ring in the one place it could ever win, and overstates it everywhere else. On a grid-connected pad the surplus is surplus all year, and the flat number is generous to the ring. On a site that lives or dies on its December output, the flat number sells the ring short. The book prints the flat number anyway, because it is the number the model can defend, and a caveat about seasons is not the same as the season-by-season table this project has not built.

Where it stops paying, said plainly

The two curves cross, and the crossing is the whole answer.

The ring costs 190 dollars a foot of diameter, so the bill grows with the size of the pad. The gain is a percentage of the solar yield, and the yield is worth the retail rate only up to what the tenant actually uses – past that, everything is export. The bigger the array, the worse the deal gets, because the surplus grows faster than the use.

The whole-shell case is the clean version. Clad all 40 panels and a fixed shell makes 1,468 kilowatt-hours a month; tracking makes 1,835. The gain is bigger – 367 kilowatt-hours – but it is *all* surplus, because the fully clad shell already makes 4.4 times what the tenant can use. You have built the most power and simultaneously the least reason to turn anything. The film's own summary says it in one line: clad one side and turning is worth 184 kilowatt-hours a month; clad all of it and you have more power than you can use, a bigger bill to buy it, and less reason to turn.

Set the ring against its honest alternative, because it is not the ring or nothing. The other way to get more power from the same pad is to hang more fixed panels, and that alternative deserves to be priced at the same export rate before the ring is praised. The fixed array already makes 734 kilowatt-hours a month against a 420 load, so another row of fixed panels adds only surplus – every one of its kilowatt-hours worth 0.045 dollars. The ring differs in exactly one respect: it lifts the *whole* existing array by 25 per cent, including the part that displaces bought power at 0.14 dollars. So a ring's marginal hour is worth slightly more than a fixed panel's marginal hour, and that slightly is the entire case for spending 7,600 dollars on it. Both additions are mostly surplus, and surplus is worth 0.045 either way. That is why the two payback numbers – 77 years and 25 years – carry the whole verdict, and why the ring loses on a grid-connected pad even though it does exactly what it promised.

There is a quieter cost the model does not put a number on, and it is the one that decides the argument for a lot of people. A rotating house rotates its *door*, its windows, its view, its shaded side. The garden faces the dome one way at breakfast and another at dinner. The wind always finds a different face. None of that is priced in the ring's bill, and all of it is real – for some people the fixed view is the entire point of the site, and the turntable has just traded it for 184 kilowatt-hours a month worth 0.045 dollars each. The sun chapter's honest total is the ring, the drive, the maintenance *and* the view you gave up, and only the first three are in the table.

So the honest verdict: on a grid-connected pad, with a tenant-sized load, tracking does not pay for its ring. Not at export rates, and barely even at retail. That is not a failure of arithmetic. It is a fact about surplus, and the model refuses to flatter it away.

Why it is still in the book

Because the failure is the lesson, and the conditions where it *wins* are specific enough to name.

Tracking pays when the power it adds displaces bought power at a high rate – which means: off-grid, or nearly off-grid; an array sized *below* what the house uses, so every gained hour is one you were going to buy; and expensive power, which off-grid power almost always is. In those conditions the 184 kilowatt-hours a month are worth the retail rate or more, and the ring stops being a sculpture and becomes a generator mount. The test is the two payback numbers: 77 years when the gain is surplus, 25 years when it displaces bought power. If your site's true number sits nearer the second figure than the first, the ring is a serious option. If it sits nearer the first, the ring is a hobby, and a fine one – just one whose price tag should be read before it is bolted to a foundation.

The verdict deserves its structure stated, because a two-sided answer is easy to misread. The ring does not fail on its own arithmetic – it makes the power it says it makes. It fails on the *value* of that power, and only under the grid's prices. That split, between a mechanism that works and a market that will not pay for the surplus, is the reason the honest verdict has

two halves instead of one. On a grid-connected pad the ring is a sculpture; off-grid it may be a generator mount. Both halves are true at once, on the same machine. The closing audit (Chapter 69) marks this chapter ‘modelled – and it loses’ for exactly that reason, and it is the cleanest row in that table: a claim tested, found false at grid prices, and printed anyway. Most ideas in this book never get a verdict that sharp.

And there is a quieter reason it stays in the book: it is a real answer to a real question. People ask whether the dome could follow the sun. It can, and the honest answer about whether it should is exactly the kind of number this book exists to print – computed, with the caveat left in, rather than waved away by a sales pitch in either direction.

The cheaper version of the same idea

Before the ring, there is a passive option the films raise and this chapter has to place: a roof coating that refuses the sun instead of chasing it.

The mechanism is a paint whose pigment reflects almost all of the sunlight that lands on it and, at the same time, radiates strongly in the one band of infrared the atmosphere lets straight out to space. Because the sky is effectively cold in that window, such a surface can sit *below* the air temperature while the sun is full on it – which no ordinary white roof does, and which is why the coating is worth discussing rather than dismissing as marketing.

What it is worth depends entirely on how thin the shell is. On an insulated assembly the heat that never enters is small, so the annual saving is small – the order of ten dollars a year on this building, which is a coat of paint’s price rather than a system’s. On a thin, uninsulated skin the same coating carries far more of the load, and the saving runs to tens of dollars a year. Either way it is a comfort decision more than an economic one: a surface that runs cooler radiates less into the room and makes the building pleasanter in August, and the air conditioner you do not buy is worth more than the kilowatt-hours on the bill.

This is also where the project caught its own mistake, and the mistake is worth more than the number. The first version of the model counted every watt of absorbed sunlight as cooling load, and the saving it produced – over a thousand dollars against a cooling bill of a few tens – is what gave the error away. Sunlight does not walk into a building. What matters is the *sol-air* temperature: the fraction of that absorbed watt that actually crosses the surface, which is of the order of one per cent on a normal assembly. A number that flattered the idea by a factor of twenty-five is a number that was never checked, and the chapter prints the corrected version because the errata page is the only place a reader can see which figure was wrong and why.

For a builder this makes the choice easy. Paint the roof pale, whatever you paint it with: a light surface is free, it lasts, and reflecting sunlight is never a mistake. If you want the radiative version for comfort, buy it for comfort. Do not buy either one expecting the ring’s numbers – the ring at least has the honesty of a payback figure, and this has the honesty of a paragraph explaining why there is no payback figure worth printing.

One Pad, Every Dome Size

An iris of hinged blades closes a single oversized pad down to whatever dome is standing on it, so one pad serves the whole catalogue instead of one model

One Pad, Every Dome Size

Pads come in steps – 36, 40, 44, 48, 64 and 80 feet – because decking comes in even lengths and a host wants the same cut list twice. Domes do not come in steps. The three sizes this site is designed around are 20.0, 32.4 and 38.8 feet across, and a dome can be any diameter the timber decides.

So a site has two honest choices. Pour a pad for the dome you have, and pour another when the dome changes. Or make one pad that can be *any* of the sizes, and let the pad adapt.

The second one is the iris: a ring of hinged blades around a fixed rim, like a camera shutter the size of a deck. Close the blades and the working pad shrinks to the small dome. Open them and the large one fits. One pad, the whole catalogue, and no concrete ever gets poured twice.

How the blades swing

The mechanism is simple enough to draw, and the numbers that describe it are solved rather than guessed, because they have to be: a blade that does not quite close is a gap a foot can fall through.

A fixed outer rim – the pad's permanent edge. Inside it, a ring of blades, each hinged at one end and lapping the next, the way the panels of the dome itself lap each other. Swing every blade the same amount and the open circle in the middle grows or shrinks, and the blades carry the difference between the rim and the dome as a deck surface.

The aperture range is the design number that matters, and it is declared rather than derived: from 19 feet – a one-person dome – up to 40 feet, the large two-person dome. Against that range sit the three sizes the site is built around:

dome	across	floor	longest member	iris covers it?
small personal	20.0 ft	314 sq ft	6.0 ft	yes
medium	32.4 ft	822 sq ft	10.0 ft	yes
large	38.8 ft	1,184 sq ft	12.0 ft	yes

All 3 sizes – one aperture range takes every size the site sells, from a pod you could put in a yard to a building two people live in with room left over.

The blades are not a lid; they are a deck, and that is what makes them buildable rather than clever. Each blade is a framed timber panel – the same joinery the dome itself teaches – lapped over the next so water runs outward and down instead of through the joints, and carried on the fixed rim at its outer edge so the *rim* takes the load, not the hinge. A camera shutter holds back light; the iris has to hold a person standing at the edge of a 38.8 foot hole, and it does it the boring way: every blade lands on the rim, and the rim lands on the ground. The hinge only swings the blade. It never carries the dome.

The closed blades drain, and the closed state is the one worth checking, because that is where two lapped edges sit tightest together. Each blade drops its outer edge below the blade it laps over, so a closed iris is a run of shallow steps from the rim down to the dome, and water that lands anywhere on it runs down those steps and off, the way it runs off the dome's own skin. A lap that caught water instead of shedding it would rot exactly where the hinge load passes through it, so the overlap is not a flourish – it is the difference between a deck that lasts and a deck that sponges.

Snow asks the same question harder, and the iris answers it the same way the dome's skin does: with laps that shed rather than flats that hold. A closed iris is open in the middle – the dome stands in the aperture – so the blades carry only the ring of deck between rim and dome, and what snow lands on those lapped steps slides down them to the ground instead of sitting until spring. In winter the aperture closes down around whatever dome is standing and stays there; the machine's only moving day is the day the dome changes. A pad that is not moving is not seizing, and a deck that sheds is not holding a winter's worth of ice. The winter argument for the iris is not that it is warm. It is that its moving parts spend the cold months doing nothing at all.

A moving deck is also a deck people walk on, and that is the safety question the brochure skips. The blades move only when the aperture changes – a few times a year, not underfoot every day – and when they do they move slowly, driven, not free. But a hinged deck is still a deck with a seam in it, and a seam is a place a toe can catch if it is not flush when it stops. The honest answer is not that the iris is as flat as a poured slab. It is that the blades are a walking surface, so they are framed flat, lapped to shed, and moved only when nobody is standing on them. That is the rule a site can keep, and it is the one thing that stops a clever deck from becoming a clever liability.

What the iris costs against what it saves

The mechanism is priced by the aperture you build it to open to, at 310 dollars a foot: 12,400 dollars at the full 40 feet.

The honest comparison is one iris pad against the alternative the site would otherwise live with – a fixed cheap pad per size:

	dollars
a cheap pad at the small size	$6,646
a cheap pad at the medium size	$13,765
a cheap pad at the large size	$18,831
three fixed pads	**$39,242**
one cheap pad at full aperture	$19,843
the iris at full aperture	$12,400
one iris pad	**$32,243**

One iris pad costs 6,999 dollars less than three fixed pads – about the price of the small pad it replaces. And that is the *worst* case for the iris, the day-one comparison where the site already knows it will pour all three. The real case is better: the site does not know. A pad poured for the small dome is a stranded asset the day the tenant wants the large one; the iris pad simply opens. The pad survives the catalogue, which is the one thing a fixed pad by definition cannot do.

The real case has a second act the one-time table cannot show: *time*. A site lives through domes the way a harbour lives through boats. This season the small dome, next season the medium, the season after that a different tenant's large one – and each arrival and departure is a trailer and a weekend, not a pour. The fixed pad earns once; the iris pad earns every time the aperture moves, and the aperture moving is the business model. That is why the comparison in the table is the floor of the argument, not the ceiling: the iris does not have to win the first day. It has to win the fifth dome, and by the fifth dome the three fixed pads have been joined by a fourth while the one iris is still the same pad, still opening.

One number in the pad chapter quietly frames the iris, so state it before the brochure does. The flagship pad of Chapter 65 takes 10 of the Creator's shipped domes. Pour that pad and you have served 10 domes with one pour and no moving parts at all. So the iris is not the only way to pour once – the big fixed pad already does that. It is the way to pour once without paying for the biggest deck under the smallest dome, and without a tenant's table standing in the middle of an empty ring of decking every night the small dome is on site. Both serve the catalogue; that was never the question. The question is what you pay for the deck you are not using, and the iris is the only answer that prices that deck at zero.

Read the cost table the right way, because it flatters the iris if you read it as a bargain. The 6,999 dollars it saves over three fixed pads is the floor of the case, and the ceiling is not a bigger number. The iris is not really a saving at all; it is insurance. A fixed pad is a bet that the dome standing on it will not change, and a wrong bet there is not an inconvenience – it is a pad that has to be abandoned or broken up. The iris is the premium paid so that no dome choice, made today or by the next tenant, can ever turn the pad into the wrong pad. Insurance is priced by the bad year – the season a tenant arrives wanting a 38.8 foot dome and finds a pad sized for 20.0. Whether the premium is worth it is not an arithmetic question – it is the question of how often the site expects its domes to change, which no pad catalogue can know in advance.

The case against it

Say the other side properly, because it is not a weak argument and the chapter would be a brochure without it.

An iris is a mechanism on a foundation. Foundations are the one part of the site whose entire virtue is that they do nothing – they sit, they bear load, they do not break. A mechanism is the opposite: it moves, so it can seize; it has joints, so it can fill with grit; it carries weight at the edge of a hole, so it needs inspecting. The cheapest pad on the site is a circle of gravel, and a circle of gravel has no failure mode at all. The iris has several, and they are all yours.

Maintenance is not a line item somebody else absorbs. A seized blade on the day a tenant arrives with a trailer is a seized *product* – the pad is the rental, and the rental is stuck. That risk is real enough that the honest recommendation is not "every pad should be an iris". It is: the pad that expects to change domes should be one, and the pad that knows its dome should be cheap, fixed, and boring.

Which is, in the end, the same rule as every other chapter in this part: the parts that move are worth spending on only where the moving is the point. The iris is the adapter between a pad catalogue and a dome catalogue, and as an adapter it is cheap at 12,400 dollars. As a foundation, it is a foundation with an instruction manual. Know which one you are buying.

CHAPTER 68 · EXPLAIN

Why a Network Beats a Park

A park is pads you rent. A network is pads that hold their value because a dome can leave one and arrive at another, and that difference is the whole resale argument

Why a Network Beats a Park

Nobody building their first dome asks the question this chapter is about. Everybody building their second one does.

What is it worth when you want out?

A park answers: whatever the pad's owner will pay to see it gone, which is whatever you will accept, which on the day you are moving is less than it cost. A dome that can only sit on the one pad it stands on is worth exactly what that pad's owner says it is worth, and the pad's owner is the only buyer in the room.

A network answers differently. If a dome can leave one pad and arrive at any other pad like it, then a used dome is not scrap tied to a postcode – it is a home that happens to be somewhere else this season. The buyer is not one landlord. The buyer is the network, and the network has a price: the cost of a new one, minus a haircut, computed here rather than waved away.

That difference is the whole resale argument, and this chapter prices both sides of it.

The line, and what runs along it

Power, water, waste and data reach a pad along a line, and the line is the only economy of scale the site has.

The site's services are not six separate connections to the road. They are one hub panel and manifold – 2,400 dollars of hardware – with a short spur of 450 dollars to each pad. A lone pad carries the whole hub: 2,400 dollars of it. Four pads – the cluster the site is planned around – split it: 600 dollars each, plus the spur.

That is the entire economy of scale in the park model, and it is worth being precise about its size because it is not a big number. The hub, shared four ways, saves 2,400 minus 600 dollars per pad beyond the first three – a real saving, and nothing like the claim a careless pitch would make. The trunk road and service drop – 6,500 dollars a pad – are amortised flat, however many pads the site has. The deck is priced by area. The permit is per pad. Nothing else on the site gets cheaper with scale, because the whole point of the earlier chapters is that the dome side does not scale that way either.

The line matters for a second reason that is not money. What runs along the line is what makes a pad *serviced*, and a serviced pad is the product. A circle of gravel is a campsite. Gravel plus the hub is a pad, and the pad is the thing with a price – 667 dollars a month, which is the number the next section hangs on.

Host and tenant, both sides of the ledger

The idea only works if both sides of it make money, so price both.

The host's side. The host builds a pad – 50,874 dollars for the loaded version – and rents it. Against that, the same money furnishing a short-let home: 68,874 dollars up front, and 6,614 dollars a year of cleaning, damage reserve, platform fees and renovation. The pad's yearly bill is 920 dollars – management, tax and insurance – because the building on it belongs to the tenant, and its wear is not the host's problem. The host's building cannot be trashed by a guest. The pad does not need cleaning between tenants. That is the host's whole pitch, and the ledger bears it out.

The tenant's side. Four ways to have a roof, all priced with power and water included: a hotel at 4,256 dollars a month, a short let at 2,888, a leased apartment at 1,520 plus 3,625 dollars to move in, and the dome on a pad at 880 dollars a month plus 18,401 dollars to arrive – of which 16,601 dollars is the dome, which you keep.

The crossover is the number that decides who this is for, and the model finds it rather than asserts it: month 3. Stay shorter than that and the honest answer is a hotel. Stay longer and the dome is the cheapest roof in the table, and it keeps getting cheaper with every month after, because the only thing you cannot take when you leave is the lease.

Two details in that table deserve their own line, because they are where the comparison could quietly cheat and refuses to. The utilities are the same for everyone – the dome's 80.2 dollars a month is the tenant's power and water at 70.5 dollars plus a small declared margin for metering, not a made-up number – and the apartment's lease is priced honestly, too: sign for 12 months and leaving early costs 2,900 dollars, while 2,175 dollars of the entry comes back. The dome's entry has no break fee because it has no lease – that is the whole point of owning the roof – and its month is the only one in the table that buys an asset instead of renting one.

The host's side has its own crossover, and it is slower. The loaded pad costs 50,874 dollars and nets its host a year like the previous chapter's – repaid in 6.8 years, not months. Hosting is a patient business. The point of the ledger is not that the pad prints money. It is that the pad's yearly bill – 920 dollars – is the *only* bill, because the tenant's building wears itself out on the tenant's side of the lease. The furnished let's 6,614 dollar year is the price of owning the furniture, the carpet and the reputation. The pad host owns a circle of gravel and a meter. That is the trade the numbers describe.

The host's ledger has one risk the table hides, and it is the mirror of the tenant's. The pad pays for itself in 6.8 years – if it is rented. An empty pad earns nothing, and unlike an empty apartment it cannot be let to a tourist for a weekend while it waits, because the product is the serviced ground, and the ground waits for a dome that has not arrived. So the host's real exposure is not the build; it is the empty months between tenants, and the honest host prices the pad against a year with some of them empty, not a year that is full. The furnished let fills those months with strangers. The pad fills them with nothing, and the nothing is the price of not owning the furniture.

Resale, with the haircut left in

Now the question the chapter opened with, computed.

A maintained dome is declared to last 30 years and never to fall below 35 per cent of its cost. But the day it becomes secondhand, it loses 15 per cent – the declared resale haircut, the thing a real network would shrink and this chapter refuses to pretend away.

So a dome that cost 16,601 dollars is worth 13,640 dollars the day you leave after a year, and 11,759 dollars after five. Walk away from a park and that is the *best* case – a buyer who exists. A dome that can only sit on one pad has no such number, because its value is whatever the pad's owner offers, and the pad's owner knows you have to leave.

That is the whole difference between a park and a network, in one sentence: a park rents pads; a network makes pads *hold their value*, because every pad in the network is a buyer for every dome on it. The network does not make the dome free to own. It makes it possible to leave with something – 11,759 dollars after five years, by this arithmetic – instead of with whatever the landlord felt like paying. And the recovery penalty that would be waved away in a pitch is printed, at 15 per cent, on the line.

Why the haircut, and why not hide it? Because the day a dome is secondhand, a buyer takes a real risk – a frame assembled by a stranger, on hardware that may have been torqued by feel – and the market prices that risk as a discount. A pitch that pretended used domes sold at new-dome prices would be the same lie as the park's. The honest pitch is better anyway: the haircut is 15 per cent *today*, when every used dome is a gamble with no track record. A network that has seen a hundred domes move between pads, with the transport damage and the wear patterns known, could price that risk down – and that is precisely the thing a network is for. The park's resale number is whatever the landlord says. The network's is 11,759 dollars and falling toward honest, because the network is the buyer that learns.

A network of one hub is still a park, whatever it calls itself. The buyer in the resale section only exists when there are enough hubs – each serving 4 pads – that a dome leaving one pad has somewhere real to arrive, and enough history that the arrival is priced from data instead of from fear. That is why the haircut is the number to watch: it is not a fee, it is a temperature. A 15 per cent haircut is the market saying "unproven." Every move between two real pads, every insurer that writes a policy, every standard the pads are built to, nudges that number down, and the day it settles is the day the network stops being a proposal and becomes the park's competitor.

This is the least-built idea in the book

Be honest about what this chapter is, because the rest of the book has been honest about everything else.

The frame is built. This book's dome is standing, and the fortnight and the hours are measured. The pad is costed – declared prices, named as such, but arithmetic against a real catalogue. The network is neither. It is arithmetic and one film, and it should be read as a proposal rather than a report.

No network of dome pads exists for these numbers to have come from. The resale haircut is declared, the crossover is computed, the host's year is a model. That does not make the chapter worthless – a proposal with its assumptions on the line is how honest ideas are made – but it makes it *conditional*, and the closing chapter of this part is exactly that condition, written out.

Here is the missing-pieces list, so "not built" means something exact. There is no insurer pricing a second-hand dome, so the buyer who risks one is self-insuring, and the risk comes straight out of the resale number. There is no written standard for a serviced pad – the pad of Chapter 65 – so "pad" means whatever the last host built, and a dome that fits one pad may not fit the next. There is no broker who moves domes for a living, so the 1,800 dollars a move costs is somebody's truck, borrowed, with nobody standing behind the frame when it arrives. Those three are the trust layer a park gets free from a lease and a landlord, and a network has to build them itself. Chapter 69 writes out what has to be true for them to exist; this chapter only says the haircut will not fall until they do.

None of the three is a dome, and none is a pad. They are the paperwork, the standards and the trucks a market runs on, and until they exist the network is a good idea with a price and no shelf.

What Would Have to Be True

The closing audit: every claim in this part restated as a condition, with the ones that are not met yet marked as not met

What Would Have to Be True

A book that only argues one way is not a book. It is a sales document, and the difference is one chapter.

Everything in this part has argued the same direction: the list is flat, the hardware is invariant, the skin is less, the pad is the product, the network holds value. Each of those is true under conditions. Each of them also has a version that is not true, and a sales document is precisely the thing that never tells you which one you are reading.

So this chapter argues the other way. It takes every claim in this part, restates it as a *condition* – "this is true if…" – and marks which conditions are met. Measured, modelled, or proposed. And where the honest number is unflattering, it prints the unflattering number, because this book has been doing that since its first chapter and it is not about to stop in the last one.

Every claim, and what it rests on

Each claim in this part, with what it stands on:

claim	rests on	status
the parts list does not grow	the flat-rate table, four diameters	measured
nine processes at any size	the process list and its counts	measured
an hour at the log pays a rate	book_math, declared prices	computed
the fuel ledger	published metabolic constants	modelled
less skin for the same floor	solved geometry, two envelopes	measured
the pony wall buys the edge back	the ladder, declared wall cost	computed
the brim catches 7,259 gallons	declared rainfall and runoff	computed
the shell ladder pays rung by rung	declared quilt constants, conduction only	modelled
one hardware set, three sizes	the Creator's own costing	measured
buy for the next two steps	declared retrofit multiple	proposed
the ground is the part you leave	foundation share, whole catalogue	measured
the pad is the product	declared pad prices	costed
rotation pays for its ring	solar layouts, export pricing	modelled – and it loses
the iris serves every size	solved blade geometry	proposed
the network holds value	declared resale haircut, arithmetic	proposed

Read the status column honestly, because it is the point of the table. One row is measured and still loses: the rotation chapter found its own claim false at grid prices, and printed it. Several rows are *proposed* – the retrofit multiple, the iris, the network – and a proposal is not a finding. The difference between those columns is the difference between a report and a plan, and this part of the book is one of each.

The three statuses are the three kinds of claim the front of this book keeps apart, under their working names. **Measured** is tested construction: the thing happened, on the standing prototype, and somebody wrote the count down. **Modelled** is known geometry carried through with declared constants: the arithmetic is real, the input is named, the thing itself has not been run. **Proposed** is a design possibility: priced or drawn, not built, and labelled as such wherever it appears. A reader who keeps those three apart is reading this table the way it was written – and the way the whole book was written.

Measured is the rarest word in the table, and it deserves a definition before the other two, because it is the one that cannot be borrowed. A row is measured when the thing exists and somebody put hands on it, counted it, and wrote the count down. The one frame that exists is measured: the 21.6 foot dome from 3 trees, the 14 days of the fortnight, the money spent against the money a bought frame would have cost. Where the flat-rate table is marked measured, it is because its counts were taken off that real frame, and the four diameters are the same counts re-derived from geometry that has already proved itself on the one that was built. Measured does not mean every row was built four times. It means the one build that exists was counted, and the rest is arithmetic sitting on that count. It is the only column you can lean your weight on, and it is short for a reason: this project has built one frame, not a development, and a book cannot measure what it has not built.

Modelled is the middle word, and it is not a polite way of saying guessed. A row is modelled when the arithmetic is real but the thing it describes has not been run. The fuel ledger is modelled because the metabolic constants are published and the tank is a listing nobody has checked. The shell ladder is modelled because the quilt constants are declared and the arithmetic is conduction only. Rotation is modelled, and it loses, because the solar layouts and the export price are real numbers that combine into a real, unflattering answer. Modelled means the input is honest and the arithmetic is done; it does not mean the thing has been stood up in weather. The distance between modelled and measured is a winter in a dome, or a full season of a pad, and that distance is the exact amount this book will not paper over. A modelled row is a promise with the arithmetic shown; it is the reader's to cash, not the author's to collect.

Proposed is the honest word for the rest, and it is the one that carries the most weight, because it is the column a less careful book would quietly promote. A row is proposed when the arithmetic says it would work and nobody has paid for it yet. Buy for the next two steps is proposed because the retrofit multiple is declared, not observed. The iris is proposed because the blade geometry is solved and no blade has been cut. The network is proposed because it is a set of prices with no pads on the ground. Proposed is not a defect; it is a label that keeps a plan from wearing a finding's coat. Every proposed row in this table is a thing this book would like to be true, and the label is the only thing stopping that wish from being printed as a fact. Most of the good ideas in this part live in this column, and they earn their place by being named, not by being dressed up.

Each row also hides an "if" that the status column cannot carry, so say them. The flat list is measured *for this frame* – a 2V wedge frame, this hardware, this assembly – and it stops being flat the moment the frequency changes, as the flat-rate chapter already admitted. The hardware invariance is measured for the Creator's designs, at three diameters, and says nothing about a bracket loaded past the section the timber allows. The pad economics are costed at declared prices, and declared prices are the author's best numbers, not market quotes. The network holds value *if* there are other pads to move to – which is exactly the thing that does not exist yet. Every row in the table is true under its condition, and the condition is the part a sales document leaves off. This book's errata pages exist because the project has been wrong before; this chapter exists so the next correction has a list to correct against.

Two numbers frame the whole audit, and they are the only counts in the chapter. The corpus this book draws on – the films, the presentations, the taxonomy of everything they teach – is 62 concepts, each with a claim, a caveat and a source. The films' own summary is 10 points. A book this careful about those numbers owes the reader the same care in reverse: this chapter is the 10 points and the 62 concepts, restated as conditions, so that nothing in the previous fifteen chapters has to be taken on trust. Trust the table, not the adjectives.

The table earns its trust the same way the rest of the book does, and it is worth saying the mechanism once, plainly, because it is the difference between this and a sales document. No figure in these chapters is typed. Every one is a token – a hole in the prose filled at export from the same arithmetic the films run – so a number cannot be correct today and silently wrong tomorrow. If a tree changes, or a price, or a panel count, the token moves and the sentence moves with it; an unknown token is a hard error, not a blank, so a mistyped token fails the build rather than shipping a wrong number. Behind the tokens sits a test suite that asserts the arithmetic agrees with itself – that the parts add to the whole, that the flat table's columns do not move, that a payback number is the ring's bill divided by the gain. The reader does not have to run any of that. They only have to know it is run, and that the printed number is the number the machine produced, not the number the argument wanted. That discipline is the reason the unflattering numbers above are worth anything at all.

The finished number nobody likes

The number this part keeps *not* quoting is the honest all-in one, so quote it here.

The ten-points list – the film's own summary – carries a finished dome at $6,943 finished: pressure treated, insulated, glassed and fitted out, at 22 dollars a square foot. That is the number the shape argument wants you to remember, and it is a real number – Pressure treated, insulated, glassed and fitted out, at $22 the square foot, with nothing salvaged and nothing donated..

Now the number next to it. The national average new home sells for 665,299 dollars, of which 428,215 is the building cost – 162 dollars a square foot over 2,647 square feet. Put the dome's 314 square foot reference floor through that rate and it comes to 50,797 dollars. The gap between the two ways of pricing one floor is 43,854 dollars, and the national rate is 7.3 times the finished figure.

Neither number is a lie, and that is exactly why both get printed. The 22 dollar figure counts the *shape* – the frame, the skin, the glass, the minimal fit-out the costing module knows about. The 162 dollar figure counts a *house* – foundations to code, mechanicals, a kitchen and a bath at market labour. The bare shell keeps getting quoted because it measures the shape; the finished number keeps getting avoided because it measures the house, and the house is where the gap is. The dome's argument is not that the gap does not exist. It is that the shape is the part this method changes, and everything else is the same bill whatever the walls are.

So what would have to be true for the dome's finished number to close the gap honestly? The same kitchen, the same bath, the same mechanicals, at the same labour rates – and then the only difference left is the shape's own: the skin, the frame, the envelope. That is the experiment this book can offer and has not run, because running it means finishing a dome to the same standard as a subdivided house and metering the result, and this project has built one frame, not a development. Until then the honest statement of the gap is this chapter's: 43,854 dollars between two real numbers, with what each one counts written next to it. The finished number nobody likes is not the dome's. It is the one that has to be printed beside the dome's.

The energy claim this book does not make

One claim is missing from the ten points on purpose, and its absence is a claim.

A dome is not automatically cheaper to heat. The envelope argument is real – 39.0 per cent less surface at the reference floor, less skin to lose heat through – but a percentage of skin is not a utility bill. The bill depends on insulation, on climate, on systems, and above all on airtightness, which is a build-quality question no geometry answers. A tightly framed box beats a leaky dome every winter, and pretending otherwise does not help the method; it just guarantees the method gets caught.

So this book makes the energy claim it can prove and refuses the one it cannot. It will tell you the envelope numbers, the ladder numbers, the honest conduction-only arithmetic with the declared constants named. It will not tell you a dome saves a specific percentage on heating, because this project has not metered a winter in one. The film that teaches this calls it what it is: the energy claim this project will not make. The number nobody wants in a pitch is the trust the whole pitch runs on.

What one person has actually done

Close with the narrowest true version of everything in this part, because it is the only version that is entirely true.

Built. A dome is standing: the 21.6 foot frame from two trees, 14 days of the fortnight, 80 hours of one person's hands on it. The frame the films show is the frame that exists.

Measured. The fortnight's money is real: 497 dollars spent against 9,798 dollars the same frame would have cost bought – 116 dollars an hour for the builder, which is what the log actually paid.

Modelled. The pad, the park, the ledger, the ladder, the iris, the network: arithmetic against the Creator's own catalogue, with every declared constant named where it was declared and every caveat left where it was found.

Still only a drawing. The network has no pads on the ground. The iris has no blades cut. The rotation chapter's own numbers say it should not be built at grid prices, and they are in here.

That is the audit, and it is the whole book in four lines: a real frame, a measured fortnight, a costed site, and a set of conditions written down – so that whatever gets built next is built by somebody who knows which parts were true, which parts were priced, and which parts would have to be true first.

What the reader does with the list is the last thing worth saying, because a list of conditions is only useful if it changes what gets built. Do not read this chapter as a verdict on domes. Read it as a checklist to carry into any build, dome or not: before you trust a claim, ask which of the three words it deserves, and refuse to let a proposal wear a measured hat. If you are about to spend money, find the row in this table that your decision rests on, and check whether that row is measured or still proposed. If it is proposed, the honest move is to say so out loud and price the test that would move it. A condition you have not yet made true is not a reason to stop. It is a thing to go and make true, in order, one at a time – which is how this part of the book got its one measured frame, its one measured fortnight, and its long honest list of everything that still has to be true. The list is not the end of the book. It is the beginning of the next build.

The ownership conditions, in one table

Four of the conditions on that list are about ownership rather than about building, and they belong together because a reader will meet them in the same year: what a dome is worth when it stops being yours, what it costs to move, what a host actually earns, and what a park's promise is worth against a host who owns their own ground.

condition	number	what would have to be true
The move haircut	15% of value lost per move	that a dome is moved rarely enough for 1,800 a move to be noise
The residual	35% kept on the books	that a used shell holds its value for 30 years
Value recovered	13,640 at twelve months, 11,759 at sixty	that the market for used domes exists at all — nobody has sold one
A host's own pad	50,874 to build, 920 a year to run	that the host can fill it: the occupancy the model cannot compute
A host who lets	68,874 to build a pad with the ring and the array, 6,614 a year	that the extra money is worth the extra machinery standing idle

Read the last two rows together and the network's real argument appears: a host who owns their own ground and a host who rents their pad are in different businesses with the same building on them, and the difference is 68,874 minus 50,874 of one-off cost against 6,614 minus 920 a year of income. The loaded pad earns more and costs more, and — as the pad chapter said — the loaded pad also *keeps* costing when nobody is standing on it.

Which is the honest form of the claim this part makes. The network is not proven; the ownership arithmetic is. A dome on somebody else's serviced pad is a building with a 18,401 entry cost that can be carried to the next pad for 1,800, and that — not a rental yield — is the thing this project has actually demonstrated: a building whose ground is separable from its shell. Everything above that line is a condition. The condition is written down, the arithmetic that would test it is on the page, and the reader who wants the network to exist is the one who has to go and make it true.

PART 3

Variations and Future Systems

“The same frame can wear more than one roof, stand on more than one ground, and hang from nothing at all.”

The manufactured version of the method, and the systems that turn it into something you can lift, hang and re-skin: the shower-cap soft shell, the mast and the floor, the floating dome, the seven ways to make a wedge, the zome and the hex, the stem cell and its core, and the seam turned into a climate machine -- then the same building as an article: what it costs to make, what it sells for, what a campaign would spend the money on, and the module catalogue that turns one frame into fifteen buildings.

The Dome That Stacks Hats

A soft shell instead of a hard one: the frame wears quilted layers of recycled clothing and a rain-slick shower cap, and each new hat is a size up from the last

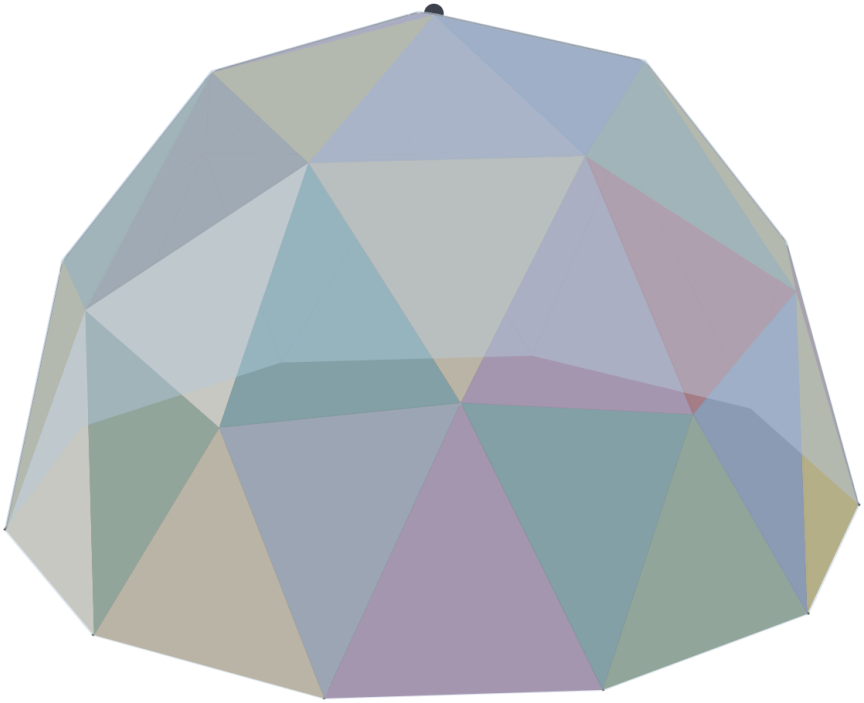
The Dome That Stacks Hats

A hard shell is made once, at one size. That sentence sounds like a fact about manufacturing and it is actually a limit on the building, and it took building this frame to see which.

The laminated hull that the manufactured version of this dome ships with is a fifty-year object – glass, resin, gelcoat, made in a mould. It is also done. The cavity behind it holds 4 quilted layers and then it is full; after that, insulating further means losing floor on the inside or building a second shell outside the first, and neither happens. So the hard shell quietly caps how good the building is ever allowed to get.

The soft shell is the other answer, and the image for it is a shower cap for the house. From the frame outwards: wood panels on the outside face, a monolithic polymer membrane over all of it, quilted layers of recycled clothing – and a rain-slick outer cap pulled over the lot and strapped down.

The point is the last layer. A cap is a bag: when the stack under it gets thicker, you buy a bigger bag. A dome that wears hats can keep getting warmer, one winter at a time, for as long as somebody owns it. A dome with a hull cannot. That is the whole argument, and the rest of this chapter is the arithmetic of it.



The frame, three blanket quilts, and the shower cap over all of it.

The cap against the hull

The numbers come out of the same model the campaign film quotes, and the table is the one worth reading slowly, because the saving is not the interesting column:

layers	cap stack	hull plus its bays	the cap saves
0	$3,265	$9,905	–
3	$3,471	$9,905	$6,434
7	$3,748	$10,510	$6,761

At three blanket quilts the cap stack is 3,471 dollars against 9,905 for the hull and the bay sandwich it needs – 6,434 dollars less, which is the six thousand dollars the campaign leads with. At seven quilts the saving is 6,761.

But look at the row the table does not show. The hull's cavity holds 4 layers and then its R-value stops moving. The cap has no cavity. Its seventh quilt is a hat 15.6 per cent bigger than the first one – area goes as the square of the radius, so every layer makes the next cap a size up – and it keeps buying R-value for as long as the owner keeps quilting.

The list price moves with the materials. The standard article with the cap lists at 12,036 dollars – 43 dollars a square foot of floor – where the same dome with the hull lists at 21,513. The cap is not a discount shell. It is the standard article, and the hull is the upgrade.

What the cap gives up

The honest list, because a cap that was only upsides would be a tarp with a marketing budget.

It is not structural. The laminated hull was a stressed skin; a strapped cap is not. Uplift is carried by the ground anchors and the frame alone, so the frame has to be good for the whole wind load rather than sharing it.

The outer layer is sacrificial. A rain-slick cap lasts 8 years and the membrane under it 12; they are meant to be replaced, and the thirty-year arithmetic prices the replacements rather than pretending a cap is bought once.

Two impermeable layers with fabric between them is a moisture trap. In a cold climate the dew point lands in the quilt, and fabric that gets damp stops insulating and starts rotting. The seam duct – the airflow through the frame's own channels – is the answer to this, and the seam duct is an experiment, not a solved problem. This chapter would rather name the dependency than hide it.

It looks like a tarp. That is a planning objection, a resale problem and a neighbour problem, and no amount of arithmetic answers it.

Why the hull becomes the upgrade

None of that makes the hull a worse thing. It makes it a *later* thing.

The frame does not know which skin it is wearing. The cap dome and the hull dome are the same 120 members, the same bays, the same column. The buyer who starts with the cap can buy the hull in year five and it fits – which is the way the whole system is meant to run: the cheap hat first, the fifty-year hat when the money exists, and the quilts carried over between them.

That is what “the dome stacks hats” means. The building is not a shell. It is a frame with a choice of skins, and the choice can be made again.

CHAPTER 71 · EXPLAIN

The Mast and the Floor

A steel-core mast through the utility column, a wooden floor that clamps to it -- the upgrade bought after the dome -- and an apex ring that hoists the whole structure

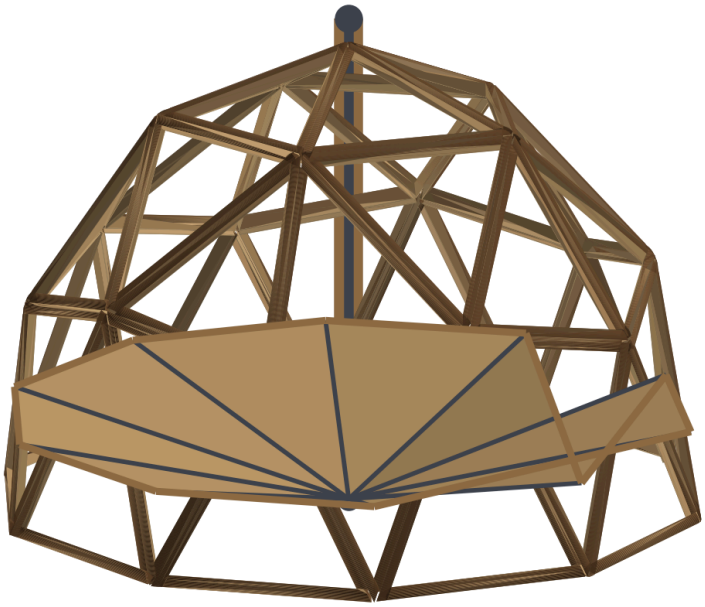
The Mast and the Floor

The dome's floor is the upgrade, and that is the sentence this chapter is built on.

On a pad, the floor is somebody else's problem – the deck the dome stands on belongs to the host and stays when the dome leaves. But a dome that wants to move, or to stand where there is no pad at all, needs a floor of its own. And the way to give it one is not four legs and a skirt. It is a mast.

The mast runs up the middle of the utility column – the insulated chase that already goes from the service port to the apex – and out through the top. It is 12.0 feet of steel core clad in timber: metal where the strength is, wood everywhere else, because everything else in the room is wood and a bare steel pipe reads as scaffolding. The services ride inside the chase around it, so the structure and the plumbing share one penetration and one object to look at.

At the top, under the seal cap, sits the lifting ring – a 240 dollar forging whose whole job is to be the point the building hangs from when it is hoisted. At the bottom, a flange bolts the mast to the service port. Between them, the floor clamps on.



The mast through the column, and the floor that clamps to it.

Why the mast runs through the column

The column was already the right shape before the mast existed.

It is floor-to-apex. It is centred. It is the one part of the building that touches the ground, the walls and the sky. The design decision is therefore not “where do we put a mast” but “the chase gets a spine” – a steel core inside the chase, so the services keep their route and the structure gets its one load path in the same twelve inches of the plan.

The floor follows the same idea. Its hub is a 310 dollar steel ring that clamps the mast; from the hub, 10 radial steel spokes run to the base ring; timber decking covers the spokes, and a low rail runs the edge. 277 square feet of floor that did not exist until the buyer wanted it, held up by the mast instead of by the ground – which is what makes the next chapter possible.

What it costs, and what it lifts

The upgrade, line by line, out of the same model the campaign quotes:

line	dollars
the mast through the column	$622
the dome's own floor	$1,899
the floating rig (next chapter)	$1,490
all three	**$4,011**

And the weight. The frame alone – the 120 members of green pine – is 2,478 pounds. The shell or cap, the column, the floor and the rig all add to it, and the honest thing has to be said before anybody ties a rope to the ring: **this book prices the parts. It does not rate them.** The mast's section, the ring's forging and the hoist's capacity are an engineer's numbers, computed for a specific building on a specific site, and nothing in these pages is a substitute for them. What the model can say is that the geometry of the lift is simple – one point, one axis, one load path – which is the kind of lift engineers like to be handed.

The floor, in other words, is the purchase that turns a dome from a thing that stands somewhere into a thing that can be picked up and put somewhere else – including somewhere with no ground under it at all.

CHAPTER 72 · EXPLAIN

The Floating Dome

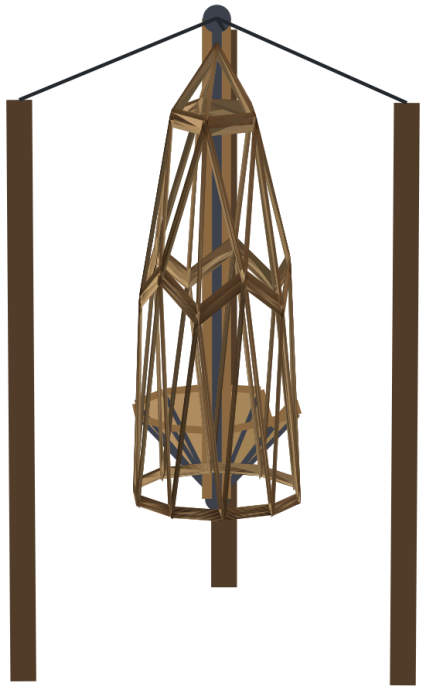
Hang the mast from three cables between trees and the dome floats with its own floor under it -- a design possibility, with the loads named as the engineer's job

The Floating Dome

The mast that hoists the dome can also hang it. That is the whole of this chapter, and it is the reason the previous one built the floor first.

Three cables run from the apex ring to three trees. The rig is 1,490 dollars of galvanized cable – three legs of 28 feet – a forged hanger that rides the ring, tree-hugger slings that put no holes in the trees, and a brake winch to do the hoisting. Winch the dome up, tie the cables off, and the building hangs in the air with its own floor clamped to the mast under it.

That is the floating dome: no pad, no piers, no ground at all between the trees – the same 120 members, the same cap, the same column, held from one point like a lamp.



The same mast and floor, hung between trees on three cables.

What changes when the ground goes away

Everything the earlier chapters assumed about the ground quietly reverses.

The base ring stops being what holds the building up; the mast is. The floor stops being something the dome stands on and becomes something the dome *carries* – clamped to the mast, hanging under the shell, level because the mast is level. The utility column keeps its job: services come up a flexible drop along one cable, into the chase at the apex, and down through the column the same way they would on a pad. The only difference is that the water line is a hose and the power line is a cable, and both arrive from above instead of below.

The trees do the work the ground did, and the deal is made without a single hole in the bark – saddles, not bolts – because a tree that is still alive is a foundation that maintains itself.

What would have to be true

Now the honest page, because this is the least-built idea in the whole book and it should be read as a drawing rather than a report.

The loads are real and none of them has been rated here: the cables in tension, the trees in bending, the mast in compression, the ring under the weight of 2,478 pounds of frame before the floor and the skin are even counted. Three points of support mean the building hangs steady in still air and *moves* in wind – a dome hung from cables is a pendulum, and the trees flex with it, and the arithmetic of that is an engineer's, not a builder's.

So the conditions this idea would have to meet are the same conditions the closing audit of the last part named, restated for the air: a site with three sound anchors and the room for them; a mast, ring and rigging sized for the lifted whole; a way in – a ladder, a stair, a bridge – that does not pull the building sideways; and an authority that will permit a dwelling hanging from living trees, which most will not without a stamp.

And yet the arithmetic worth keeping is the one this chapter opened with. The same mast, the same floor, the same building: hoisted onto a pad one month, hung between trees the next, winched onto a trailer the month after that. The floating dome is not a product. It is the same product with the ground removed – and the ground, as the pad chapters spent a part of this book proving, was always the part you could not take with you anyway.

Seven Ways to Make a Wedge

The solid split-log wedge is the reference, not the whole idea: hollow, laminated, log-derived, manufactured and hybrid members all exist, and each one trades something

Seven Ways to Make a Wedge

Every member in this book so far has been one solid eighth of a log — the reference member, chosen because it needs nothing but a saw and a tree. This chapter is the rest of the idea: the other six ways to arrive at a triangular-section member, each one trading something the reference member has for something it has not.

The seven are not rivals fighting for one crown. They are a ladder of capability, and the book's method sits at the bottom of it on purpose — the rung that costs the least and asks the least, and the one from which every other rung is reachable later. A builder who starts with split logs and later wants a factory member has not changed methods; they have climbed this chapter's ladder, one rung at a time, and each rung is listed here with what it buys and what it gives up.

The solid split wedge

The reference member, stated as the baseline the other six are measured against.

What it is good at: it exists the moment the saw finishes the seventh cut. No shop, no glue, no kiln, no supply chain — the tree supplies the material, the saw supplies the shape, and the harvest advantage of Chapter 13 applies in full, because nothing is ever squared, edged or planed. It is also the member that grades itself: bark out, pith in, the orientation cue of Chapter 14.

What it cannot do: it cannot be bigger than the tree it came from, straighter than the tree grew, or drier than the day allows. Its section is whatever the butt diameter and the split count decide, its quality is whatever the forest produced, and its perfection is whatever the grading passes let through. The reference wedge is the *available* member, and availability is its entire virtue — and its entire ceiling.

Every other entry in this chapter exists to raise one part of that ceiling, at the price of giving up another part of the floor.

Hollow members

Stiffness lives near the surface: in bending, the material at the middle of a section does almost nothing, which is why a hollow triangle can use roughly half the material for most of the stiffness. The frame's own weight — the green pine the campaign computed — falls by a like fraction, and what two people can lift changes, which changes how the dome gets built.

The argument against, which the wedge book's own page calls decisive: you cannot make one with a saw. A hollow triangular member is four thin pieces, ripped accurately, joined along four long glue lines, in a jig, by somebody with a shop — the exact work the wedge method exists to avoid. And the whole harvest advantage disappears: a hollow member is made from milled boards, so it starts from the mill's recovery rate, not the split's.

Where it would make sense: a manufactured dome, in a factory, where the jig is a fixed cost spread over hundreds of frames and the weight saving compounds through shipping. Not for somebody with a chainsaw and 3 trees.

The middle option nobody tries, noted because it is the bridge: solid wedges with the middle bored out along the length. It keeps the sawn faces, keeps the continuous grain at the surface, removes some weight — and it is fiddly, it needs a long auger, and nobody has measured what it does. Listed as an idea rather than a recommendation.

Laminated members

Glued strips buy what the tree refuses to sell: size beyond the tree, straightness on demand, and defect-spreading — a knot in one strip is surrounded by clean wood in its neighbours, where a knot in a solid wedge is a knot in the member.

The price is the shop, again: laminating is a glue, clamp and press operation, it starts from milled boards, and it inherits the mill's recovery rate and the mill's supply chain — the whole middlemen ladder of Chapter 2, paid back in exchange for straightness. Lamination is the answer when the build needs members the local woodlot cannot supply in one piece, or when the frame must be *predictable* to a degree rough timber cannot promise. It is not the answer for the first dome, and the chapter says so plainly: the first dome's job is to prove the method, and the method's proof is that it needs nothing the tree did not already provide.

Log-derived, manufactured, and hybrid

The middle three rungs, grouped because each one trades the tree's own shape for a different kind of leverage.

Log-derived. The films' variant of the reference: buck the log into longer sections, split each into eight, and crosscut every wedge in the middle — two struts per section instead of one. It is the same solid wedge, organised differently, and it suits the sixty-foot trunk the pine film prices: five long sections, eighty struts a tree. It is not a different member; it is a different *schedule* for the same member, and the book keeps both schedules visible so the two trees the films count and the 3 trees this book fells can be reconciled honestly.

Manufactured. The wedge, moulded in a factory: screw holes, threaded inserts and the spline ridge already in the part, a steel core where the strength must be, a load rating off a test rig instead of an argument. It is a better product and a worse argument — better because consistency, grading and repeatability arrive with the part; worse because it costs a factory, a tooling run and a supply chain, and the whole point of the method was that a building could be made where those things do not exist. The manufactured member is the method's *export* version, the one the campaign's tooling budget exists to develop — and Chapter 78 is where the campaign's version of this rung lives.

Hybrid metal and wood. Steel where the load is a joint, wood where the load is a span: a steel base ring under the frame's own compression, a steel mast through the column, steel apex and base connectors where the wood's weakest directions meet. The book's later chapters already assume this rung in places — the mast of Chapter 71, the core of Chapter 79 — and the honest note is the one those chapters carry: metal in a wooden frame is a *decision* about thermal movement, galvanic contact and connection detail, made with the engineer's sentence from Chapter 10 in force.

Choosing one

The decision tree that sends each builder to one of the seven, and the two questions that do most of the sorting.

The first question: **where does the wood come from?** If it is standing on the site, the solid wedge — log-derived or bucked-short — is the whole ladder you need, and every other rung is an upgrade to be bought later, if ever. If the wood must be bought, the ladder inverts: the manufactured or laminated member stops being an extravagance and starts being the honest comparison against the lumberyard, with the split's harvest advantage already forfeited.

The second question: **who is building it?** One person with a saw and a fortnight — the reference wedge, full stop. A shop with clamps and patience — laminated or hollow. A company amortising a mould — manufactured, with the hybrid's steel at the joints. The method has never been one member; it has been the *triangle*, and the seven ways to fill one are this chapter's inventory of how far the triangle travels before it stops being the book's cheap, tree-shaped, saw-cut friend.

Every Other Way of Doing It

Eight frame materials, eight strut sections, sixteen panel types, nine claddings and eleven foundations -- all priced on this dome, so the choice is a comparison instead of a preference

Every Other Way of Doing It

This book argues for one combination of material, section, panel, cladding and foundation — the combination Chapters 12 through 40 built. This chapter is the counter-weight: all the others, priced on the same dome, from the wedge book's own catalogues, so the book's choice is a comparison instead of a preference.

The five tables below are the breadth of the whole geodesic world, compressed onto one page: eight frame materials from whole-tree trunks to structural steel, eight strut sections from round tube to the quarter wedge, sixteen panel types from open frames to precast concrete, nine claddings from nothing to a poured shell, eleven foundations from gravel to a treehouse platform. Every option is costed on the reference build's own quantities — the same member count, the same surface, the same span — which is the only way two options can honestly be compared at all.

The breadth tables

The page is the argument. The reference row — the solid split wedge, bark out — is not the cheapest row in every table and the book does not pretend it is. Whole-tree trunks are cheaper as material; gravel is cheaper as a foundation; an open frame is cheaper as a panel. What the reference combination wins is the *combination*: the only row across all five tables that needs no shop, no mill, no second trade and no supply chain — which is the one property the tables cannot print but the chapter can state, and it is the property the whole method was built to buy.

Every other way of doing it, priced on this dome

option	price	note

Prices are the wedge book's catalogue on the reference quantities -- one dome, every alternative, so the choice is a comparison instead of a preference. The reference row is the one this book builds.

Eight materials, eight sections, sixteen panels, nine claddings, eleven foundations, priced on one dome.

What the prices say

The pattern across all five catalogues, read once and stated plainly.

The industrial rows — steel, aluminium, precast — win on strength and lose on *access*: their prices assume a buyer who can order them, a truck that can deliver them, and a building that needs what they sell. The natural rows — trunks, split wedges, bark — lose on uniformity and win on exactly what the industrial rows cannot buy: they are already on the site, they cost what the fuel costs, and they ask for no permission. The prices are not a ranking of quality; they are a ranking of *position*, and the builder's position decides which row is right before any preference does.

When a losing row is the right row

The chapter's closing honesty, because a table of options is useless if it cannot crown its losers.

Every option in these tables is somebody's right answer. A foundation for frost, a steel frame for a permanent public structure, a concrete shell for a bunker, a treehouse platform for a site with no ground to spare — the rows lose on this dome and win on theirs, and the table exists so that choice is made with the numbers up, the way this book makes every choice. The reference combination is the book's answer to *3 trees, one saw, forty hours*. The tables are the same question, asked eleven ways, with the whole catalogue priced — and the honest close: the book's way is not the only way, it is the way the numbers say fits the person this book was written for, and every other row is here so that sentence could be checked.

Hubs, or No Hubs

The trade this method answers, counted across the whole catalogue rather than argued: hubs buy a familiar kit and charge for it in parts, no-hubs buys the woodpile and charges in wood

Hubs, or No Hubs

Every geodesic dome in the world faces one decision before any other: do the struts meet at a connector, or at each other?

The hub is the familiar answer — the machined star at every vertex, the kit's photograph, the thing every dome catalogue shows. The hubless answer is this book's — the pinwheel of Chapter 15, where each member's butt lands on the next member's side and the vertex is air. This chapter is the trade, counted across the whole catalogue rather than argued, because the two answers are not rivals: they are the same dome sold to two different builders, and the numbers say which builder each one is for.

What a hub buys

A hub makes every joint the same part. One machined connector — drilled to the frame's angles, plated, catalogued — replaces every question about how three, five or six struts meet at a vertex. The struts all end at the same point, each end cut to the same pattern, each bolt in the same hole. For manufactured struts from a factory, where every stick arrives identical and every angle is already known, the hub is the right answer: it is the cheapest way to join parts that were made to be joined by it, and it makes the dome a *kit* — ship the parts, follow the drawing, bolt the stars.

The hub also buys familiarity, which is a real asset with a real price tag: the inspector has seen one, the appraiser's comparable has one, the spare part exists in a catalogue. A hubbed dome speaks the language the building world already speaks, and for a manufactured product sold to people who will never cut a stick, that language is worth the hub's cost.

What a hub costs

The hub's bill arrives in three places, and the third is the one this book exists because of.

First, the parts: a hub per vertex, plus its bolts, washers and drilling — a real line item, paid per corner, for the life of the build. Second, the precision: a hub assumes its struts all arrive at the same point at the same angle, and every strut that does not — every bowed stick, every drifted cut — fights the hub instead of fitting it. The kit economy buys its sameness from a factory; the hub inherits the factory's requirements.

Third, and the decisive one for this book: the hub cannot absorb what rough timber *is*. The frankendome's hub experiments — Chapter 4 — were this cost, paid in afternoons: shims, washers, longer bolts, every joint a negotiation between a machined star and a stick that never agreed to the drawing. The hub is perfect for identical struts and useless

the moment the struts differ — which is exactly the moment this method begins, because the struts are split logs, and the split log is the one member the hub economy was never asked to join.

What no-hubs buys and costs

The pinwheel deletes the hub by letting each member end land on the side of the next, and it pays for the deletion in wood — the 1.85 members per unique edge that Chapter 18 accounted for. That ratio is the no-hub bill: a frame that owns every edge in its own panels carries more timber than a frame that shares struts between neighbours, and the extra timber is the price of never needing a connector.

What the wood buys, in order: a joint that tolerates the stick's own shape, a panel that can be built flat on a bench, a seam with room for a key and a gasket, and a frame in which every member is a member you can watch and swap — the redundancy Chapter 19 spent on grading. The no-hub dome is not cheaper in timber; it is cheaper in *machines*, and it is the only version a woodpile can be built into.

How to choose

The three questions that sort it, stated as the chapter's whole conclusion.

Who is cutting the wood? A factory — hubs. A chainsaw — no-hubs, and the chapter's answer is not a preference but a consequence: the hub assumes the stick, and the saw cannot supply the assumption.

Who is assembling it? A buyer following a drawing — hubs, the kit's instructions are the hub's. A builder who cut the pile and built the bench — no-hubs, the jig is the instruction.

Will the dome ever be taken apart? A kit that ships and re-ships wants its connectors back — hubs. A frame that moves whole on a truck, or stays where the tree stood — no-hubs, where the only hardware is the screws and the keys, and nothing was ever bolted to a star.

The chapter closes with the book's own position, stated the way the breadth tables stated it: hubs are a fine building system, the right one for most of the dome world — and this book is the book for the rest of it, the part with 3 trees and a saw, where the question was never "which connector" but "what would it take for there to be no connector at all". The answer is the pinwheel. The bill is the 1.85. The whole account is Chapters 15 and 18, and this chapter was the summary sheet.

A Zome Is Not a Piece of a Sphere

Sweep a star of directions instead of subdividing a solid and every panel is flat by construction -- the zome, the one-panel golden zome, and what the flatness costs

A Zome Is Not a Piece of a Sphere

The geodesic dome starts from a solid — the icosahedron of Chapter 8 — and projects its faces onto a sphere. A zome starts from directions and sweeps. The difference is one sentence, and it decides everything about the panels: **a geodesic panel is a flat triangle standing in for a curved surface; a zome panel is flat because the whole zome is flat, panel by panel, by construction.**

The zome is in this book because the project's films teach it and the wedge book gives it a chapter, and because it is the method's natural cousin: same members, same seams, same jig — a different *shape family*, with one property the geodesic dome has to work for and the zome gets for free. This chapter is the sweep that makes the property, the famous one-panel zome that follows from it, and the honest bill the flatness charges at the joints.

The sweep, in four steps

From a star of directions to a frame of flat parallelogram panels, with the guarantee built in at every step.

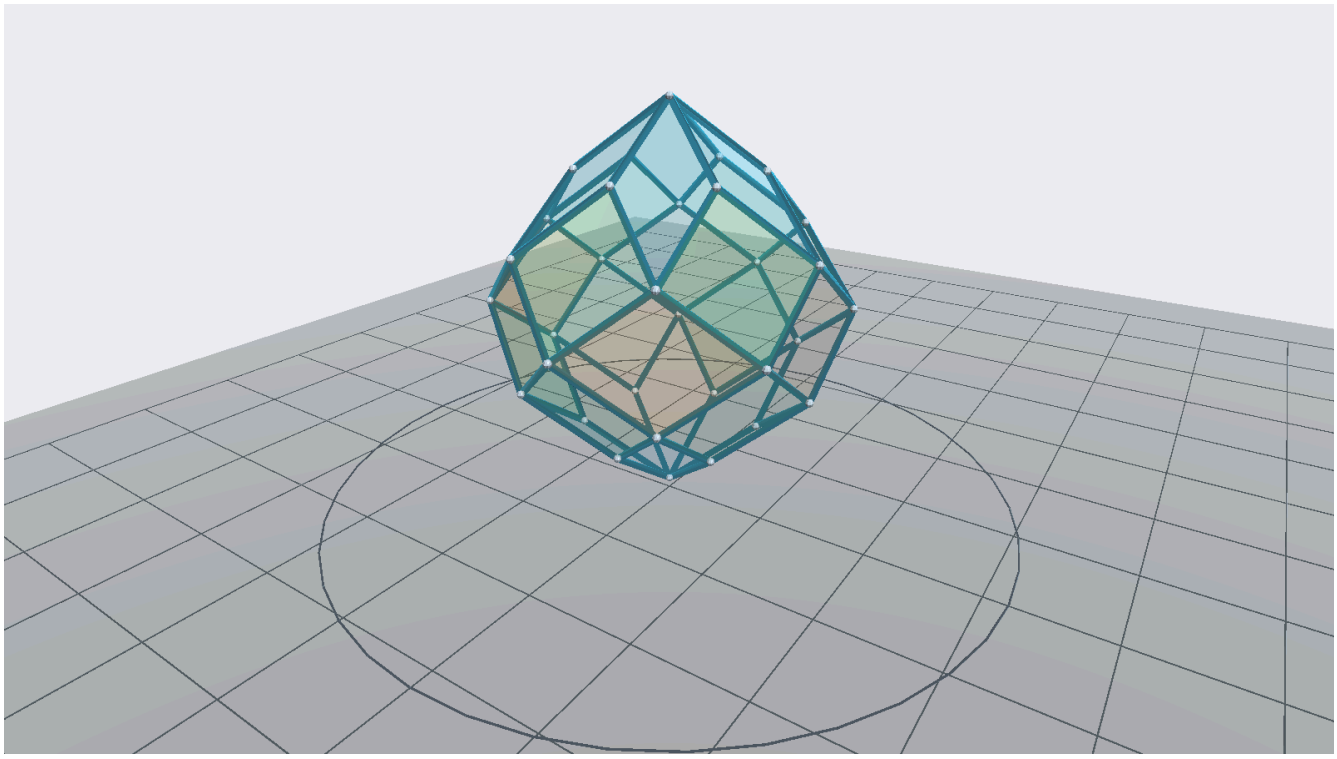
One: choose a star. Pick a handful of directions in space — a star of lines through the origin, like the ribs of an umbrella opened in three dimensions. These directions are the zome's *edges*, and they are chosen, not inherited from a solid.

Two: draw the planes. For every pair of directions, the two lines span a plane. Each such plane is the face of a panel, and because it is spanned by two straight lines, it is perfectly flat — not approximately flat, not projected flat: *flat*, the way a sheet of paper is flat.

Three: bound the panels. In each plane, the panel's edges are the points where the star's lines meet the design's envelope — cut the plane into a parallelogram or a triangle using the neighbouring directions as the other two sides.

Four: assemble. The panels meet along the star's directions, each edge shared by two flat panels that were flat before they met. The frame that results is a zome: a shell of flat panels hanging on a star of directions, with no curved surface anywhere in its ancestry.

The guarantee is the whole point. A geodesic dome's panels are flat *approximations* of a sphere, and the sphere keeps its curve in the fold angles between them — the seam schedule of Chapter 16 is the book's payment of that debt. A zome's panels are flat because the geometry never asked them to be anything else, and the fold angles fall out of the star instead of the sphere. Nothing was projected. Nothing was approximated. The flatness is not a manufacturing achievement; it is a property of the construction.



A zome is not a piece of a sphere.

The zome

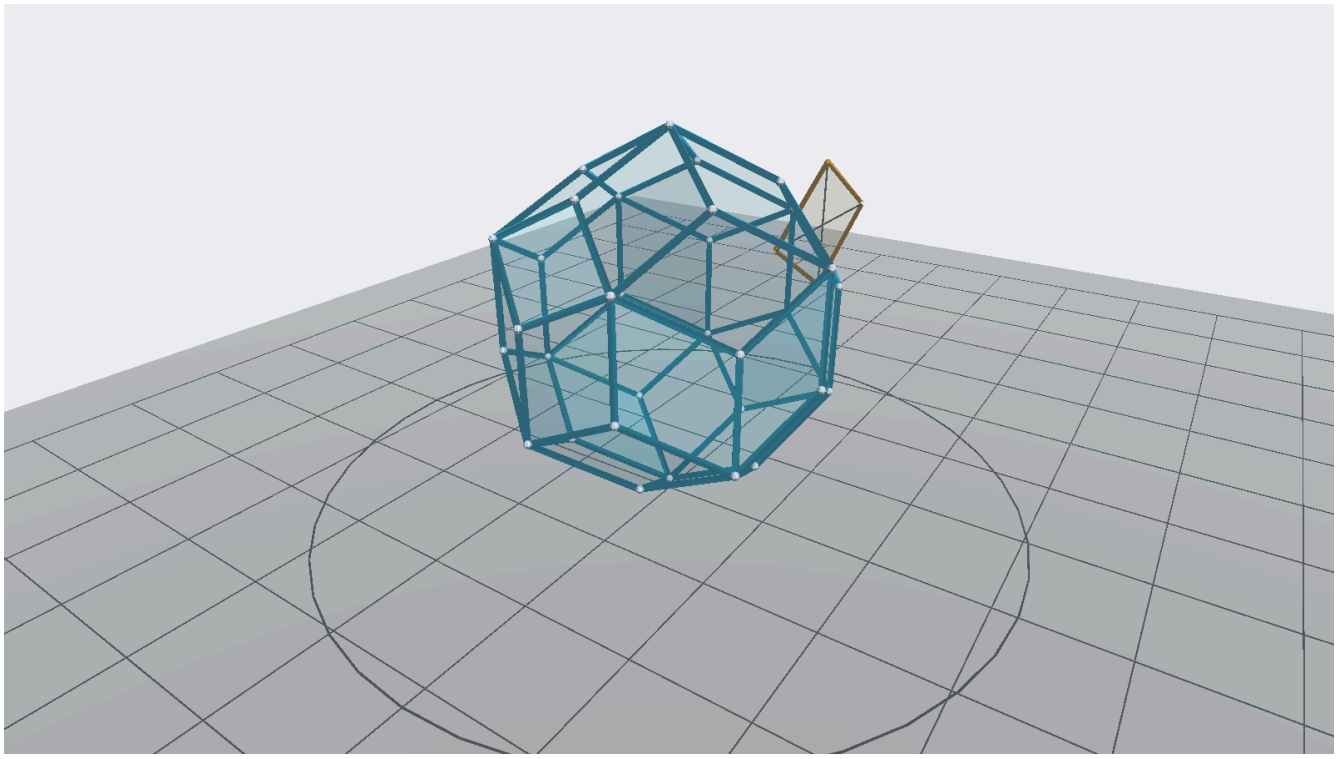
The plate is the zome standing: a point on top, level rings of hubs where the directions cross, and every panel a flat parallelogram hung between them. Read it against the geodesic plate of Chapter 8 and the difference is the chapter: no two triangle shapes, no fold schedule, no sphere under the surface — a shell that looks like a crystal because it is built the way one is, face by flat face.

The zome's counts are the sweep's counts: the directions decide the hubs, the hub pairs decide the panels, and the whole thing can be counted before any wood is bought — which is exactly the discipline the book's method chapters applied to the geodesic frame, transferred to the star. The top comes to a point because the star's directions meet there; the hubs land on level rings because the sweep is symmetric; and the panels hang flat because they were always going to.

The famous one-panel zome

The star's freedom has a limit case, and the limit case is the famous one: choose the directions *equal* — every direction in the star at the same angle to its neighbours — and every panel the sweep produces is the same panel.

One strut length, one panel template, ninety repetitions. The one-panel zome is the flat-panel world's answer to the geodesic's two-lengths-two-shapes economy, and it is genuinely famous among zome builders for the same reason this book loves the 2V dome: a frame of one repeated thing is a frame a jig can build, and the jig does not care whether the thing is a triangle or a parallelogram.



The famous one-panel zome.

What the golden zome charges for its single panel is the same bill every simplification charges, moved to the next line: the joints get particular, the floor meets the frame on a slope that wants its own treatment, and the roof's point concentrates the water the geodesic's crown was spreading. The one-panel zome is not the better dome; it is the *simpler one*, and the simplicity is bought with the star's freedom — which is the chapter's closing distinction, stated once more: the geodesic dome pays for the sphere at every seam, and the zome pays for flatness at every corner. Pick which bill you want, and the method — the split, the jig, the seam — builds either one.

Zome against geodesic dome

The comparison table's prose, because the chapter owes the reader the choice the films make side by side.

A geodesic dome is rounder: its panels approximate a sphere, so its skin is the smoother weather surface and its floor is the larger circle. A zome is flatter: its panels need no approximation, so its joints are the simpler ones and its glazing — the one place flatness pays the builder directly — is a matter of cutting rectangles instead of solving triangles. The counts differ the way the constructions differ: the zome's template count is set by the star's symmetry, the geodesic's by the frequency. And the method's answer to both is the same sentence the book has been repeating since the tree: **the wedge does not care which shape family it frames.** It cares that the frame is made of triangles or of flat panels a jig can repeat — and both families are, which is why the zome gets a chapter instead of a footnote.

Exactly Twelve Pentagons, Always

A sheet of hexagons will not curve, curvature is bought with missing angle, and the missing angle always costs exactly twelve pentagons -- the hexagonal dome, and where it stops being flat

Exactly Twelve Pentagons, Always

The hexagon is the panel a factory loves: one shape, flat sheets, six identical sides, and a tiling that covers a floor with no gaps and no thinking. The question the hexagonal dome asks is what it takes to make a *sheet* of hexagons into a dome — and the answer is a theorem so tidy it sounds invented: **exactly twelve pentagons. Always.** No more, no fewer, at every size, for every hexagonal dome that ever closes.

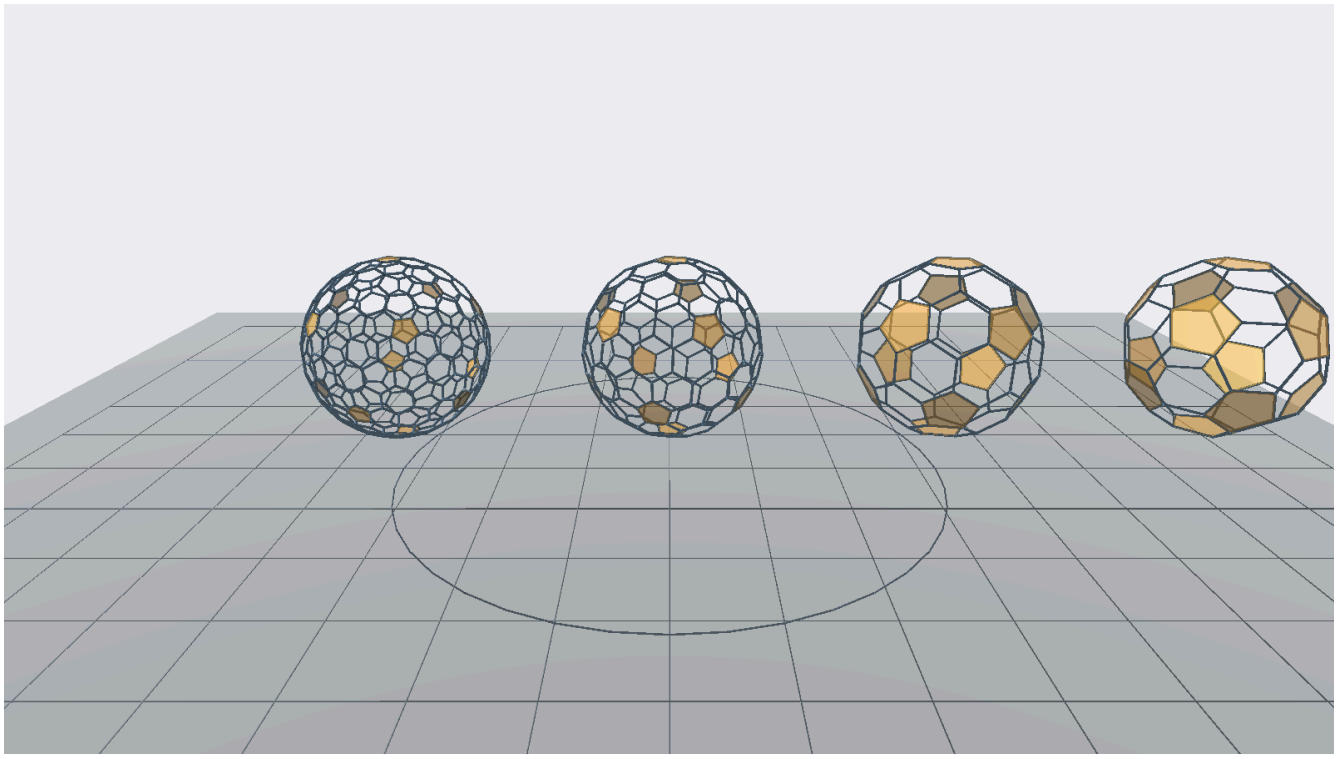
This chapter is the theorem, the dome it builds, and the place the hexagon's luck runs out — because the hexagonal dome is the project's other cousin, the one the hex film teaches, and its lesson is the geodesic's lesson from the opposite side: the dome pays for its curve somewhere, and the somewhere is never optional.

Curvature is bought with missing angle

Flat hexagons tile a plane exactly — six sides, three meeting at every corner, the angles adding to a full circle at every vertex, the sheet lying flat because the angles *agree* to lie flat. To make the tiling curve, some of that agreement has to be broken: somewhere, the angles at a vertex must add to *less* than a full circle, and the missing angle is what makes the sheet buckle upward into a bowl.

Where does the missing angle go? It goes where the tiling changes shape — a pentagon. Five sides instead of six means one sixth of the circle is missing at that vertex, and the pentagon is where the sheet's flatness is spent. The theorem is the count of how many pentagons the curve demands, and the count is the one fact about hexagons worth memorising: a closed surface of hexagons needs exactly twelve pentagons, every time. Not because twelve is a nice number — because the total missing angle the sphere demands is one full turn's worth, and each pentagon supplies exactly one sixth of a turn. Twelve sixths make the sphere. The football's pattern is the same theorem; so is the buckyball; so is every hexagonal dome the world has ever built.

The dome's version is the same tiling with the same bill: twelve pentagons pay for the curve, the hexagons fill the space between, and the whole thing closes into a hemisphere with one strut length running through it — which is the hexagonal dome's gift to the builder, stated next.

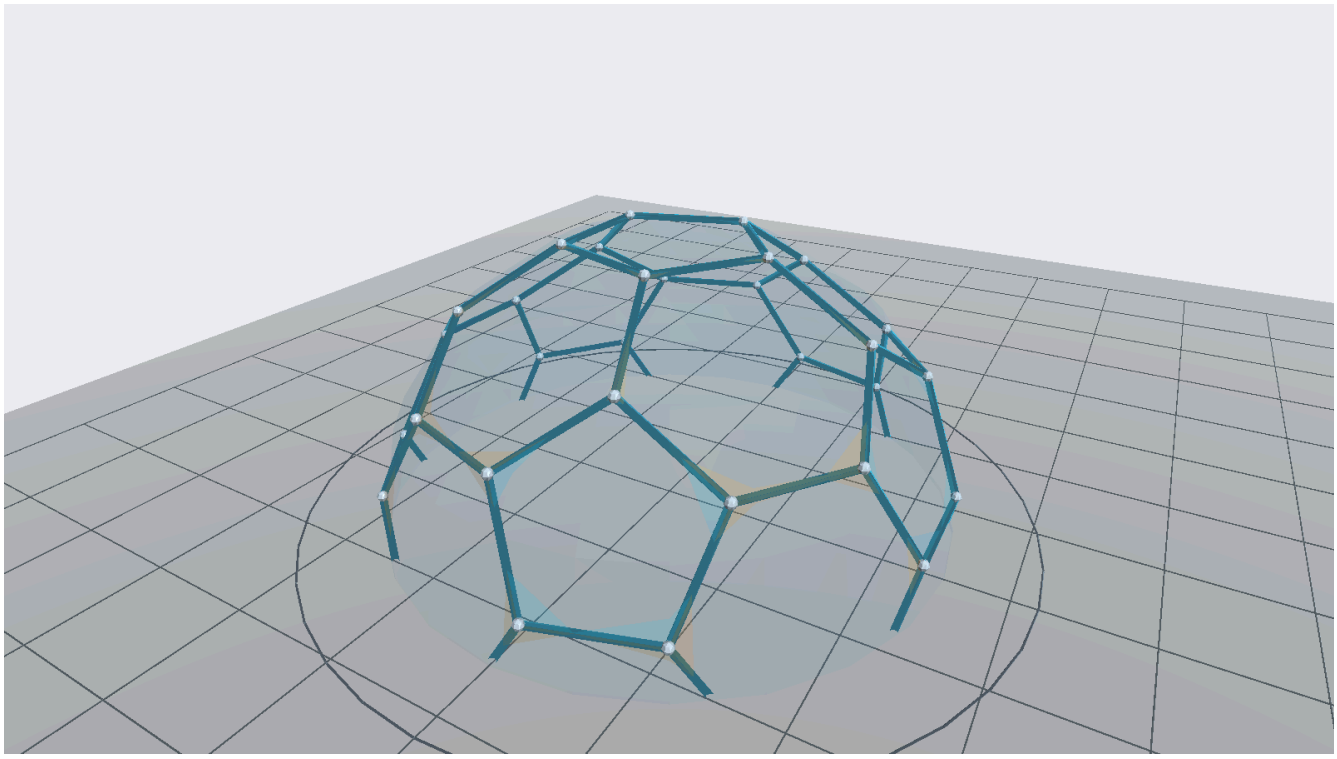


Exactly twelve pentagons. Always.

Twelve pentagons

The plate is the hexagonal dome standing: a field of hexagons, twelve pentagons hidden in it like the theorem promised, and one strut length cut ninety times.

The one-strut economy is the hexagonal dome's whole appeal. Where the 2V geodesic cuts two edge families and the pinwheel turns them into four lengths, the hex dome cuts one — every edge the same, every panel the same sheet, the factory's dream made into a frame. The pentagons are the small print: twelve panels of a different shape, cut from the same stock, whose only difference is one fewer side. The cut list is one length and two panel templates, which is the shortest cut list in this book — shorter than the geodesic's, shorter than the zome's — and the reason the hex dome is the one a manufacturer would pick first.



The single-hexagon dome.

What the one-strut economy costs is the skin's flatness, and the next section is where the cost lands.

Where the panels stop being flat

The hexagon's luck runs out the moment the dome stops being a hint and starts being a building.

The twelve-pentagon tiling curves, but its *panels* are flat only while they are small enough that the curve is someone else's problem. Step the hexagon count up — a bigger dome, the same one strut length — and the panels themselves begin to carry the curve: a hexagon spanning a curving surface is a hexagon whose six corners no longer lie in one plane. The panel *warps*, and a warped panel is a sheet that must be bent, or cut into triangles, or replaced by the geodesic's honest approximation — which is the point the hex film makes with the two domes side by side.

The rule the chapter leaves on the bench: the hexagonal dome is the cheapest dome in the book while its panels stay flat, and the geodesic dome is the honest one after they stop. The method — the wedge, the jig, the seam — builds either; the split log never cared what shape its frame was pretending to. But the *choice* belongs to the geometry, and the geometry's sentence is the chapter's title: twelve pentagons, always, and beyond a certain size, triangles — because triangles are the only panel that can never be asked to warp, and the geodesic dome is the building made entirely of that refusal.

Why We Call It a Stem Cell

One undifferentiated dome body, one cut list, and the decision of what the building is for deferred to a set of panels -- fifteen buildings from one frame

Why We Call It a Stem Cell

A stem cell is a body that has not decided what it will be. The frame this book has been building is the building version of that — and the name is a discipline, not a slogan.

The discipline: the frame, the skin, the cap and the core are the same in every version of the building, and the decision of what the building *is* — a home, a gym, a guest house, a workshop — is deferred to a set of panels that can be decided later, swapped later, and taken to the next dome later. The name forbids building the decision in. A house with its use poured into the walls is a house that can only ever be that use; the stem cell is the frame that refused, and the refusal is what the catalogue of Chapter 81 prices.

The undifferentiated body

The body that has not decided, listed the way the campaign lists it, because the list is the argument.

One 2V frame — the 120 members, the 40 panels, the seams, the jig — identical whether the building will hold a bed or a bench press. One skin, one cap, one utility core, one cut list. The only thing that differs between the fifteen buildings the campaign names is which *panels* the walls are wearing: the windows, the doors, the benches, the beds, the hooks, the sinks — a set of interchangeable faces hung on the frame's own openings. The undifferentiated body is the frame; the differentiation is the fit-out; and the fit-out is panels, which is the one part of a building that bolts on and off without asking the structure anything.

The name does the work a name should: it stops the argument before it starts. A "dome" is a shape; a "stem cell" is a promise about *deferral* — the promise that the building's purpose is the cheapest thing to change, because the design put the purpose where change is cheap. The shape was always the frame's; the name is the method's, extended to the building's whole life.

One body, many buildings.

The stem cell

The plate is the body before the decision: the frame standing, its openings still wearing the blank panels they will keep wearing until somebody chooses what the building is for. Read it as the book's own dome, one step later — the fortnight done, the shell on, and the single most expensive question in building left deliberately unanswered.

That question — *what is this building for?* — is the one every other construction method answers first, with concrete and studs. The stem cell answers it last, with panels. The difference is the difference between a decision and a *preference*: a building that decides its purpose in its walls has bought the purpose for its whole life; a building that defers the purpose to its panels can re-buy it in an afternoon, and the second owner inherits the option the first owner paid for.

The seeds

The fifteen named versions, because "any building" is a claim until the catalogue says which buildings.

Home, food, storage, guest, gym, studio, workshop, garage, nursery — and the rest of the campaign's list — each one a *set of panels*, not a new building. The gym is the body with its walls wearing racks; the guest house is the body with a bed wall; the nursery is the body with light panels and a low threshold. The frame never changes. The core never changes. The cut list never changes — which is the catalogue's entire economic point: the expensive, hard, structural parts of the building are bought once, and the cheap, easy, panelled parts are what make fifteen buildings out of one.

What the name buys

The chapter's close: what calling it a stem cell actually buys the builder, beyond a better metaphor.

It buys a resale argument — the building that can change its use is worth more to its second owner, and the network chapters lean on exactly this. It buys an upgrade path — the fit-out can grow with its owner's money the way the shell ladder grows with its owner's winters. And it buys a manufacturing argument — one body, one cut list, one jig, and fifteen products, which is what turns a building method into a product line and a fortnight's dome into the campaign's catalogue. The name is not the project's poetry. It is the project's economics, wearing a word it earned.

Sink the Money Into the Part That Moves

The utility core is the one component that outlives its shell, so it is the one worth buying once: fourteen parts, five services, one penetration

Sink the Money Into the Part That Moves

Every dome this method builds will be outlived by one part of itself. Put the money there.

The part is the utility core: the column through the centre of the dome, carrying power, water, waste and air in one object that unbolts from the building and moves to the next one. Every other component of the dome is either cheap enough to leave behind or bolted to a frame that will stay — the core is the one purchase that survives its building, which makes it the one purchase the book tells you to make properly, once. The chapter is the core: what it is, the order it is built in, and the accounting that turns a purchase into an asset.

What the core is

The column, stated as the campaign states it: fourteen parts, five services, one penetration.

The dome's centre is the one point every radius meets, and the column that stands there is the building's backbone for services — the riser that carries power up from the pad, the chase that carries water and waste down, the mounting point for the fan, the light and the cooling that the room's centre naturally serves. Five services, one object, and one hole in the floor — the penetration that would otherwise be five holes scattered through a structure whose whole argument is closure. The column is the dome's answer to the utility wall, and the answer is better than the question: a wall divides a room, the column *defines* it, and the round room's centre — the one place the furniture chapter kept empty — is where the services already are.

The top of the column is the socket: the seal cap, the mast of Chapter 71, the vent, the lightning's path to ground — the apex is an *interface*, not a roof, and the interface is standard across every building the catalogue sells, which is the first half of why the core moves.

The top of the dome is a socket.

Built on a bench, proved before it stands

The build order, which is the module-build film's whole teaching and the core's quality system: drain first, because it is the one you cannot re-route; water second, and proved dry before anything electrical exists; power last, into a core already known to be dry.

The order is not caution; it is sequencing the *checks*. A drain tested before the walls close is a drain that can be fixed with a wrench; a drain tested after is a renovation. Water is proved by pressure before the first wire is pulled, because water and electricity in one column are only ever friends in the order they were tested. And power lands last, in a core whose wet services have already passed — which is why the module-build chapter can say the core is *proved before it stands*, and why the campaign sells it as the one part the owner does not make: the part that carries five services is the part whose failure mode is the house, and the house is not where that test belongs.

A purchase that becomes an asset

The accounting, and the chapter's title made into a rule.

The core is the most expensive single component of the dome's services, and the only one that transfers: unbolt the column, cap the pad, and the next dome's core is already bought. A purchase that outlives its building is the definition of an asset, and the campaign's arithmetic rests on exactly that — the core's share of the dome's cost, justified by the domes it will serve rather than the one it sits in. The rule for the builder is the chapter's title: sink the money into the part that moves. The frame is splendid and cheap and made of trees; the core is unglamorous and precise and made of decisions — and when the building is eventually left behind, the core is what you carry to the next one. Everything else in the dome is a building. The core is the project's *continuity*, and continuity is the one thing worth buying once.

What the core costs, and what it is worth over a life

The chapter's claim is that one component should be bought well. Here is the arithmetic that decides how well, and it is the only place in this book where a purchase is justified by buildings that do not exist yet.

the core	number
Parts	1,405 for 23 items
Labour to build it	201 at 7.2 hours once practised
The first one	17.2 hours, which is why the first is expensive
What it costs to move	260 a move

Cost. 1,405 of parts and about 7.2 hours of bench work, against a frame whose whole structure costs less than that. The core is the most expensive thing per pound in the building and the cheapest thing per service it carries, because nothing else in the building carries six services through one hole.

Life. It is the part that moves: unbolted from four joints, lifted out and landed on the next pad's port. The frame around it is expected to be cut from whatever trees are on the next site, and the core is expected to be the *same core* — which makes its cost per building fall every time it moves, and makes its real comparison not "what does the core cost" but "what does the core cost divided by the buildings it serves".

And the upgrade rule. Because the core is the interface, it is also the thing to buy *ahead* of the building rather than behind it: a bigger panel than this dome needs, a spare breaker, four capped tails for plumbing a future build might want. The network's own model prices that habit at 2.0 times — retrofitting a service later costs about twice what fitting it during the build costs — and the difference is entirely access. The core is where that difference is paid or avoided.

Which is why this chapter's advice contradicts the rest of the book on purpose. Everywhere else the rule is to buy the rung you need and no further: one layer of quilt, one pad, the panels your views want. Here the rule inverts, because the core is the one purchase whose value is measured in buildings rather than in years, and a reader who spends 1,405 on it has bought the only part of this building that goes on to the next one.

The Boat-Hull Shell

Four ways to skin the dome, priced by the pound of glass and resin each one puts on: the hard shell as the fifty-year upgrade over the cap

The Boat-Hull Shell

The cap keeps the weather off. A hull *becomes* the weather — one structural skin, laminated over the frame, that sheds water, carries its own loads and refuses to come off. Four systems build one, and their prices disagree on purpose: each system buys a different number of years for a different number of pounds, and the chapter's job is to make the disagreement readable.

The hull is in this book as the fifty-year option — the upgrade the cap of Chapter 70 deliberately defers. The cap's whole argument is that the hard shell can wait; this chapter is the hard shell, priced, weighed, and sliced so that when the waiting is over the choice is a comparison instead of a guess.

Four ways to skin it

The four systems, each priced and weighed on the same dome, with the honest note about what the prices rest on.

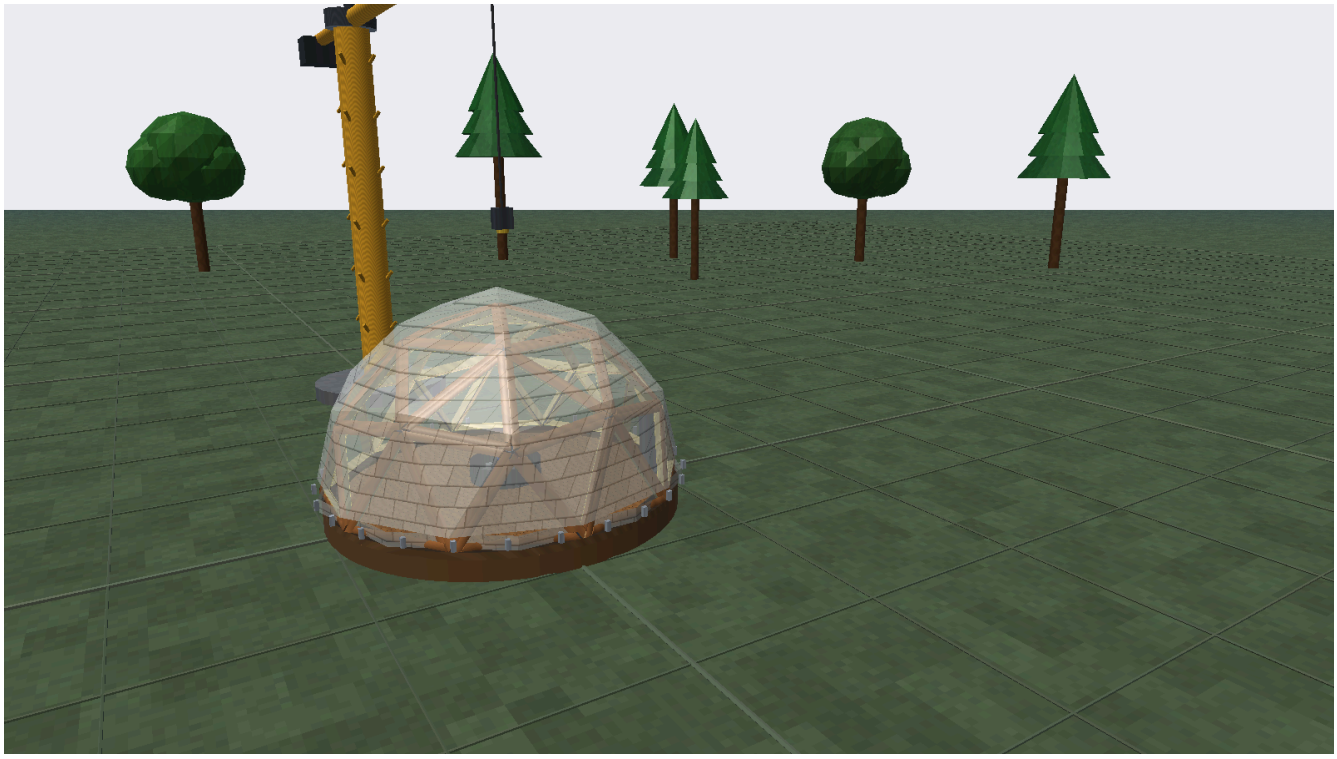
Sheathed plywood. Sheets over the frame, seams taped, painted. The cheapest hull, the heaviest per pound of weather protection, and the shortest life — a wooden skin that needs its paint on schedule. It is the hull for the builder who wants the dome closed *now* and upgraded later.

Boatyard polyester. Fibreglass cloth over a plywood or foam core, wet out with the polyester resin the boat-repair trade uses every day. The workhorse hull: familiar materials, forgiving process, decades of life, and the price per square foot the campaign quotes as its middle system.

Premium marine. Epoxy instead of polyester, better cloth, better bonds — the hull for a dome that will be a *home*, where the skin's life and the frame's life should match and the cost difference is spread over fifty years.

Vinyl ester. The premium's premium: the resin class that resists water absorption best, chosen where the skin sits wet or the budget simply wants the longest interval between decisions.

The prices the chapter quotes are the campaign's own, computed from the shell's square footage and the glass-and- resin weights each system puts on — and the honest note, kept because the film that discovered it kept it: the weights are hand-layup ratios, the prices rest on declared material costs, and a real hull's final number is the builder's own lamination quality. The table ranks the systems; the quality decides the years.



Four ways to skin it, and what each one costs.

Slices instead of a monolith

The one design decision that makes the hull an upgrade path instead of a commitment: the shell is built in slices.

A one-piece hull can never come off — it is the building, and every later change is surgery on the building. Four slices with an S-shaped lip between them can: each slice unbolts at its lip, the roof comes apart too, and the hull that was the standard article becomes a *set* of replaceable panels — the same panel logic the frame was built on, carried out to the skin. The campaign's numbers put the slice's cost against the monolith's and the difference is small; what it buys is the stem cell's deferral, one layer further out. A slice can be repaired alone, upgraded alone, and carried to the next dome alone — which is what turns a fifty-year skin into fifty small decisions instead of one large one.

Which one to pick

The ranking, stated without romance.

The cheapest hull is not the cheapest to own — plywood needs its paint, and paint is a subscription. The premium hull is a fifty-year decision the cap lets you defer, and deferral is the book's whole method: build the cap, live a winter, and let the dome's own behaviour tell you whether the hull is ever needed — and if it is, which system the site's weather actually earns. The chapter's last sentence is the cap chapter's first, closed into a loop: **the hard shell is the upgrade you buy after the soft one has taught you what you need.** The hull is waiting. The cap is working. And the four systems above are the comparison for the day the waiting ends.

The four systems, priced and weighed

The chapter has described the schedules. Here is what the model says they cost and what they weigh on this shell's own core — 660 square feet of it, 25.2 sheets, 957 pounds of core before anything is wet out, held down by 26 latches and 63 feet of rim gasket.

system	what it is	total	resin	laminate weight
Sheathed ply	light cloth in epoxy, both faces	5,395	22.1 gal	265 lb
Boatyard polyester	mat, then stitched biaxial, gelcoat	6,045	66.0 gal	889 lb
Marine polyester	the same schedule, better resin	6,668	66.0 gal	889 lb
Vinyl ester	the below-the-waterline schedule	9,144	66.0 gal	889 lb

Four systems, 5,395 to 9,144 — a spread of 3,749 — and a weight spread of 3.4 times between the lightest and the heaviest. Read that next to the chapter above and the choice stops being a matter of taste:

The cheapest schedule is also the lightest, by a wide margin. Sheathed ply is cloth and epoxy over the core: 265 pounds, the system a boatbuilder would use on a dinghy. It seals the wood and takes abrasion, with the most expensive resin on the list used in the smallest quantity — 22.1 gallons against 66.0.

The production schedule is the heavy one. Mat as a bond coat, stitched biaxial over it, gelcoat on the weather face: 889 pounds, 3.4 times the sheathed laminate, for the middle price on the board. It is bought for that weathered face — the gelcoat at 806 is what lets a hull survive ultraviolet light — and the weight is what the protection weighs.

Vinyl ester is bought for water rather than for strength. It is the dearest schedule at 9,144, and the reason is on the label: it is what goes below the waterline on a boat expected to stay there. For a shell that lives outdoors that is exactly the argument — and the chapter's honest note is that it is also an argument the *cap* makes for 5,395 by keeping the water off the shell altogether. The cheap laminate and the soft cap are competing answers to the same question, and the cap is the one you can try first.

The wet-out area is 1,325 square feet once inner face, outer face and skirt are counted, against 660 square feet of core. That difference is why laminating is priced by the pound of glass and resin rather than by the sheet of plywood: every square foot of this shell gets covered twice, and the model bills it that way.

CHAPTER 81 · REFERENCE

Fifteen Buildings From One Body

The fit-out catalogue, priced cheapest first: the same frame as gym, guest house, workshop, garage, studio and nursery

Fifteen Buildings From One Body

The stem cell's promise, priced: the same body, fifteen fit-outs, and the ladder that orders them. This chapter is the catalogue Chapter 78 promised — the reference page a builder or a backer reads to see what the one frame actually becomes, with every row priced by the same model so the rows can be compared.

The catalogue is ordered cheapest first, and the ordering is the argument: the differences between the rows are panels and services, not structure, so the ladder's rungs move in small steps — a gym is the body plus racks, a guest house the body plus a bed wall — and the whole ladder is reachable from the same undifferentiated frame the fortnight built.

The catalogue

The table is the complete list: every named fit-out, the panels it changes, and its price from the stem-cell model's own quotes. The frame under every row is the same body, the core is the same core, and only the walls' faces differ — which is why the prices cluster the way they do, and why the cheapest rung is not a smaller building but a *barer* one.

Fifteen buildings from one body

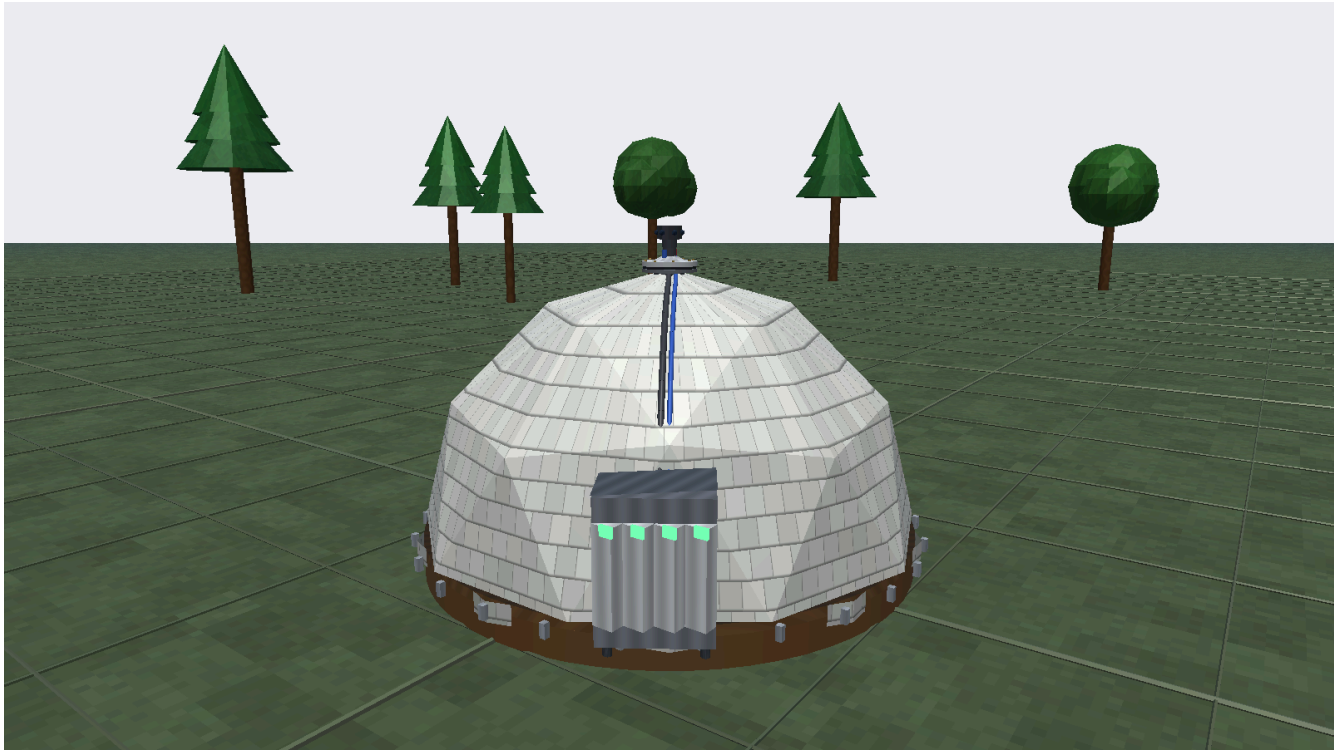
fit-out	price	panels changed
Stem cell	--	{} panels
Hothouse seed	--	{'window': 6, 'skylight': 2, 'door': 1, 'vent_window': 2, 'stove_flue': 1} panels
Food seed	--	{'serving': 2, 'window': 4, 'exhaust': 2, 'door': 1, 'louvre': 2} panels
Advertiser seed	--	{} panels
Cold storage seed	--	{'bay_door': 1, 'exhaust': 2} panels
Bunker seed	--	{} panels
Treehouse seed	--	{} panels
Sauna seed	--	{'louvre': 1, 'door': 1, 'window': 1} panels
Gym seed	--	{'mirror': 5, 'acoustic': 8, 'window': 4, 'door': 1, 'louvre': 2} panels
Guest house seed	--	{'window': 6, 'skylight': 3, 'door': 1, 'vent_window': 2} panels
Workshop seed	--	{'skylight': 6, 'louvre': 4, 'window': 3, 'door': 1, 'solar': 4} panels
Garage seed	--	{'bay_door': 1, 'skylight': 4, 'louvre': 3, 'window': 2, 'solar': 4} panels
Studio seed	--	{'acoustic': 12, 'window': 3, 'skylight': 2, 'door': 1} panels
Nursery seed	--	{'window': 5, 'skylight': 3, 'acoustic': 6, 'door': 1} panels
Jacuzzi seed	--	{} panels

Prices are the stem-cell model's quotes on the reference dome, cheapest first. The frame under every row is the same body; only the panels and the services change.

The fit-out catalogue, priced cheapest first.

Snap-on services

The plate shows the discipline that makes the ladder real: everything else snaps onto the outside. A fit-out never plumbs into a wall, never wires through a member, never drills the frame for a decision the next owner may reverse — the services hang off the frame's own faces and the column's own ports, the way Chapter 79 built them to, and the fit-out is the set of things that bolt on. That is the difference between the catalogue and a showroom of fifteen houses: a house has its services buried in its walls, and this frame has fifteen faces it can wear *without* burying anything.



The services snap onto the outside.

Reading the ladder

What moves between rungs, and what the ladder is actually telling the builder.

Between the cheapest rung and the dearest, the frame does not change by one member. What moves is panels — the windows, the walls, the benches — and services — the sink, the rack, the bed — and the ladder's lesson is the stem cell's lesson made countable: **the building's purpose is its cheapest part.** The builder who reads the ladder correctly stops asking "which dome should I build" and starts asking "which rung can I afford now" — because the rung is a purchase, not a construction, and the next rung is always one set of panels away. The catalogue closes the stem-cell

chapters the way the shell ladder closed the warmth chapters: the body is built once, the decisions come one at a time, and the book has just priced the whole staircase.

The same engine, twelve finished designs

The fit-outs above are one body wearing fifteen purposes. The same software also ships 12 *designs* — complete buildings, framed, paneled, fitted and priced — and the table is worth reading because it is the fastest way to see what the geometry does and does not decide.

The catalogue, priced

design	freq	diameter ft	floor sqft	struts	cost	$/sqft
Rebar Garden Dome	3V	36.1	991	165	$7,002	$7
Grow Dome	3V	32.8	819	165	$11,489	$14
Split-Log Homestead	2V	39.4	1,217	120	$18,067	$15
Timber Workshop	3V	32.8	819	165	$19,514	$24
Whole Trunk Lodge - 20 ft	2V	64.6	3,281	65	$22,747	$7
Glass Studio Loft	4V	32.8	845	250	$24,670	$29
Treehouse Canopy Dome	2V	27.6	597	65	$27,243	$46
Woodland Square Mirror	4V	36.1	1,023	250	$50,968	$50
Concrete Monocoque Form	3V	39.4	1,179	165	$58,661	$50
Hex Cell Pavilion	3V	39.4	1,179	165	$61,355	$52
Woodland Hex Mirror	3V	36.1	991	165	$67,346	$68
Continuous Steel Arc Hangar	2V	52.5	2,164	65	$100,454	$46
all 12			**597-3,281**		7,002 – **100,454**	7 – **68**

Sorted by cost. One geometry engine builds all of them: Rebar Garden Dome is the cheapest at $7,002, Whole Trunk Lodge - 20 ft has the biggest floor at 3,281 sq ft, and the spread in cost per square foot is the covering rather than the frame. Every figure is read off the model the tool loads.

The catalogue, priced.

Read it in four passes.

The floors span 5.5 times. From 597 square feet in the smallest to 3,281 in the largest: 5.5 times the building. Nothing in the frame's list of operations changes across that span — which is the flat-rate chapter's argument, restated as a price list.

The prices span 14.3 times. From 7,002 for the Rebar Garden Dome to 100,454 for the Continuous Steel Arc Hangar. That is the widest spread in the book, and it is not the structure: it is steel arcs, mirror tile and poured concrete, all of it skin and hardware.

Per square foot it runs 7 to 68. Look at which designs sit at the cheap end — the rebar garden dome and the whole-trunk lodge with canvas over it — and at the dearest: a shell wearing sixty-eight dollars of mirror tile per square foot. The cheapest floor in the catalogue is the one with the least building on it, which is the same conclusion the envelope chapter reached from the other direction.

Only 3 frequencies are used, and 4 of the twelve designs are two-frequency domes — including the one this book builds. A reader who has followed the method to this point has not learned the beginner's version of a bigger catalogue. They have learned the version 4 of the tool's own products are made from.

Between the lightest shell in that table and the heaviest there is a factor of 83,576 against 2,379 pounds, and the whole of the difference is materials rather than geometry. A foot of dome is a foot of dome; what you hang on it is a decision, and the catalogue is what that decision looks like when somebody has already costed it.

CHAPTER 82 · EXPLAIN

The Seam That Does Four Jobs

The gap between two panels is not a joint waiting to leak. It is a channel: gutter, vent, dehumidifier and air barrier in one, and the services ride in it

The Seam That Does Four Jobs

The seam this book closed with a key has a second life. Open it deliberately — plan the gap instead of merely filling it — and the most feared line in the dome becomes the most useful void in the building.

The 55 seams between the panels are a connected network: every seam meets its neighbours at a vertex, and the vertices join at the rim and the crown, so the whole shell's seam system is one continuous channel running through the entire frame. The wedge book's climate chapters — and the films this chapter draws on — engineer that channel to do four jobs at once, and the jobs are compatible for one elegant reason: the channel runs on positive pressure, so a leak stops being a leak and becomes a vent. This chapter is the channel's mechanics; the next one is its climate.

Gutter, vent, dehumidifier, air barrier

The four jobs, and the sentence that makes them one system.

Gutter. Water that reaches the seam — rain past the skin's shoulder, condensation off the inside face — runs down the channel's walls to the rim, where the lowest course drains it. The seam does not merely resist water; it *carries* the water that got in, to the one place it is allowed to leave.

Vent. The channel breathes: air moves along it from the rim to the crown and back, pushed by the stack effect — the same warm-air-rises engine that drives every chimney. The vent's job is to keep the cavity's air moving, so nothing that should dry ever sits still long enough to rot.

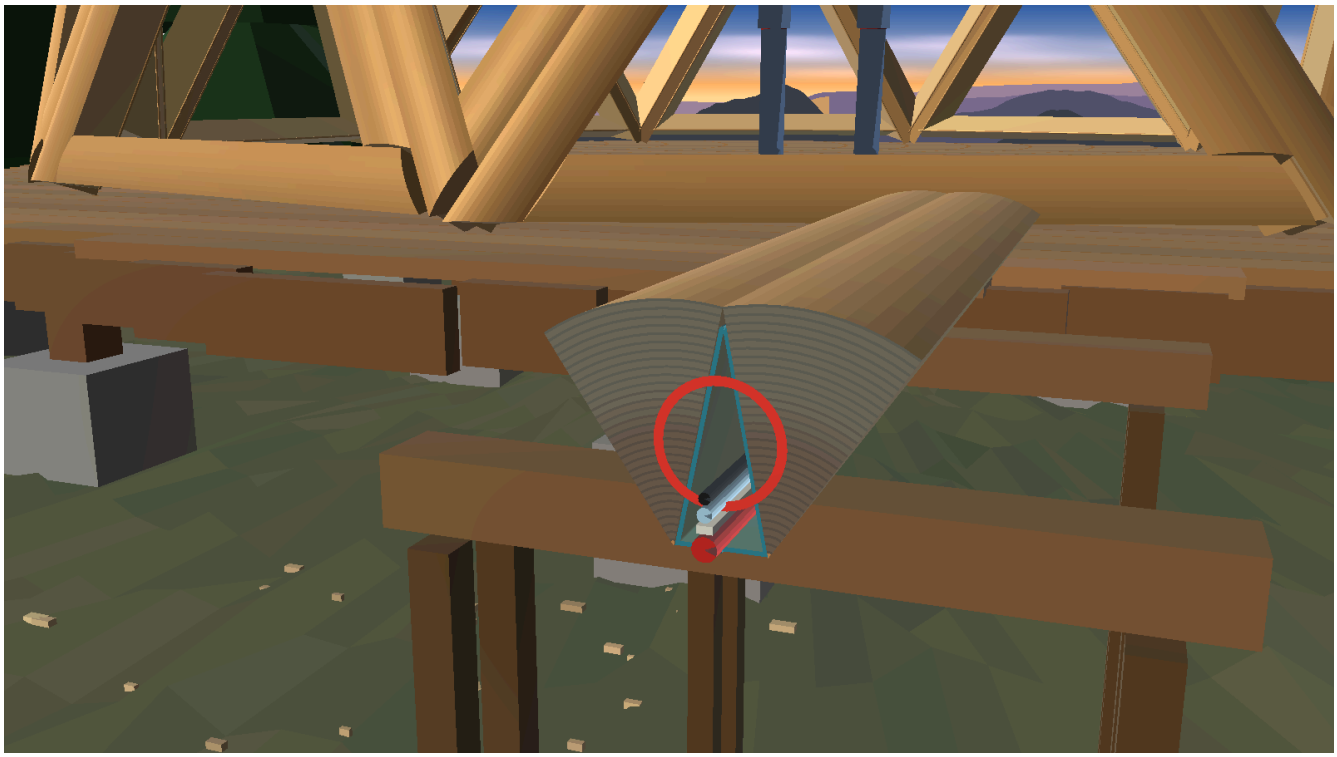
Dehumidifier. Because the channel runs on positive pressure — air pushed through it, not pulled — moisture in the cavity is carried out rather than drawn in, and a leak in the channel pushes air *out* instead of weather *in*. The sentence from the opener, stated as physics: the channel's pressure makes its failures vent, which turns the dome's worst-case leak into an air change.

Air barrier. The channel's walls are the panels' edges, and the edges, sealed along their length, are the building's airtightness line — the continuous break that stops the room's air from short-circuiting through the shell. One system, four jobs, and the jobs are not stacked; they are the same airflow, doing four things on its way past.

What fits in the channel

The channel is the building's service corridor, and the question of what fits is answered by the channel's own cross-section: water and wire fit; a round duct does not.

The V between two wedges is a long triangular void — wide at the outside face, narrow at the key. A small pipe, a cable, a sensor line: all fit, following the seam from rim to crown the way the services chapter of Part One prescribed. A round duct — the diameter a ventilation system wants — does not fit, and the honest answer to that refusal is the chapter's most practical trick: the key becomes a spacer. Open the seam wider, print the key in two profiles that hold the gap apart, and the channel *becomes* the duct — a triangular duct, sized by the gap, running where the frame was already going.



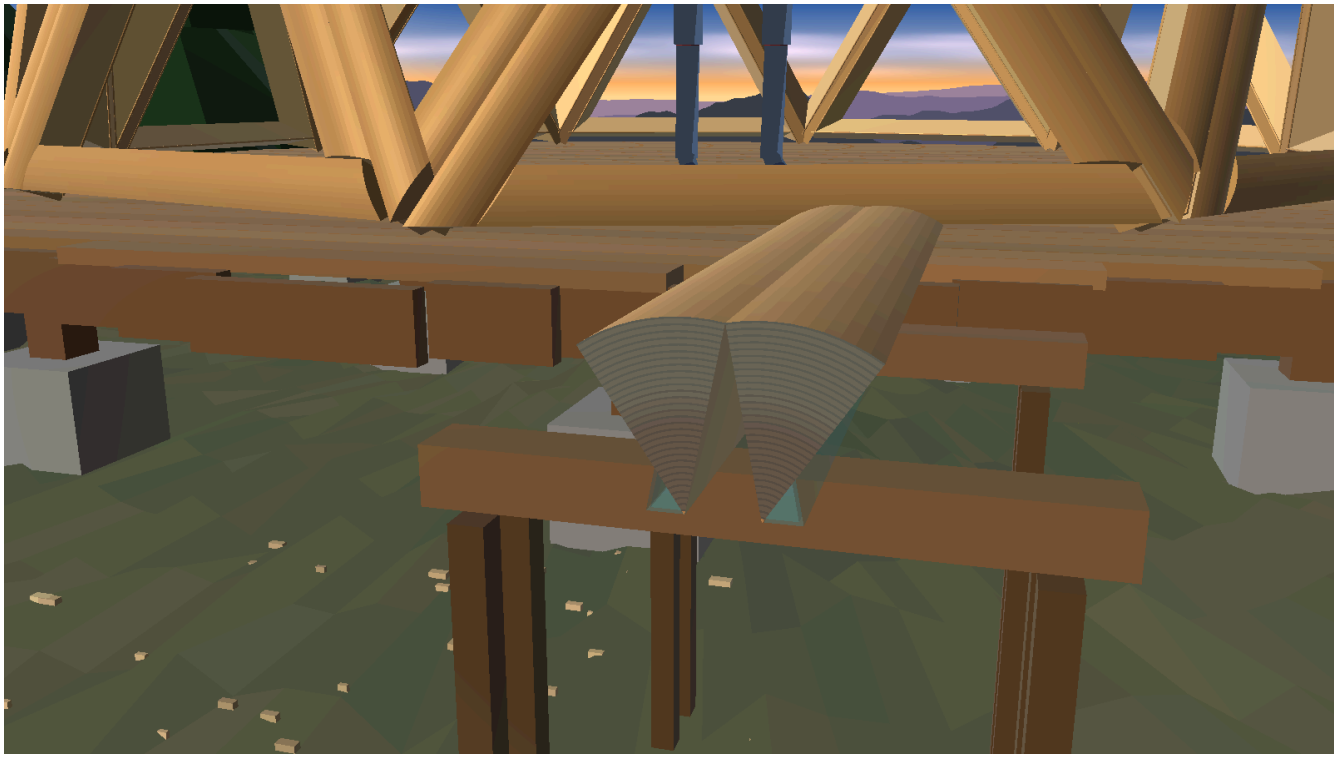
Water and wire fit. Round ducts do not.

The spacer's cost is the dome's own: opening the seams grows the shell slightly, and the film that worked this through priced the growth in panels and dome diameter — the honest note that the duct is not free, it is *bought with radius*, and the builder who wants a round duct can pay in size instead of in pipes.

Keys in halves, and rosettes

The two details that make the channel buildable, borrowed from the printing chapter of the film.

The key, printed in halves — two profiles, fed in from both ends of the seam — because a long seam cannot be threaded through the frame's closed corners, and a key you can push in from either end is a key you can fit after the dome is up. The halves meet in the middle, the gasket hides the joint, and the seam's structural key and its channel walls become one printed part — the film's answer to the hand-cut key of Part One, for the builder whose patience ends where a printer begins.



Printing the key in halves.

The rosette is the vertex's answer: where five or six seams meet at a corner, their channels converge on one point, and the rosette is the fitting that joins them — the channel system's hub, doing at the vertices what the key does along the seams. With rosettes in, the channel is one continuous network top to bottom, and the four jobs of the first section run through the whole shell instead of dying at every corner.

Two rules

The chapter's closing discipline, the two lines that keep a clever channel from becoming a buried hazard.

No pressure fitting inside a seam. The channel's pipes carry water only where the joint is accessible — a pressure fitting buried in a seam is a leak the building hides until the wall tells on it. The channel carries *chases*; the pressure parts live at the core and the rim, where Chapter 79 put them.

The cable in its chase is the inspector's call. Wire in a seam channel is a legitimate routing and a code question, and the book's rule is the same as every load in Part One: ask, before the skin closes over it, because the answer afterward costs a seam.

The seam that does four jobs closes the chapter the way it closes the building: gutter, vent, dehumidifier, barrier — one channel, 55 lines of it, and the next chapter is the air it carries.

The seams are not all the same seam

55 seams sounds like one material repeated fifty-five times. It is not: sort them by height and they fall into 5 bands, and the bands do different jobs because of one number — the slope they rise at.

band	seams	feet	slope
The lower band, rim to the belt	20	113	55° and up
The belt	10	53	8.3°
The upper band, belt to the shoulder	15	87	32-48°
The pentagon ring below the cap	5	30	0.0° — level
The cap	5	27	15.9°

Three consequences, and they are the whole reason the bands are worth sorting.

Water runs where the slope is. The lower band rises at 55 degrees, so anything that gets into it leaves fast. That is the band to run the gutter in: a channel that is steep, long — 113 feet of it — and directly above the rim drains for the building.

The belt is the collector. At 8.3 degrees the belt is nearly level, and it zigzags, so it has low points that are natural drains. Water that reaches the belt will pool somewhere specific, which is either the best place to put a port or the worst place to ignore one, and the chapter would rather name the spot than leave a reader to find it in the first storm.

The pentagon ring cannot drain. 5 seams at 0.0 degrees — dead level, 30 feet of it. Water that gets in there stands in the wood. So that band is never asked to carry water: it is the exhaust manifold, air only, and the difference between “the seam is a gutter” and “this seam is a gutter” is exactly this table.

Read together, the bands turn one channel into five, each with a job it can do and a job it cannot. The climate chapter’s controller is written against these bands rather than against an average, for the same reason a builder measures each ring instead of trusting the rope: the geometry was never uniform, and the book’s numbers were never an average.

How much room is actually in there

Four numbers decide what a channel can carry, and the films’ channel model computes all four off the same two seam types this book solves. The wide seam (fold 18.03°, 25 of them) opens 2.80 inches at the key. The tight one (fold 22.46°, 30 of them) opens 2.35 inches.

the channel	wide seam	tight seam
Inside area, key in place	6.51 in²	5.31 in²
Largest round thing that passes	1.97 in	1.69 in
Round duct of equal area	2.9 in	2.6 in
The key’s own wall, in section	1.60 in²	—

Now the honest test, and it is the reason this section exists. Run the services a building actually needs — a half-inch water line, a fifteen-amp circuit, the pair of wires to the plates and the condensate drain — through the tight seam and they fill 8.6 per cent of it, against the 40 per cent the conduit rule allows. The water and the wire fit with room to pull them. That is the film’s first conclusion and the book’s: **the channel carries services without being opened up for them.**

But look at the third row of that table again: 2.9 inches is the round duct of *equal area*. No ordinary ducting is that small — the smallest catalogue duct is three inches, and a three-inch duct does not pass a 1.97-inch opening. Which brings the chapter to the option that costs money, and to the reason the book prints it as a cost rather than as a feature.

The key can stop being a filler and become a spacer. Move every panel outward along its own face and the seam opens; the shift is what it costs.

to carry	panels shift	the dome grows	seam then passes
A 3 in duct	6.97 in	6.0% radius	5.11 in
A 4 in duct	15.55 in	13.3% radius	—
A 4 in duct plus the services	16.31 in	14.0% radius	—

Read the middle column and the argument makes itself. A three-inch duct costs 6.0 per cent more radius — more skin, more members, more pad, more heat — to carry air that the *plain channel* already carries 1.97 inches' worth of, straight up the seam, without any of it. The films' conclusion is not that the spacer is impossible; it is that a building should not pay six per cent of its envelope to move air it could move for nothing.

Which leaves the rule this chapter has been building towards, stated once: **services go in the channel; ducts stay out of it.** Water, wire, drain and sensor — yes, and here is the arithmetic that says they fit. Round metal — only if you have decided the building is a duct system first and a house second, and you are willing to pay the shift on every panel to prove it.

CHAPTER 83 · EXPLAIN

Compare Dew Points, Not Humidity

Never ask which air is more humid; ask which is drier, where it will condense, and which way the water moves -- the seam's climate, run by one controller

Compare Dew Points, Not Humidity

Humidity numbers lie about moisture. Dew points do not. This chapter is the rule, the levels, and the drawer — the seam-climate system that turns the channel of Chapter 82 into the building's moisture brain, run by one controller that decides where the water goes before the water decides for itself.

The chapter's material comes from the wedge book's five climate chapters and the film that built them, and its central claim is a rule of thumb with real physics under it: **never compare relative humidities; compare dew points.** The rest of the chapter is that rule, spent — the three questions it asks, the levels it sorts the seams into, the metal that moves the heat, and the desiccant that lives in a drawer instead of a seam.

The rule

Two airs compared by relative humidity can point the wrong way; compared by dew point they cannot — because relative humidity is a ratio against *temperature*, and temperature changes the denominator.

Warm air at fifty per cent humidity carries far more water than cold air at fifty per cent; a number that ignores the temperature has quietly answered the wrong question. The dew point is the question done right: the temperature at which the air's water begins to condense out. Two airs compared by dew point rank by how *dry* they are — the property that actually decides whether water appears on a surface — and the comparison never flips the way humidity's can.

The rule produces the three questions the rest of the chapter answers, and the film's phrasing is worth keeping whole: **which air is drier? Will it condense? And which way does the water move?** Every decision in the climate system — where to vent, where to drain, what the metal does, where the desiccant lives — is an answer to one of the three, and the three are asked in that order, every time, by one controller.

The four levels and the ring that cannot drain

The dome's seams do not all slope alike, and the slope decides the water's fate — so the system sorts the shell into bands.

Near the crown, the seams run steep: water moves fast, dwells nowhere, and the channel's job is simple — carry it down. Near the rim, the seams run shallow, and one band is the troublemaker: the pentagon ring runs **dead level**, a horizontal seam all the way around the dome, and a level seam cannot drain — water that reaches it sits there, against the key, against the skin, waiting for a gap.

The answer is the chapter's second rule: **condense low, drain at the rim.** The system arranges for the moisture to appear where it is welcome — the coolest, lowest, most reachable part of the channel — and gives that band a deliberate drain to the outside, so the water the ring catches is the water the drain releases. The dead-level ring stops being a trap and becomes the system's collection point, and the stack effect — warm air rising through the channel, pascals of it — keeps the whole loop moving in the direction the drain expects.

The dome's levels, from the solver.

A plate is a heat pump

Metal in the seam runs both ways, and the direction is the climate's lever.

A Peltier plate — the solid-state heat pump the film demonstrates — sits in the channel with one face to the room and one to the channel. In winter it moves heat *in*, warming the seam's inside face so nothing condenses where the room can see it; in summer it moves heat *out*, chilling the channel's surfaces to a steady fifteen or twenty degrees below the room, so the moisture condenses *there, on purpose* — on the plate's face, at the dew point, where the drain of the last section is waiting. The plate turns the chapter's threat into its instrument: the water appears exactly where the design wants it, on a surface built to receive it, and the frost that forms at the dew point in the coldest weather is the system's own thermometer, not its failure.

A plate is a heat pump.

Two disciplines keep the metal honest. **Plate first, wood second:** the plate is the condensing surface, sized and finned for it, so the wood behind it never becomes the condensing surface by accident. And the galvanic table: the channel mixes metals — aluminium plate, copper wiring, steel screws — and dissimilar metals in a damp channel make a battery that eats one of them; the table decides which metals may touch and which need a barrier between them. Moisture, in this chapter, is a *designed* resident of the channel — and the design extends to the metals it lives with.

The desiccant belongs in a drawer

The last component is the one the chapter's title sentences to the furniture, and the reason is two measurements.

A desiccant — molecular sieve, the beads that hold a fifth of their weight in water — looks like the channel's perfect resident: pack the seam, let the beads drink. The measurements say no. A packed seam passes almost no air — the beads' resistance rises steeply with the bed's length, which is the Ergun equation doing what it does — so the packed channel stops being the vent the whole system depends on. And the beads cannot be regenerated where they sit: drying them needs heat and airflow the seam cannot supply, so a saturated seam is a wet seam that stays wet.

Two drawers and a gate.

The answer is the drawer: two desiccant beds at the rim, in removable drawers, with a gate that sends the channel's air through one while the other dries out — a twin-tower system, small, accessible, and *fails safe*, because a drawer left out is a bypass, not a blockage. The stove's heat never shares the stove's air: the combustion circuit is sealed, and the desiccant dries on waste heat the building was already paying for.

Dew point decides

The chapter's close is the controller, because the system's parts only cohere when one thing holds the rule.

Eleven weathers — the film's own list of the conditions a year throws at a shell — one controller, and one decision procedure: compare dew points, ask the three questions, choose between the loop and the purge. The loop dries the *structure* — air around the frame, through the channel, slow, continuous. The purge dries the *room* — air through the house, fast, wasteful if misused. The controller's whole job is to know which drying the building needs, and the knowing is the rule, coded: **dew point decides.** The seam climate ends where the book began — with a number computed, a decision made from the number, and nothing left to the weather's mood.

The idea this chapter will not sell you

One more thing belongs in this chapter, and it is the thing the book's predecessor film was most tempted by, so it is worth refusing in writing.

There is a seductive idea that a dome's whole wall could work as an air filter: push air in at the base ring, let it cross the entire surface, and the shell becomes a distributed plenum that filters, warms and dries the incoming air in one pass. Run

it the other way and the wall becomes *dynamic insulation* — outside air drawn inward through the structure, picking up heat on the way in, so the timber runs warmer and drier than the room. Both directions are the same machine: a fan, a ring main at the base, and a shell that is porous on purpose.

The airflow side of that idea is computable, and the films computed it. The volumetric rate needed for ventilation is small, and spread over a shell this large it crosses the surface at a crawl rather than a draught — which is the appealing part, because moving air slowly enough is air you cannot feel.

But the parts that decide whether the idea is *good* are not computable, and this book's rule is to say so rather than to let a clever airflow figure imply the rest:

The wall is untested as a plenum. Nobody has built a strut lattice at building scale and measured how air actually distributes through it. A model that spreads flow evenly across a hundred and twenty members is an assumption, not a measurement, and the first place it will be wrong is where the members meet.

Direction is the whole risk. Air moving outward through a warm, damp structure carries moisture *into* the wood rather than out of it, and the wall that dries in one season rots in the next. A dynamic-insulation wall works because it is engineered to be warm and dry on the outside face — and that is a designed assembly with a vapour profile, not a gap between wedges.

A sealed skin cannot breathe. A membrane that keeps rain out also keeps this idea from working, so the wall-as-filter requires a shell that is deliberately permeable — which is the opposite of what the weather chapter spent its length achieving.

An unreachable filter is never changed. If the shell is the filter, the filter is the building. Nobody takes a dome apart to clean it, and a filter that is never cleaned is a blockage with a fan behind it.

So the honest position: the airflow arithmetic stands, the idea is not refuted, and the project has not built one. It is the difference between a chapter that proves something and a chapter that computes something and says what it does not know — and for a shell somebody is going to live inside, it is the second kind that deserves the room to say so.

It also leaves a reader with a usable test. If you want to try the breathing wall, do it in a building you can afford to fix: instrument the members, give the shell a vented cavity rather than a sealed one, put the filter somewhere you can reach, and treat the first winter as the experiment. The idea is worth trying. It is not worth building a whole house on the assumption that it works.

The House as a Product

What the finished article costs to build, what it sells for, and the difference between a target and a quote

Everything up to this point has been a build. You fell the trees, you split the wedges, you cut the compound ends, you bolted forty panels into a shell and you laid a floor in it. That is the method.

This chapter is the article. One standard dome, priced the way a manufacturer prices anything: materials, labour, overhead, margin, and a price at the end with a number on it. It exists because the films quote that price — 12,036 — and a book that teaches the method and then leaves the arithmetic of the product to the marketing has broken its own rule. Every figure here comes out of the same cost stack the campaign's own films read. Change the model and this chapter changes with it.

It also exists because of a specific honesty this project has had to practise twice already, in the money chapter and in the hours chapter: state the number that does not help the argument. The product's price is one of those numbers. It is 43.45 a square foot finished, which sounds reasonable until you remember that this book's whole claim is that building it yourself costs a fraction of that — and both statements are true, because they are prices for different things. One is a building you make. The other is a building somebody else makes for you, and ships.

One standard article

The article is one dome of the reference size, complete and habitable: the frame, the watertight cap, the utility column with its seal cap, the panels, the fittings, and the labour to put it together. Frame, cap, column, services, utility panels and build labour, in the order the money is actually spent.

The frame first, because it is the cheapest part of the product and the most expensive part of the method. The standard article's frame is 692 of the build. That is not a typo: the entire structure — one hundred and twenty split members, the hardware that joins them, the hardware that is a set rather than a pile — costs less than a tenth of what the finished article costs. This book has spent fifty chapters teaching you the hardest and cheapest part of the job.

The cap is the expensive layer. 3,265 of the build goes into the watertight shell that goes over the top, and it is the correct place for the money to go. The frame is allowed to be imperfect because the skin closes the shell; the skin is not allowed to be imperfect because there is nothing above it. A building whose cheapest component carries the load and whose most expensive component keeps the weather out is a building whose cost distribution tells you exactly where the risk lives.

The column, 1,405, is the one part an owner cannot make. Water manifold, drain stack, electrical sub-panel, riser, seal cap: the centre of the building arrives as a tested object, which is why the campaign in the next chapter spends its largest single line on tooling for it.

Light, fan and cooling, 485, and the **utility panels**, 491 — the framed inserts that turn a bay into something that does a job. Between them they are under a thousand dollars, and they are what make the difference between a shell and a room.

Build labour, 1,884, at a burdened wage rather than a friend's afternoon. This is the line a self-builder deletes entirely, and it is worth knowing how much they are deleting: about a fifth of the build.

Add those and you have the direct cost, 8,221. Add overhead and the warranty provision and you have what it actually costs to deliver the thing: **10,030**.

The standard article, priced

what	usd	share of build
Wedge-cut frame	$692	6.9%
Shower-cap shell	$3,265	32.5%
Utility column and seal cap	$1,405	14.0%
Light, fan and cooling	$485	4.8%
Utility panels	$491	4.9%
Build labour	$1,884	18.8%
Materials and labour, direct	$8,221	
Overhead	$1,480	
Warranty	$329	
Built, all in	$10,030	100.0%
Margin at 20%	**$2,006**	
Price, shipped	**$12,036**	
Ground, priced separately	**$6,240**	
Dome standing on its own ground	**$18,276**	

Built $10,030, sold at $12,036: $43.45 a square foot of floor. The ground is quoted separately and carries no markup. Read from the campaign's own cost stack, so the chapter and the film quote the same account.

What the standard article costs, and what it sells for.

The price is that, plus the margin: 2,006 at 20 per cent, for a shipped price of **12,036** — 43.45 a square foot of floor, finished, on a pad the customer built or bought separately.

Two remarks on the margin, because a reader is entitled to hear them from the person taking it. First, twenty per cent on a manufactured article is not greedy; it is at the thin end of what small manufacturing runs on, and it is stated here as a fraction rather than buried in a price list. Second, it is a *planning* margin on a product that has been built once or twice, not a production margin on a product built a thousand times. The honest comparison is not against a furniture factory. It is against the fact that the same money in a conventional house buys roughly a third of the square footage, because that money is carrying land, a builder's overhead, a seller's commission and a mortgage.

The price the films quote

12,036. It is worth walking that number back to its parts, because a price that cannot be taken apart is a price that cannot be checked.

10,030 to make, 2,006 of margin at 20 per cent, and 12,036 shipped. Over the area of the standard article that is 43.45 a square foot.

Now the comparisons the films make, all of them computed rather than asserted:

Against a car. The completed article costs less than a new mid-range pickup, and it is a building. That is not a boast, it is a category observation: the reason a manufactured home can be priced like a vehicle is that both are products made in small factories in low volume, and the reason this one can be priced *below* most of them is that its structure costs almost nothing.

Against the same floor in a conventional house. At the published cost of a new site-built house per square foot, 43.45 buys a fraction — and the difference is not the shape. It is land, financing, labour at market rates, and the chain of hands the second chapter counted.

Against building it yourself. That is the whole book. The price of the article is the price of *not* building it, and the difference is the reason the method exists.

And the two numbers a buyer should ask for and rarely gets: what the article costs to *run*, and what it costs to *change*. Running costs are in the climate chapter, computed at seventeen cents a kilowatt hour. Change costs are the next chapter but one, and they are the product's real feature — the modules are priced separately precisely so that a building can be upgraded rather than replaced.

The target, and the gap

The campaign states a design target for the barest version: the trailer model, 5,000, relying on the pad for services the way a caravan relies on a campsite. It is the number the project would like the standard article to reach, and it is a good target because it is specific: 5,000, not “affordable”.

The computed price of the standard article is 7,036 above that target, and this book prints both numbers side by side on purpose. A target that is never compared with the real price is not a target, it is a slogan; and a project that quotes the target in public while computing the real number in private has started lying to its own readers. So: the target is 5,000, the achieved price is 12,036, the gap is 7,036, and the ways to close it are visible in the cost stack — the frame is already negligible, so the gap lives in the cap, the column, and the labour of assembling them. That is a design problem with three named parts, which is a much better position than a vague promise of cheapness.

Three points about the target, in the order they matter:

1. **A target is a decision; a price is an account.** The 5,000 was decided by somebody who had not yet priced a moulded strut. Deciding is legitimate. Quoting the decision as though it were the account is not.
2. **The ground is real money and it is priced separately.** 6,240 of the 18,276 it takes to stand the article on its own serviced pad. That is more than a fifth of the whole proposition, it is on the pad chapters’ arithmetic rather than this chapter’s, and it is marked up by nobody — the ground section is quoted at cost, which is a deliberate choice and a good one.
3. **None of this is a quote.** Local labour rates, freight to your site, permits, the engineer’s stamp on your wind load and your snow load: all outside this account and all somebody else’s numbers. A price in a book is a planning figure. A price for your building is a conversation with a builder who has seen your ground.

What the price does not include

Read the exclusions before the headline. That is the rule this project has applied to every claim it has made, and the product is not exempt from it.

The ground. Priced separately, at 6,240 on the reference pad, and not marked up. If you have a level site with power and water, that number shrinks; if you have a slope and no access, it grows past the price of the dome.

The modules. The catalogue in chapter 86 is priced module by module, and the standard article ships with one blank utility panel. A building’s fit-out is the customer’s list, not the product’s: the advertiser’s forty lit triangles and the sauna’s heater cost different amounts of money and neither is included in 12,036.

The engineer. The frame’s wind and snow loads, the mast, and any hoist carrying people are somebody else’s signature. The campaign funds that signature as a line item rather than pretending the arithmetic in this book is a structural rating — and this book has said so since page one.

The appliances, the furniture and the site work. Everything that makes a shell into a home is priced in the chapter on the round room, in the money chapter, and in the catalogue, and none of it is in the article’s price.

What that leaves is a claim that survives all the exclusions: a complete, habitable shell of 40 panels and 120 members, made of 3 trees’ worth of split timber or of bought members, for 12,036 delivered as a product — or for the price of your

own labour if you take the method the rest of this book teaches, which is the comparison that made the project worth doing in the first place.

CHAPTER 85 · STORY

What the Campaign Is For

The 110,000 list, the rungs on offer, and the promises the money is meant to keep

A campaign document is the most honest thing a project produces, because it has to name what it does not have. A finished building hides its gaps; a budget cannot. The films put this list on screen — 12 lines, 110,000 — and every line is a thing that does not exist yet.

This chapter is that list, with the arithmetic around it. It is in the book for two reasons. The first is completeness: the films state these numbers, so a book that claims to be the master repository of what this project has said has to carry them. The second is more useful to a reader who has no interest in a campaign at all: a line-item budget is a map of where a project like this is actually hard, and the map is instructive whether or not you ever fund one.

The goal, line by line

12 lines, adding to 110,000. The largest single line is 18,000, and it is worth pausing on what it buys: Tooling to mould a strut with its hardware in it. Not software, not marketing, not salaries — tooling. The largest thing this project needs money for is the ability to make the same part again, correctly, a hundred times.

The goal, line by line

what the line buys	usd	share
A full test platform and one standing dome	$4,200	3.8%
Sensors in the wall, logging for a year	$1,850	1.7%
Utility hub: three iterations to a design that ships	$9,500	8.6%
CNC router, 4x8 bed	$14,500	13.2%
CNC plasma table	$11,800	10.7%
Laser cutter, for gaskets and templates	$8,900	8.1%
Tooling to mould a strut with its hardware in it	$18,000	16.4%
Composite triangle: steel core, moulded shell	$12,500	11.4%
Automated processing for the repetitive work	$7,600	6.9%
Seeding the quilt network	$5,000	4.5%
An engineer, for the things that carry people	$6,800	6.2%
Making and shipping what backers are owed	$9,350	8.5%
The whole ask	**$110,000**	**100.0%**

Every line is a thing the project does not have yet: a year of weather on a standing dome, the machines that make the parts, the engineer, and the first quilters. A line item list is the most honest document a campaign produces.

Every line the campaign asks for.

Read the list as a reader rather than a backer and it divides into four honest kinds.

The measurements. 4,200 buys a serviced pad and a complete standard dome standing on it through a full winter, with the instruments to log it. This project is almost entirely solved and almost entirely unmeasured: the geometry is proved, the arithmetic closes, and nobody has watched a real one go through a wet season with sensors on it. That line is the project admitting it in public, and it is the single most useful thing on the list.

The machines. A CNC router, a plasma table, a laser cutter, tooling to mould a member with its hardware already in it, and the automation to feed, index, cut and repeat. A dome is one hundred and twenty members, a hundred and sixty threaded inserts and forty panels, and every one of them is the same operation done many times. There is no version of this product that is made by hand, and the list says so with prices rather than adjectives.

The parts that carry people. 6,800 for an engineer to rate the wind and snow loads, the mast, and any hoist that lifts a person. This book has refused to rate those things on every page and pointed the reader at a licensed engineer each time. Here is the cost of that refusal, and putting it on the list means nobody has to wonder whether it was quietly skipped.

The people. 5,000 to seed the quilt network: machines, shipping, and paying the first quilters for the first layers. The insulation in this building is a waste stream and somebody's evening, which only works if the network exists *before* the first dome needs quilting. That ordering is the whole reason the line is in the goal rather than in the delivery budget.

And one line that is unusually candid: some thousands of dollars budgeted for fulfilment at *cost* rather than at the reward price, because a campaign that funds its tooling out of its shipping money delivers neither the tooling nor the shipping.

The line that buys a year of weather

Singled out, because it is the one line a sceptic should fund first.

4,200 buys a platform and a dome on it, and instrumentation in the wall, logged for a year: temperature and humidity at each layer of the cap stack, inside and out, recorded continuously. What that measurement decides is the moisture question — the one thing in this method that is argued rather than computed. The films have said, in public, that they will publish the result whichever way it comes out. That promise is the reason to trust the rest of the arithmetic, and it is also the reason the number is in this book rather than only in a film.

A reader building their own version can do a small piece of this for almost nothing: put a cheap logging thermometer under the cap and another inside the room, and write down the days the two disagree. The project's own version costs 4,200 because it buys a building and a year, not because the sensors are expensive.

The rungs, and what they cost to deliver

8 rungs, from 35 to 19,800, and each one has a delivered cost printed beside its price. That is unusual and it is deliberate: a reward priced without its delivered cost is a promise to lose money slowly, and a backer deserves to see the margin on what they are buying rather than read it later in a post about a campaign that overran.

The rungs, and what they cost to deliver

rung	pledge	delivered	left	left %
The drawings and the numbers	$35	$2	$33	94%
Quilter's kit	$95	$58	$37	39%
One bay's hardware set	$180	$112	$68	38%
A shower cap for your dome	$1,450	$980	$470	32%
The utility hub	$2,400	$1,620	$780	32%
Dome kit, frame included	$19,800	$14,900	$4,900	25%
Dome kit, bring your own trees	$11,400	$8,100	$3,300	29%
Pad host's pack	$640	$395	$245	38%

A reward priced without its delivered cost is a promise to lose money slowly. The two kits are the campaign's whole argument in two numbers: the difference between them is the timber, and the buyer who has trees keeps it.

What each rung is, and what it costs to deliver.

The rungs fall into three groups, and the middle one is the interesting one.

The drawings and the numbers, at 35: every table in this book, the cut list for any diameter, and the solver that made them. It is a download, and the price is card fees and hosting. A reader who buys nothing else can build the whole frame from this rung.

The objects. A quilter's kit. One bay's hardware set. The watertight cap, made to the reference diameter, hemmed and grommeted — the only layer in the building you cannot sew at home. The utility column, assembled and tested, which is the part an owner cannot make and the part the goal exists to get right. These rungs are the building sold as components, in the order a self-builder actually needs them.

The buildings. The two kit rungs, and they are the campaign's whole argument expressed as a price difference. 11,400 buys everything except the frame: you fell, split and cut your own members from your own trees, which is the entire premise of this book, and the kit sends the rest. 19,800 buys the same thing *with* the timber included — the expensive version, honestly priced at 8,400 more. Nobody is pretending the difference is anything other than the wood, and nobody is pretending the wood is the cheapest way to get it.

The dearest rung is Dome kit, frame included, at 19,800 against a delivered cost of 14,900. Look at that pair: the most expensive thing on offer is a building with somebody else's timber in it, which is precisely the thing this method exists to avoid.

Bring your own trees, priced

The two kit prices are worth one more paragraph, because the 8,400 between them is the book's thesis with a dollar sign on it.

Everything in the product that is not the frame — the cap, the column, the hardware sets, the panels, the fittings — is the same in both kits. The difference is whether a pallet of graded, dried, dimensioned members arrives at your site, or whether you walk into your own woodlot with a chainsaw and make them. That is 8,400. Not a discount, not a subsidy: the price of the timber plus the price of the work the factory would have done to it.

A reader who has trees and a saw has just been told what their trees are worth in this building, and it is the same number the value ladder in the earlier chapters reaches from the other direction. A reader who has neither has been told what the convenience costs. Both of those are useful, and neither of them is hidden.

The promises worth holding it to

Three promises are made in the campaign, and a reader should keep them where they can be checked.

Open source. The geometry, the solver and the tables are published, which is why this book can print every number and point at the code that made it.

A model anybody can run. Every figure in this book comes out of the same programs, with the same commands, and the software chapter at the back lists them. A reader does not have to take a number on faith: change the tree, re-run the build, and the book's tables describe their tree. That promise is the difference between a book and a brochure, and it is why this project's corrections can be checked by strangers.

Publish the measurement. The winter of logged temperature and humidity under the cap, whichever way it comes out. Of the three, this is the one that costs the most credibility if it is broken, because everything else here is arithmetic that anybody can re-run and this one is a fact that only the project can produce.

A campaign that promises a number and then buries it has spent the only credibility it had. That is the standard this chapter holds the project to, and it is the standard a reader is entitled to hold this book to as well.

What the earlier cuts asked for

There is an earlier version of this ask, and the book prints both because the difference between them is instructive.

The first campaign montage asked for 50,000, and it asked for it in one sentence: tooling rather than salary, every dollar of it turning back into machines. No line items, no delivered costs, no tiers — an argument that a workshop is the thing standing between the design and the product.

The line-item goal in this chapter is 110,000, which is 60,000 more than that first number. The extra is not inflation and not scope creep in the ordinary sense. It is what happens when you cost the same intention line by line and discover the parts the sentence omitted: a year of weather on a standing dome, the instruments, the engineer's signature, the quilters, and the shipping of what backers are owed. A single figure states a need. A line-item list states a *plan*, and the second is always larger, because the first was guessing.

Of the 110,000, **82,800 is machines** — 75 per cent of the ask, and every line of it an object that still exists on day three hundred: the router, the plasma table, the laser, the tooling that moulds a member with its hardware already in it, and the automation to feed, index, cut and repeat. That share is the honest answer to what this project needs money *for*, and it is why the goal is described as tooling rather than as funding a build.

What it is up against, stated plainly

A campaign film is allowed to be cheerful. This book is not, and a chapter about a product launch that did not describe the opponent would be a brochure.

The opponent is not a competing dome. It is a market that buys houses at scale. Institutional capital buys single-family housing by the thousand, with financing, appraisals, insurance and title work all designed around a rectangular building on a permanent foundation. This project arrives with one prototype, a chainsaw, a building that can be taken apart, and a set of costs that are honest about being small. That is not a fair fight in either direction, and the sensible response is not to pretend it is one.

The arithmetic has already stopped working for somebody the reader knows. That is the argument the films make and the one this chapter will not sentimentalise: there are people who will not buy a house, not from laziness or bad decisions but because the numbers no longer close where they live. A method that turns three trees and an autumn into a shell is an answer to that specific problem — and it is an answer that a mortgage, a permit schedule and an appraisal process can each refuse independently.

The ask on a reader is small and specific. Not to buy anything: to send it to one person who can carry it further. One share from somebody with reach does more than a month in a field with a chainsaw, which is the most honest thing a small project can say about the difference between working and being heard.

And the reason underneath all of it, which deserves a paragraph because it predates every film and every number in this book. Buckminster Fuller wanted housing cheap enough that a mother would not be so daily burdened by it that she could not get on with raising her child. That is a domestic, unglamorous reason to care about a structural system — and it is the reason this project has measured everything it could rather than asserting what it could not. The arithmetic is the respect the idea deserves, and printing the numbers that do not help the argument is how a reader knows the rest were not chosen for effect.

CHAPTER 86 · REFERENCE

The Module Catalogue

25 snap-in modules, 15 named seeds, and five places on a dome where anything can attach

The frame is the product. Everything else is a module.

That sentence is the design position of this whole project, and this chapter is what it looks like when it is finished: a catalogue of 25 parts that snap onto the building, 15 named buildings assembled from them, and only 5 places on a dome where anything can attach at all.

The reason it matters to a builder is not commerce. It is that every one of these parts is a *decision deferred*. A reader who builds the frame in this book does not have to know, on the day they bolt the tenth panel, whether the building will end up a workshop, a sauna, a food stall or somebody's bedroom. The frame takes all of them, and the difference is a shopping list. That is what a kit of parts buys you, and it is why this chapter sits at the end of the book rather than at the beginning: you have to believe in the frame first.

Five kinds of mounting point

25 modules mount at 5 places, and the short list is the important part.

The apex. The top of the dome is a single point where every seam arrives, which makes it the one place in the building where a hole is structurally free: the ring around the crown is already carrying itself. The camera ring looks out from here and the bunker's only above-ground object is a vent.

The utility column. The centre of the floor, where the services arrive. Anything needing water, waste or significant power lands here, because this is where the manifolds and the drain stack are — shower, sink, three-bay sink, grey tank, the electrical sub-panel.

A bay panel. Forty triangles, forty places to put something flat. Windows, insulated panels, a fridge, a cooktop, a lit advert, a roller shutter, a sauna heater. The panel is the building's real estate and the catalogue treats it that way.

The floor. Shelving, a stair, a tub, a tree saddle. Anything whose load wants to go straight down rather than into the shell.

A utility panel. The framed insert the column carries — where the exhaust fan, the tankless heater, the dehumidifier and the power wall live, because they want outside air, a drain, or both.

That is the whole interface. Five places, and because they are fixed, the catalogue can grow forever without the building changing.

The module catalogue

module	mounts at	usd	draws	water
360 camera ring	apex	$470	15 W	--
Bunker ventilation	apex	$780	120 W	--
Shower module	column	$610	--	yes
Sink and drain	column	$340	--	yes
Three-bay sink	column	$610	--	yes
Exhaust fan	polyp	$180	45 W	--
Tankless water heater	polyp	$520	1,800 W	yes
Solar and battery	polyp	$1,650	--	--
Vented grease hood	polyp	$940	300 W	--
Advert controller	polyp	$690	--	--
Dehumidifier	polyp	$330	300 W	--
Cooktop	panel	$260	1,500 W	--
Fridge	panel	$430	60 W	--
Interior screen	panel	$380	55 W	--
Serving hatch	panel	$780	--	--
Prep counter	panel	$520	--	yes
Backlit advert panel	panel	$118	18 W	--
Roller shutter door	panel	$860	--	--
Bunker liner and collar	panel	$2,100	--	--
Sauna heater and benches	panel	$1,250	4,500 W	--
Grey water tank	floor	$290	--	--
Shelving bay	floor	$145	--	--
Tree saddle set	floor	$1,450	--	--
Access stair	floor	$620	--	--
Tub liner, pump and heater	floor	$2,400	1,500 W	yes
All 25 modules, once each		**$18,723**		

Five kinds of mounting point hold the whole catalogue: the apex, the utility column, a utility panel, a bay panel and the floor. Once those interfaces are fixed, a building's use is a shopping list rather than a redesign.

Every module, by mounting point and price.

From Backlit advert panel to Tub liner, pump and heater

The catalogue runs from 118 to 2,400, and buying every module once would cost 18,723 — which is more than the building, and exactly the point. Nobody buys the catalogue. A building buys the two or three things it is for, which is why the catalogue is priced per module rather than as packages with names.

Three notes on reading the prices.

They are delivered prices, not installed ones. A module is a part; the labour to fit it is your afternoon, counted in the hours chapter's terms rather than in dollars here.

The draw is part of the price. The heaviest electrical load in the catalogue is Sauna heater and benches, at 4,500 watts, and a sauna or a tankless heater changes what the electrical panel and the battery bank have to be. A cheap module that needs a bigger system is not a cheap module.

Water implies drain. Every module that wants water also wants a drain, and both arrive on the column. That is a design rule rather than a catalogue rule: nothing in this building takes water anywhere that cannot also take it away, which is the difference between a wet room and a rot problem.

15 seeds, one body

47 module slots across 15 named seeds. A seed is a complete shopping list on the standard frame: a name, a shape, the modules it carries, and the panels it swaps for bay windows, skylights, louvres and doors. The fullest is the Home seed, at 8 modules.

Fifteen seeds, one body

seed	shell	modules	panels	bay panels	first module
Stem cell	hemisphere	0	1	0	the bare article
Gym seed	hemisphere	1	1	20	exhaust fan
Studio seed	hemisphere	2	1	18	2 modules
Nursery seed	hemisphere	1	1	15	exhaust fan
Guest house seed	hemisphere	6	2	12	6 modules
Workshop seed	hemisphere	2	2	18	2 modules
Garage seed	hemisphere	1	1	14	solar battery
Home seed	hemisphere	8	2	12	8 modules
Sauna seed	hemisphere	2	1	3	2 modules
Food seed	hemisphere	7	3	11	7 modules
Advertiser seed	hemisphere	3	2	0	3 modules
Cold storage seed	hemisphere	3	1	3	3 modules
Bunker seed	buried	4	1	0	4 modules
Treehouse seed	treed	4	1	0	4 modules
Jacuzzi seed	inverted	3	2	0	3 modules

Every seed is the same solved dome. The bunker buries it, the jacuzzi turns it over and uses the shell as the vessel, the treehouse hangs it in a trunk, and the advertiser lights every one of its forty triangles and sells them.

The seeds, side by side.

The list is worth reading as a measure of what one frame can be:

Household. The home seed: shower, sink, grey tank, cooktop, fridge, tankless heater, camera ring, interior screen. A starter domicile, which is what this book has been building towards since the first chapter. Guest house for visitors; nursery for a room that attaches to a house and then becomes something else when it is outgrown; studio and gym as outbuildings for the things that do not fit indoors.

Working. Workshop, garage, cold storage. Light, air and power in a shell where skylights do the work a strip light would, because on a dome the upper ring *is* the roof. Food service — hatch, hood, three bays of sink and a cold box — which is a food truck's fit-out in a building that never moves.

Commercial. The advertiser seed is the blunt one: forty lit advert panels, one per triangle, controller included. Park it beside a road and the shell is the product. Every one of the forty triangles faces somewhere, and the catalogue is honest about what that is worth.

The strange ones, which are where the argument gets interesting. The Bunker seed buries the dome and leaves one riser above grade: the same frame, the same port, the shell now doing what a basement does. The treehouse hangs it in a trunk on a saddle. The jacuzzi turns the dome over and uses the shell as the vessel and the frame as its cradle. In each case the interface did the work and the building did not change — which is the payoff for having fixed the interfaces in the first place.

What the catalogue admits

The bare standard article is the stem cell: frame, floor, shell, column, seal cap, and one blank utility panel. Everything a reader recognises as a building — a kitchen, a bedroom, a workshop, a shop — is an add-on, and the catalogue says so in its own prices.

That is an unusual thing for a product catalogue to admit, and it is worth naming why the book prints it. A building whose use is a module is a building that can change its mind. The chapter on the round room said the same thing from the inside: headroom and floor area decide what furniture fits, and furniture is what a room is *for*. The catalogue is that idea, priced: the frame is the permanent decision, and almost everything else is reversible.

Which brings the book back to where it started. Three trees, forty panels, one hundred and twenty members, forty hours or less — and then a lifetime of shopping lists, in the best sense: a building you can keep changing without ever taking apart the part that was hard to make.

CHAPTER 87 · EXPLAIN

Sixteen Billion Shells

Every building the menus can make, counted from the menus themselves -- and the honest deduction that half of them are the same building

The Dome Creator offers 13 menus and 4 dials. Multiply them honestly and the thing you have is not a catalogue at all: it is a space so large that “which one should I build” stops being a search and becomes an argument.

This chapter counts that space, off the tool’s own menu code rather than off a list kept in a book — the same discipline as every table in the book: read the code, print what it says, and let a reader run it. And then it does the part that almost nobody does, which is the deduction: a great many of those combinations are the same building twice, and the honest count is smaller than the impressive one. Both numbers are printed here, which is the point.

The menus, and how many options each one has

11 menus change the building you look at, and 2 change what the floor is for. Between them they carry 98 options.

Every menu, and what it multiplies by

menu	group	options	running total
how many times each icosahedron face is divided	shell	4	4
how the sticks meet each other	shell	6	24
pucks or bracket plates, where there are hubs at all	shell	2	48
the cross section every member is cut to	shell	8	384
what the frame is made of, with its density and price	shell	8	3,072
finish on the frame	shell	8	24,576
what fills the triangles between the struts	shell	16	393,216
tint over a colourable panel	shell	8	3,145,728
one of three stacked layers over the sheathing	shell	9 x 3 slots	2,293,235,712
what the dome stands on	shell	7	16,052,649,984
which way a split log's bark faces	shell	2	32,105,299,968
markings, low walls, full walls, or nothing	fitout	4	128,421,199,872
what each of the ten floor sections is for	fitout	16 x 10 settings	41,208,602,512,209,762,844,672
Shell combinations			**32,105,299,968**
Floor layouts		**4 x 16^10**	**4,398,046,511,104**
Dial settings		**4 dials, stepped**	**446,823**
Distinct shells after the deduction			**16,554,295,296**

Read from the tool's own menu code, not from a list kept in the book. The running total is the honest product: 32,105,299,968 shells before the deduction, 16,554,295,296 after it, and 48.4 per cent of the menu doing

nothing at all.

Every menu, and how many options it carries.

The shell menus are the ones this book has spent its length on, and the list is worth reading as an inventory of everything a reader has already decided by the time they open the tool:

- **Frequency**, 4 options — how many times each icosahedron face is divided. This is the one that changes the part count, and it is the only menu that does.
- **Frame style**, 6 ways the sticks can meet, from hub and strut through hubless doubled to poured concrete formwork.
- **Hub style**, 2 — and note that this one does nothing at all on a frame that has no hubs, which is where the deduction starts.
- **Strut shape**, 8 cross-sections; **frame material**, 8 materials; **frame colour**, a finish that changes nothing structural.
- **Panel type**, 16 types, from open to precast concrete; **panel colour**; and the **cladding layer**, which is not one choice but 3 stacked slots.
- **Foundation**, 7 kinds, from bare ground to a treehouse platform.
- **Wedge curve**, two ways round — the rotation this book gave a whole chapter to, reduced here to a single switch that changes which face the weather lands on.

And the two fit-out menus: **partitions**, and the **room type** assigned to each of the 10 floor sections — which is where the four-trillion number comes from, because ten sections each taking 16 values multiply out very fast.

Multiply it honestly

The shell space is 32,105,299,968 combinations. Not a typo, not a rounding: multiply every shell menu out, counting the cladding as the 3 stacked slots it actually is, and that is the number.

On top of the shell, the floor layout gives 4,398,046,511,104 arrangements. And the dials — radius, strut width, recess depth and foundation size — are continuous in principle and countable in practice, because the tool moves them in fixed steps: 446,823 settings between them. That the tool quantises its own sliders is what makes this chapter possible: a continuum cannot be counted, and a tool that steps its dials has already turned the continuum into a list.

Three numbers, then, and they multiply:

```
32,105,299,968  shell combinations
x 4,398,046,511,104  floor layouts
x 446,823  dial settings
```

Which is a number too large to be useful, and that is the honest conclusion of this section rather than a failure of it. A space that size does not need a catalogue. It needs a *decision procedure*, and the rest of this book is one: choose the frequency that fits your trees, the material your land gives you, the foundation your ground will take, and let the arithmetic of the previous chapters fix everything else. The tool exists to show you what you chose, not to choose for you.

The deduction that matters

Now the part that most product catalogues would leave out.

A large fraction of those combinations change nothing you can see. A hub style on a frame that has no hubs. A strut cross-section under a different material that overrides it anyway. A panel colour on a panel that is defined as open. The tool's own arithmetic notices, counts the combinations that are genuinely a different building, and gets 16,554,295,296.

That is 48.4 per cent of the shell menu doing nothing at all — very nearly half. It is not a bug and it is not a scandal; it is what happens when you offer a colour for a thing that has no surface, and the honest response is to count it, print it, and let the reader see which menus are decoration. A catalogue that quietly reported the bigger number would be selling browsing rather than building.

Two lessons for a reader who is actually building something:

Count the decisions, not the options. Four menus matter to this book's frame — frequency, material, section, foundation — and every other menu is a finish or a fit-out. That is why the earlier chapters spent their length on those four and treated colour as a paragraph.

A menu can lie about its own size. 16 panel types sounds like sixteen buildings. It is one building and sixteen surfaces, and the chapter on skins said so in its own terms: the shape is fixed, the skin is chosen.

Why the book does not describe sixteen billion buildings

At 3 seconds each, showing every distinct shell would take 1,574 years of film.

That number is worth sitting with, because this project has made films — a great many of them, over a long time — and the honest total of what has been shown on screen is a rounding error against it. Which is exactly why the films are organised the way they are: not a walkthrough of a catalogue, but a handful of decisions, each demonstrated once, so that a viewer can rebuild any of the 16,554,295,296 themselves from a small number of rules.

Three points for a reader deciding what to build:

1. **A space this large is an argument against shopping and for deciding.** Nobody can browse sixteen billion options. Everybody can answer four questions about their own land.
2. **The menus that cost nothing are the ones to change your mind about.** Frame colour, panel colour and the wedge curve cost nothing and can be reconsidered on the day; the material, the section and the foundation cannot.
3. **The three expensive menus are the ones this book already chose.** The earlier chapters did the collapsing — one frequency, one section, one foundation, one frame whose parts list does not grow with the floor — and this chapter is the arithmetic that shows why that was a kindness rather than a limitation.

What the count is for

Not to impress anybody with a number. The purpose of counting the space is to show *where the decisions are*, which is the one thing a catalogue can never do.

A reader who knows there are 16 panel types and exactly one geometry can stop looking for a better shape and start choosing a skin. A reader who knows the hub style is inert on a hubless frame stops agonising over hubs. A reader who knows that half the menu is decoration stops treating the tool as an authority and starts treating it as a calculator — which is what it is, and what this book has been all along.

The dome this book builds is one shell out of 16,554,295,296. It is not the only good one. It is the one that three trees and a small chainsaw can make, and being able to say that — as an arithmetic result rather than a preference — is the whole reason there is a count at all.

CHAPTER 88 · REFERENCE

The Tried and True Dome

The popular three-frequency panel dome, solved by this book's own code and set beside the wedge dome for comparison

One of the most popular small domes people build is a three-frequency shell, cut five-eighths of the way down, framed as panels of sawn sticks on a jig. It is tried and true: a great many of them are standing, as cabins, studios, greenhouses and guest rooms. That makes it the best comparison this book's method has, and this chapter sets it beside the wedge dome.

Nothing here is copied from anybody's plans. The dome is solved by this project's own code (`panel_method.py`) at a 16 foot base, a size chosen for this book, and the drawings are this book's own.

The dome, solved

A published 3V plan set, read as data

what	how much	note
The design	4,695 mm across	15.4 ft, three-frequency
Panels, hexagons	73	4 of them glazed
Panels, pentagons	30	the twelve-corner geometry's five-sided faces
Panels, assembled	103	against the solver's 105 for a hemisphere
Saw setting, hexagon long edge	31.4 deg	off square
Saw setting, hexagon short edge	29.3 deg	off square
Saw setting, pentagon	19.3 deg	off square
Panel strut	32 x 38 mm	sawn stock, not a split wedge
Base polygon, inside to inside	4,507 mm	
Base polygon, across points	4,570 mm	63 mm wider than its flats
Sheet goods	20 of 4x8	or 14 of 4x12
Cover	65 sq m of film	greenhouse plastic, taped at the laps
Framing timber	27 three-metre lengths	plus sheet goods and the door set
Door	832 x 1,840 mm	framed into two half hexagons
Floor enclosed	17.3 sq m	at the plan set's own diameter

Every figure here is a declared input taken from a plan set published by other people (Golden Trillium Geodesics, 2020) and is named as theirs. The drawings in this book are the book's own; this table is the data in them, restated so a reader can compare it with the method this book teaches.

reader can compare it with the method this book teaches.

The popular 3V dome, solved.

Read as a census, that is the whole design: a 4.88 metre shell of **105 panels** – 75 that belong to hexagons and 30 that belong to the twelve pentagons – standing 2.94 metres tall, its height 0.594 of its width. Every panel is one of two triangles, and every stick is a 38 by 30 millimetre half of a ripped 2x3.

Three figures in that table decide the work.

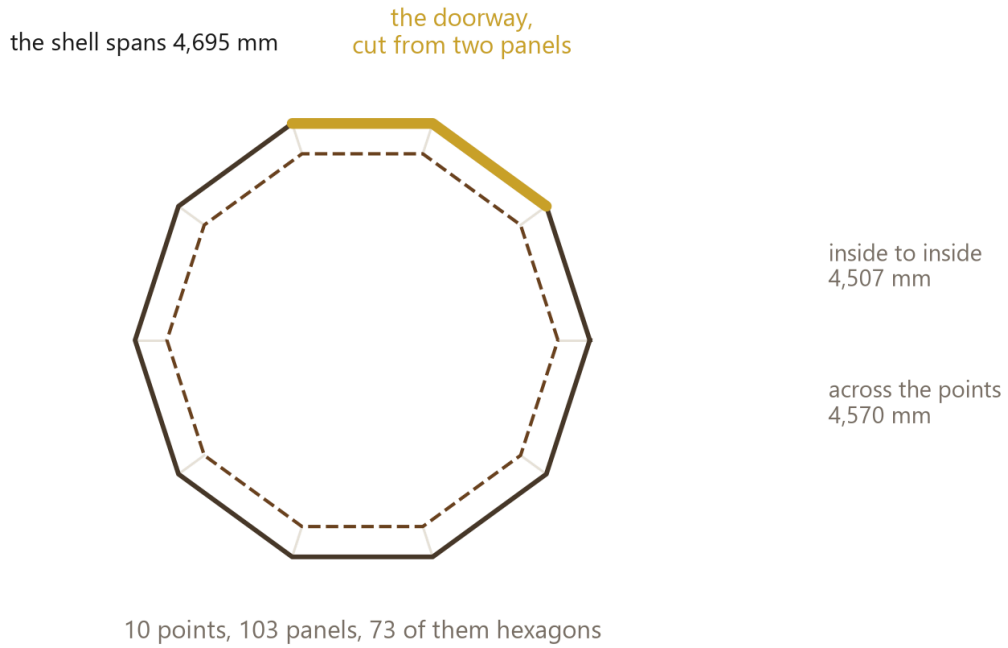
The sticks are pinwheeled, so equal sides are not equal sticks. Each stick runs to one corner and butts the next at the other. That gives the dome 6 distinct sticks, not three – 315 in all – with tail cuts of 29.3 and 31.4 degrees on the hexagon panels and 19.3 and 35.4 on the pentagon panels.

One bevel cannot be exact. The panels turn by different amounts where they meet, so the exact bevel on a stick runs from 5.7 to 7.2 degrees. A shop rips every stick at one setting, and a bead of sealant or a gasket between frames takes up the difference.

And the skin goes on in sections. 6 pentagons and 10 hexagons come out whole at this cut, each skinned in one sheet, so a torn section is one section to replace.

A base that zig-zags

The base polygon, and where the door goes



Ten sides, because a three-frequency split of an icosahedron lands its lowest ring on ten points. The published plan set measures it two ways -- a shade over four and a half metres inside to inside, and sixty-odd millimetres wider across the points -- and frames a doorway into two half hexagons rather than cutting a hole in one panel. This drawing is the book's own; the dimensions are the plan set's and are named as theirs.

Fifteen base edges on two heights.

The geometry worth pausing on is the base. A three-frequency sphere has no level ring of points at its middle, so the cut goes lower, and the bottom edge it leaves – 15 edges – does not lie on one level. Its corners alternate between two heights 40 millimetres apart. So **10 base sections are tapered** and 5 sit level.

The census checks itself: the cut keeps 105 faces, 165 edges and 61 points, and the whole sphere has only 3 chord lengths. The pentagons are where the icosahedron’s twelve corners were, as they are in every geodesic dome.

What this method buys and what it costs

The honest comparison, in the terms this book has used throughout.

What it buys. A method you can start tomorrow. The stock is at the yard, the tool list is a table saw, a miter saw, a drill and clamps, and the panels are built indoors at a bench, finished and skinned before they leave the shop. A first-time builder can make a dry 18.1 square metre room with it. That is why so many have.

What it costs. The wood is bought, so this book's whole recovery argument does not apply: every stick is sawn, dried, graded and delivered. There are many more pieces – 315 sticks against the wedge dome's members – and more kinds of them. And a film-covered shell is a greenhouse in July unless somebody details it.

And what it shares with this book's method. The panel is the unit of work, the jig holds the geometry so the builder does not have to, the angles live in the parts rather than in a hub, and the frame is allowed to be rough because the skin closes it. A reader who has understood those four sentences has understood the important part of either method.

What a reader may take from it

Take the jig discipline. Drawing the panel full size from three lengths, and letting rails hold every stick on its line, is the same idea as this book's own jig, applied to sawn stock.

Take the sections. Skinning a whole pentagon or hexagon in one sheet is a good habit wherever a skin goes on in pieces.

Take the loose assembly. A shell screwed together loosely until the last panel closes it, then tightened, is easier to build and truer when done.

And keep this book's method where it differs. The wedge dome starts from trees, carries its angles in the shape of the stick, and has fewer, bigger pieces. The tried-and-true dome starts from a lumber yard. Both end in a dome; choose by what you have.

CHAPTER 89 · HOWTO

The Non-Beveled Method

How a zome is actually framed: two customary methods, one of which needs no compound cuts at all, and the sequence that makes it work

The zome chapter described the shape. This chapter is the *method*: how a working shop actually frames one, and why the version that looks lazier is the better one. The example is this book's own nine-sided zome, solved by `panel_method.py`: 36 panels in 4 rows of 9, 81 struts and 46 corners, every panel side 35.4 inches.

Two ways to frame a zome, weighed

Two ways to frame a zome, weighed

at the saw and at the jointnon-beveled		beveled
Cutting	miters only, one per panel type	a mitre and a bevel on every strut
Jigs	none	one per panel type
Compound angles	none	many, each with its own orientation
Stock	any depth	limited by the saw's cutting depth
Joint	reciprocal overlap, fastened through	beveled faces closed against each other
Outside face	small gaps, wider with deeper stock	closed, flush
Interior face	identical to the beveled method	identical to the non-beveled method
Stability	needs a centre strut, a wedge block or a cover	the bevel closes the frame in itself
Above 20 ft	beveled blocks advised between panels	not required
Published advice	recommended for all builds	not recommended; no significant gain

Restated from a published zome method (Trillium Domes, 2025), in this book's own words and order. The recommendation is theirs, quoted as theirs; the reason it holds is in the second and third rows, which are arithmetic rather than preference.

The two methods, weighed.

The two customary methods differ in one decision: whether the struts are beveled where they meet, or simply mitered and overlapped.

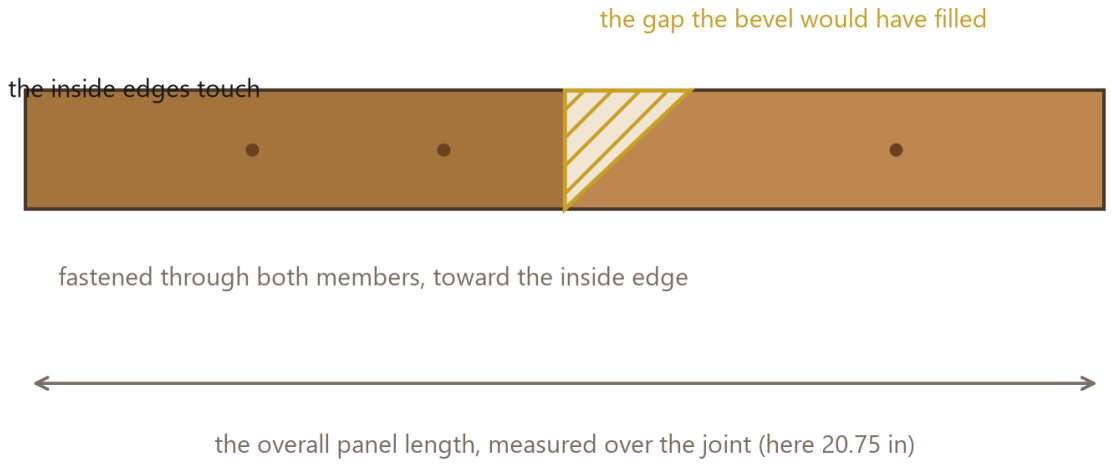
The non-beveled method leaves the members square in section. Two struts meet at a miter, overlap, and are fastened through one another — a reciprocal joint. Because no bevel is cut, the member keeps its full depth at the joint, the saw work is one miter per panel type, and no jig is needed beyond a stop block. The cost appears on the outside face: where the two members rotate into their panel angle, their outer edges stand apart by the bevel that was never cut, leaving a small gap that widens with the depth of the member.

The beveled method rips the bevel onto one side of every strut so the members close against each other. The outside face comes out flush and the joint is closed in itself, at the price of a compound cut at every intersection with its own orientation, a jig per panel type, and a member depth limited by what the saw can cut.

Builders who have framed zomes both ways overwhelmingly choose the non-beveled method, and the arithmetic supports them: the difficulty of the beveled method buys a flush outside face and nothing else, while its depth limit costs strength. The second and third rows of that table are the whole argument, and they are not a matter of taste.

The reciprocal joint

The reciprocal joint



Two struts overlap at a miter and are fastened through each other. The panel's overall length is what gets measured -- not a cut list of angles -- so the saw is set once per panel type and every strut of that type is cut to the same stop. Because no bevel is cut, the full depth of the member survives and the joint leaves a small gap on the outside face, which is why the published method adds beveled blocks between panels above twenty feet.

The reciprocal joint, and its outside gap.

The joint is the method, so it is worth describing precisely. Two struts are mitered at their ends, laid so they overlap, and fastened through both members with structural screws sized for the load.

What is *measured* is the panel's overall length — 35.4 inches on this book's zome — taken across the overlap rather than along one member. The angles are **scribed** rather than calculated: the first miter is cut on two struts, they are butted together on the bench, the overall length is marked from the plan, and the second miter is marked off that line. Then a stop block is set and every strut of that panel type comes off the saw identical.

Which is the whole reason the method is fast. A cut list of angles is replaced by one measurement and one stop block per panel type; the fabricator is not solving a compound angle, they are copying a mark.

The gap that result leaves is worth understanding rather than hiding, because its size is predictable: it is the bevel the joint is missing, twice the depth of the member times the tangent of half the fold, so it grows with the depth. In a 3.5 inch deep member on this book's zome it opens from 1.19 to 1.73 inches. On a small zome that is a seam a covering laps over without noticing; on a large one, bevelled blocks between the panels fill it.

The sequence, as a shop would run it

In the order this book would teach it, with the reasoning attached to each:

1. **Clear the bench and rip the stock.** Rip first, because every later measurement is taken from a face. Rough-cut to the panel length if it helps; the joints will trim a little off each piece anyway.
2. **Cut the first miters.** Two struts, one panel type, the angle set with a digital finder. If the saw will not reach the angle, an auxiliary fence bolted to the bench gives the rest of it — the same remedy this book's cut chapter recommends when a miter saw runs out of scale.
3. **Mark the overall length over the joint.** Struts butted and overlapping, the plan's panel length marked from the correct reference, never from the end of one member.
4. **Scribe the second miter.** Long point to short point, on the mark, with the angle finder.
5. **Cut to a stop block.** Set the stop once and cut every strut of that type the same way, front to back, out of full-length stock.
6. **Check the overall dimension before cutting the batch.** Assemble two struts, measure the panel length, adjust the stop, recut. This check comes *before* the pile is cut – the same discipline as this book's dry-fit-one-triangle rule.
7. **Build pairs, then halves, then panels.** Pairs make panels; partial panels need halves, and a half is made by cutting the middle strut and lining it up with the corners of a full panel to mark it.
8. **Add the middle strut to each panel.** This is the step that turns a parallelogram into two triangles. It is optional once a structural covering is on, and not before.
9. **Assemble with the joinery loose until it sets itself.** Two fasteners per strut, placed toward the inside edge, not overtightened: the panels are meant to hinge a little as they are lined up, because the bevel angle between panels then sets itself. Bevel blocks, where they are used, act as the gauge.
10. **Then go back and finish the fastening.** Once everything is closed, the rest of the screws go in, evenly along every strut.

Two of those steps are the method's real content, and both are habits rather than tricks: **measure the assembly rather than the part** (step 3), and **leave the frame loose until the geometry closes it** (step 9). A builder who understands why those two work can frame a zome from any set of panel dimensions, which is a more transferable possession than any particular plan.

The stability the shape does not have

A zome's panels are parallelograms, and a parallelogram is a mechanism: it changes shape without any of its sides changing length. This book's rigidity argument — the one it makes about triangles on its second page — is exactly why a zome is *less* stable than a geodesic dome of the same size.

The remedies are the ones the geometry implies:

A centre strut across each panel. One member from corner to corner converts the parallelogram into two triangles and the mechanism into a frame. It costs a strut per panel and adds nothing to the shell's area.

Or a diaphragm. A structural covering — plywood, polycarbonate, even a tightly laced fabric — does the same job and is often cheaper than the strut, which is why either will do. The cover is part of the structure in a zome, not decoration on it, and a reader who takes one thing from this chapter should take that.

And bevel blocks on a big zome. The blocks that close the outside gap also brace the joint, so the detail that is cosmetic on a small zome becomes structural on a large one. That is a rare and satisfying thing in building: the same small piece of wood doing two jobs at once.

Covering, and the one rule

Covering a zome reduces to a single rule that matters more than the material: **every panel laps over the one below it.** A zome's panels share a consistent up-and-down orientation, so a shell can be shingled with almost anything — rigid polycarbonate overlapping at each seam, sheet metal, membrane, or conventional roofing over a plywood deck. A conventional assembly runs framing, deck, flashing, underlayment, roofing; an insulated one can have its gaps between panels foamed, which air-seals and insulates in the same pass.

Read against this book's weather chapter, the zome is the easier shell: no seam key, no channel, no fold angle varying across the building. Every panel edge is the same lap detail at the same angle, all the way up. That is a genuine advantage, and it is why a zome takes an opening as a whole panel left out and framed, with the lap continuing around it.

Where this leaves the book's own method

Two variation chapters, one conclusion, stated without favouritism.

A zome makes a better room. Its walls are vertical where a dome's are not, its parallelograms take furniture and windows and doors the way a wall should, its panels are flat so a covering laps without geometry fights, and its hubs land on level rings by construction.

A dome makes a better shell. Its triangles are rigid without a centre strut or a diaphragm, its struts are one length and one miter for the whole building, and its load path is closed by geometry rather than by the covering.

Which one a reader wants depends on which half of that sentence matters more, and the honest answer for most small buildings is that a zome is the easier first build while this book's wedge dome is the better second one. The method in this book exists because its author had trees; the method in this chapter exists because its author had a saw and a yard. Both are legitimate reasons to start, and neither is a reason to sneer at the other.

That is the whole of what this pair of chapters has to say about other ways of doing it. The rest of this book is one method, taught properly, with its numbers computed. These pages are the map of where it sits.

CHAPTER 90 · REFERENCE

One Sheet, Forty Panels

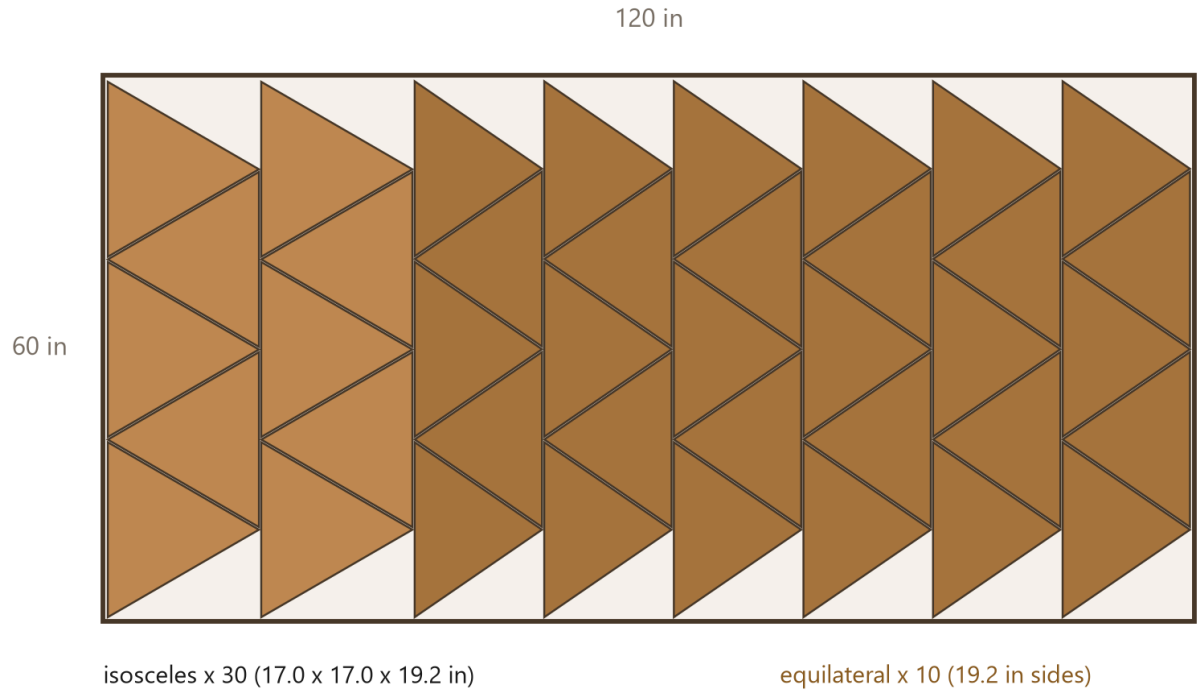
What one 120 x 60 inch sheet of plate becomes when a plasma torches the whole shell's panels out of it: a 5.18 foot vessel, forty panels, 237 gallons, and one afternoon of welding

Every chapter before this one has treated the frame as the building and the skin as something bought. This chapter reverses the habit: the panels are the structure, the sheet is the material, and the cutting table turns one into the other.

It works because the arithmetic of the shell does not change. Forty panels is forty panels whether they are wedge-cut from a tree or torched from a sheet: thirty isosceles triangles and ten equilateral ones, the same two frequencies making the same hemisphere. What changes is the material and the joint – a welded butt instead of a seam key, a sheet instead of a tree – and it happens that the sheet is the same size as the table’s bed.

The nest

One sheet, forty panels



A 120 by 60 inch sheet of plate, nested with the 40 panels of a 5.18 foot shell, in 8 strips of triangles pointing alternately up and down. 78 per cent of the plate is inside a part; every part is at least 0.25 in from its neighbours and 0.5 in from the edge, checked part by part.

One sheet, forty panels.

The nest is this book's own computation, and it is built the way a person with soapstone and a straight edge would lay a sheet out: in **strips** of triangles pointing alternately up and down. A strip of alternating triangles tiles with nothing between neighbours but the cutting gap, so the only waste is at the ends of the strips and above the shorter ones.

Every panel is slid along its strip until the real distance to its neighbour is exactly the gap – not a guess at the spacing, the distance itself – and when the sheet is full the layout is checked part by part: inside the margin, and no two parts closer than the gap. The panels come from the solved dome, not from a table of factors, so a change to the geometry changes the nest.

At the largest size that passes, the numbers are:

the sheet	
Sheet	120 x 60 inches
Panels cut from it	40
Long side	19.2 inches
Short side	17.0 inches
Strips	8
Plate inside a part	78.1 per cent
Torch path	181 feet, about 24 minutes
Parts, in 3 mm steel	188 lb
The whole sheet weighs	241 lb
Offcut	53 lb

That is the honest reading of *the largest forty panels one sheet can carry*: a shell **5.18 feet across and 2.59 feet high**, with 19.7 square feet inside its base and – welded into a bowl – 237 gallons. Floor and volume are measured off the faceted shell itself, not off a sphere, which would flatter both.

Almost fits

The one way to go bigger from the same sheet is to let each panel's point be cut back a little. The trimmed points leave a small hole where five or six panels meet, and a round cap welded over each vertex closes it. On this sheet the best trimmed nest needs only 0.30 inches off each point and 16 caps at least 0.40 inches across – and buys a shell **1.4 per cent** bigger, 5.25 feet across. That is a real option and a small one: the strips are already full, and the sheet's own area is not far off. Most shops will cut the panels whole.

Three caveats, because a nest is a claim about cutting.

It is a bench layout, not a nesting program. Nesting software that lets strips interlock in two dimensions can do better. The number here is what a person can lay out by hand, which is the standard this book holds itself to.

Kerf and gap are allowances. The model leaves a quarter-inch gap between parts and half an inch at the sheet's edge. A plasma torch's kerf depends on nozzle, amperage and plate; measure your own on scrap and change the gap.

Thickness is a decision. The table above is 3 mm steel. The same nest in 2 mm weighs 126 lb of parts: light enough to lift a panel alone, thin enough that anything big wants ribs.

What the sheet becomes

Four things one sheet makes

plate	panels weigh	the sheet	left over	what it is for
1.5 mm plate	94 lb of panels	121 lb sheet	26 lb offcut	a tub liner, upside down or right way up
2.0 mm plate	126 lb of panels	161 lb sheet	35 lb offcut	a seat shell, a planter, a fire bowl
3.0 mm plate	188 lb of panels	241 lb sheet	53 lb offcut	a wood burner's body and baffle
5.0 mm plate	314 lb of panels	402 lb sheet	88 lb offcut	a stove that outlives its owner
the shell itself	**5.18 ft across**	**2.59 ft high**	**237 gal**	**19.7 sq ft of floor upside down**

The same forty panels, cut from one sheet and welded at 55 seams. Weight decides which thickness is honest for which job: a tub liner at 1.5 mm, a seat or planter at 2 mm, a wood burner's body at 3 mm, and a stove you intend to leave to somebody at 5 mm. At this size it is not a shelter: it stands 2.59 ft tall.

Four things one sheet makes.

The same forty panels become different objects according to the plate they are cut from, and the weight column decides which is honest.

A tub. In thin plate the shell stands as a bowl 5.18 feet across and 2.59 feet deep, holding 237 gallons. Turned over, the same shell is a hood, a hide, a kiln cover or a chicken tractor, because a hemisphere does not care which way up it is used.

A wood burner. In 3 mm plate the shell is a stove body: a flat panel on hinges for the door, a baffle inside, and the flue leaving the apex, where the heat already wants to go. Its seams are welded joints in a hot part, so they want full penetration and a test fire before anyone relies on it.

A chair, a planter, a fire bowl. The small end of the same idea: one sheet, forty triangles, an afternoon of welding, and an object nobody would guess was a building component. This is what to make while learning the machine – which is why the first thing the table cuts is a vessel rather than a building.

Not a storm shelter – not at one sheet. A shell 2.59 feet tall holds nobody upright. A shelter is the next section's size, cut from several sheets.

The welding, and the honest limit

The vessel is not free: 55 seams, 82 feet of weld and about 2.1 hours at a comfortable hand speed. Each seam is a butt weld, ground flush on the face that shows and left proud on the face that does not. The book would rather print the hours than pretend a plasma table welds anything.

Bigger domes, in plate

The same table cuts the triangles of a building. A three-frequency dome cut five-eighths of the way down has two panel shapes, and the same strip nest tells you how many sheets it takes. For a 16.0-foot 3V dome: **23 sheets**, only 41 per cent of them in parts, 1,522 lb of steel in 2 mm. The waste has one cause – the big panels are too tall for two strips to share a five-foot sheet.

So size the dome to the sheet. Below a certain size two strips of the big panels fit, and the count drops sharply: the largest whole-foot 3V dome that nests well is **13 feet**, in **9 sheets** at 69 per cent, 1,005 lb of steel in 2 mm. One foot bigger, 14 feet, takes 16. Past that, a panel is cut in two along its height and welded on the bench, with the joints staggered so no two run in a line – which is how large plate domes are built.

Which is the chapter's claim, stated plainly: **the frame does not care what it is made of, and the shop does not care what the frame is drawn from.** Split wedges, sawn members and welded plate are three materials for one geometry, and the cutting table is what makes the third available at a size one person can handle.

The Shop the Campaign Buys

The machine those lines pay for, built rather than bought: a 10 x 5 foot table for 4,151 dollars, and the router conversion that shares its frame

The campaign's goal has two machine lines in it: 11,800 for a CNC plasma table and 14,500 for a CNC router. Both are honest numbers for *buying* those machines at the size this work needs.

This chapter is the other option, and it is the one a shop with a welder should take: build the table from a published set of plans, then put a router head on the same frame. The arithmetic lands at 4,151 for the plasma table and 990 more for the router head – 5,141 for both machines, against the two lines' 11,800 and 14,500.

The design is the Homemade Madness DIY CNC plasma table, whose plans and cut files this shop is working from, and the credit for it is theirs. The plans are drawn for a two-by-one-metre sheet and say the machine can be built as big as you like by two rules: the base frame is 250 millimetres longer and wider than the largest sheet, and the gantry tube is 253 millimetres longer than the cut width. Everything in this chapter is those two rules applied to a 10 x 5 foot sheet, and then counted.

The table, at ten feet

The machine's bill, line by line

part	quantity	each	line
Gear rack, mod 1.5, 15 x 15 x 500 mm (paid)	19 lengths	$29.39	$558.41
Pinion, mod 1.5, 15 T, 10 mm bore (paid)	4 gears	$12.23	$48.92
HGR20 rail kit, 2 x 1500 mm + 4 HGH20CA blocks (paid)	5 kits	$109.99	$549.95
Stepper, NEMA 23, 3 Nm, 10 mm shaft (quoted)	4 motors	$48.39	$193.56
Stepper driver, DM556T (quoted)	4 drivers	$33.08	$132.32
Power supply, 48 V 10 A (quoted)	1 supply	$46.99	$46.99
Mach3 licence (quoted)	1 licence	$218.73	$218.73
Plasma cutter, 60 A pilot arc, CNC port (quoted)	1 unit	$699.00	$699.00
Square tube, 2 x 2 x 11 ga, 12 ft (quoted)	7 sticks	$51.98	$363.86
Breakout board (the guide's red board) + relay for the torch (allowance)	1 set	$45.00	$45.00
Ballscrew SFU1605, 300 mm, with nut and end supports (allowance)	1 set	$40.00	$40.00
Rect. tube, 2 x 4 x 11 ga, gantry (allowance)	6 ft	$9.00	$54.00
Plate, 3/16 in (5 mm), gantry sides and carriage (allowance)	1 2 x 4 ft piece	$110.00	$110.00
Flat bar, 2 x 3/16 in, 20 ft, slats (allowance)	15 sticks	$38.00	$570.00
Sheet, 16 ga, 4 x 8 ft, water pan bottom (allowance)	3 sheets	$80.00	$240.00
Drag chain, cable, e-stop, connectors (allowance)	1 lot	$160.00	$160.00
Floating-head switch, Z coupler, mounts, bolts, paint (allowance)	1 lot	$120.00	$120.00
The plasma table, built			$4,150.74
Spindle, 2.2 kW water-cooled, with VFD and pump (allowance)	1 set	$260.00	$260.00
Z axis, ballscrew, 200 mm travel, rigid plate (allowance)	1 set	$180.00	$180.00
Bed: support grid + MDF spoilboard (allowance)	1 lot	$220.00	$220.00
Dust shoe and extraction hose (allowance)	1 set	$90.00	$90.00
Collets, end mills, v-bits (allowance)	1 set	$180.00	$180.00
Fifth driver and wiring for the spindle's control (allowance)	**1 lot**	**$60.00**	**$60.00**
The router conversion			**$990.00**
Both machines, one frame			**$5,140.74**

Each line says what kind of price it is. 'paid': the rack, the 15-tooth pinion and the HGR20 rail kits, at what this shop paid. 'quoted': a current retail listing (cnc_table.SOURCES names each). 'allowance': an estimate for a part

not yet priced -- a decision, labelled as one, to be replaced by a quote.

The machine's bill, line by line.

The frame. A base frame 3,298 x 1,774 millimetres, in the plans' 50 x 50 x 3 box section (2 x 2 x 11 gauge), with legs and a water pan the size of its inside. The plans' table is two metres long; this one is half again as long, so it gets a leg pair and a cross member at mid-length – the one change this book makes to the design, so no rail span goes unsupported for more than about 1.7 metres. A plasma table's accuracy is mostly its frame: square, flat, parallel, and checked by its diagonals before anything is bolted to it.

The gantry. One 1,777 millimetre length of 100 x 50 x 3 tube between two 5 mm plate sides, which the plans give as a cut list with paper templates – cut them by hand, or have them cut. Two HGR20 rails on its face carry the torch carriage.

The rails. 10 rails in all – a joined line along each long side and two on the gantry – which is 5 of the two-rail kits already bought. The long sides take three rails each, joined end to end on a straight edge. The plans are explicit that joined rails are fine and that alignment is everything: mount the top rail first, then set the second parallel to it with the moving carriage. The floating head's short rail is an offcut from the gantry.

The rack. Both long sides are driven, so both need rack: 7 half-metre lengths a side, four on the gantry, and a spare – **19 lengths**. Each joint is cut at exactly the right tooth and aligned with a short spare piece held across it. Racks sit 50 mm in from the leg on the machine's home side.

The pinions, and the number that has to be right. Four pinions: two on the long axis, one on the gantry, one spare. A 15-tooth mod 1.5 pinion has a 22.5 millimetre pitch diameter, so one motor turn moves an axis 70.69 millimetres – with eight microsteps on a two-hundred-step motor, **22.64 steps per millimetre**. That is the number Mach3 needs before anything is cut; get it wrong and every part comes out the wrong size in a way that looks like a drawing error. The plans prefer a 13-tooth pinion for torque and say plainly that whatever you find, you set the ratio in software. Fifteen teeth is what was available; the machine does not care.

The motors. The pinions bought have a 10 mm bore, so the motor's shaft must be 10 mm: a NEMA 23 of 3 newton-metres with a 10 mm shaft fits with no sleeve. (A NEMA 34's 14 mm shaft would not.) Two of them on the long axis can push the gantry – about 40 kilograms, estimated – at 6.7 metres per second squared using only half their holding torque, which is far more than a plasma, with no cutting force at all, ever asks for.

The control and the torch. Four DM556-class drivers, a 48-volt supply, the plans' breakout board with a relay on the torch trigger, a Mach3 licence, and a **pilot-arc** plasma with a CNC port: the plans warn that a high-frequency start puts noise into the control wiring. A torch height controller is left out on purpose – the plans' board cannot run one, and the alternative they describe, a floating head that touches off the plate again partway through a long cut, costs nothing.

What it costs, and what kind of number each price is

The bill keeps three kinds of price apart. **1,157 dollars** is what this shop actually paid for rack, pinions and rail kits. **1,654** is current retail listings, each named in the model. **1,339** is still allowances – estimates for steel, plate, the ballscrew and the wiring – labelled as allowances until they are quoted.

part	qty	each	total	basis
Gear rack, mod 1.5, 15 x 15 x 500 mm	19	29.39	558.41	paid
Pinion, mod 1.5, 15 T, 10 mm bore	4	12.23	48.92	paid
HGR20 rail kit, 2 x 1500 mm + 4 HGH20CA blocks	5	109.99	549.95	paid
Stepper, NEMA 23, 3 Nm, 10 mm shaft	4	48.39	193.56	quoted
Stepper driver, DM556T	4	33.08	132.32	quoted
Power supply, 48 V 10 A	1	46.99	46.99	quoted
Mach3 licence	1	218.73	218.73	quoted
Plasma cutter, 60 A pilot arc, CNC port	1	699.00	699.00	quoted
Square tube, 2 x 2 x 11 ga, 12 ft	7	51.98	363.86	quoted
Breakout board (the guide's red board) + relay for	1	45.00	45.00	allowance
Ballscrew SFU1605, 300 mm, with nut and end suppor	1	40.00	40.00	allowance
Rect. tube, 2 x 4 x 11 ga, gantry	6	9.00	54.00	allowance
Plate, 3/16 in (5 mm), gantry sides and carriage	1	110.00	110.00	allowance
Flat bar, 2 x 3/16 in, 20 ft, slats	15	38.00	570.00	allowance
Sheet, 16 ga, 4 x 8 ft, water pan bottom	3	80.00	240.00	allowance
Drag chain, cable, e-stop, connectors	1	160.00	160.00	allowance
Floating-head switch, Z coupler, mounts, bolts, pa	1	120.00	120.00	allowance
			----------	
the plasma table			4,150.74	

	budgeted	built
CNC plasma table	11,800	4,151
CNC router	14,500	990, as a head on the same frame
Both	**11,800 + 14,500**	**5,141**

Read as a shop decision rather than a saving, that difference is 21,159 dollars, and the campaign's own lines say where it could go: the year of weather on a standing dome, the engineer, the first quilters. Both choices are legitimate. The campaign's 11,800 buys a machine with support, a warranty and a phone number, which is worth real money to somebody who is not a machinist. A built table takes weeks of evenings, has no warranty, and its first troubleshooter is its owner.

The router head

990 on the same frame buys the machine this book's own method wants. Everything between the floor and the gantry is shared – the long axes, the rack, the rails, the motors, the control. What changes is what touches the work, and the bed under it:

- **A rigid Z** with a ballscrew in place of the floating head, because a router is asked to hold a few thousandths where a torch holds a sixteenth.
- **A spindle**, water-cooled, with its VFD – quieter and stiffer than a trim router, and able to run for an hour without stopping.

- **A flat bed.** The water pan's slats lift out, and a support grid with an MDF spoilboard goes in, skimmed flat by the machine's own cutter so the bed is true to the gantry rather than to a level.
- **Dust extraction**, because wood dust and plasma sparks must never share a shop at the same time.
- **Cutters:** end mills for the jigs, V-bits for marking, and collets.

Is the plasma frame stiff enough to rout? The gantry tube, simply supported over its length, sags less than a few hundredths of a millimetre under a 50 newton side load at its middle – less than a router's own runout. The weak point is the rack's backlash, a tenth of a millimetre or so: fine for jigs and panels in plywood, not for steel parts that fit like bearings.

What the machine makes possible

Put this chapter next to the one before it and the argument completes itself.

Steel vessels and steel shells. Forty panels from one sheet make a 5.18 foot vessel holding 237 gallons; a 13 foot 3V dome in welded plate takes 9 sheets. That is a product the campaign can sell, and it exists because a plasma table cuts plate the way a printer prints paper.

And the jigs for the wooden version. Every fixture this book describes – the bench jig, the stop blocks, the sled fence, the panel templates – is a flat part. A router cuts them accurately, repeatably, and in the time it takes to explain what they should look like. The machine that cuts the steel also makes the shop that cuts the wood.

The last word belongs to the plans the shop is working from, because their advice is the advice this book gives about building: start with the frame, make it straight, and expect the first machine to be a teacher rather than a product.

The Master Tables

Four tables carry the whole build, and they are the ones to come back to at the saw. Every number in them is generated at build time by the same solver that drew the 3-D model, so a table and a drawing can never disagree. Print these pages and take them to the bench.

Full cut list

The cut list

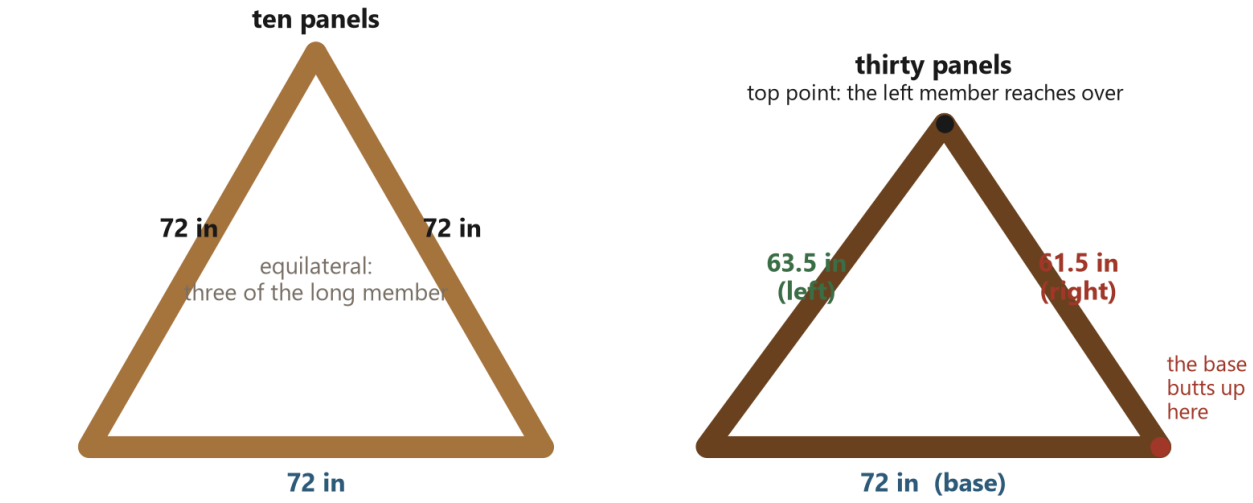
mark	class	count	length in	bearing in	inset in
M1	SHORT	30	62.731	4.103	7.950
M2	SHORT	30	63.267	4.640	7.413
M3	LONG	30	71.241	4.640	8.686
M4	LONG	30	72.000	4.419	7.927
	total	**120**	**8,077**		

Cut to the length column. The bearing column is how much of the end actually lands on the next stick's side; the inset is how far the member sits back from the panel's own corner.

The complete cut list.

The whole frame in one table: every member, by class, length and count. The frame has 120 members in 4 distinct lengths – two classes, short and long, each cut once per panel. The table is sorted the way the work is sorted: by class first, then by length, so a member on the pile can be found from the table in one look. Mark each class with its own colour at the saw, and keep the classes in separate stacks; the raise is where a mixed pile turns into a slow afternoon.

The master cut list



These are resultant lengths: cut to these and the panel is the size the dome wants. An equilateral panel is three of the long member. An isosceles panel, laid with its base nearest you, takes the long member as the base, the middle length on the left and the short one on the right -- the left member is the longer of the two and so reaches over to the top point, while the base butts up to the short right side. That single sentence is the bench rule; nothing about the pair is obvious by eye, so the sides are picked from their own stacks in order.

The master cut list, and which side it goes on.

The list the saw is set to, as this build finished it. 3 lengths, and every stick in the frame is one of them:

length	what it is	where it goes
72.0 in	the long member	all three members of an equilateral panel; the **base** of an isosceles panel
63.5 in	the middle member	the **left** side of an isosceles panel
61.5 in	the short member	the **right** side of an isosceles panel

These are resultant lengths: cut a member to one of them and the panel is the size the dome wants, with nothing left to trim.

The rule, stated once and followed at the bench: **shortest on the right, middle on the left, longest at the base**, laid with the base nearest you. The left member is the longer of the two sides, so it reaches over to the top point, which sits off the middle of the base; the right member meets it there, and the base butts up to that short right side. The two sides of an isosceles panel differ by two inches over five feet, so nothing about the pair is obvious by eye – pick them from their own stacks, in order, and the panel is right. Equilateral panels have no rule to remember: three long members.

The fully seated schedule the pinwheel model prints is in the table above. Where the two disagree the master list is the report of the build and the model is the model of the joint.

Full seam schedule

The seam schedule: how many of each key

key	edge	seams	fold deg	raw gap deg	length ft
K1	A	25	18.029	26.971	150.0
K2	B	30	22.459	22.541	159.2
	all	**55**			**309.2**

2 distinct keys close all 55 seams. Make one of each to fit, then repeat it. The raw gap column is the angle the key actually has to fill when both faces are left exactly as split. Fold plus raw gap is 45° at every one of them -- the sector angle. The key never fills more than one wedge's worth of angle. Cross-check: the pinwheel model reports 55 interior seams and 10 rim edges over 410 feet.

The complete seam schedule.

Every seam in the frame, with its fold angle and its key. The frame’s 55 seams fold at only two angles – 18.03 degrees, 25 of them, and 22.46 degrees, 30 of them – so the brake is set twice and the schedule is how you know which setting a given

seam wants. The key column is the gasket: 0.75 inches of compressible strip in every seam, which is what lets a frame of green timber close and stay closed while the wood moves.

Butt cut setups

Distinct butt-cut setups in the whole dome

#	members	bevel deg	mitre deg
1	-	22.50	34.43
2	-	22.50	21.14
3	-	22.50	30.00
	3 setups	**120 members**	

3 setups for 120 members. That ratio is the whole argument for the pinwheel: you change the saw a handful of times, not a hundred and twenty.

Every distinct setup.

Every distinct saw setup in the build, in the order the work meets them. The butt cut is the one compound angle in the frame, and this table is the one place its setups live: stock left long by 6 inches at the head end, allowance of 8 inches past the cut, and the angle itself. The rule from the cutting chapters applies here in table form: the angle lives in the tool, not in the operator. Set the jig once per row, run every member of that row, and never measure an angle at the saw.

Declared constants

Declared constants: inputs, not results

what	value	unit	kind
saw displacement cc	38.20	cc	measured
saw bar in	16.00	in	measured
saw price usd	199.00	usd	priced
chain kerf in	0.25	in	measured
two by four price usd	8.88	usd	priced
build days	14.00	days	decided
cutting afternoon struts	30.00	struts	measured
cutting afternoon hours	6.00	hours	measured
gasket thickness in	0.75	in	decided
head overfit in	6.00	in	decided
butt allowance in	8.00	in	decided
saw fuel tank l	0.28	L	estimated
saw fuel tank high l	0.37	L	estimated
brief strut value usd	10.00	usd	estimated

Everything else in this book is derived from geometry. These 14 could not be, so they are declared here, with what kind of number each one is. 'measured' was read off a real object; 'priced' came from a shelf; 'decided' is a build choice you may make differently; 'estimated' is a guess, and is labelled so.

choice you may make differently, "estimated" is a guess, and is labelled so you can discount it.

Declared inputs.

The inputs the whole book is built on, repeated here for reference. These are the numbers that are *declared* rather than derived – the gasket thickness, the handling limit, the prices and rates that could not be computed from geometry because they are facts about the world. Every one of them is stated with its reason where it is used, and this table collects them so a reader can change one and see what moves. That is the promise the front of this book makes and the arithmetic of this book keeps: no number is typed into a sentence. Change a constant here and every figure that depends on it – in the chapters, in the films, in the 3-D model – changes with it.

From Number to Pixel

How a strut length becomes a picture: what the films do thirty times a second, and why each step is the only one available

The films that come with this project are not recordings of a building. They are drawn: the dome on screen is the same solved model the tables in this book are computed from, turned into triangles by the same programs and photographed by a camera that exists only as three matrices. That is worth a chapter of its own, for two reasons.

The first is that a reader who can see how the picture is made can check it. When a film says a member is 72 inches, the model on screen is that member; when it says the seam is 27 degrees, the picture is the geometry that says so. Nothing in a film is an artist's impression, because there is no artist.

The second reason is more useful to a builder than it looks. Every step below is a version of a step this book has already taught you. A strut is a line with no thickness that has to be given a body — the same problem as a member that has to be given a joint. A surface has to be told which way it faces, or it disappears — the same problem as a pinwheel that has to run the same way round every panel. A frame that has to be rebuilt from scratch every frame, with nothing carried over, is the same discipline as a build whose every measurement comes off the same model. This is the method, applied to pictures.

A strut is not a line

A graphics card draws triangles and nothing else. It cannot draw a line with thickness, and a member in a dome is exactly that: two points with no width between them. So the line has to be given a body.

Take the strut's two ends, `a` and `b`. Its direction is the difference between them, normalised. Now pick any direction that is not parallel to the strut, and take two cross products:

```
d = (b - a) / |b - a|        the strut's direction
s = d x t                    a direction at right angles to d
```

```
u = d x s                    and a third at right angles to both
```

The cross product of two directions returns a third at right angles to both, which is exactly what a ring needs: `s` and `u` span the plane the tube's cross-section lives in. Walk a ring of 8 points around the axis at each end, stitch the two rings together, and the line has become a tube.

One strut is 28 triangles: 8 side quads, each split into two triangles, plus two end caps. The hubbed frame on screen is 65 strut tubes, which is 1,820 triangles of frame. Every visible member in every film in this project is a tube of 8 flat sides built by two cross products, and it is flat-sided rather than round on purpose: 8 sides is enough to read as a round member at film resolution, and the lighting on a faceted tube is what makes the member look like timber rather than plastic.

Which way round the ring is wound decides which way each triangle faces, and that is the next problem.

Which way a triangle faces

A triangle has a front and a back, and the difference is the order its corners are given in. Take the two edges that meet at the first corner, cross one with the other, and you get a vector at right angles to the face: the *normal*. Its length, halved, is the triangle's area. Its direction says which side is front.

This matters because a graphics card is allowed to throw away any triangle facing away from the camera — that is how it draws a closed surface at half the cost — and it decides by watching the winding order after projection. Get the winding wrong on a panel and the panel either vanishes or turns the dome inside out. Get it wrong on one panel of forty and you have a hole in the building that only shows up from certain angles.

So every face in the model is written with its corners in the same order, and the films' own model checks the normals against the outward direction: for the reference triangle the dot product of the normal with the outward direction is better than 0.99, which is the arithmetic way of saying "this face is looking outwards". That is the same idea as the pinwheel chapter's rule that every panel runs the same way round: a building whose members disagree about which way is out is a building that will not close, and a model whose triangles disagree is a picture with holes in it.

One frame, counted

By this point the model is not points and edges. It is one flat list of numbers, and there are ten of them for every corner:

```
position  x y z     where the corner is
normal    x y z     which way its surface faces
colour    r g b a   what shade it is; a is opacity
```

Ten floats at four bytes each is 40 bytes a vertex. The dome standing in the films is 117,000 floats, which is 11,700 vertices, which is 3,900 triangles, which is 468,000 bytes — 457.0 kilobytes. Smaller than one photograph.

No corner is shared between triangles: three fresh vertices per face. That costs memory and buys the freedom to give every face its own flat normal and its own colour, which is why the members in the films are flat-shaded and why each bay can be lit differently. And that whole buffer is rebuilt and re-uploaded 30 times a second, because nothing is stored

between frames: every scene is a pure function of its chapter and its progress, so the same second of film renders identically on any machine, every time.

One frame, counted

what	how much	where it comes from
Frame	1,920 x 1,080	30 fps
Pixels a frame	2,073,600	fragment shader runs, worst case
Floats in the buffer	117,000	10 per vertex
Vertices	11,700	three per triangle
Triangles in the model	3,900	no corner shared
Bytes a vertex	40	10 floats x 4 bytes
Bytes in the buffer	468,000	457.0 KB
Sides on a strut tube	8	8 flat sides
Triangles a strut	28	2 x 8 sides + two caps
Triangles of frame	1,820	65 strut tubes
Budget a frame	33.33 ms	at 30 fps
Budget a triangle	8.547 us	if all were visible
Shader runs a second	62.2 M	full coverage
Lens	48 degrees	vertical field of view
Near and far planes	0.08 / 120	world units
Specular exponent	42	rim exponent 2.5
Ambient floor	0.3	before any light lands

Every number here is counted off the buffer the renderer actually builds, not quoted from a specification: the vertex list is the film's own mesh, and the tube count follows from the ring the renderer walks. A reader who changes the model changes this table.

model changes this table.

One frame, counted off the renderer's own buffer.

The camera is a matrix

There is no camera on a graphics card. There is a fixed eye at the origin looking down one axis, and everything the card draws is drawn there. A "camera" is therefore an illusion maintained by two matrices that move the *world* instead of the eye, and a third that turns depth into size.

Placing the eye. 48 degrees of vertical field of view, a yaw, a pitch and a distance:

```
eye = target + d (cos p cos y, cos p sin y, sin p)
```

The view matrix builds an orthonormal frame from that eye — forward, right and up — and moves the world so that the eye sits at the origin looking down −z. Three rows of three numbers and a translation, and the camera exists.

The projection matrix arranges the divide that makes distance real. For a vertical field of view of 48 degrees:

```
f = 1 / tan(fov / 2)
```

and the matrix carries `f`, the aspect ratio, and the near and far planes — 0.08 and 120 world units in these films. Its last row is the important one: it copies −z into `w`. Nothing has been divided yet. The matrix has only arranged for the division to happen, which is the single cleverest thing in the whole pipeline.

The divide is where perspective comes from. Divide x, y and z by w, and w *is* depth, so anything twice as far away comes out half as big. What falls out is normalised device coordinates in a −1..+1 cube, and anything outside the cube is not drawn — a frustum, a pyramid with its tip cut off, which is what "the camera can see" means.

Landing on a pixel. Add one, halve, multiply by the frame's width; and for the vertical axis, subtract from one first, because screens count rows downward:

```
px = (ndc_x * 0.5 + 0.5) * width
py = (1 - (ndc_y * 0.5 + 0.5)) * height
```

The apex of the reference dome lands at a specific pixel — not approximately, exactly — and the films' own self-test checks that number.

What hides what, and why that is a buffer

A building drawn as triangles has a front and a back, and the triangles arrive in whatever order the model lists them. Something has to decide which surface is in front of which.

The films do not sort anything. Every pixel remembers how deep the nearest thing drawn there was, and a new fragment is kept only if it is nearer. The result is order-independent — the picture is right no matter what order the triangles arrive in — and it costs one number per pixel.

Two consequences worth knowing, because both are visible in the films if you look for them. **Z-fighting**: the divide by w squashes depth non-linearly, so almost all of the precision in a 0.08-to-120 range is spent on the first fraction of a unit, and two far surfaces can round to the same depth and flicker against each other. Pushing the near plane further out is the fix. And **transparency**, which needs the order back: a transparent surface has to be mixed with what is behind it, so the films draw the solid building first, then the transparent parts with depth-writing turned off, then swap the buffers.

Why the lighting is three dot products

A frame has 2,073,600 pixels and 30 frames a second, which is up to 62.2 million shader runs a second. A frame's whole budget is 33.33 milliseconds, which is 8.547 microseconds for each of 3,900 triangles if every one were visible. There is no room for anything clever, so the shading is three dot products and an ambient floor:

Diffuse. `max(n · l, 0)`: how squarely the surface faces the light. A member's flat sawn face catches the light differently from its bark side, and that difference is what tells a reader which way a member is turned.

Specular. `max(n · h, 0)` raised to 42, with `h` the direction halfway between the light and the eye. A high exponent means a small, hard highlight — the shine that makes a cut face read as cut.

Rim. `(1 - max(n · v, 0))` raised to 2.5: bright at the silhouette, which separates one member from the one behind it without any outline drawing at all.

Plus an ambient floor of 0.3 so nothing is ever pure black. No shadows, no bounced light, no ray tracing. Every one of those would be better and none of them fits in 33.33 milliseconds at 30 frames a second, which is the sort of trade this book makes about wood as well: the simplest thing that is honest about the shape.

What this has to do with building

Everything in this section is a solution to the same class of problem the rest of the book solves in timber.

A member with no thickness needs a body — and so does a joint with no connector. A face has to declare which way is out — and so does a wedge in a pinwheel. A depth buffer makes order irrelevant so that a device with no patience can draw a building correctly — and so does a jig, which makes the order you cut things in stop mattering. Thirty frames a second with nothing carried over is a pure function of the model, which is the same guarantee the book's tables give: change the tree, re-run the build, get *your* numbers rather than the author's.

The films are not decoration on the method. They are the method, rendered — which is why this book's figures could all be regenerated by a reader with the software chapter's commands, and why a film's number and a table's number are the same number.

The Composer

The tool that furnishes a dome: a placement rule rather than a drawing, a cast that obeys its own rules, and what the shell refuses

The chapters on the round room described what it is like to live inside a dome: headroom falls as you walk out from the middle, so what fits where is a function of where it stands. The project has a program that does that reasoning, and it furnishes the building piece by piece.

This section is about how it decides. It is in the back matter rather than in the book because it is machinery rather than method — but it belongs in the book at all because it is the one place where the shape's real cost to a furnished home is stated as arithmetic instead of as an impression.

A placement rule, not a drawing

Nothing in the catalogue carries a list of legal places to stand. A sofa does not know that it fits here and not there, and no file in the project says where a bookshelf is allowed. Instead, every piece carries a footprint and a height, and the program checks both against the shell before it will place it:

Walk the piece's footprint outward from where it wants to stand, and find the height of the shell above every point of it. If any part of the piece is taller than the shell at that point, it does not fit.

The piece being carried is drawn in one colour where it fits and another where it does not, and the colour is a *measurement*, not a warning animation. A reader who has read the round-room chapter has already done this by eye at a window; the composer does it for forty pieces of furniture at once, and the answer is always the same answer geometry would give: the closer to the wall, the lower the ceiling, and therefore the shorter the thing that can stand there.

Two consequences that are worth carrying back into the building:

A fitting pod stops the editor. A two point two metre fitting pod — a shower, a wardrobe, a changing room — has nearly eight metres of headroom over it at the middle of a thirteen-metre dome, and walked outward the shell comes down to meet it. There is a line on the floor past which the pod cannot go, and the program finds where it is rather than approximating it. That is the round room's real argument in one tool: the perimeter of a dome is not small floor, it is *low* floor.

Nothing is placed twice. Each placed piece is remembered by what it is and where it stands, so a composition can be walked, turned, dropped, re-walked, and the whole arrangement remains a list of named objects on a solved floor — which is also how the plans in the round-room chapter are drawn.

The cast, and the rules it obeys

The people in the films are drawn by the same program, and they arrive with rules.

There are eight of them, each with her own skin tone, hair colour, hair style and stance, and the loader refuses a cast in which two share a tone or a hair style: the code raises rather than quietly drawing a room of near-identical people. That is a small rule with a large effect on what the films look like, and it is the sort of rule this project has applied elsewhere for

the same reason — the fixture that refuses a wrongly turned wedge, the marking scheme that makes a backwards panel announce itself. A rule enforced by the tool is worth a rule remembered by a person.

Hair is geometry rather than colour. Each strand leaves the scalp along its growth direction, hands over to gravity about a head's radius later, and carries a wave or a helix by style — which means a haircut is a set of parameters rather than a texture, and the same head can be drawn wet, tied back or cut short without redrawing anything. In a project whose whole subject is members that are given their shape by geometry, a haircut is not an odd thing to find in the back matter.

Landmarks, not measurements

Clothing in the composer is a list, not a model. An outfit is a bodice, a skirt, a sleeve, a heel — each piece lofted through the *body's* own cross-section, so a garment cannot come out the wrong size for the person wearing it: it is generated from her, in her dimensions.

And the lengths are landmarks rather than numbers. Hair is "mid-thigh", "knee", "waist", "floor". A hem is "above the knee". Because those are places on a body rather than measurements in inches, one wardrobe serves a cast of any height, and a garment does what a real one does: it falls to a place rather than to a figure.

There is a fifth mode with no garment at all: the silhouette in skin tone only, like a shop's dress form. It exists to show what the clothes are doing — a lofted ellipse carries no anatomy, so the form mode is the honest answer to "what does this outfit actually add". Every catalogue in this book has a bare version for the same reason. The stem cell is the form mode of a building.

What the dome refuses

The set the composer works in is a thirteen-metre dome, warmer and lower than the reference articles, furnished rather than fitted out: a sofa whose seat height comes from the seated pose in the same way a garment's hem comes from a landmark, and a stove whose flue wants the crown, because that is where a dome's heat already is.

Walking that plan outward is an education. The middle of a dome is a room with more headroom than any house — and the edge of it is a place where a person cannot stand up, all the way around, in a ring whose width does not change with the size of the building. That ring is the cheapest floor in the building and it is also unusable floor until something is done about it, which is what the chapter on the pony wall priced: lift the shell on a short wall and the dead ring becomes furniture height, storage height, worktop height.

The composer and the pony wall are the same conclusion reached from two directions. One is a rule that refuses to place a wardrobe, the other is a course of studs that stops the refusal being necessary. A building this shape asks that question once, in every room, and the answer is always the same three options: accept the low ring, lift the shell, or put the bed in the middle and the storage round the edge.

Which is what the composer is for. Not to draw a room, but to make the building's own geometry answer for what can be put in it.

Glossary

The words this book uses in the senses it uses them, for the terms a reader may not have met.

Board foot. A volume of wood: one foot by one foot by one inch. The unit mills sell by.

Bucking. Cutting a felled trunk into sections of working length, cross grain.

Butt cut. The one compound-angled cut in the frame, made on the butt end of a member so it seats flat on the side of its neighbour. Once per member, off the jig, at a handful of fixed settings.

Channel. The void inside a seam, run as a connected network of gutters, vents and service chases through the whole shell.

Chord. The straight line between two points on the circle; the edge a member spans. A pinwheel member is always shorter than its chord, because it is inset from the vertices it spans.

Declared constant. A number the book states as an input rather than derives: a price, a rate, a handling limit. Every declared constant is named with its reason where it is used, and collected in the master tables.

Dihedral angle. The angle between two faces meeting at an edge; in this frame, the fold between two panels at a seam.

Frequency. How many times a geodesic edge is subdivided: 1V, 2V, 3V. Higher frequency means more, shorter members for the same dome. The flat parts list holds across diameter, not across frequency.

Gasket. The compressible strip in every seam – 0.75 inches – that lets a green-timber frame close and stay closed as the wood moves.

Hat. The upper part of the shell, the twenty upper triangles, whose edge is the brim. The hat's overhang throws water clear and collects it.

Hub. A junction where several members meet through one piece of hardware. The wedge frame is hubless: members lap past the vertices instead of butting into a hub.

Iris. The ring of hinged blades that closes one oversized pad down to whatever dome stands on it.

Jig. A purpose-built guide that holds a tool or a workpiece at one setting, so a cut or a fold repeats exactly. The rule of the whole build: the angle lives in the tool, not in the operator.

Kerf. The width of the cut a saw removes.

Key. The strip of timber or compressible material seated in a seam between two panels, cut to the seam's fold angle. The key locates the geometry; the gasket beside it seals the gap.

Limbing. Cutting the branches off a felled tree.

Member. One stick of the frame, cut and dressed for its place. The frame has 120 of them.

Pad. A serviced deck of a given diameter – deck, power, water, drainage, anchorage – the unit a dome site is built from.

Pinwheel. The frame's corner pattern: members lap past the vertices rather than butting into them, so no two members fight for the same end point. The inset the lapping requires is why a member is shorter than its chord.

Pith. The soft core of a trunk at the tree's centre. Splitting runs with the pith, and a wedge's point is the pith side.

Pony wall. A short straight wall stood under the rim of a dome, lifting the shell and buying back the low-ceiling ring of floor.

Rip. To cut along the grain, the length of the member.

Runoff coefficient. The share of rain that sheds off a roof rather than evaporating or splashing: 85 per cent for a metal roof.

Seam. The joint between two adjacent panels; a fold angle plus a key.

Sector. One wedge split out of a round log section, pith to bark.

Setback. The inset of a pinwheel member from the vertex it laps past.

Shell. The enclosed skin of the dome, frame and covering together.

Shell ladder. The sequence of quilted insulation layers hung under the shell, one winter at a time.

Strand. The reading track a chapter belongs to: story, howto, explain or reference. The contents page promises each track reads on its own.

Strut. A frame member loaded along its length – the load a triangulated frame puts on every stick in it.

Token. A figure in double braces in the manuscript that resolves, at build time, to a number computed by the software. The reason no number in this book was ever typed into a sentence.

Vertex. A corner point of the frame where members converge.

Wedge. A member split from a round log so its cross-section tapers, bark face wide, pith edge narrow. The wedge is the method: a structural section no mill sells, cut to be loaded along its length.

Wedge method. The whole system this book documents: split logs into wedges, lap them into a pinwheel frame with no hubs, and let the triangles do the structural work.

Worked example. A page that carries one number all the way through a calculation, showing every step rather than asserting the answer.

Boat hull. The laminated hard shell – fibreglass and resin over the frame – the fifty-year alternative to the soft cap, built in four detachable slices.

Dew point. The temperature at which air's moisture begins to condense. The book's climate rule: compare dew points, never relative humidities.

Desiccant drawer. A removable bed of drying beads at the rim, in a twin-drawer pair with a gate, so one bed dries while the other works.

Fit-out. One of the stem cell's fifteen named purposes – gym, guest house, workshop – carried by panels and services hung on the frame, never built into it.

Overfit. The deliberate extra length left on a member's head end so error leaves as offcut when the flush cut is made against the jig.

Peltier plate. A solid-state heat pump set in the seam channel, moving heat into the room in winter and out in summer, and condensing moisture deliberately at the dew point.

Rosette. The fitting where five or six seam channels meet at a vertex, joining them into one continuous network.

Seam channel. The connected void running through every seam: the building's gutter, vent, dehumidifier and air barrier in one.

Stem cell. The undifferentiated dome body – one frame, one cut list, one core – whose purpose is deferred to a swappable set of panels: fifteen buildings from one body.

Utility core. The column through the dome's centre carrying power, water, waste and air in one object that unbolts and moves to the next dome.

Value ladder. The six rungs on which one tree can be priced – standing, firewood, sawn, split, structure, financed – and the ratios between them.

Yakisugi. The char-and-oil finish for the bark face only, leaving the sawn faces bare so the seam channel can breathe.

Zome. A dome built by sweeping a star of directions into flat panels, so every panel is flat by construction rather than by approximation.

The Software in This Book

What generated these numbers

Every figure in this book comes from one of the programs below, and each program is included with the book. They are named here so a reader can find the exact line behind any number printed in these pages.

`geodesic_raw_wedge_dome_dihedral.py` – the live solver and 3-D world. This is the heart of the project: it solves the raw-wedge dome, its seams, its dihedral angles and its jig, and it is where the standing dome's own mesh lives. The book's figures of the frame are this program's mesh, not a sketch of it.

`two_v_demo/book_math.py` – the book's arithmetic. The two sizing methods, the tree model, the fortnight, the round trip and the declared constants. Where a chapter works a number through by hand, this module is the hand.

`two_v_demo/wedge_geometry.py` – the tree and pinwheel arithmetic the book and the films share: member classes, splits, yield, recovery.

`two_v_demo/book_tokens.py` – the 91-chapter book's living figures. Every figure a sentence quotes – a token in double braces – resolves here, from the modules above, at export time.

`two_v_demo/book.py` – the outline: the parts, the chapters, the pages and where each figure sits.

`two_v_demo/book_plots.py` and `two_v_demo/book_diagrams.py` – the charts and tables, and the line drawings. Both draw from the same geometry, never from a number typed beside the drawing.

`two_v_demo/book_figures.py` – the renderers, and the rule that a re-rendered figure gets a new name rather than replacing the old one.

`two_v_demo/book_manuscript.py` and `two_v_demo/book_export.py` – the manuscript itself, and the readable forms: this HTML, the PDF, and the Markdown source. The diagrams on this page are drawn by that exporter.

`two_v_demo/book_app.py` – Book Studio, the desk the book is written at, including the reader built into it.

Nothing in this list restates a figure another module already derives, and nothing draws a dome that the solver has not solved. That is what "the numbers are computed, not typed" means in practice: every one of them traces to a function in one of these files.

How to get it, and start it

The project is a repository, and this is the whole install path, from nothing:

```text

# 1. Python 3.10-3.13 (not 3.14 – pygame and moderngl have no wheels yet)

# https://www.python.org/downloads/

# 2. the code

git clone https://github.com/dkzeanah/DomeSimulator.git cd DomeSimulator
```

3. the three libraries the core tools need, and the front door

py -3.12 -m pip install pygame moderngl numpy py -3.12 launcher.py ```

That is enough to open every tool in the project, including this book: `launcher.py` is a window with a tab per tool and no command-line flags to remember. Two optional extras turn on the parts of the project that make films and narration:

```text

# video export (the films, the narration, the MP4s)

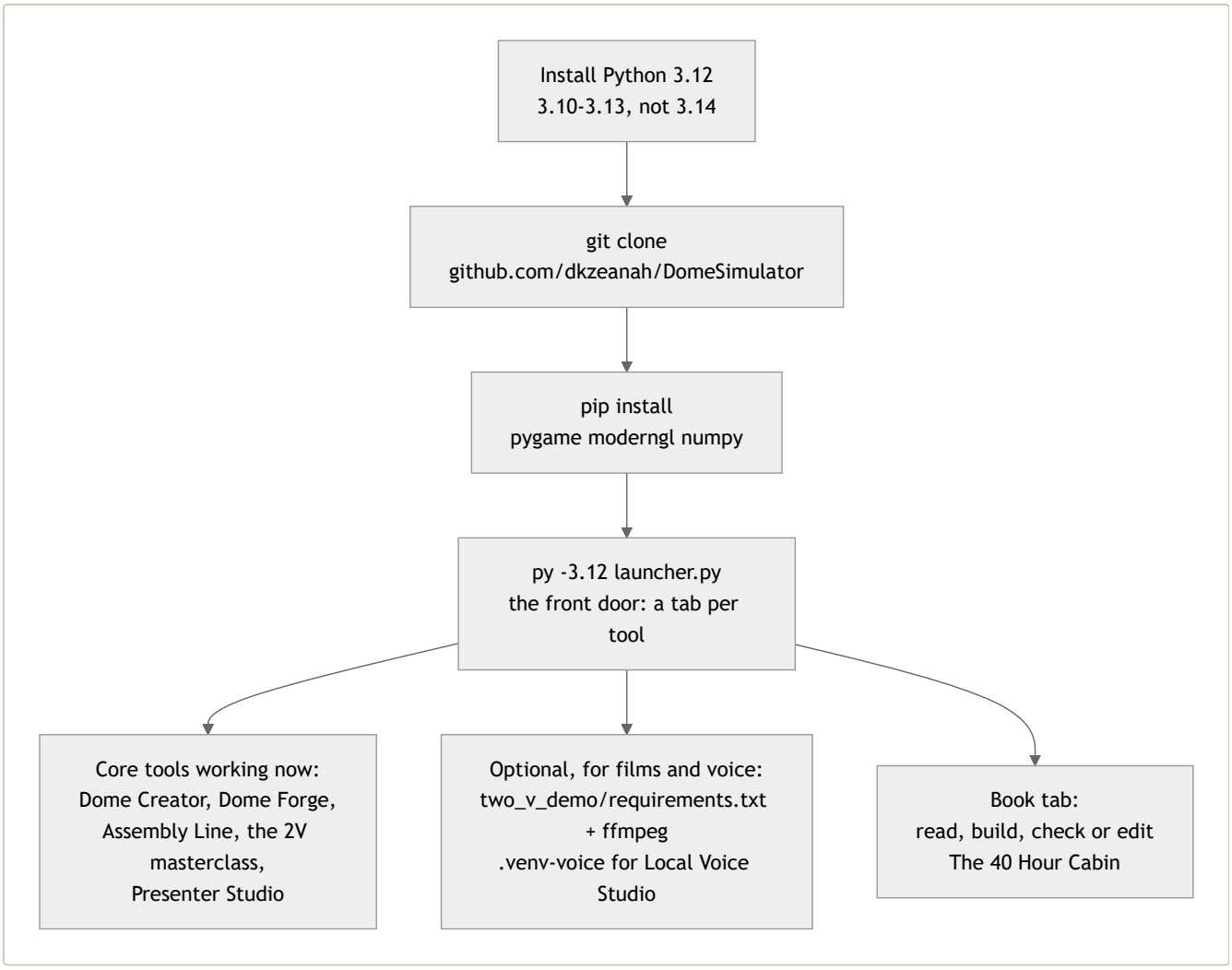
py -3.12 -m pip install -r two_v_demo/requirements.txt winget install ffmpeg # or brew/apt install ffmpeg ```

```text

# local voice cloning and transcription (heavier AI packages, Python 3.11)

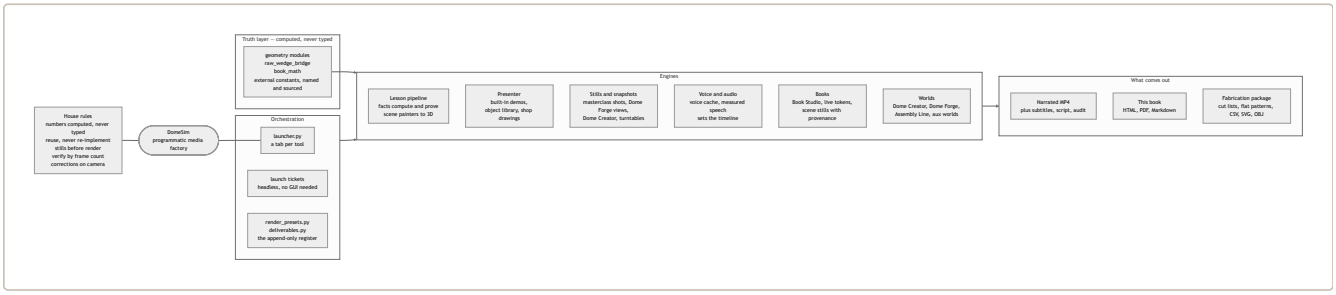
py -3.11 -m venv .venv-voice ..venv-voice\Scripts\python.exe -m pip install -r .\local_voice_studio\requirements-core.txt
```

The full setup guide, including what each optional tool needs and what it reports when something is missing, is `SETUP.md` beside the code.
```



The map of the code

One picture of the whole factory, because the project is larger than the book and the book is one of its products. Read it left to right: the truth layer at the bottom is what everything computes from, the middle is the engines that make things, and the right is what comes out.

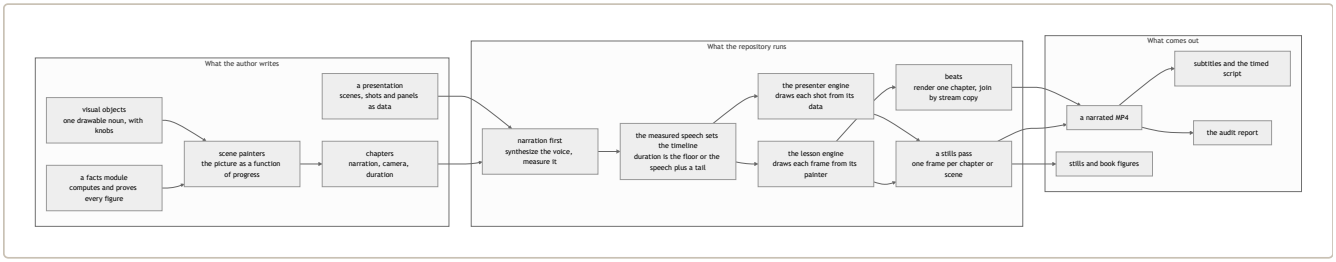


The house rules in that diagram are not decoration: they are the checks that run before a render or an export, and the reason a figure in this book can be traced back to a line of code. The truth layer is the same geometry the films draw,

which is why the book and the films cannot disagree about the dome.

The video and presenter pipelines

The factory makes two kinds of film, and the two engines are worth knowing before anything else on this page, because every rule below is the same rule twice: **a film is data plus painters, and nothing else.**



Lesson films (`two_v_demo/`) bind copy to painters. A `Lesson` carries no logic at all: it holds chapters, a dictionary of scene painters, a selftest and a report. A chapter carries its spoken narration, its fixed equation lines, a duration *floor*, a camera and the name of the painter that draws it. The painter is a pure function of progress – `def scene(app, opaque, transparent, p)` with `p` running 0 to 1 across the chapter – so the same frame is drawn at the same `p` on every machine, and a film can be re-rendered from its own source years later.

Presenter films (`presenter/`) are even more data: a `Presentation` holds scenes, a scene holds shots, and a shot holds a camera, a lens, a focus target, narration, a caption and an overlay panel. Animation is interpolation of one named parameter, and the camera follows any moving focus target for free. Both engines take their timing the same way – narration is synthesized and measured **first**, and the chapter or shot duration is the larger of its floor and the measured speech plus a tail – which is why a voice change is a timing change, and why the house voice is stated in the presets rather than left to whatever the tab was showing.

The bridges between them are three, and they are what make the two engines one factory:

- **Visual objects** (`two_v_demo/visual_objects.py`) are the elements either engine can draw by name. Written once, they are available to the lessons as `draw(stage, "key", ...)` and to the presenter as `vo:<key>` – see the next section.
- **Dome Forge layers** are available to the presenter as `forge:<layer>`, so a film about a water-harvesting dome shows the modelled dome rather than a picture of one.
- **The raw-wedge solver** reaches the lessons through `two_v_demo/raw_wedge_bridge.py`, which converts the solver's own meshes into lesson geometry. This is the bridge behind the rule at the top of this book: a film that talks about the wedge dome shows the simulator's dome.

Long films are cut into **beats** (`two_v_demo/beats.py`): a chapter, or a claim and the math screen that proves it, rendered as its own file and joined by stream copy. A beat that is already on disk is skipped, so fixing one chapter costs one beat re-render instead of a whole film, and `verify_renders.py` is the check that says whether what came out of the encoder is the film the timeline promised.

Adding to the pipeline

Four kinds of addition, and the same four-step shape every time: *write the thing, register it, check it with a still, then render it*. The table is the whole answer to “which files do I touch”:

you want to	you create	you alter	the check that proves it
a new element (a noun the renderer can draw)	`two_v_demo/visuals_<group>.py`	nothing, unless a film should use it	`validate_visual_objects()` then a still
a new scene and chapter in a film	–	`two_v_demo/lesson_<key>.py` (`SCENES` + chapters)	the lesson’s own `selftest`, then a still per chapter
a whole new film	`two_v_demo/lesson_<key>.py` and `<key>_facts.py` (make them with `scaffold_lesson.py`)	`two_v_demo/lesson_registry.py` , `render_presets.py` , `two_v_demo/deliverables.py`	`validate_render_presets()` then `verify_renders.py` on the MP4
a presenter object or demo	`presentations/<name>.py` , or an object in `presenter/library.py`	`presenter/library.py` for the catalogue	`PresenterApp(..., headless=True).screenshot(...)`

1. A new element the renderer can draw

This is the addition most people mean by “a new element”, and it is the best-supported one: write the noun once, and both pipelines can use it.

Where it goes. `two_v_demo/visual_objects.py` holds the contract and the registry. The objects themselves live in their own modules beside it – `two_v_demo/visuals_forest.py` is the worked example, holding the pine, the bucked log and the figures that go with them – and each module registers its objects on import, so nothing else needs editing. `registry()` imports the known groups and returns everything.

The contract, in the code’s own words:

```python VisualObject( key=”log_sections”, # the name a film draws by label=”Log, bucked and split”, # the name a person reads category=”wood”, # must be a taxonomy key blurb=”The usable trunk on the ground, cut into the plan’s ” “sections and every section split into the plan’s sectors…”, knobs=(…), # the dials, with ranges draw=_draw_log_sections, # drawer(stage, knobs) -> anchors source=”new”, # or the dotted path it wraps reach=18.0, # how far it extends, for cameras words=(“tree”, “log”, “section”, “sector”, “split”), )

register(VisualObject(…)) # refuses a duplicate key ```
```

A **knob** is one dial, declared with its range so it can never be set to something undrawable:

```
python _k("fell", "Felled", 1.0, 0.0, 1.0, "x", "0 standing, 1 on the ground", animate=True)
```

The drawer's signature is `draw(stage, knobs) -> anchors`: it builds triangles on the stage and returns named points – an anchor called `top`, say – which the caller can aim a camera at or hang a label on. A knob marked `animate=True` is one the caller may sweep across a chapter, which is how an element performs rather than just appears. `category` must be one of the taxonomy's keys, because the catalogue sorts the nouns by it: `structure, members, wood, envelope, base, geometry, math, measure, tools, processes, forces, money, buildings, land, services, materials, people, body, time, media, computing, ideas, quantities, story, names`.

What to draw with. Inside the drawer the vocabulary is the engine's own: `TriangleBatch` primitives from `two_v_demo/render_kit.py` (`cylinder` – where `cylinder(a, b, r, colour, 3)` is a wedge and `4` is square, `box`, `sphere`, `triangle`, `cone`, `arrow`, `disc`), procedural wood from `two_v_demo/timber.py` (`draw_timber`, `wood_colour`, `draw_glue`, `draw_patch`), pictograms from `two_v_demo/icons.py` (`draw_icon`), and the easing helpers in `render_kit` (`clamp`, `smoothstep`, `ease_in_out`) so movement in a frame has the same feel as movement in every other film. Figures that must be *spoken* rather than drawn belong in the chapter's callouts, not in the element: `two_v_demo/callouts.py` times them to the measured voice.

How a film calls it. Two lines, in the scene painter:

```
python def scene_hv_fell(app, opaque, transparent, p: float) -> None: stage = stage_for(app, opaque, transparent, origin=ORIGIN) anchors = draw(stage, "pine_tree", fell=p, units_per_ft=UPF)
```

`stage_for` puts the element in the world at an origin; `draw` resolves the knobs – defaults, overridden by what you gave, clamped to what is drawable, and a misspelt knob is a hard error rather than a silent nothing – and returns the anchors.

It reaches the presenter for free. `presenter_emitters()` wraps every registered object as `vo:<key>` with five placement knobs (across, along, up, heading, size), and `presenter_object_specs()` lists the same objects in the presenter library's own terms. So a new element is usable in a presenter shot the moment it exists, under a prefixed name that cannot collide with the presenter's own objects.

The check. One command, then one still:

```
text py -3.12 -c "from two_v_demo.visual_objects import validate_visual_objects; validate_visual_objects()"
```

which draws every object at its defaults and at both ends of every knob – a cheap way to find the element that collapses at its own limit, before a render finds it for you.

2. A new scene, and the chapter that shows it

A scene is a painter; a chapter is the copy, the camera and the clock. In the lesson module you are changing, three edits, in this order:

Write the painter. It takes the app, two triangle batches (opaque and transparent) and `p`, and it must be a pure function of `p` – no clocks, no randomness that is not seeded, no state kept between frames. Draw into the batches; put text on screen with the engine's label list; return nothing.

Register it in the lesson's `SCENES` dictionary under the name the chapter will use:

```
python SCENES = { "wg_open": scene_wg_open, "wg_split": scene_wg_split, ... }
```

Add the chapter, giving it that scene name, its narration, its camera and its duration floor:

```
python Chapter( "split", "07", "Split like a cake", "Three passes and the stick is already the right shape", ("Spoken narration line one.", "Spoken narration line two."), ("fixed equation line",), # only when the stage is a math screen 12.0, # duration FLOOR, not a promise (34.0, 24.0, 16.0), # camera: yaw, pitch, distance "wg_split", # the SCENES key that paints it None, # or "math" for a derivation screen callouts= (...), # figures cued to the measured voice )
```

Two things are worth stating because they are the whole reason this pipeline is reproducible. The duration is a **floor**: the real length is the measured speech plus a tail, so a chapter never races its own narrator. And every figure the chapter shows should come out of the lesson's **facts module** rather than the painter, because that module is what the selftest proves – which is why the generated pair from `scaffold_lesson.py` is a facts module *and* a lesson module, never just a lesson.

Look at a still before you render. A staging bug – a camera pointing at the back of an element, a label under the horizon – is invisible in code and obvious in one frame. The headless way, with no window open:

```
text py -3.12 -c "import launcher_common as lc; lc.write_config('two_v_masterclass', {'action':'shots','lesson':'wedge','shots':'8,24,40','no_narration':True})" py -3.12 two_v_masterclass.py
```

which writes `two_v_demo_output/<lesson>/<prefix>_<seconds>s.png` for each second you asked for.

3. A whole new film

Start with the generator, because the boilerplate it writes is boilerplate that already passes:

```
text py -3.12 scaffold_lesson.py trusses "Truss Masterclass"
```

That writes a facts module and a lesson module that render, prove themselves and pass a selftest, laid out along the right axis because this renderer's cameras sit on +Y. The first real work is deleting the placeholder facts and putting real ones in – which is the point.

Then register it in **three** places, because a film is a lesson, a preset and a deliverable, and the project refuses to let them disagree:

file	what you add
`two_v_demo/lesson_registry.py`	an import, and the lesson in the `LESSONS` tuple – this is what makes it launchable by key
`render_presets.py`	a preset built with the `_video(key, lesson, filename, summary)` helper, so the exact settings that produced the film are one click away
`two_v_demo/deliverables.py`	a `Deliverable(lesson, filename, note, compose=True)` – the rendered output and the lesson that reproduces it

`validate_render_presets()` is the gate, and its last assertion is the one that keeps the three in step: *deliverables with no render preset* is a failure, not a warning. The check runs both ways, which is the part people miss – `validate_deliverables()` also refuses a lesson that has been registered and never given a deliverable, so a new film cannot be half-registered and still pass a selftest. The preset's summary must be a real sentence – there is a length assertion on it, because a preset nobody can read is a preset nobody will use.

Then render, headless, and check what came out:

```text py -3.12 -c "import launcher_common as lc; lc.write_config('two_v_masterclass', {'action':'selftest','lesson':'trusses'})" py -3.12 two_v_masterclass.py

py -3.12 -c "import launcher_common as lc; lc.write_config('two_v_masterclass', {'action':'export_video','lesson':'trusses','export_video':'deliverables/masterclass/truss-masterclass.mp4','orientation':'both'})" py -3.12 two_v_masterclass.py

py -3.12 verify_renders.py deliverables/masterclass/truss-masterclass.mp4 ```

`verify_renders.py` fails when the video and audio disagree by two seconds or more, when the frame count is under about 29 frames a second, or when the duration is zero. That is the house check – a film is not finished because the encoder exited cleanly, it is finished because the frame count agrees with the timeline. The export also writes the `.srt`, the timed script and the release folder (landscape cut, phone cut, thumbnails and platform copy), because one render is a release rather than a file.

### 4. Adding to the presenter instead

The presenter is the faster path when the film is about *showing* rather than *deriving*: its objects are already placed and lit, and its shots are data.

**A new object** is an `ObjectSpec` in `presenter/library.py` – a key, a label, a category, a blurb and a tuple of `ParamSpec`s (the same parameter description Dome Forge uses, imported from `dome_forge.layers`, so one vocabulary serves both). Before writing one, check the bridge: if the element exists in `two_v_demo/visual_objects.py`, the presenter already has it as `vo:&lt;key&gt;`, and you do not need to write anything.

**A new demo** is a module in `presentations/` – `dome_accessibility.py` and its neighbours are the thirteen worked examples – built out of `Presentation`, `Scene`, `Shot` and `OverlayPanel` from `presenter/script.py`. Keep the numbers in `presentations/_numbers.py`, which computes them once from `al_build` and `two_v_demo.geometry`, so a presenter film and a lesson film quote the same arithmetic.

**The check** is a screenshot, not a render:

`text py -3.12 -c "import launcher_common as lc; lc.write_config('presenter', {'action':'shots','demo':'dome_accessibility','shots':'4,20,38'})" py -3.12 presenter_studio.py`

or, programmatically, `presenter.engine.PresenterApp(presentation, headless=True)` and then `.render(t, present=False)` followed by `.screenshot(path)` – a frame for the price of a frame, which is what you want while placing an object. When the film is ready, the same class exports it:
```

```
text py -3.12 -c "import launcher_common as lc; lc.write_config('presenter', {'action':'export','demo':'dome_accessibility','export':'deliverables/presenter/accessibility.mp4'})" py -3.12 presenter_studio.py
```

`PresenterApp.export(path, fps=None, narration=True, ffmpeg=None, overlay=None)` is the whole film export: narration measured first, frames rendered deterministically, audio muxed by stream copy. The `overlay` option is the presenter's version of the lesson styles – `full`, `no_captions`, `titles_only` or `clean`.

The studio itself (`presenter/studio.py`) is a real editing suite – library, viewport, timeline, inspector – over the same pure-functional edit API, and it saves and reopens a film as JSON, so a presentation can be drafted by a tool and finished by hand.

Adding an element to a world instead

The visual-object route is the right one for an element a *film* needs. An element that only makes sense inside a world – a forge layer, a jig stage, an assembly-line station, a piece of furniture – is added where that world keeps its registry, and the world's own viewer is how you look at it. The table is the whole map:

world	the element	where it is declared	what else changes
Dome Forge	a **layer** (anything from the frame to the rain)	`emit_<kind>(...)` in `dome_forge/build.py`, then the `EMITTERS` dictionary	a `LayerKind` with its `ParamSpec`s in `dome_forge/layers.py` – and it then appears in the presenter as `forge:<key>` automatically
Dome Forge	a **jig stage**	`STAGE_ORDER` in `dome_forge/jigs.py`	the drawing inside `emit_jig`, gated on the stage's position in that order
Dome Forge	a **camera view** for the stills pass	the views list in `dome_forge/cli.py`	nothing
Assembly Line	a **station**	a `StageDef` in `_STAGE_LIST` (`al_build.py`)	a key in `HOME_STAGE_KEYS`, an entry in `CATEGORY_ECON` and `CATEGORY_LABEL`, and an `_emit_<stage>` called from `build_dome_catalog`
Assembly Line	a **part**	`cat.begin("<category>")` / `cat.end("<stage>")` inside the relevant `_emit_*` (`al_build.py`)	the `CATEGORY_ECON` triple – material cost, labour minutes, weight per element
Dome Composer	a **prop or piece of furniture**	`dome_interiors.py`: a `_prop(...)` plus its builder in the `BUILDERS` table	a placement rule only if it needs one
Dome Composer	a **placement rule**	a branch in `check(...)` and its constant, in `two_v_demo/scene_composer.py`	an assertion in `validate_scene_composer()` so the rule is proved
Dome Creator	a **preset building**	`(name, config_dict)` appended to `PRESETS` in `presets.py`	nothing – it registers itself in the menus
Dome Creator	a **model option**	a field on `DomeConfig` in `dome_model.py`	its `to_dict` / `from_dict`, the menu entry, and `build_dome_mesh` if it changes geometry

Two of those rows are worth the extra sentence. An Assembly Line part is declared between `begin` and `end`, and `end` sizes its cost, labour and weight from the category's per-element figures unless you override them – which is why the parts catalogue and the economics cannot drift apart. And a Dome Forge layer is the cheapest way to get a real, modelled *thing* into a presenter film: write the layer once and it is both an option in the forge's own viewer and an object in every presenter shot.

The other worlds, and their stills

The remaining three worlds are where an element is checked in three dimensions before it ever appears in a film, and each answers the same ticket:

```text
py -3.12 -c "import launcher_common as lc; lc.write_config('dome_forge', {'action':'shots'})"
py -3.12 dome_forge.py
```

py -3.12 -c “import launcher_common as lc; lc.write_config(‘assembly_line’, {‘action’:’shots’,’shots’:‘74,97’,’shot_speed’:6})” py -3.12 assembly_line.py

py -3.12 -c “import launcher_common as lc; lc.write_config(‘dome_composer’, {‘action’:’shots’})” py -3.12 dome_composer.py ```

Dome Forge writes exterior, cutaway, top and ground views to `dome_forge_shots/`; the Assembly Line writes `shots/shot_<t>s.png`; the Composer writes a four-frame turntable of the scene as PNGs. A fifth world, the Dome Creator, has no stills action on purpose – its check is the smoke test (`DOME_SMOKE` for a frame count, `DOME_SHOT_DIR` to ask for screenshots along the way), because the thing being checked there is the world itself rather than a frame of it. That smoke test still opens a real graphics context, so it is a check to run while watching, not a background job.

Every ticket above lands in `.launcher_configs/` and is read, then archived, when the tool starts: the tool is launched with no arguments, which is also how the launcher itself launches it. The archive is the point – the exact input a run used stays readable in `.launcher_configs/used/`, so a film can be reproduced from what was actually asked for rather than from what somebody remembers asking for. The last check of all is the repository’s own: `LAUNCHER_SMOKETEST=1 py -3.12 launcher.py` builds every tab, every launch button and every render preset without opening a window, and it is an environment variable rather than a flag.

The commands, in order

A new element, from nothing to used in a film, in the order the project expects:

```text

# 1. write the object and register it, then prove every knob draws

py -3.12 -c “from two_v_demo.visual_objects import validate_visual_objects; validate_visual_objects()”

# 2. put it in a scene painter, then look at a still before rendering anything

py -3.12 -c “import launcher_common as lc; lc.write_config(‘two_v_masterclass’, {‘action’:’shots’,’lesson’:’wedge’,’shots’:‘8’,’no_narration’:True})” py -3.12 two_v_masterclass.py
```

3. run the lesson’s selftest: the numbers before the pixels

py -3.12 -c “import launcher_common as lc; lc.write_config(‘two_v_masterclass’, {‘action’:’selftest’,’lesson’:’wedge’})” py -3.12 two_v_masterclass.py

4. render, then verify by frame count

py -3.12 -c “import launcher_common as lc; lc.write_config(‘two_v_masterclass’, {‘action’:’export_video’,’lesson’:’wedge’,’export_video’:’deliverables/masterclass/wedge-v2.mp4’,’orientation’:’both’})” py -3.12 two_v_masterclass.py py -3.12 verify_renders.py deliverables/masterclass/wedge-v2.mp4

5. register the film so it stays reproducible, and run the repository’s own gate

py -3.12 -c “import render_presets as rp; rp.validate_render_presets()” LAUNCHER_SMOKETEST=1 py -3.12 launcher.py ```

The rules a new element has to obey

Four, and they are the book’s rules applied to code:

Its numbers are computed, or declared. An element that shows a figure takes it from a facts module; if the figure cannot be derived, it is a declared constant with a stated reason. No number is ever typed into a painter, a caption or a knob’s default.

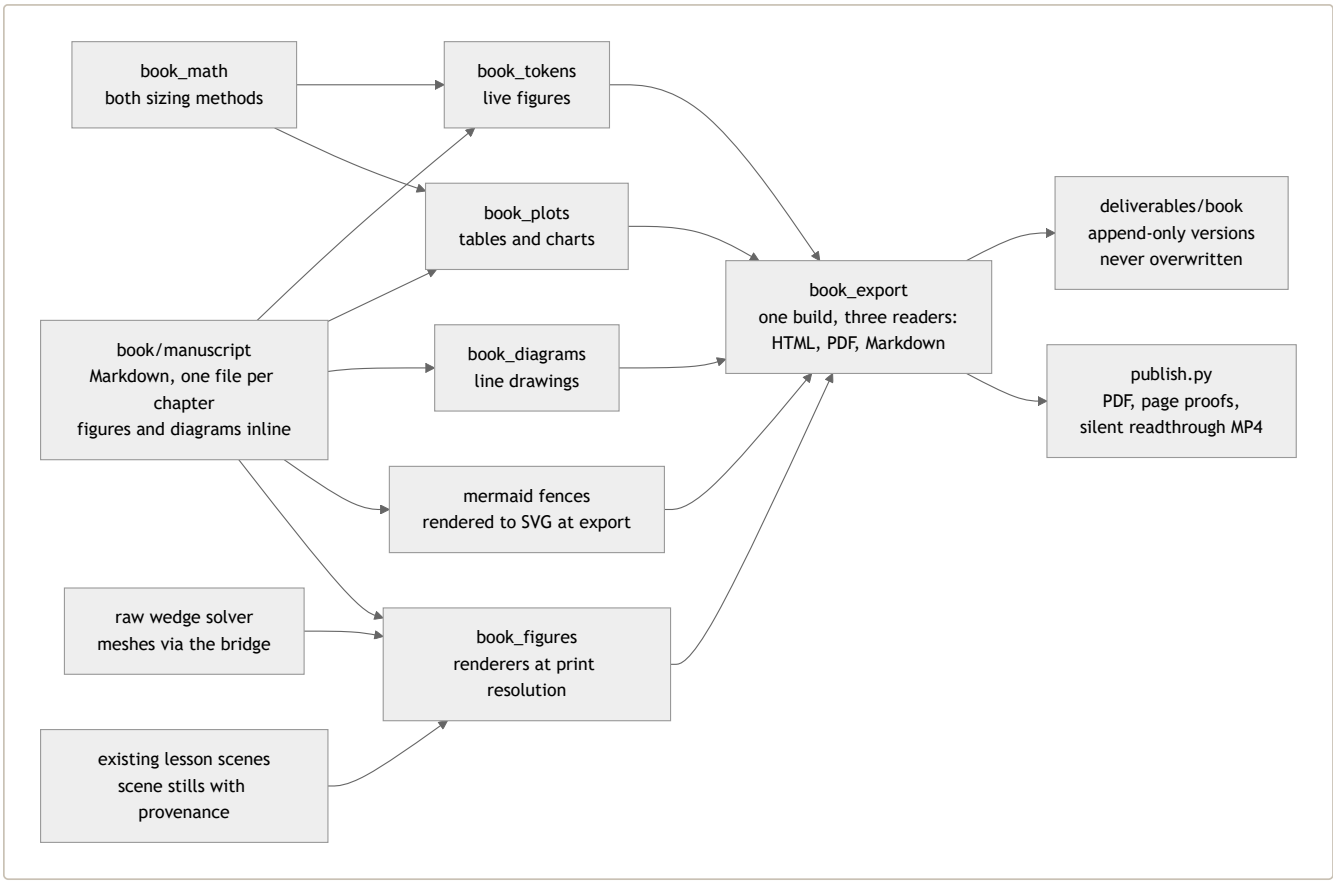
It reuses the geometry. An element that depicts the dome asks `raw_wedge_bridge` for the solver’s own mesh. An element that depicts wood uses `timber.py`. A second drawing of the same thing is a second version of the truth.

It is proved before it is rendered. `validate_visual_objects` for an element, the lesson’s `selftest` for a chapter, `verify_renders.py` for a film. The order matters: the cheapest check that can fail is the one to run first.

It never overwrites. A re-render writes `-v2`. An element that changes what a published film looks like is a new film, not a quieter edit of the old one – which is the same rule the book keeps about its own chapters.

The chain that made this book

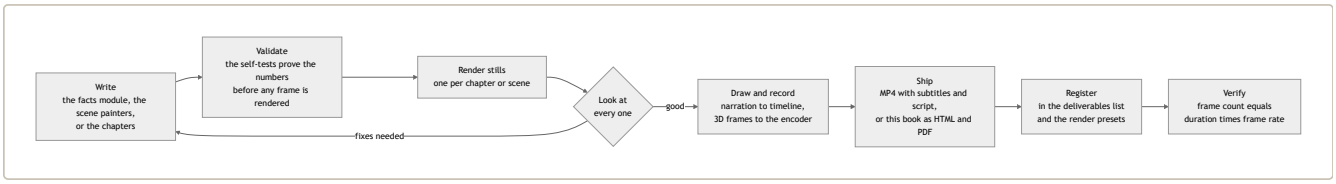
This book is one of the factory’s outputs, and its own chain is worth drawing because it is the shortest way to see how the “no number is typed” rule is enforced:



Read the arrows into `book_export` as the gate: that is where the tokens are resolved, the figures are found, and the book either builds or fails. A token that does not resolve is a hard error, not a blank, which is the reason a typo in a sentence cannot reach a reader.

The loop an agent runs

The project is written so that a person or an agent does the same thing, in the same order, and can check every step:



The loop is the method this book teaches, applied to software: measure, build, look at the result, and write down what went wrong. The render presets and the deliverables register exist so that the last two boxes are not optional — an unregistered or unverified artifact does not pass as finished in this repository.

Reproducing every figure in this book

The whole book rebuilds from these commands, run from the project's folder with Python 3.12:

```
text py -3.12 -c "from two_v_demo.book import validate_everything; validate_everything()"
```

checks the arithmetic, the outline, every token, every figure and the manuscript machinery, and reports what passed. It is the gate the book must pass before it is exported.

```
text py -3.12 launcher.py
```

opens the project's launcher. Its **Book** tab carries the actions: `read_html` and `read_pdf` build the book you are reading; `export` writes the Markdown source; `audit` prints every number the book states and the calculation behind it; `progress` shows what is written; `render_figures` redraws every illustration. Each build writes a new, versioned file – nothing in this project is ever overwritten by a rebuild, so the copy you have and the copy a build makes cannot be the same file by accident.

The 3-D figures are rendered by the solver's own renderers; the film-still figures come from the project's films through the launcher's Masterclass tab; the diagrams on this page are drawn from the manuscript's own mermaid source at export time, by the mermaid CLI the repository already carries. The photographs that remain are the author's to take; until then the book simply leaves their pages out.

The ten commands worth knowing

Everything in the factory is reachable from the launcher, and everything the launcher does is reachable without it. These are the commands a person or an agent actually types:

```text

# the front door, and a new lesson to hang a film on (key, title)

py -3.12 launcher.py py -3.12 scaffold_lesson.py trusses “Truss Masterclass”

# headless: any launcher action, as a ticket, with no GUI open

py -3.12 -c “import launcher_common as lc; lc.write_config(‘two_v_masterclass’, {‘action’:’selftest’,’lesson’:’wedge’})” py -3.12 -c “import launcher_common as lc; lc.write_config(‘two_v_masterclass’, {‘action’:’shots’,’lesson’:’look’,’shots’:‘8,24,40’,’no_narration’:True})”
```

the book you are reading: check it, write it, rebuild it

py -3.12 -c “from two_v_demo.book import validate_everything; validate_everything()” py -3.12 book_studio.py py -3.12 -c “from two_v_demo import book_export; print(book_export.export_html().path)”

find the stills that match a passage, then draw one

py -3.12 two_trees_codex/scene_catalog.py –query “eight wedges from six six-foot trunk sections”

check a finished film before it is called finished

py -3.12 verify_renders.py deliverables/masterclass/whey-wedges-no-sawmill.mp4 ```

A ticket is the launch configuration without the window: it writes the same settings the GUI would, so a film or a book can be rebuilt from a script, a test or another tool. That is how this repository keeps its films and its book reproducible by somebody who never clicks anything.

The launcher

Every tool in this project opens from one window, `launcher.py`:

```
text py -3.12 launcher.py
```

Its tabs are the project’s own: the simulator, the lessons and their films, the dome park, the presenter, and the book this page belongs to. The book tab is the one to use for anything in this book: read it, build it, check it, or sit down and change it – the same files this book was written in are the files you would write in, plain Markdown in `book/manuscript/`, one file per chapter.

The launcher also runs the films’ smoketests, and the book’s, which is the last line of this section and the one worth repeating: a copy of this book exists only if every check passed.

Index and Colophon

Colophon

This book was not typeset from a manuscript that could drift from the things it describes. It was built, in one step, from the same software that draws the models and solves the geometry, and that is worth one page of explanation because it is the book’s one unusual promise.

Every figure in these pages is computed at the moment the book is built. When a chapter says the dome is 21.6 feet across, the number came out of the solver a second before it reached the page; when a table prints the fold angles of 91 chapters' worth of claims, each one is resolved against the geometry, and a misspelt figure stops the build instead of printing as a blank. The arithmetic is a test suite, and it runs before any copy of this book is allowed to exist.

The book itself is the same material in three forms: the HTML you can read in a browser and print from, the PDF typeset from that same HTML, and the Markdown source. All three are produced by one function, so the version you read and the version you send cannot be different books.

The tools that made it are named in the software section, and they are included with this book. Everything here can be regenerated, and the declared constants can be changed – a different tree, a different handling limit, a different rainfall – and the book rebuilt around them. That is what makes this a manual rather than a memoir of one build: the same geometry that produced this dome will produce yours.